100% Construction Documents

Project Manual
Volume 1 of 1

Jefferson County – Boettcher Mansion

Chiller Replacement

Spacek Timbie and Blevins Engineering Inc

September 22, 2026

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SECTION 00 0107
PROFESSIONAL SEALS

PROJECT

NAME: CHILLER REPLACEMENT
LOCATION: LOOKOUT MOUNTAIN, 900 COLOROW RD., GOLDEN, CO 80401

MECHANICAL ENGINEER OF RECORD

TODD SPACEK, PE
523 PARK POINT DR., SUITE 320
GOLDEN, COLORADO 80401
TSPACEK@STB.ENGINEERING



09/22/2026

MECHANICAL ENGINEER OF RECORD **DATE**

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09/22/2026

ELECTRICAL ENGINEER OF RECORD **DATE**

SECTION 01 1000
SUMMARY

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. Project information.
   2. Work covered by Contract Documents.
   3. Phased construction.
   4. Work by Owner.
   5. Access to site.
   6. Coordination with occupants.
   7. Work restrictions.
   8. Specification and Drawing conventions.
   9. Miscellaneous provisions.

B. Related Requirements:
   1. Section 015000 "Temporary Facilities and Controls" for limitations and procedures governing temporary use of Owner's facilities.

1.03 PROJECT INFORMATION

A. Project Identification: Jefferson County - Boettcher Mansion - Chiller Replacement
   1. Project Location: Lookout Mountain, 900 Colorow Rd, Golden, CO 80401

B. Engineer: Spacek Timbie and Blevins Engineering, Inc, 523 Park Point Drive, suite 320, Golden, Colorado 80401

1.04 WORK COVERED BY CONTRACT DOCUMENTS

A. The Work of the Project is defined by the Contract Documents and consists of the following:
   1. Work for this project includes demolition, materials, labor, transportation, security, temporary construction, and other items identified in or reasonably inferable from the Construction Drawings and Project Manual and other Work indicated in the Contract Documents.
   2. The Jefferson County - Boettcher Mansion - Chiller Replacement project includes replacement in kind of the existing air-cooled chiller and chilled water system pumps, including associated appurtenances and piping between the chiller and pumps. A new chilled water buffer tank shall be provided to increase system volume and reduce chiller short cycling during periods of low demand. The existing glycol feeder system, air separator, expansion tank, and associated system components shall also be replaced.

B. Type of Contract:
   1. Project will be constructed under a single prime contract.

1.05 PHASED CONSTRUCTION

A. Before commencing Work of each phase, submit an updated copy of Contractor's construction schedule showing the sequence, commencement and completion dates, and anticipated dates and durations of utility interruption for coordination with Owner's personnel for all phases of the Work.

1.06 WORK BY OWNER

A. Concurrent Work: Owner will perform the following operations at Project site. Those operations will be conducted simultaneously with Work under this Contract.

1. The Owner will maintain and operate Power, Domestic hot and cold water, Heating Water and Chilled water system. Required shutdowns will be jointly managed by contractor and owner for integrity of these services.

1.07 ACCESS TO SITE

A. General: The contractor shall have limited use of the project site for construction operations as indicated in the drawings by the Contract limits and as indicated by requirements of this Section.

B. Use of Site: Limit the use of the Project site to Work in areas indicated. Do not disturb portions of the project site beyond areas in which the Work is indicated.
   1. Driveways, Walkways and Entrances: Keep driveways, loading areas, and entrances serving premises clear and available to Owner, Owner's employees, and emergency vehicles at all times. Do not use these areas for parking or for storage of materials.
      a. Schedule deliveries to minimize use of driveways and entrances by construction operations.
      b. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.

C. Condition of Existing Building: Maintain portions of existing building affected by construction operations in a weathertight condition throughout construction period. Repair damage caused by construction operations.

D. Condition of Existing Grounds: Maintain portions of existing grounds, landscaping, and hardscaping affected by construction operations throughout construction period. Repair damage caused by construction operations.

1.08 COORDINATION WITH OCCUPANTS

A. Full Owner Occupancy: Owner will occupy site and building(s) during entire construction period. Cooperate with Owner during construction operations to minimize conflicts and facilitate Owner usage. Perform the Work so as not to interfere with Owner's day-to-day operations. Maintain existing exits unless otherwise indicated.
   1. Maintain access to existing walkways, corridors, and other adjacent occupied or used facilities. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from Owner and approval of authorities having jurisdiction.
   2. Be responsible to coordinate with the local Fire Department and maintain Fire Truck access acceptable to that Authority throughout the entire project duration.
   3. Notify Owner not less than 72 hours in advance of activities that will affect Owner's operations.

1.09 WORK RESTRICTIONS

A. Work Restrictions, General: Comply with restrictions on construction operations.
   1. Comply with limitations on use of public streets and with other requirements of authorities having jurisdiction.

B. On-Site Work Hours: Limit work in the existing building to normal business working hours of 7 a.m. to 5 p.m., Monday through Friday, unless otherwise indicated.
   1. Weekend Hours: In the event weekend work is desired, coordinate with owner. Obtain owner's written permission before proceeding.
   2. Hours for Utility Shutdowns: Coordinate with Owner.
   3. Hours for Core Drilling, Concrete Jackhammering, concerte sawing, steel cutting, miscellaneous noisy activities: Noisy work must occur before or after normal facility business hours. Coordinate and schedule with owner.

C. No Work Dates: No work dates occur on Holidays such as Memorial Day, Fourth of July, etc. In the event Holiday falls on a weekend, the No Work date shall be observed the preceding Friday for dates occurring on a Saturday or the following Monday for dates occurring on a Sunday.

D. Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by Owner or others unless permitted under the following conditions and then only after providing temporary utility services according to requirements indicated:
   1. Notify Engineer not less than three days in advance of proposed utility interruptions.
   2. Obtain Owner's written permission before proceeding with utility interruptions.

E. Noise, Vibration, and Odors: Coordinate operations that may result in high levels of noise and vibration, odors, or other disruption to Owner occupancy with Owner.
   1. Notify Owner not less than two days in advance of proposed disruptive operations.
   2. Obtain Owner's written permission before proceeding with disruptive operations.

F. Restricted Substances: Use of tobacco products and other controlled substances on Project site is not permitted.

G. Employee Identification: Owner will perform background checks on contractor personnel and issue contractor badges. Identification tags required for all Contractor personnel working on Project site. Coordinate with Owner.

1.10 SPECIFICATION AND DRAWING CONVENTIONS

A. Specification Content: The Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:
   1. Imperative mood and streamlined language are generally used in the Specifications. The words "shall," "shall be," or "shall comply with," depending on the context, are implied where a colon (:) is used within a sentence or phrase.
   2. Specification requirements are to be performed by Contractor unless specifically stated otherwise.

B. Division 01 General Requirements: Requirements of Sections in Division 01 apply to the Work of all Sections in the Specifications.

C. Drawing Coordination: Requirements for materials and products identified on Drawings are described in detail in the Specifications. One or more of the following are used on Drawings to identify materials and products:
   1. Terminology: Materials and products are identified by the typical generic terms used in the individual Specifications Sections.
   2. Abbreviations: Materials and products are identified by abbreviations and scheduled on Drawings.
   3. Keynoting: Materials and products are identified by reference keynotes referencing Specification Section numbers found in this Project Manual.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 011000

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SECTION 01 2500
SUBSTITUTION PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for substitutions.

B. Related Requirements:
   1. Section 016000 "Product Requirements" for requirements for submitting comparable product submittals for products by listed manufacturers.

1.03 DEFINITIONS

A. Substitutions: Changes in products, materials, equipment, and methods of construction from those required by the Contract Documents and proposed by Contractor.
   1. Substitutions for Cause: Changes proposed by Contractor that are required due to changed Project conditions, such as unavailability of product, regulatory changes, or unavailability of required warranty terms.
   2. Substitutions for Convenience: Changes proposed by Contractor or Owner that are not required in order to meet other Project requirements but may offer advantage to Contractor or Owner.

1.04 ACTION SUBMITTALS

A. Substitution Requests During the Bid Phase: Submit a copy of each request for consideration. Identify product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.
   1. Substitution Request Form: Use CSI Form 1.5C.
   2. Documentation: Show compliance with requirements for substitutions and the following, as applicable:
      a. Statement indicating why specified product or fabrication or installation method cannot be provided, if applicable.
      b. Coordination of information, including a list of changes or revisions needed to other parts of the Work and to construction performed by Owner and separate contractors that will be necessary to accommodate proposed substitution.
      c. Detailed comparison of significant qualities of proposed substitutions with those of the Work specified. Include annotated copy of applicable Specification Section. Significant qualities may include attributes, such as performance, weight, size, durability, visual effect, sustainable design characteristics, warranties, and specific features and requirements indicated. Indicate deviations, if any, from the Work specified.
      d. Product Data, including drawings and descriptions of products and fabrication and installation procedures.
      e. Samples, where applicable or requested.
      f. Certificates and qualification data, where applicable or requested.
      g. List of similar installations for completed projects, with project names and addresses as well as names and addresses of architects and owners.
      h. Material test reports from a qualified testing agency, indicating and interpreting test results for compliance with requirements indicated.
      i. Research reports evidencing compliance with building code in effect for Project, from ICC-ES.
      j. Detailed comparison of Contractor's construction schedule using proposed substitutions with products specified for the Work, including effect on the overall

Contract Time. If specified product or method of construction cannot be provided within the Contract Time, include letter from manufacturer, on manufacturer's letterhead, stating date of receipt of purchase order, lack of availability, or delays in delivery.

k. Cost information, including a proposal of change, if any, in the Contract Sum.

l. Contractor's certification that proposed substitution complies with requirements in the Contract Documents, except as indicated in substitution request, is compatible with related materials and is appropriate for applications indicated.

m. Contractor's waiver of rights to additional payment or time that may subsequently become necessary because of failure of proposed substitution to produce indicated results.

3. Engineer's Action: If necessary, Engineer will request additional information or documentation for evaluation within seven 7 days of receipt of a request for substitution. Engineer will notify Contractor of acceptance or rejection of proposed substitution within 15 days of receipt of request, or seven 7 days of receipt of additional information or documentation, whichever is later.

a. Forms of Acceptance: Change Order, Construction Change Directive, or Engineer's Supplemental Instructions for minor changes in the Work.

b. Use product specified if Engineer does not issue a decision on use of a proposed substitution within time allocated.

1.05 QUALITY ASSURANCE

A. Compatibility of Substitutions: Investigate and document compatibility of proposed substitution with related products and materials. Engage a qualified testing agency to perform compatibility tests recommended by manufacturers.

1.06 PROCEDURES

A. Coordination: Revise or adjust affected work as necessary to integrate work of the approved substitutions.

1.07 SUBSTITUTIONS

A. Substitutions for Cause: Submit requests for substitution immediately on discovery of need for change, but not later than 15 fifteen days prior to time required for preparation and review of related submittals.

1. Conditions: Engineer will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Engineer will return requests without action, except to record noncompliance with these requirements:

a. Requested substitution is consistent with the Contract Documents and will produce indicated results.

b. Substitution request is fully documented and properly submitted.

c. Requested substitution will not adversely affect Contractor's construction schedule.

d. Requested substitution has received necessary approvals of authorities having jurisdiction.

e. Requested substitution is compatible with other portions of the Work.

f. Requested substitution has been coordinated with other portions of the Work.

B. Substitutions for Convenience: Engineer will consider requests for substitution if received within 20 twenty days after commencement of the Work . Requests received after that time may be considered or rejected at discretion of Engineer.

1. Conditions: Engineer will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Engineeer will return requests without action, except to record noncompliance with these requirements:

a. Requested substitution offers Owner a substantial advantage in cost, time, energy conservation, or other considerations, after deducting additional responsibilities Owner must assume. Owner's additional responsibilities may include compensation

to Engineer for redesign and evaluation services, increased cost of other construction by Owner, and similar considerations.

b. Requested substitution does not require extensive revisions to the Contract Documents.
c. Requested substitution is consistent with the Contract Documents and will produce indicated results.
d. Substitution request is fully documented and properly submitted.
e. Requested substitution will not adversely affect Contractor's construction schedule.
f. Requested substitution has received necessary approvals of authorities having jurisdiction.
g. Requested substitution is compatible with other portions of the Work.
h. Requested substitution has been coordinated with other portions of the Work.
i. Requested substitution provides specified warranty.
j. If requested substitution involves more than one contractor, requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all contractors involved.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 012500

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SECTION 01 2600
CONTRACT MODIFICATION PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for handling and processing Contract modifications.

B. Related Requirements:
   1. Section 01 2500 "Substitution Procedures" for administrative procedures for handling requests for substitutions made after the Contract award.

1.03 MINOR CHANGES IN THE WORK

A. Engineer will issue supplemental instructions authorizing minor changes in the Work, not involving adjustment to the Contract Sum or the Contract Time, on AIA Document G710 .

1.04 PROPOSAL REQUESTS

A. Owner-Initiated Proposal Requests: Engineer will issue a detailed description of proposed changes in the Work that may require adjustment to the Contract Sum or the Contract Time. If necessary, the description will include supplemental or revised Drawings and Specifications.
   1. Work Change Proposal Requests issued by Engineer are not instructions either to stop work in progress or to execute the proposed change.
   2. Within time specified in Proposal Request or 20 days, when not otherwise specified, after receipt of Proposal Request, submit a quotation estimating cost adjustments to the Contract Sum and the Contract Time necessary to execute the change.
      a. Include a list of quantities of products required or eliminated and unit costs, with total amount of purchases and credits to be made. If requested, furnish survey data to substantiate quantities.
      b. Indicate applicable delivery charges, equipment rental, and amounts of trade discounts.
      c. Include costs of labor and supervision directly attributable to the change.
      d. Include an updated Contractor's construction schedule that indicates the effect of the change, including, but not limited to, changes in activity duration, start and finish times, and activity relationship. Use available total float before requesting an extension of the Contract Time.
      e. Quotation Form: Use CSI Form 13.6D, "Proposal Worksheet Summary," and Form 13.6C, "Proposal Worksheet Detail." .

B. Contractor-Initiated Proposals: If latent or changed conditions require modifications to the Contract, Contractor may initiate a claim by submitting a request for a change to Engineer.
   1. Include a statement outlining reasons for the change and the effect of the change on the Work. Provide a complete description of the proposed change. Indicate the effect of the proposed change on the Contract Sum and the Contract Time.
   2. Include a list of quantities of products required or eliminated and unit costs, with total amount of purchases and credits to be made. If requested, furnish survey data to substantiate quantities.
   3. Indicate applicabl delivery charges, equipment rental, and amounts of trade discounts.
   4. Include costs of labor and supervision directly attributable to the change.
   5. Include an updated Contractor's construction schedule that indicates the effect of the change, including, but not limited to, changes in activity duration, start and finish times,

and activity relationship. Use available total float before requesting an extension of the Contract Time.

6. Comply with requirements in Section 012500 "Substitution Procedures" if the proposed change requires substitution of one product or system for product or system specified.
7. Proposal Request Form: Use Proposal Request Form: Use CSI Form 13.6A, "Change Order Request (Proposal)," with attachments CSI Form 13.6D, "Proposal Worksheet Summary," and Form 13.6C, Proposal Worksheet Detail. .

1.05 CHANGE ORDER PROCEDURES

A. On Owner's approval of a Work Change Proposal Request, Owner will issue a Change Order for signatures of Owner and Contractor on their "Authorization to Proceed" order form.

1.06 CONSTRUCTION CHANGE DIRECTIVE

A. Construction Change Directive: Engineer may issue a Construction Change Directive on AIA Document G714 . Construction Change Directive instructs Contractor to proceed with a change in the Work, for subsequent inclusion in a Change Order.
   1. Construction Change Directive contains a complete description of change in the Work. It also designates method to be followed to determine change in the Contract Sum or the Contract Time.

B. Documentation: Maintain detailed records on a time and material basis of work required by the Construction Change Directive.
   1. After completion of change, submit an itemized account and supporting data necessary to substantiate cost and time adjustments to the Contract.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 012600

SECTION 01 2900
PAYMENT PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements necessary to prepare and process Applications for Payment.

B. Related Requirements:
   1. Section 012600 "Contract Modification Procedures" for administrative procedures for handling changes to the Contract.

1.03 DEFINITIONS

A. Schedule of Values: A statement furnished by Contractor allocating portions of the Contract Sum to various portions of the Work and used as the basis for reviewing Contractor's Applications for Payment.

1.04 SCHEDULE OF VALUES

A. Coordination: Coordinate preparation of the schedule of values with preparation of Contractor's construction schedule. Cost-loaded Schedule may serve to satisfy requirements for the schedule of values.
   1. Coordinate line items in the schedule of values with items required to be indicated as separate activities in Contractor's construction schedule.
   2. Submit the schedule of values to Owner and Engineer at earliest possible date, but no later than seven days before the date scheduled for submittal of initial Applications for Payment.
   3. Subschedules for Phased Work: Where the Work is separated into phases requiring separately phased payments, provide subschedules showing values coordinated with each phase of payment.
   4. Subschedules for Separate Elements of Work: Where the Contractor's construction schedule defines separate elements of the Work, provide subschedules showing values coordinated with each element.

B. Format and Content: Use Project Manual table of contents as a guide to establish line items for the schedule of values. Provide at least one line item for each Specification Section.
   1. Identification: Include the following Project identification on the schedule of values:
      a. Project name and location.
      b. Name of Engineer.
      c. Engineer's Project number.
      d. Contractor's name and address.
      e. Date of submittal.
   2. Arrange schedule of values consistent with format of AIA Document G703 .
   3. Arrange the schedule of values in tabular form, with separate columns to indicate the following for each item listed:
      a. Related Specification Section or Division.
      b. Description of the Work.
      c. Name of subcontractor.
      d. Name of manufacturer or fabricator.
      e. Name of supplier.
      f. Change Orders (numbers) that affect value.

g. Dollar value of the following, as a percentage of the Contract Sum to nearest one-hundredth percent, adjusted to total 100 percent. Round dollar amounts to whole dollars, with total equal to Contract Sum.
   1) Labor.
   2) Materials.
   3) Equipment.

4. Provide a breakdown of the Contract Sum in enough detail to facilitate continued evaluation of Applications for Payment and progress reports. Provide multiple line items for principal subcontract amounts in excess of five percent of the Contract Sum.
5. Provide a separate line item in the schedule of values for each part of the Work where Applications for Payment may include materials or equipment purchased or fabricated and stored, but not yet installed.
   a. Differentiate between items stored on-site and items stored off-site.
6. Overhead Costs: Include total cost and proportionate share of general overhead and profit for each line item.
7. Overhead Costs: Show cost of temporary facilities and other major cost items that are not direct cost of actual work-in-place as separate line items.
8. Closeout Costs. Include separate line items under Contractor and principal subcontracts for Project closeout requirements in an amount totaling five percent of the Contract Sum and subcontract amount.
9. Schedule of Values Revisions: Revise the schedule of values when Change Orders or Construction Change Directives result in a change in the Contract Sum. Include at least one separate line item for each Change Order and Construction Change Directive.

1.05 APPLICATIONS FOR PAYMENT

A. Each Application for Payment following the initial Application for Payment shall be consistent with previous applications and payments as certified by Engineer and paid for by Owner.

B. Payment Application Times: The date for each progress payment is indicated in the Agreement between Owner and Contractor. The period of construction work covered by each Application for Payment is the period indicated in the Agreement.

C. Application for Payment Forms: Use AIA Document G702 and AIA Document G703 as form for Applications for Payment.
   1. Other Application for Payment forms proposed by the Contractor shall be acceptable to Engineer and Owner. Submit forms for approval with initial submittal of schedule of values.

D. Application Preparation: Complete every entry on form. Notarize and execute by a person authorized to sign legal documents on behalf of Contractor. Engineer will return incomplete applications without action.
   1. Entries shall match data on the schedule of values and Contractor's construction schedule. Use updated schedules if revisions were made.
   2. Include amounts for work completed following previous Application for Payment, whether or not payment has been received. Include only amounts for work completed at time of Application for Payment.
   3. Include amounts of Change Orders issued before last day of construction period covered by application. Must be fully executed. Proceed Orders must be in Change Order before billing.
   4. Indicate separate amounts for work being carried out under Owner-requested project acceleration.

E. Stored Materials: Include in Application for Payment amounts applied for materials or equipment purchased or fabricated and stored, but not yet installed. Differentiate between items stored on-site and items stored off-site. Owner will only pay stored materials at invoice price, if in bonded warehouse or the jobsite and if owner can verify.

1. Provide certificate of insurance, evidence of transfer of title to Owner, and consent of surety to payment for stored materials.
2. Provide supporting documentation that verifies amount requested, such as paid invoices. Match amount requested with amounts indicated on documentation; do not include overhead and profit on stored materials.
3. Provide summary documentation for stored materials indicating the following:
    a. Value of materials previously stored and remaining stored as of date of previous Applications for Payment.
    b. Value of previously stored materials put in place after date of previous Application for Payment and on or before date of current Application for Payment.
    c. Value of materials stored since date of previous Application for Payment and remaining stored as of date of current Application for Payment.

F. Transmittal: Submit three signed and notarized original copies and/or electronic version of each Application for Payment to Engineer by a method ensuring receipt within 24 hours. One copy shall include waivers of lien and similar attachments if required.
1. Transmit each copy with a transmittal form listing attachments and recording appropriate information about application.

G. Initial Application for Payment: Administrative actions and submittals that must precede or coincide with submittal of first Application for Payment include the following:
1. List of subcontractors.
2. Schedule of values.
3. Contractor's construction schedule (preliminary if not final).
4. Combined Contractor's construction schedule (preliminary if not final) incorporating Work of multiple contracts, with indication of acceptance of schedule by each Contractor.
5. Products list (preliminary if not final).
6. Sustainable design action plans, including preliminary project materials cost data.
7. Schedule of unit prices.
8. Submittal schedule (preliminary if not final).
9. List of Contractor's staff assignments.
10. List of Contractor's principal consultants.
11. Copies of building permits.
12. Copies of authorizations and licenses from authorities having jurisdiction for performance of the Work.
13. Initial progress report.
14. Report of preconstruction conference.
15. Certificates of insurance and insurance policies. Insurance and bonds required prior to Notice to Proceed being issued.
16. Performance and payment bonds.
17. Data needed to acquire Owner's insurance.

H. Application for Payment at Substantial Completion: After Owner issues the Certificate of Substantial Completion, submit an Application for Payment showing 100 percent completion for portion of the Work claimed as substantially complete.
1. Include documentation supporting claim that the Work is substantially complete and a statement showing an accounting of changes to the Contract Sum.
2. This application shall reflect Certificate(s) of Substantial Completion issued previously for Owner occupancy of designated portions of the Work.

I. Final Payment Application: After completing Project closeout requirements, submit final Application for Payment with releases and supporting documentation not previously submitted and accepted, including, but not limited, to the following:
1. Evidence of completion of Project closeout requirements. Warranties, Conset of Surety, O&Ms, owner training, extra stock and materials.

2. Insurance certificates for products and completed operations where required and proof that taxes, fees, and similar obligations were paid.
3. Updated final statement, accounting for final changes to the Contract Sum.
4. AIA Document G706.
5. AIA Document G706A.
6. AIA Document G707.
7. Evidence that claims have been settled.
8. Final meter readings for utilities, a measured record of stored fuel, and similar data as of date of Substantial Completion or when Owner took possession of and assumed responsibility for corresponding elements of the Work.
9. Final liquidated damages settlement statement.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 012900

SECTION 01 3100
PROJECT MANAGEMENT AND COORDINATION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:
   1. General coordination procedures.
   2. Coordination drawings.
   3. RFIs.
   4. Digital project management procedures.
   5. Project meetings.

B. Each contractor shall participate in coordination requirements. Certain areas of responsibility are assigned to a specific contractor.

C. Related Requirements:
   1. Section 013200 "Construction Progress Documentation" for preparing and submitting Contractor's construction schedule.
   2. Section 017300 "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
   3. Section 017700 "Closeout Procedures" for coordinating closeout of the Contract.
   4. Section 019113 "General Commissioning Requirements" for coordinating the Work with Owner's Commissioning Authority.

1.03 DEFINITIONS

A. BIM: Building Information Modeling.

B. RFI: Request for Information. Request from Owner, Engineer, or Contractor seeking information required by or clarifications of the Contract Documents.

1.04 INFORMATIONAL SUBMITTALS

A. Subcontract List: Prepare a written summary identifying individuals or firms proposed for each portion of the Work, including those who are to furnish products or equipment fabricated to a special design. Include the following information in tabular form:
   1. Name, address, telephone number, and email address of entity performing subcontract or supplying products.
   2. Number and title of related Specification Section(s) covered by subcontract.
   3. Drawing number and detail references, as appropriate, covered by subcontract.

B. Key Personnel Names: Within 15 days of starting construction operations, submit a list of key personnel assignments, including superintendent and other personnel in attendance at Project site. Identify individuals and their duties and responsibilities; list addresses and cellular telephone numbers and e-mail addresses. Provide names, addresses, and telephone numbers of individuals assigned as alternates in the absence of individuals assigned to Project.
   1. Post copies of list in project meeting room, in temporary field office, and in prominent location in each built facility. Keep list current at all times.

1.05 GENERAL COORDINATION PROCEDURES

A. Coordination: Coordinate construction operations included in different Sections of the Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations included in different Sections that depend on each other for proper installation, connection, and operation.

1. Schedule construction operations in sequence required to obtain the best results where installation of one part of the Work depends on installation of other components, before or after its own installation.
2. Coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair.
3. Make adequate provisions to accommodate items scheduled for later installation.

B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.
1. Prepare similar memoranda for Owner and separate contractors if coordination of their Work is required.

C. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to, the following:
1. Preparation of Contractor's construction schedule.
2. Preparation of the schedule of values.
3. Installation and removal of temporary facilities and controls.
4. Delivery and processing of submittals.
5. Progress meetings.
6. Preinstallation conferences.
7. Project closeout activities.
8. Startup and adjustment of systems.

1.06 COORDINATION DRAWINGS

A. Coordination Drawings, General: Prepare coordination drawings according to requirements in individual Sections, and additionally where installation is not completely indicated on Shop Drawings, where limited space availability necessitates coordination, or if coordination is required to facilitate integration of products and materials fabricated or installed by more than one entity.
1. Content: Project-specific information, drawn accurately to a scale large enough to indicate and resolve conflicts. Do not base coordination drawings on standard printed data. Include the following information, as applicable:
a. Use applicable Drawings as a basis for preparation of coordination drawings. Prepare sections, elevations, and details as needed to describe relationship of various systems and components.
b. Coordinate the addition of trade-specific information to coordination drawings by multiple contractors in a sequence that best provides for coordination of the information and resolution of conflicts between installed components before submitting for review.
c. Indicate functional and spatial relationships of components of architectural, structural, civil, mechanical, and electrical systems.
d. Indicate space requirements for routine maintenance and for anticipated replacement of components during the life of the installation.
e. Show location and size of access doors required for access to concealed dampers, valves, and other controls.
f. Indicate required installation sequences.
g. Indicate dimensions shown on Drawings. Specifically note dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternative sketches to Engineer indicating proposed resolution of such conflicts. Minor dimension changes and difficult installations will not be considered changes to the Contract.

B. Coordination Drawing Organization: Organize coordination drawings as follows:

1. Mechanical Rooms: Provide coordination drawings for mechanical rooms showing plans and elevations of mechanical, plumbing, fire-protection, fire-alarm, and electrical equipment.
2. Mechanical and Plumbing Work: Show the following:
    a. Sizes and bottom elevations of ductwork, piping, and conduit runs, including insulation, bracing, flanges, and support systems.
    b. Dimensions of major components, such as dampers, valves, diffusers, access doors, cleanouts and electrical distribution equipment.
    c. Fire-rated enclosures around ductwork.
3. Electrical Work: Show the following:
    a. Runs of vertical and horizontal conduit 1-1/4 inches in diameter and larger.
    b. Light fixture, exit light, emergency battery pack, smoke detector, and other fire-alarm locations.
    c. Panel board, switch board, switchgear, transformer, busway, generator, and motor-control center locations.
    d. Location of pull boxes and junction boxes, dimensioned from column center lines.
4. Review: Engineer will review coordination drawings to confirm that in general the Work is being coordinated, but not for the details of the coordination, which are Contractor's responsibility. If Engineer determines that coordination drawings are not being prepared in sufficient scope or detail, or are otherwise deficient, Engineer will so inform Contractor, who shall make suitable modifications and resubmit.
5. Coordination Drawing Prints: Prepare coordination drawing prints according to requirements in Section 013300 "Submittal Procedures."

C. Coordination Digital Data Files: Prepare coordination digital data files according to the following requirements:
1. File Submittal Format: Submit or post coordination drawing files using format same as file preparation format PDF format.
2. Engineer will furnish Contractor one set of digital data files of Drawings for use in preparing coordination digital data files.
    a. Engineer makes no representations as to the accuracy or completeness of digital data files as they relate to Drawings.

1.07 REQUEST FOR INFORMATION (RFI)

A. General: Immediately on discovery of the need for additional information, clarification, or interpretation of the Contract Documents, Contractor shall prepare and submit an RFI in the form specified.
1. Engineer will return without response those RFIs submitted to Engineer by other entities controlled by Contractor.
2. Coordinate and submit RFIs in a prompt manner so as to avoid delays in Contractor's work or work of subcontractors.

B. Content of the RFI: Include a detailed, legible description of item needing information or interpretation and the following:
1. Project name.
2. Project number.
3. Date.
4. Name of Contractor.
5. Name of Engineer.
6. RFI number, numbered sequentially.
7. RFI subject.
8. Specification Section number and title and related paragraphs, as appropriate.
9. Drawing number and detail references, as appropriate.
10. Field dimensions and conditions, as appropriate.

11. Contractor's suggested resolution. If Contractor's suggested resolution impacts the Contract Time or the Contract Sum, Contractor shall state impact in the RFI.
12. Contractor's signature.
13. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.
    a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.

C. RFI Forms: Software-generated form with substantially the same content as indicated above, acceptable to Engineer.
1. Attachments shall be electronic files in PDF format.

D. Engineer's Action: Engineer will review each RFI, determine action required, and respond. Allow seven working days for Engineer's response for each RFI. RFIs received by Engineer after 1:00 p.m. will be considered as received the following working day.
1. The following Contractor-generated RFIs will be returned without action:
    a. Requests for approval of submittals.
    b. Requests for approval of substitutions.
    c. Requests for approval of Contractor's means and methods.
    d. Requests for coordination information already indicated in the Contract Documents.
    e. Requests for adjustments in the Contract Time or the Contract Sum.
    f. Requests for interpretation of Engineer's actions on submittals.
    g. Incomplete RFIs or inaccurately prepared RFIs.
2. Engineer's action may include a request for additional information, in which case Engineer's time for response will date from time of receipt by Engineer of additional information.
3. Engineer's action on RFIs that may result in a change to the Contract Time or the Contract Sum may be eligible for Contractor to submit Change Proposal according to Section 012600 "Contract Modification Procedures."
    a. If Contractor believes the RFI response warrants change in the Contract Time or the Contract Sum, notify Engineer in writing within 10 days of receipt of the RFI response.

E. RFI Log: Prepare, maintain, and submit a tabular log of RFIs organized by the RFI number. Submit log weekly . Use software log that is part of web-based Project software. Include the following:
1. Project name.
2. Name and address of Contractor.
3. Name and address of Engineer.
4. RFI number including RFIs that were returned without action or withdrawn.
5. RFI description.
6. Date the RFI was submitted.
7. Date Engineer's response was received.
8. Identification of related Minor Change in the Work, Construction Change Directive, and Proposal Request, as appropriate.
9. Identification of related Field Order, Work Change Directive, and Proposal Request, as appropriate.

F. On receipt of Engineer's action, update the RFI log and immediately distribute the RFI response to affected parties. Review response and notify Engineer within seven days if Contractor disagrees with response.

1.08 DIGITAL PROJECT MANAGEMENT PROCEDURES

A. Use of Engineer's Digital Data Files: Digital data files of Engineer's CAD drawings will be provided by Engineer for Contractor's use during construction.

1. Digital data files may be used by Contractor in preparing coordination drawings, Shop Drawings, and Project record Drawings.
2. Engineer makes no representations as to the accuracy or completeness of digital data files as they relate to Contract Drawings.

B. PDF Document Preparation: Where PDFs are required to be submitted to Engineer, prepare as follows:
1. Assemble complete submittal package into a single indexed file incorporating submittal requirements of a single Specification Section and transmittal form with links enabling navigation to each item.
2. Name file with submittal number or other unique identifier, including revision identifier.
3. Certifications: Where digitally submitted certificates and certifications are required, provide a digital signature with digital certificate on where indicated.

1.09 PROJECT MEETINGS

A. General: Schedule and conduct meetings and conferences at Project site unless otherwise indicated.
1. Attendees: Inform participants and others involved, and individuals whose presence is required, of date and time of each meeting. Notify Owner and Engineer of scheduled meeting dates and times a minimum of 10 working days prior to meeting.
2. Agenda: Prepare the meeting agenda. Distribute the agenda to all invited attendees.
3. Minutes: Entity responsible for conducting meeting will record significant discussions and agreements achieved. Distribute the meeting minutes to everyone concerned, including Owner and Engineer, within three days of the meeting.

B. Preconstruction Conference: Schedule and conduct a preconstruction conference before starting construction, at a time convenient to Owner and Engineer, but no later than 15 days after execution of the Agreement.
1. Attendees: Authorized representatives of Owner , Owner's Commissioning Authority, Engineer, and their consultants; Contractor and its superintendent; major subcontractors; suppliers; and other concerned parties shall attend the conference. Participants at the conference shall be familiar with Project and authorized to conclude matters relating to the Work.
2. Agenda: Discuss items of significance that could affect progress, including the following:
a. Responsibilities and personnel assignments.
b. Tentative construction schedule.
c. Phasing.
d. Critical work sequencing and long lead items.
e. Designation of key personnel and their duties.
f. Lines of communications.
g. Use of web-based Project software.
h. Procedures for processing field decisions and Change Orders.
i. Procedures for RFIs.
j. Procedures for testing and inspecting.
k. Procedures for processing Applications for Payment.
l. Distribution of the Contract Documents.
m. Submittal procedures.
n. Sustainable design requirements.
o. Preparation of Record Documents.
p. Use of the premises and existing building.
q. Work restrictions.
r. Working hours.
s. Owner's occupancy requirements.
t. Responsibility for temporary facilities and controls.

u. Procedures for moisture and mold control.
v. Procedures for disruptions and shutdowns.
w. Construction waste management and recycling.
x. Parking availability.
y. Office, work, and storage areas.
z. Equipment deliveries and priorities.
aa. First aid.
bb. Security / background checks.
cc. Progress cleaning.

3. Minutes: Entity responsible for conducting meeting will record and distribute meeting minutes.

C. Preinstallation Conferences: Conduct a preinstallation conference at Project site before each construction activity when required by other sections and when required for coordination with other construction.
1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by the installation and its coordination or integration with other materials and installations that have preceded or will follow, shall attend the meeting. Advise Engineer , and Owner's Commissioning Authority of scheduled meeting dates.
2. Agenda: Review progress of other construction activities and preparations for the particular activity under consideration, including requirements for the following:
a. Contract Documents.
b. Options.
c. Related RFIs.
d. Related Change Orders.
e. Purchases.
f. Deliveries.
g. Submittals.
h. Sustainable design requirements.
i. Review of mockups.
j. Possible conflicts.
k. Compatibility requirements.
l. Time schedules.
m. Weather limitations.
n. Manufacturer's written instructions.
o. Warranty requirements.
p. Compatibility of materials.
q. Acceptability of substrates.
r. Temporary facilities and controls.
s. Space and access limitations.
t. Regulations of authorities having jurisdiction.
u. Testing and inspecting requirements.
v. Installation procedures.
w. Coordination with other work.
x. Required performance results.
y. Protection of adjacent work.
z. Protection of construction and personnel.
3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.
4. Reporting: Distribute minutes of the meeting to each party present and to other parties requiring information.

5. Do not proceed with installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to performance of the Work and reconvene the conference at earliest feasible date.

D. Project Closeout Conference: Schedule and conduct a project closeout conference, at a time convenient to Owner and Engineer, but no later than 90 days prior to the scheduled date of Substantial Completion.
   1. Conduct the conference to review requirements and responsibilities related to Project closeout.
   2. Attendees: Authorized representatives of Owner, Owner's Commissioning Authority, Engineer, and their consultants; Contractor and its superintendent; major subcontractors; suppliers; and other concerned parties shall attend the meeting. Participants at the meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
   3. Agenda: Discuss items of significance that could affect or delay Project closeout, including the following:
      a. Preparation of Record Documents.
      b. Procedures required prior to inspection for Substantial Completion and for final inspection for acceptance.
      c. Procedures for completing and archiving web-based Project software site data files.
      d. Submittal of written warranties.
      e. Requirements for completing sustainable design documentation.
      f. Requirements for preparing operations and maintenance data.
      g. Requirements for delivery of material samples, attic stock, and spare parts.
      h. Requirements for demonstration and training.
      i. Preparation of Contractor's punch list.
      j. Procedures for processing Applications for Payment at Substantial Completion and for final payment.
      k. Submittal procedures.
      l. Coordination of separate contracts.
      m. Owner's partial occupancy requirements.
      n. Installation of Owner's furniture, fixtures, and equipment.
      o. Responsibility for removing temporary facilities and controls.
   4. Minutes: Entity conducting meeting will record and distribute meeting minutes.

E. Progress Meetings: Conduct progress meetings at weekly intervals.
   1. Coordinate dates of meetings with preparation of payment requests.
   2. Attendees: In addition to representatives of Owner , Owner's Commissioning Authority, and Engineer, each contractor, subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented at these meetings. All participants at the meeting shall be familiar with Project and authorized to conclude matters relating to the Work.
   3. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to status of Project.
      a. Contractor's Construction Schedule: Review progress since the last meeting. Determine whether each activity is on time, ahead of schedule, or behind schedule, in relation to Contractor's construction schedule. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Contract Time.
         1) Review schedule for next period.
      b. Review present and future needs of each entity present, including the following:
         1) Interface requirements.

2) Sequence of operations.
3) Resolution of BIM component conflicts.
4) Status of submittals.
5) Status of sustainable design documentation.
6) Deliveries.
7) Off-site fabrication.
8) Access.
9) Site use.
10) Temporary facilities and controls.
11) Progress cleaning.
12) Quality and work standards.
13) Status of correction of deficient items.
14) Field observations.
15) Status of RFIs.
16) Status of Proposal Requests.
17) Pending changes.
18) Status of Change Orders.
19) Pending claims and disputes.
20) Documentation of information for payment requests.

4. Minutes: Entity responsible for conducting the meeting will record and distribute the meeting minutes to each party present and to parties requiring information.
   a. Schedule Updating: Revise Contractor's construction schedule after each progress meeting where revisions to the schedule have been made or recognized. Issue revised schedule concurrently with the report of each meeting.

F. Coordination Meetings: Conduct Project coordination meetings at weekly intervals. Project coordination meetings are in addition to specific meetings held for other purposes, such as progress meetings and preinstallation conferences.
   1. Attendees: In addition to representatives of Owner , Owner's Commissioning Authority and Engineer, each contractor, subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented at these meetings. All participants at the meetings shall be familiar with Project and authorized to conclude matters relating to the Work.
   2. Agenda: Review and correct or approve minutes of the previous coordination meeting. Review other items of significance that could affect progress. Include topics for discussion as appropriate to status of Project.
      a. Combined Contractor's Construction Schedule: Review progress since the last coordination meeting. Determine whether each contract is on time, ahead of schedule, or behind schedule, in relation to combined Contractor's construction schedule. Determine how construction behind schedule will be expedited; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities will be completed within the Contract Time.
      b. Schedule Updating: Revise combined Contractor's construction schedule after each coordination meeting where revisions to the schedule have been made or recognized. Issue revised schedule concurrently with report of each meeting.
      c. Review present and future needs of each contractor present, including the following:
         1) Interface requirements.
         2) Sequence of operations.
         3) Status of submittals.
         4) Deliveries.
         5) Off-site fabrication.
         6) Access.

7) Site use.
8) Temporary facilities and controls.
9) Work hours.
10) Hazards and risks.
11) Progress cleaning.
12) Quality and work standards.
13) Status of RFIs.
14) Proposal Requests.
15) Change Orders.
16) Pending changes.

3. Reporting: Record meeting results and distribute copies to everyone in attendance and to others affected by decisions or actions resulting from each meeting.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 013100

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SECTION 01 3200
CONSTRUCTION PROGRESS DOCUMENTATION

PART 1 GENERAL

1.01 COORDINATE WITH CONTRACT LANGUAGE

1.02 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, other Division 01 Specification Sections, apply to this Section.

1.03 SUMMARY

A. Section includes administrative and procedural requirements for documenting the progress of construction during performance of the Work, including the following:
   1. Startup construction schedule.
   2. Contractor's Construction Schedule.
   3. Construction schedule updating reports.
   4. Daily construction reports.
   5. Material location reports.
   6. Site condition reports.
   7. Unusual event reports.

1.04 DEFINITIONS

A. Activity: A discrete part of a project that can be identified for planning, scheduling, monitoring, and controlling the construction Project. Activities included in a construction schedule consume time and resources.
   1. Critical Activity: An activity on the critical path that must start and finish on the planned early start and finish times.
   2. Predecessor Activity: An activity that precedes another activity in the network.
   3. Successor Activity: An activity that follows another activity in the network.

B. Cost Loading: The allocation of the schedule of values for completing an activity as scheduled. The sum of costs for all activities must equal the total Contract Sum.

C. CPM: Critical path method, which is a method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.

D. Critical Path: The longest connected chain of interdependent activities through the network schedule that establishes the minimum overall Project duration and contains no float.

E. Event: The starting or ending point of an activity.

F. Float: The measure of leeway in starting and completing an activity.
   1. Float time is not for the exclusive use or benefit of either Owner or Contractor, but is a jointly owned, expiring Project resource available to both parties as needed to meet schedule milestones and Contract completion date.
   2. Free float is the amount of time an activity can be delayed without adversely affecting the early start of the successor activity.
   3. Total float is the measure of leeway in starting or completing an activity without adversely affecting the planned Project completion date.

G. Resource Loading: The allocation of manpower and equipment necessary for completing an activity as scheduled.

1.05 INFORMATIONAL SUBMITTALS

A. Format for Submittals: Submit required submittals in the following format:
   1. Working electronic copy of schedule file, where indicated.
   2. PDF file.

B. Startup construction schedule.
   1. Submittal of cost-loaded, startup construction schedule will not constitute approval of schedule of values for cost-loaded activities.

C. Startup Network Diagram: Of size required to display entire network for entire construction period. Show logic ties for activities.

D. Contractor's Construction Schedule: Initial schedule, of size required to display entire schedule for entire construction period.
   1. Submit a working digital copy of schedule, using software indicated, and labeled to comply with requirements for submittals.

E. Construction Schedule Updating Reports: Submit with Applications for Payment.

F. Daily Construction Reports: Submit at weekly intervals.

G. Material Location Reports: Submit at weekly intervals.

H. Site Condition Reports: Submit at time of discovery of differing conditions.

I. Unusual Event Reports: Submit at time of unusual event.

J. Qualification Data: For scheduling consultant.

1.06 QUALITY ASSURANCE

A. Prescheduling Conference: Conduct conference at Project site to comply with requirements in Section 013100 "Project Management and Coordination." Review methods and procedures related to the preliminary construction schedule and Contractor's Construction Schedule, including, but not limited to, the following:
   1. Review software limitations and content and format for reports.
   2. Verify availability of qualified personnel needed to develop and update schedule.
   3. Discuss constraints, including phasing work stages area separations interim milestones and partial Owner occupancy.
   4. Review delivery dates for Owner-furnished products.
   5. Review schedule for work of Owner's separate contracts.
   6. Review submittal requirements and procedures.
   7. Review time required for review of submittals and resubmittals.
   8. Review requirements for tests and inspections by independent testing and inspecting agencies.
   9. Review time required for Project closeout and Owner startup procedures , including commissioning activities.
   10. Review and finalize list of construction activities to be included in schedule.
   11. Review procedures for updating schedule.

1.07 COORDINATION

A. Coordinate Contractor's Construction Schedule with the schedule of values, list of subcontracts, submittal schedule, progress reports, payment requests, and other required schedules and reports.
   1. Secure time commitments for performing critical elements of the Work from entities involved.
   2. Coordinate each construction activity in the network with other activities and schedule them in proper sequence.

1.08 CONTRACTOR'S CONSTRUCTION SCHEDULE, GENERAL

A. Computer Scheduling Software: Prepare schedules using current version of a program that has been developed specifically to manage construction schedules.

B. Time Frame: Extend schedule from date established for the Notice to Proceed to date of Substantial Completion .

1. Contract completion date shall not be changed by submission of a schedule that shows an early completion date, unless specifically authorized by Change Order.

C. Activities: Treat each floor or separate area as a separate numbered activity for each main element of the Work. Comply with the following:
   1. Activity Duration: Define activities so no activity is longer than 30 days, unless specifically allowed by Engineer.
   2. Procurement Activities: Include procurement process activities for the following long lead items and major items, requiring a cycle of more than 60 days, as separate activities in schedule. Procurement cycle activities include, but are not limited to, submittals, approvals, purchasing, fabrication, and delivery.
      a. Chiller
   3. Submittal Review Time: Include review and resubmittal times indicated in Section 013300 "Submittal Procedures" in schedule. Coordinate submittal review times in Contractor's Construction Schedule with submittal schedule.
   4. Startup and Testing Time: Include no fewer than 15 days for startup and testing.
   5. Commissioning Time: Include no fewer than 15 days for commissioning.
   6. Substantial Completion: Indicate completion in advance of date established for Substantial Completion, and allow time for Engineer's administrative procedures necessary for certification of Substantial Completion.
   7. Punch List and Final Completion: Include not more than 30 days for completion of punch list items and final completion.

D. Constraints: Include constraints and work restrictions indicated in the Contract Documents and as follows in schedule, and show how the sequence of the Work is affected.
   1. Phasing: Arrange list of activities on schedule by phase.
   2. Work Restrictions: Show the effect of the following items on the schedule:
      a. Coordination with existing construction.
      b. Limitations of continued occupancies.
      c. Uninterruptible services.
      d. Partial occupancy before Substantial Completion.
      e. Use-of-premises restrictions.
      f. Provisions for future construction.
      g. Seasonal variations.
      h. Environmental control.
   3. Work Stages: Indicate important stages of construction for each major portion of the Work, including, but not limited to, the following:
      a. Subcontract awards.
      b. Submittals.
      c. Purchases.
      d. Mockups.
      e. Fabrication.
      f. Sample testing.
      g. Deliveries.
      h. Installation.
      i. Tests and inspections.
      j. Adjusting.
      k. Curing.
      l. Building flush-out.
      m. Startup and placement into final use and operation.
      n. Commissioning.
   4. Construction Areas: Identify each major area of construction for each major portion of the Work. Indicate where each construction activity within a major area must be sequenced or integrated with other construction activities to provide for the following:

a. Temporary enclosure and space conditioning.
b. Permanent space enclosure.
c. Completion of mechanical installation.
d. Completion of electrical installation.
e. Substantial Completion.

E. Milestones: Include milestones indicated in the Contract Documents in schedule, including, but not limited to, the Notice to Proceed, Substantial Completion, and final completion .
1. Temporary enclosure and space conditioning.

F. Cost Correlation: Superimpose a cost correlation timeline, indicating planned and actual costs. On the line, show planned and actual dollar volume of the Work performed as of planned and actual dates used for preparation of payment requests.
1. See Section 012900 "Payment Procedures" for cost reporting and payment procedures.

G. Upcoming Work Summary: Prepare summary report indicating activities scheduled to occur or commence prior to submittal of next schedule update. Summarize the following issues:
1. Unresolved issues.
2. Unanswered Requests for Information.
3. Rejected or unreturned submittals.
4. Notations on returned submittals.
5. Pending modifications affecting the Work and the Contract Time.

H. Contractor's Construction Schedule Updating: At monthly intervals, update schedule to reflect actual construction progress and activities. Issue schedule one week before each regularly scheduled progress meeting.
1. Revise schedule immediately after each meeting or other activity where revisions have been recognized or made. Issue updated schedule concurrently with the report of each such meeting.
2. Include a report with updated schedule that indicates every change, including, but not limited to, changes in logic, durations, actual starts and finishes, and activity durations.
3. As the Work progresses, indicate final completion percentage for each activity.

I. Recovery Schedule: When periodic update indicates the Work is 14 or more calendar days behind the current approved schedule, submit a separate recovery schedule indicating means by which Contractor intends to regain compliance with the schedule. Indicate changes to working hours, working days, crew sizes, equipment required to achieve compliance, and date by which recovery will be accomplished.

J. Distribution: Distribute copies of approved schedule to Engineer Owner, separate contractors, testing and inspecting agencies, and other parties identified by Contractor with a need-to-know schedule responsibility.
1. Post copies in Project meeting rooms and temporary field offices.
2. When revisions are made, distribute updated schedules to the same parties and post in the same locations. Delete parties from distribution when they have completed their assigned portion of the Work and are no longer involved in performance of construction activities.

1.09 STARTUP CONSTRUCTION SCHEDULE

A. Gantt-Chart Schedule: Submit startup, horizontal, Gantt-chart-type construction schedule within seven days of date established for the Notice to Proceed .

B. Preparation: Indicate each significant construction activity separately. Identify first workday of each week with a continuous vertical line. Outline significant construction activities for first 90 days of construction. Include skeleton diagram for the remainder of the Work and a cash requirement prediction based on indicated activities.

1.10 GANTT-CHART SCHEDULE REQUIREMENTS

A. Gantt-Chart Schedule: Submit a comprehensive, fully developed, horizontal, Gantt-chart-type, Contractor's Construction Schedule within 30 days of date established for the Notice to Proceed .
   1. Base schedule on the startup construction schedule and additional information received since the start of Project.

B. Preparation: Indicate each significant construction activity separately. Identify first workday of each week with a continuous vertical line.
   1. For construction activities that require three months or longer to complete, indicate an estimated completion percentage in 10 percent increments within time bar.

1.11 SCHEDULE REQUIREMENTS

A. Contract Modifications: For each proposed contract modification and concurrent with its submission, prepare a time-impact analysis using a network fragment to demonstrate the effect of the proposed change on the overall Project schedule.

B. Initial Issue of Schedule: Prepare initial network diagram from a sorted activity list indicating straight "early start-total float." Identify critical activities. Prepare tabulated reports showing the following:
   1. Contractor or subcontractor and the Work or activity.
   2. Description of activity.
   3. Main events of activity.
   4. Immediate preceding and succeeding activities.
   5. Early and late start dates.
   6. Early and late finish dates.
   7. Activity duration in workdays.
   8. Total float or slack time.
   9. Average size of workforce.
   10. Dollar value of activity (coordinated with the schedule of values).

C. Schedule Updating: Concurrent with making revisions to schedule, prepare tabulated reports showing the following:
   1. Identification of activities that have changed.
   2. Changes in early and late start dates.
   3. Changes in early and late finish dates.
   4. Changes in activity durations in workdays.
   5. Changes in the critical path.
   6. Changes in total float or slack time.
   7. Changes in the Contract Time.

D. Value Summaries: Prepare two cumulative value lists, sorted by finish dates.
   1. In first list, tabulate activity number, early finish date, dollar value, and cumulative dollar value.
   2. In second list, tabulate activity number, late finish date, dollar value, and cumulative dollar value.
   3. In subsequent issues of both lists, substitute actual finish dates for activities completed as of list date.
   4. Prepare list for ease of comparison with payment requests; coordinate timing with progress meetings.
      a. In both value summary lists, tabulate "actual percent complete" and "cumulative value completed" with total at bottom.
      b. Submit value summary printouts one week before each regularly scheduled progress meeting.

1.12 REPORTS

A. Daily Construction Reports: Prepare a daily construction report recording the following information concerning events at Project site:
   1. List of subcontractors at Project site.
   2. List of separate contractors at Project site.
   3. Approximate count of personnel at Project site.
   4. Equipment at Project site.
   5. Material deliveries.
   6. High and low temperatures and general weather conditions, including presence of rain or snow.
   7. Testing and inspection.
   8. Accidents.
   9. Meetings and significant decisions.
   10. Unusual events.
   11. Stoppages, delays, shortages, and losses.
   12. Meter readings and similar recordings.
   13. Emergency procedures.
   14. Orders and requests of authorities having jurisdiction.
   15. Change Orders received and implemented.
   16. Construction Change Directives received and implemented.
   17. Services connected and disconnected.
   18. Equipment or system tests and startups.
   19. Partial completions and occupancies.
   20. Substantial Completions authorized.

B. Material Location Reports: At weekly intervals, prepare and submit a comprehensive list of materials delivered to and stored at Project site. List shall be cumulative, showing materials previously reported plus items recently delivered. Include with list a statement of progress on and delivery dates for materials or items of equipment fabricated or stored away from Project site. Indicate the following categories for stored materials:
   1. Material stored prior to previous report and remaining in storage.
   2. Material stored prior to previous report and since removed from storage and installed.
   3. Material stored following previous report and remaining in storage.

C. Site Condition Reports: Immediately on discovery of a difference between site conditions and the Contract Documents, prepare and submit a detailed report. Submit with a Request for Information. Include a detailed description of the differing conditions, together with recommendations for changing the Contract Documents.

D. Unusual Event Reports: When an event of an unusual and significant nature occurs at Project site, whether or not related directly to the Work, prepare and submit a special report. List chain of events, persons participating, responses by Contractor's personnel, evaluation of results or effects, and similar pertinent information. Advise Owner in advance when these events are known or predictable.
   1. Submit unusual event reports directly to Owner within one day(s) of an occurrence. Distribute copies of report to parties affected by the occurrence.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 013200

SECTION 01 3233
PHOTOGRAPHIC DOCUMENTATION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for the following:
   1. Preconstruction photographs.
   2. Periodic construction photographs.
   3. Final completion construction photographs.

B. Related Requirements:
   1. Section 017700 "Closeout Procedures" for submitting photographic documentation as Project Record Documents at Project closeout.
   2. Section 017900 "Demonstration and Training" for submitting video recordings of demonstration of equipment and training of Owner's personnel.
   3. Section 024119 "Selective Demolition" for photographic documentation before selective demolition operations commence.

1.03 INFORMATIONAL SUBMITTALS

A. Key Plan: Submit key plan of Project site and building with notation of vantage points marked for location and direction of each photograph . Indicate elevation or story of construction. Include same information as corresponding photographic documentation.

B. Digital Photographs: Submit image files within three days of taking photographs.
   1. Submit photos by uploading to web-based project software site. Include copy of key plan indicating each photograph's location and direction.
   2. Identification: Provide the following information with each image description in web-based project software site:
      a. Name of Project.
      b. Date photograph was taken.
      c. Description of location, vantage point, and direction.
      d. Unique sequential identifier keyed to accompanying key plan.

1.04 QUALITY ASSURANCE

1.05 FORMATS AND MEDIA

A. Digital Photographs: Provide color images in JPG format, produced by a digital camera with minimum sensor size of 12 megapixels, and at an image resolution of not less than 3200 by 2400 pixels, and with vibration-reduction technology. Use flash in low light levels or backlit conditions.

B. Digital Images: Submit digital media as originally recorded in the digital camera, without alteration, manipulation, editing, or modifications using image-editing software.

C. Metadata: Record accurate date and time and GPS location data from camera.

D. File Names: Name media files with date Project area and sequential numbering suffix.

1.06 CONSTRUCTION PHOTOGRAPHS

A. General: Take photographs with maximum depth of field and in focus.
   1. Maintain key plan with each set of construction photographs that identifies each photographic location.

B. Preconstruction Photographs: Before commencement of demolition / starting construction, take photographs of Project site and surrounding properties, including existing items to remain during construction, from different vantage points, as directed by Engineer.
   1. Flag any excavation areas construction limits before taking construction photographs.
   2. Take 20 photographs to show existing conditions adjacent to property before starting the Work.
   3. Take 20 photographs of existing buildings either on or adjoining property to accurately record physical conditions at start of construction.
   4. Take additional photographs as required to record settlement or cracking of adjacent structures, pavements, and improvements.

C. Periodic Construction Photographs: Take a minimum of 20 photographs monthly. Select vantage points to show status of construction and progress since last photographs were taken.

D. Final Completion Construction Photographs: Take a minimum of 50 photographs after date of Substantial Completion for submission as Project Record Documents. Engineer will inform photographer of desired vantage points.

E. Additional Photographs: Engineer may request photographs in addition to periodic photographs specified.
   1. Three days' notice will be given, where feasible.
   2. In emergency situations, take additional photographs within 24 hours of request.
   3. Circumstances that could require additional photographs include, but are not limited to, the following:
      a. Substantial Completion of a major phase or component of the Work.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 013233

SECTION 01 3300
SUBMITTAL PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. Submittal schedule requirements.
   2. Administrative and procedural requirements for submittals.

B. Related Requirements:
   1. Section 012900 "Payment Procedures" for submitting Applications for Payment and the schedule of values.
   2. Section 013100 "Project Management and Coordination" for submitting coordination drawings and subcontract list and for requirements for web-based Project software.
   3. Section 013200 "Construction Progress Documentation" for submitting schedules and reports, including Contractor's construction schedule.
   4. Section 013233 "Photographic Documentation" for submitting preconstruction photographs, periodic construction photographs, and final completion construction photographs.
   5. Section 014000 "Quality Requirements" for submitting test and inspection reports, and schedule of tests and inspections.
   6. Section 017700 "Closeout Procedures" for submitting closeout submittals and maintenance material submittals.
   7. Section 017823 "Operation and Maintenance Data" for submitting operation and maintenance manuals.
   8. Section 017839 "Project Record Documents" for submitting record Drawings, record Specifications, and record Product Data.
   9. Section 017900 "Demonstration and Training" for submitting video recordings of demonstration of equipment and training of Owner's personnel.

1.03 DEFINITIONS

A. Action Submittals: Written and graphic information and physical samples that require Engineer's responsive action. Action submittals are those submittals indicated in individual Specification Sections as "action submittals."

B. Informational Submittals: Written and graphic information and physical samples that do not require Engineer's responsive action. Submittals may be rejected for not complying with requirements. Informational submittals are those submittals indicated in individual Specification Sections as "informational submittals."

1.04 SUBMITTAL SCHEDULE

A. Submittal Schedule: Submit, as an action submittal, a list of submittals, arranged in chronological order by dates required by construction schedule. Include time required for review, ordering, manufacturing, fabrication, and delivery when establishing dates. Include additional time required for making corrections or revisions to submittals noted by Engineer and additional time for handling and reviewing submittals required by those corrections.
   1. Coordinate submittal schedule with list of subcontracts, the schedule of values, and Contractor's construction schedule.
   2. Initial Submittal: Submit concurrently with startup construction schedule. Include submittals required during the first 60 days of construction. List those submittals required

to maintain orderly progress of the Work and those required early because of long lead time for manufacture or fabrication.

3. Final Submittal: Submit concurrently with the first complete submittal of Contractor's construction schedule.
    a. Submit revised submittal schedule to reflect changes in current status and timing for submittals.
4. Format: Arrange the following information in a tabular format:
    a. Scheduled date for first submittal.
    b. Specification Section number and title.
    c. Submittal Category: Action; informational.
    d. Name of subcontractor.
    e. Description of the Work covered.
    f. Scheduled date for Engineer's final release or approval.
    g. Scheduled dates for purchasing.
    h. Scheduled date of fabrication.
    i. Scheduled dates for installation.
    j. Activity or event number.

1.05 SUBMITTAL FORMATS

A. Submittal Information: Include the following information in each submittal:
    1. Project name.
    2. Date.
    3. Name of Engineer.
    4. Name of Contractor.
    5. Name of firm or entity that prepared submittal.
    6. Names of subcontractor, manufacturer, and supplier.
    7. Unique submittal number, including revision identifier. Include Specification Section number with sequential alphanumeric identifier; and alphanumeric suffix for resubmittals.
    8. Category and type of submittal.
    9. Submittal purpose and description.
    10. Number and title of Specification Section, with paragraph number and generic name for each of multiple items.
    11. Drawing number and detail references, as appropriate.
    12. Indication of full or partial submittal.
    13. Location(s) where product is to be installed, as appropriate.
    14. Other necessary identification.
    15. Remarks.
    16. Signature of transmitter.

B. Options: Identify options requiring selection by Engineer.

C. Deviations and Additional Information: On each submittal, clearly indicate deviations from requirements in the Contract Documents, including minor variations and limitations; include relevant additional information and revisions, other than those requested by Engineer on previous submittals. Indicate by highlighting on each submittal or noting on attached separate sheet.

D. PDF Submittals: Prepare submittals as PDF package, incorporating complete information into each PDF file. Name PDF file with submittal number.

E. Submittals for Web-Based Project Software: Prepare submittals as PDF files, or other format indicated by Project software website.

1.06 SUBMITTAL PROCEDURES

A. Prepare and submit submittals required by individual Specification Sections. Types of submittals are indicated in individual Specification Sections.

1. Web-Based Project Software: Prepare submittals in PDF form, and upload to web-based Project software website. Enter required data in web-based software site to fully identify submittal.
    a. Engineer will return annotated file. Contractor shall annotate and retain one copy of file as a digital Project Record Document file.

B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities.
1. Coordinate each submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.
2. Submit all submittal items required for each Specification Section concurrently unless partial submittals for portions of the Work are indicated on approved submittal schedule.
3. Submit action submittals and informational submittals required by the same Specification Section as separate packages under separate transmittals.
4. Coordinate transmittal of submittals for related parts of the Work specified in different Sections so processing will not be delayed because of need to review submittals concurrently for coordination.
    a. Engineer reserves the right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

C. Processing Time: Allow time for submittal review, including time for resubmittals, as follows. Time for review shall commence on Engineer's receipt of submittal. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing, including resubmittals.
1. Initial Review: Allow (10) consecutive working days from when submittal is recieved by design team for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required. Engineer will advise Contractor when a submittal being processed must be delayed for coordination. Additional days will be required for the review of any shop drawings and other submittals requiring review by the design team if received in quantity equal to or less than fifty (50) sheets during five (5) consecutive working days. For each sheet or other item in excess of over fifty (50) sheets received in five (5) consecutive working days, additional time will be required for review time. The Engineer will advise the Contractor of additional time required.
2. Intermediate Review: If intermediate submittal is necessary, process it in same manner as initial submittal.
3. Resubmittal Review: Allow (10) consecutive working days from when submittal is recieved by design team for review of each resubmittal.
4. Concurrent Consultant Review: Where concurrent review of submittals by Engineer's consultants, Owner, or other parties is required, allow twenty (20) working days for initial review of each submittal.
5. No extension of the Contract Time will be authorized because of:
    a. Failure to comply with approved Submittal Schedule.
    b. Failure to transmit submittals enough in advance of the Work to permit processing.

D. Resubmittals: Make resubmittals in same form and number of copies as initial submittal.
1. Note date and content of previous submittal.
2. Note date and content of revision in label or title block and clearly indicate extent of revision.
3. Resubmit submittals until they are marked with approval notation from Engineer's action stamp.
4. Resubmittals shall include complete submittal information regardless if certain items within the previous submittal did not require a resubmittal. Partial resubmittals are not acceptable. The resubmittal shall serve a complete consolidated submittal of all items being provided.

5. Engineer review of shop drawings include up to (1) resubmittal of the initial submittal. Engineer reserves the right to bill for additional review time at current hourly rates for any additonal resubmittals. Contractor shall be responsible for any additional expenses due to additional review / resubmittal efforts by the Engineer.

E. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, authorities having jurisdiction, and others as necessary for performance of construction activities. Show distribution on transmittal forms.

F. Use for Construction: Retain complete copies of submittals on Project site. Use only final action submittals that are marked with approval notation from Engineer's action stamp.

1.07 SUBMITTAL REQUIREMENTS

A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.
   1. If information must be specially prepared for submittal because standard published data are unsuitable for use, submit as Shop Drawings, not as Product Data.
   2. Mark each copy of each submittal to show which products and options are applicable.
   3. Include the following information, as applicable:
      a. Manufacturer's catalog cuts.
      b. Manufacturer's product specifications.
      c. Testing by recognized testing agency.
      d. Notation of coordination requirements.
   4. For equipment, include the following in addition to the above, as applicable:
      a. Wiring diagrams that show factory-installed wiring.
      b. Printed performance curves.
      c. Operational range diagrams.
      d. Clearances required to other construction, if not indicated on accompanying Shop Drawings.
   5. Submit Product Data before Shop Drawings, and before or concurrent with Samples.

B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data.
   1. Preparation: Fully illustrate requirements in the Contract Documents. Include the following information, as applicable:
      a. Identification of products.
      b. Schedules.
      c. Compliance with specified standards.
      d. Notation of coordination requirements.
      e. Notation of dimensions established by field measurement.
      f. Relationship and attachment to adjoining construction clearly indicated.
      g. Seal and signature of professional engineer if specified.

C. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other materials.
   1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
   2. Identification: Permanently attach label on unexposed side of Samples that includes the following:
      a. Project name and submittal number.
      b. Generic description of Sample.
      c. Product name and name of manufacturer.
      d. Sample source.
      e. Number and title of applicable Specification Section.
      f. Specification paragraph number and generic name of each item.

3. Web-Based Project Software: Prepare submittals in PDF form, and upload to web-based Project software website. Enter required data in web-based software site to fully identify submittal.
4. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout the course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
    a. Samples that may be incorporated into the Work are indicated in individual Specification Sections. Such Samples must be in an undamaged condition at time of use.
    b. Samples not incorporated into the Work, or otherwise designated as Owner's property, are the property of Contractor.
5. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing the full range of colors, textures, and patterns available.
    a. Number of Samples: Submit one full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Engineer will return submittal with options selected.

D. Product Schedule: As required in individual Specification Sections, prepare a written summary indicating types of products required for the Work and their intended location. Include the following information in tabular form:
1. Type of product. Include unique identifier for each product indicated in the Contract Documents or assigned by Contractor if none is indicated.
2. Manufacturer and product name, and model number if applicable.
3. Number and name of room or space.
4. Location within room or space.

E. Qualification Data: Prepare written information that demonstrates capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, contact information of architects and owners, and other information specified.

F. Design Data: Prepare and submit written and graphic information indicating compliance with indicated performance and design criteria in individual Specification Sections. Include list of assumptions and summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Number each page of submittal.

G. Certificates:
1. Certificates and Certifications Submittals: Submit a statement that includes signature of entity responsible for preparing certification. Certificates and certifications shall be signed by an officer or other individual authorized to sign documents on behalf of that entity. Provide a notarized signature where indicated.
2. Installer Certificates: Submit written statements on manufacturer's letterhead certifying that Installer complies with requirements in the Contract Documents and, where required, is authorized by manufacturer for this specific Project.
3. Manufacturer Certificates: Submit written statements on manufacturer's letterhead certifying that manufacturer complies with requirements in the Contract Documents. Include evidence of manufacturing experience where required.
4. Material Certificates: Submit written statements on manufacturer's letterhead certifying that material complies with requirements in the Contract Documents.
5. Product Certificates: Submit written statements on manufacturer's letterhead certifying that product complies with requirements in the Contract Documents.
6. Welding Certificates: Prepare written certification that welding procedures and personnel comply with requirements in the Contract Documents. Submit record of Welding Procedure Specification and Procedure Qualification Record on AWS forms. Include names of firms and personnel certified.

H. Test and Research Reports:

1. Material Test Reports: Submit reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in the Contract Documents.
2. Product Test Reports: Submit written reports indicating that current product produced by manufacturer complies with requirements in the Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.

1.08 DELEGATED-DESIGN SERVICES

A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.
   1. If criteria indicated are insufficient to perform services or certification required, submit a written request for additional information to Engineer.

B. Delegated-Design Services Certification: In addition to Shop Drawings, Product Data, and other required submittals, submit digitally signed PDF file paper copies of certificate, signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional.
   1. Indicate that products and systems comply with performance and design criteria in the Contract Documents. Include list of codes, loads, and other factors used in performing these services.

1.09 CONTRACTOR'S REVIEW

A. Action Submittals and Informational Submittals: Review each submittal and check for coordination with other Work of the Contract and for compliance with the Contract Documents. Note corrections and field dimensions. Mark with approval stamp before submitting to Engineer.

B. Contractor's Approval: Indicate Contractor's approval for each submittal with a uniform approval stamp indication in web-based Project software. Include name of reviewer, date of Contractor's approval, and statement certifying that submittal has been reviewed, checked, and approved for compliance with the Contract Documents.
   1. Engineer will not review submittals received from Contractor that do not have Contractor's review and approval.

1.10 ENGINEER'S ACTION

A. Action Submittals: Engineer will review each submittal, indicate corrections or revisions required, and return it.
   1. PDF Submittals: Engineer will indicate, via markup on each submittal, the appropriate action .
   2. Submittals by Web-Based Project Software: Engineer will indicate, on Project software website, the appropriate action.

B. Informational Submittals: Engineer will review each submittal and will not return it, or will return it if it does not comply with requirements. Engineer will forward each submittal to appropriate party.

C. Partial submittals prepared for a portion of the Work will be reviewed when use of partial submittals has received prior approval from Engineer.

D. Incomplete submittals are unacceptable, will be considered nonresponsive, and will be returned for resubmittal without review.

E. Engineer will return without review submittals received from sources other than Contractor.

F. Submittals not required by the Contract Documents will be returned by Engineer without action.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 013300

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SECTION 01 3516
ALTERATION PROJECT PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes special procedures for alteration work.

1.03 DEFINITIONS

A. Alteration Work: This term includes remodeling, renovation, repair, and maintenance work performed within existing spaces or on existing surfaces as part of the Project.

B. Consolidate: To strengthen loose or deteriorated materials in place.

C. Design Reference Sample: A sample that represents the Engineer's prebid selection of work to be matched; it may be existing work or work specially produced for the Project.

D. Dismantle: To remove by disassembling or detaching an item from a surface, using gentle methods and equipment to prevent damage to the item and surfaces; disposing of items unless indicated to be salvaged or reinstalled.

E. Match: To blend with adjacent construction and manifest no apparent difference in material type, species, cut, form, detail, color, grain, texture, or finish; as approved by Engineer.

F. Refinish: To remove existing finishes to base material and apply new finish to match original, or as otherwise indicated.

G. Repair: To correct damage and defects, retaining existing materials, features, and finishes. This includes patching, piecing-in, splicing, consolidating, or otherwise reinforcing or upgrading materials.

H. Replace: To remove, duplicate, and reinstall entire item with new material. The original item is the pattern for creating duplicates unless otherwise indicated.

I. Replicate: To reproduce in exact detail, materials, and finish unless otherwise indicated.

J. Reproduce: To fabricate a new item, accurate in detail to the original, and from either the same or a similar material as the original, unless otherwise indicated.

K. Retain: To keep existing items that are not to be removed or dismantled.

L. Strip: To remove existing finish down to base material unless otherwise indicated.

1.04 COORDINATION

A. Alteration Work Subschedule: A construction schedule coordinating the sequencing and scheduling of alteration work for entire Project, including each activity to be performed, and based on Contractor's Construction Schedule. Secure time commitments for performing critical construction activities from separate entities responsible for alteration work.
   1. Schedule construction operations in sequence required to obtain best Work results.
   2. Coordinate sequence of alteration work activities to accommodate the following:
      a. Owner's continuing occupancy of portions of existing building.
      b. Owner's partial occupancy of completed Work.
      c. Other known work in progress.
      d. Tests and inspections.
   3. Detail sequence of alteration work, with start and end dates.
   4. Utility Services: Indicate how long utility services will be interrupted. Coordinate shutoff, capping, and continuation of utility services.

5. Equipment Data: List gross loaded weight, axle-load distribution, and wheel-base dimension data for mobile and heavy equipment proposed for use in existing structure. Do not use such equipment without certification from Contractor's professional engineer that the structure can support the imposed loadings without damage.

B. Pedestrian and Vehicular Circulation: Coordinate alteration work with circulation patterns within Project building(s) and site. Some work is near circulation patterns and adjacent to restricted areas . Circulation patterns cannot be closed off entirely and in places can be only temporarily redirected around small areas of work. Access to restricted areas may not be obstructed. Plan and execute the Work accordingly.

1.05 PROJECT MEETINGS FOR ALTERATION WORK

A. Preliminary Conference for Alteration Work: Before starting alteration work, conduct conference at Project site .
   1. Attendees: In addition to representatives of Owner, Engineer, and Contractor, testing service representative, specialists, and chemical-cleaner manufacturer(s) shall be represented at the meeting.
   2. Agenda: Discuss items of significance that could affect progress of alteration work, including review of the following:
      a. Alteration Work Subschedule: Discuss and finalize; verify availability of materials, specialists' personnel, equipment, and facilities needed to make progress and avoid delays.
      b. Fire-prevention plan.
      c. Governing regulations.
      d. Areas where existing construction is to remain and the required protection.
      e. Hauling routes.
      f. Sequence of alteration work operations.
      g. Storage, protection, and accounting for salvaged and specially fabricated items.
      h. Existing conditions, staging, and structural loading limitations of areas where materials are stored.
      i. Qualifications of personnel assigned to alteration work and assigned duties.
      j. Requirements for extent and quality of work, tolerances, and required clearances.
      k. Embedded work such as flashings and lintels, special details, collection of waste, protection of occupants and the public, and condition of other construction that affects the Work or will affect the work.
   3. Reporting: Record conference results and distribute copies to everyone in attendance and to others affected by decisions or actions resulting from conference.

B. Coordination Meetings: Conduct coordination meetings specifically for alteration work at weekly intervals. Coordination meetings are in addition to specific meetings held for other purposes, such as progress meetings and preinstallation conferences.
   1. Attendees: In addition to representatives of Owner, Engineer, and Contractor, each specialist, supplier, installer, and other entity concerned with progress or involved in planning, coordination, or performance of alteration work activities shall be represented at these meetings. All participants at conference shall be familiar with Project and authorized to conclude matters relating to alteration work.
   2. Agenda: Review and correct or approve minutes of previous coordination meeting. Review other items of significance that could affect progress of alteration work. Include topics for discussion as appropriate to status of Project.
      a. Alteration Work Subschedule: Review progress since last coordination meeting. Determine whether each schedule item is on time, ahead of schedule, or behind schedule. Determine how construction behind schedule will be expedited with retention of quality; secure commitments from parties involved to do so. Discuss whether schedule revisions are required to ensure that current and subsequent activities are completed within the Contract Time.

b. Schedule Updating: Revise Contractor's Alteration Work Subschedule after each coordination meeting where revisions to schedule have been made or recognized. Issue revised schedule concurrently with report of each meeting.
c. Review present and future needs of each entity present, including review items listed in the "Preliminary Conference for Alteration Work" Paragraph in this article and the following:
1) Interface requirements of alteration work with other Project Work.
2) Status of submittals for alteration work.
3) Access to alteration work locations.
4) Effectiveness of fire-prevention plan.
5) Quality and work standards of alteration work.
6) Change Orders for alteration work.

3. Reporting: Record meeting results and distribute copies to everyone in attendance and to others affected by decisions or actions resulting from each meeting.

1.06 INFORMATIONAL SUBMITTALS

A. Alteration Work Subschedule:
1. Submit alteration work subschedule within seven days of date established for commencement of alteration work .

B. Preconstruction Documentation: Show preexisting conditions of adjoining construction and site improvements that are to remain, including finish surfaces, that might be misconstrued as damage caused by Contractor's alteration work operations.

C. Fire-Prevention Plan: Submit 30 days before work begins.

1.07 QUALITY ASSURANCE

A. Specialist Qualifications: An experienced firm regularly engaged in specialty work similar in nature, materials, design, and extent to alteration work as specified in each Section and that has completed a minimum of five recent projects with a record of successful in-service performance that demonstrates the firm's qualifications to perform this work.
1. Field Supervisor Qualifications: Full-time supervisors experienced in specialty work similar in nature, material, design, and extent to that indicated for this Project. Supervisors shall be on-site when specialty work begins and during its progress. Supervisors shall not be changed during Project except for causes beyond the control of the specialist firm.

B. Fire-Prevention Plan: Prepare a written plan for preventing fires during the Work, including placement of fire extinguishers, fire blankets, rag buckets, and other fire-control devices during each phase or process. Coordinate plan with Owner's fire-protection equipment and requirements. Include fire-watch personnel's training, duties, and authority to enforce fire safety.

C. Safety and Health Standard: Comply with ANSI/ASSE A10.6.

1.08 STORAGE AND HANDLING OF SALVAGED MATERIALS

A. Existing Materials to Remain: Protect construction indicated to remain against damage and soiling from construction work. Where permitted by Engineer, items may be dismantled and taken to a suitable, protected storage location during construction work and reinstalled in their original locations after alteration and other construction work in the vicinity is complete.

B. Storage: Catalog and store items within a weathertight enclosure where they are protected from moisture, weather, condensation, and freezing temperatures.
1. Identify each item for reinstallation with a nonpermanent mark to document its original location. Indicate original locations on plans, elevations, sections, or photographs by annotating the identifying marks.
2. Secure stored materials to protect from theft.
3. Control humidity so that it does not exceed 85 percent. Maintain temperatures 5 deg F or more above the dew point.

1.09 FIELD CONDITIONS

A. Survey of Existing Conditions: Record existing conditions that affect the Work by use of measured drawings preconstruction photographs and .
   1. Comply with requirements specified in Section 013233 "Photographic Documentation."

B. Discrepancies: Notify Engineer of discrepancies between existing conditions and Drawings before proceeding with removal and dismantling work.

PART 2 PRODUCTS - (NOT USED)

PART 3 EXECUTION

3.01 PROTECTION

A. Protect persons, motor vehicles, surrounding surfaces of building, building site, plants, and surrounding buildings from harm resulting from alteration work.
   1. Use only proven protection methods, appropriate to each area and surface being protected.
   2. Provide temporary barricades, barriers, and directional signage to exclude the public from areas where alteration work is being performed.
   3. Erect temporary barriers to form and maintain fire-egress routes.
   4. Erect temporary protective covers over walkways and at points of pedestrian and vehicular entrance and exit that must remain in service during alteration work.
   5. Contain dust and debris generated by alteration work, and prevent it from reaching the public or adjacent surfaces.
   6. Provide shoring, bracing, and supports as necessary. Do not overload structural elements.
   7. Protect floors and other surfaces along hauling routes from damage, wear, and staining.
   8. Provide supplemental sound-control treatment to isolate demolition work from other areas of the building.

B. Temporary Protection of Materials to Remain:
   1. Protect existing materials with temporary protections and construction. Do not remove existing materials unless otherwise indicated.
   2. Do not attach temporary protection to existing surfaces except as indicated as part of the alteration work program.

C. Comply with each product manufacturer's written instructions for protections and precautions. Protect against adverse effects of products and procedures on people and adjacent materials, components, and vegetation.

D. Existing Drains: Prior to the start of work in an area, test drainage system to ensure that it is functioning properly. Notify Engineer immediately of inadequate drainage or blockage. Do not begin work in an area until the drainage system is functioning properly.
   1. Prevent solids such as adhesive or mortar residue or other debris from entering the drainage system. Clean out drains and drain lines that become sluggish or blocked by sand or other materials resulting from alteration work.
   2. Protect drains from pollutants. Block drains or filter out sediments, allowing only clean water to pass.

3.02 PROTECTION FROM FIRE

A. General: Follow fire-prevention plan and the following:
   1. Comply with NFPA 241 requirements unless otherwise indicated. Perform duties titled "Owner's Responsibility for Fire Protection."
   2. Remove and keep area free of combustibles, including rubbish, paper, waste, and chemicals, unless necessary for the immediate work.
      a. If combustible material cannot be removed, provide fire blankets to cover such materials.

B. Heat-Generating Equipment and Combustible Materials: Comply with the following procedures while performing work with heat-generating equipment or combustible materials, including welding, torch-cutting, soldering, brazing, removing paint with heat, or other operations where open flames or implements using high heat or combustible solvents and chemicals are anticipated:
   1. Obtain Owner's approval for operations involving use of open-flame or welding or other high-heat equipment. Notify Owner at least 72 hours before each occurrence, indicating location of such work.
   2. As far as practicable, restrict heat-generating equipment to shop areas or outside the building.
   3. Do not perform work with heat-generating equipment in or near rooms or in areas where flammable liquids or explosive vapors are present or thought to be present. Use a combustible gas indicator test to ensure that the area is safe.
   4. Use fireproof baffles to prevent flames, sparks, hot gases, or other high-temperature material from reaching surrounding combustible material.
   5. Prevent the spread of sparks and particles of hot metal through open windows, doors, holes, and cracks in floors, walls, ceilings, roofs, and other openings.
   6. Fire Watch: Before working with heat-generating equipment or combustible materials, station personnel to serve as a fire watch at each location where such work is performed. Fire-watch personnel shall have the authority to enforce fire safety. Station fire watch according to NFPA 51B, NFPA 241, and as follows:
      a. Train each fire watch in the proper operation of fire-control equipment and alarms.
      b. Prohibit fire-watch personnel from other work that would be a distraction from fire-watch duties.
      c. Cease work with heat-generating equipment whenever fire-watch personnel are not present.
      d. Have fire-watch personnel perform final fire-safety inspection each day beginning no sooner than 30 minutes after conclusion of work in each area to detect hidden or smoldering fires and to ensure that proper fire prevention is maintained.
      e. Maintain fire-watch personnel at each area of Project site until 60 minutes after conclusion of daily work.

C. Fire-Control Devices: Provide and maintain fire extinguishers, fire blankets, and rag buckets for disposal of rags with combustible liquids. Maintain each as suitable for the type of fire risk in each work area. Ensure that nearby personnel and the fire-watch personnel are trained in fire-extinguisher and blanket use.

3.03 PROTECTION DURING APPLICATION OF CHEMICALS

A. Protect motor vehicles, surrounding surfaces of building, building site, plants, and surrounding buildings from harm or spillage resulting from applications of chemicals and adhesives.

B. Cover adjacent surfaces with protective materials that are proven to resist chemicals selected for Project unless chemicals being used will not damage adjacent surfaces as indicated in alteration work program. Use covering materials and masking agents that are waterproof and UV resistant and that will not stain or leave residue on surfaces to which they are applied. Apply protective materials according to manufacturer's written instructions. Do not apply liquid masking agents or adhesives to painted or porous surfaces. When no longer needed, promptly remove protective materials.

C. Do not apply chemicals during winds of sufficient force to spread them to unprotected surfaces.

D. Neutralize alkaline and acid wastes and legally dispose of off Owner's property.

E. Collect and dispose of runoff from chemical operations by legal means and in a manner that prevents soil contamination, soil erosion, undermining of paving and foundations, damage to landscaping, or water penetration into building interior.

3.04 GENERAL ALTERATION WORK

A. Have specialty work performed only by qualified specialists.

B. Ensure that supervisory personnel are present when work begins and during its progress.

C. Record existing work before each procedure (preconstruction), and record progress during the work. Use digital preconstruction documentation photographs . Comply with requirements in Section 013233 "Photographic Documentation."

D. Perform surveys of Project site as the Work progresses to detect hazards resulting from alterations.

E. Notify Engineer of visible changes in the integrity of material or components whether from environmental causes including biological attack, UV degradation, freezing, or thawing or from structural defects including cracks, movement, or distortion.
   1. Do not proceed with the work in question until directed by Engineer.

END OF SECTION 013516

SECTION 01 4000
QUALITY REQUIREMENTS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for quality assurance and quality control.

B. Testing and inspection services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with the Contract Document requirements.
   1. Specific quality-assurance and quality-control requirements for individual work results are specified in their respective Specification Sections. Requirements in individual Sections may also cover production of standard products.
   2. Specified tests, inspections, and related actions do not limit Contractor's other quality-assurance and quality-control procedures that facilitate compliance with the Contract Document requirements.
   3. Requirements for Contractor to provide quality-assurance and quality-control services required by Engineer, Owner, Commissioning Authority, or authorities having jurisdiction are not limited by provisions of this Section.
   4. Specific test and inspection requirements are not specified in this Section.

1.03 DEFINITIONS

A. Experienced: When used with an entity or individual, "experienced" unless otherwise further described means having successfully completed a minimum of five previous projects similar in nature, size, and extent to this Project; being familiar with special requirements indicated; and having complied with requirements of authorities having jurisdiction.

B. Field Quality-Control Tests: Tests and inspections that are performed on-site for installation of the Work and for completed Work.

C. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, assembly, and similar operations.
   1. Use of trade-specific terminology in referring to a trade or entity does not require that certain construction activities be performed by accredited or unionized individuals, or that requirements specified apply exclusively to specific trade(s).

D. Product Tests: Tests and inspections that are performed by a nationally recognized testing laboratory (NRTL) according to 29 CFR 1910.7, by a testing agency accredited according to NIST's National Voluntary Laboratory Accreditation Program (NVLAP), or by a testing agency qualified to conduct product testing and acceptable to authorities having jurisdiction, to establish product performance and compliance with specified requirements.

E. Source Quality-Control Tests: Tests and inspections that are performed at the source; for example, plant, mill, factory, or shop.

F. Testing Agency: An entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.

G. Quality-Assurance Services: Activities, actions, and procedures performed before and during execution of the Work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.

H. Quality-Control Services: Tests, inspections, procedures, and related actions during and after execution of the Work to evaluate that actual products incorporated into the Work and completed construction comply with requirements. Contractor's quality-control services do not include contract administration activities performed by Engineer.

1.04 DELEGATED-DESIGN SERVICES

A. Performance and Design Criteria: Where professional design services or certifications by a design professional are specifically required of Contractor by the Contract Documents, provide products and systems complying with specific performance and design criteria indicated.
   1. If criteria indicated are not sufficient to perform services or certification required, submit a written request for additional information to Engineer.

1.05 CONFLICTING REQUIREMENTS

A. Conflicting Standards and Other Requirements: If compliance with two or more standards or requirements are specified and the standards or requirements establish different or conflicting requirements for minimum quantities or quality levels, comply with the most stringent requirement. Refer conflicting requirements that are different, but apparently equal, to Engineer for direction before proceeding.

B. Minimum Quantity or Quality Levels: The quantity or quality level shown or specified shall be the minimum provided or performed. The actual installation may comply exactly with the minimum quantity or quality specified, or it may exceed the minimum within reasonable limits. To comply with these requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Engineer for a decision before proceeding.

1.06 ACTION SUBMITTALS

A. Delegated-Design Services Submittal: In addition to Shop Drawings, Product Data, and other required submittals, submit a statement signed and sealed by the responsible design professional, for each product and system specifically assigned to Contractor to be designed or certified by a design professional, indicating that the products and systems are in compliance with performance and design criteria indicated. Include list of codes, loads, and other factors used in performing these services.

1.07 INFORMATIONAL SUBMITTALS

A. Schedule of Tests and Inspections: Prepare in tabular form and include the following:
   1. Specification Section number and title.
   2. Entity responsible for performing tests and inspections.
   3. Description of test and inspection.
   4. Identification of applicable standards.
   5. Identification of test and inspection methods.
   6. Number of tests and inspections required.
   7. Time schedule or time span for tests and inspections.
   8. Requirements for obtaining samples.
   9. Unique characteristics of each quality-control service.

B. Reports: Prepare and submit certified written reports and documents as specified.

C. Permits, Licenses, and Certificates: For Owner's record, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents established for compliance with standards and regulations bearing on performance of the Work.

1.08 CONTRACTOR'S QUALITY-CONTROL PLAN

A. Quality-Control Plan, General: Submit quality-control plan within 10 days of Notice to Proceed, and not less than five days prior to preconstruction conference. Submit in format acceptable to Engineer. Identify personnel, procedures, controls, instructions, tests, records, and forms to be

used to carry out Contractor's quality-assurance and quality-control responsibilities. Coordinate with Contractor's Construction Schedule.

B. Quality-Control Personnel Qualifications: Engage qualified personnel trained and experienced in managing and executing quality-assurance and quality-control procedures similar in nature and extent to those required for Project.
   1. Project quality-control manager may also serve as Project superintendent .

C. Submittal Procedure: Describe procedures for ensuring compliance with requirements through review and management of submittal process. Indicate qualifications of personnel responsible for submittal review.

D. Testing and Inspection: In quality-control plan, include a comprehensive schedule of Work requiring testing or inspection, including the following:
   1. Contractor-performed tests and inspections including Subcontractor-performed tests and inspections. Include required tests and inspections and Contractor-elected tests and inspections. Distinguish source quality-control tests and inspections from field quality-control tests and inspections.
   2. Special inspections required by authorities having jurisdiction and indicated on the Statement of Special Inspections.
   3. Owner-performed tests and inspections indicated in the Contract Documents , including tests and inspections indicated to be performed by Commissioning Authority.

E. Continuous Inspection of Workmanship: Describe process for continuous inspection during construction to identify and correct deficiencies in workmanship in addition to testing and inspection specified. Indicate types of corrective actions to be required to bring work into compliance with standards of workmanship established by Contract requirements and approved mockups.

F. Monitoring and Documentation: Maintain testing and inspection reports including log of approved and rejected results. Include work Engineer has indicated as nonconforming or defective. Indicate corrective actions taken to bring nonconforming work into compliance with requirements. Comply with requirements of authorities having jurisdiction.

1.09 REPORTS AND DOCUMENTS

A. Test and Inspection Reports: Prepare and submit certified written reports specified in other Sections. Include the following:
   1. Date of issue.
   2. Project title and number.
   3. Name, address, telephone number, and email address of testing agency.
   4. Dates and locations of samples and tests or inspections.
   5. Names of individuals making tests and inspections.
   6. Description of the Work and test and inspection method.
   7. Identification of product and Specification Section.
   8. Complete test or inspection data.
   9. Test and inspection results and an interpretation of test results.
   10. Record of temperature and weather conditions at time of sample taking and testing and inspection.
   11. Comments or professional opinion on whether tested or inspected Work complies with the Contract Document requirements.
   12. Name and signature of laboratory inspector.
   13. Recommendations on retesting and reinspecting.

B. Manufacturer's Technical Representative's Field Reports: Prepare written information documenting manufacturer's technical representative's tests and inspections specified in other Sections. Include the following:

1. Name, address, telephone number, and email address of technical representative making report.
2. Statement on condition of substrates and their acceptability for installation of product.
3. Statement that products at Project site comply with requirements.
4. Summary of installation procedures being followed, whether they comply with requirements and, if not, what corrective action was taken.
5. Results of operational and other tests and a statement of whether observed performance complies with requirements.
6. Statement whether conditions, products, and installation will affect warranty.
7. Other required items indicated in individual Specification Sections.

C. Factory-Authorized Service Representative's Reports: Prepare written information documenting manufacturer's factory-authorized service representative's tests and inspections specified in other Sections. Include the following:
1. Name, address, telephone number, and email address of factory-authorized service representative making report.
2. Statement that equipment complies with requirements.
3. Results of operational and other tests and a statement of whether observed performance complies with requirements.
4. Statement whether conditions, products, and installation will affect warranty.
5. Other required items indicated in individual Specification Sections.

1.10 QUALITY ASSURANCE

A. General: Qualifications paragraphs in this article establish the minimum qualification levels required; individual Specification Sections specify additional requirements.

B. Manufacturer Qualifications: A firm experienced in manufacturing products or systems similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units. As applicable, procure products from manufacturers able to meet qualification requirements, warranty requirements, and technical or factory-authorized service representative requirements.

C. Fabricator Qualifications: A firm experienced in producing products similar to those indicated for this Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

D. Installer Qualifications: A firm or individual experienced in installing, erecting, applying, or assembling work similar in material, design, and extent to that indicated for this Project, whose work has resulted in construction with a record of successful in-service performance.

E. Professional Engineer Qualifications: A professional engineer who is legally qualified to practice in jurisdiction where Project is located and who is experienced in providing engineering services of the kind indicated. Engineering services are defined as those performed for installations of the system, assembly, or product that are similar in material, design, and extent to those indicated for this Project.

F. Testing Agency Qualifications: An NRTL, an NVLAP, or an independent agency with the experience and capability to conduct testing and inspection indicated, as documented according to ASTM E 329 ; and with additional qualifications specified in individual Sections; and, where required by authorities having jurisdiction, that is acceptable to authorities.

G. Manufacturer's Technical Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to observe and inspect installation of manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

H. Factory-Authorized Service Representative Qualifications: An authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of

manufacturer's products that are similar in material, design, and extent to those indicated for this Project.

1.11 QUALITY CONTROL

A. Owner Responsibilities: Where quality-control services are indicated as Owner's responsibility, Owner will engage a qualified testing agency to perform these services.
   1. Owner will furnish Contractor with names, addresses, and telephone numbers of testing agencies engaged and a description of types of testing and inspection they are engaged to perform.
   2. Costs for retesting and reinspecting construction that replaces or is necessitated by work that failed to comply with the Contract Documents will be charged to Contractor.

B. Contractor Responsibilities: Tests and inspections not explicitly assigned to Owner are Contractor's responsibility. Perform additional quality-control activities, whether specified or not, to verify and document that the Work complies with requirements.
   1. Unless otherwise indicated, provide quality-control services specified and those required by authorities having jurisdiction. Perform quality-control services required of Contractor by authorities having jurisdiction, whether specified or not.
   2. Engage a qualified testing agency to perform quality-control services.
   3. Contractor shall not employ same entity engaged by Owner, unless agreed to in writing by Owner.
   4. Notify testing agencies at least 24 hours in advance of time when Work that requires testing or inspection will be performed.
   5. Where quality-control services are indicated as Contractor's responsibility, submit a certified written report, in duplicate, of each quality-control service.
   6. Testing and inspection requested by Contractor and not required by the Contract Documents are Contractor's responsibility.
   7. Submit additional copies of each written report directly to authorities having jurisdiction, when they so direct.

C. Retesting/Reinspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and reinspecting, for construction that replaced Work that failed to comply with the Contract Documents.

D. Testing Agency Responsibilities: Cooperate with Engineer , Commissioning Authority and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.
   1. Notify Engineer , Commissioning Authority, and Contractor promptly of irregularities or deficiencies observed in the Work during performance of its services.
   2. Determine the locations from which test samples will be taken and in which in-situ tests are conducted.
   3. Conduct and interpret tests and inspections and state in each report whether tested and inspected work complies with or deviates from requirements.
   4. Submit a certified written report, in duplicate, of each test, inspection, and similar quality-control service through Contractor.
   5. Do not release, revoke, alter, or increase the Contract Document requirements or approve or accept any portion of the Work.
   6. Do not perform duties of Contractor.

E. Manufacturer's Field Services: Where indicated, engage a factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing as specified in Section 013300 "Submittal Procedures."

F. Manufacturer's Technical Services: Where indicated, engage a manufacturer's technical representative to observe and inspect the Work. Manufacturer's technical representative's services include participation in preinstallation conferences, examination of substrates and

conditions, verification of materials, observation of Installer activities, inspection of completed portions of the Work, and submittal of written reports.

G. Associated Contractor Services: Cooperate with agencies and representatives performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide the following:
   1. Access to the Work.
   2. Incidental labor and facilities necessary to facilitate tests and inspections.
   3. Adequate quantities of representative samples of materials that require testing and inspection. Assist agency in obtaining samples.
   4. Facilities for storage and field curing of test samples.
   5. Delivery of samples to testing agencies.
   6. Preliminary design mix proposed for use for material mixes that require control by testing agency.
   7. Security and protection for samples and for testing and inspection equipment at Project site.

H. Coordination: Coordinate sequence of activities to accommodate required quality-assurance and quality-control services with a minimum of delay and to avoid necessity of removing and replacing construction to accommodate testing and inspection.
   1. Schedule times for tests, inspections, obtaining samples, and similar activities.

I. Schedule of Tests and Inspections: Prepare a schedule of tests, inspections, and similar quality-control services required by the Contract Documents. Coordinate and submit concurrently with Contractor's Construction Schedule. Update as the Work progresses.
   1. Distribution: Distribute schedule to Owner, Engineer, Commissioning Authority, testing agencies, and each party involved in performance of portions of the Work where tests and inspections are required.

1.12 SPECIAL TESTS AND INSPECTIONS

A. Special Tests and Inspections: Conducted by a qualified testing agency as required by authorities having jurisdiction, as indicated in individual Specification Sections, and as follows:
   1. Verifying that manufacturer maintains detailed fabrication and quality-control procedures and reviewing the completeness and adequacy of those procedures to perform the Work.
   2. Notifying Engineer , Commissioning Authority, and Contractor promptly of irregularities and deficiencies observed in the Work during performance of its services.
   3. Submitting a certified written report of each test, inspection, and similar quality-control service to Engineer and Commissioning Authority with copy to Contractor and to authorities having jurisdiction.
   4. Submitting a final report of special tests and inspections at Substantial Completion, which includes a list of unresolved deficiencies.
   5. Interpreting tests and inspections and stating in each report whether tested and inspected work complies with or deviates from the Contract Documents.
   6. Retesting and reinspecting corrected work.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.01 TEST AND INSPECTION LOG

A. Test and Inspection Log: Prepare a record of tests and inspections. Include the following:
   1. Date test or inspection was conducted.
   2. Description of the Work tested or inspected.
   3. Date test or inspection results were transmitted to Engineer.
   4. Identification of testing agency or special inspector conducting test or inspection.

B. Maintain log at Project site. Post changes and revisions as they occur. Provide access to test and inspection log for Engineer's, Commissioning Authority's, reference during normal working hours.
   1. Submit log at Project closeout as part of Project Record Documents.

3.02 REPAIR AND PROTECTION

A. General: On completion of testing, inspection, sample taking, and similar services, repair damaged construction and restore substrates and finishes.
   1. Provide materials and comply with installation requirements specified in other Specification Sections or matching existing substrates and finishes. Restore patched areas and extend restoration into adjoining areas with durable seams that are as invisible as possible. Comply with the Contract Document requirements for cutting and patching in Section 017300 "Execution."

B. Protect construction exposed by or for quality-control service activities.

C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

END OF SECTION 014000

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SECTION 01 4200
REFERENCES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, Division 01 Specification Sections, apply to this Section.

1.02 DEFINITIONS

A. General: Basic Contract definitions are included in the Conditions of the Contract.

B. "Approved": When used to convey Engineer's action on Contractor's submittals, applications, and requests, "approved" is limited to Engineer's duties and responsibilities as stated in the Conditions of the Contract.

C. "Directed": A command or instruction by Engineer. Other terms including "requested," "authorized," "selected," "required," and "permitted" have the same meaning as "directed."

D. "Indicated": Requirements expressed by graphic representations or in written form on Drawings, in Specifications, and in other Contract Documents. Other terms including "shown," "noted," "scheduled," and "specified" have the same meaning as "indicated."

E. "Regulations": Laws, ordinances, statutes, and lawful orders issued by authorities having jurisdiction, and rules, conventions, and agreements within the construction industry that control performance of the Work.

F. "Furnish": Supply and deliver to Project site, ready for unloading, unpacking, assembly, installation, and similar operations.

G. "Install": Unload, temporarily store, unpack, assemble, erect, place, anchor, apply, work to dimension, finish, cure, protect, clean, and similar operations at Project site.

H. "Provide": Furnish and install, complete and ready for the intended use.

I. "Project Site": Space available for performing construction activities. The extent of Project site is shown on Drawings and may or may not be identical with the description of the land on which Project is to be built.

1.03 INDUSTRY STANDARDS

A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into the Contract Documents to the extent referenced. Such standards are made a part of the Contract Documents by reference.

B. Publication Dates: Comply with standards in effect as of date of the Contract Documents unless otherwise indicated.

1.04 ABBREVIATIONS AND ACRONYMS

A. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities indicated in Gale's "Encyclopedia of Associations: National Organizations of the U.S." or in Columbia Books' "National Trade & Professional Associations of the United States."

B. Code Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in the following list. This information is believed to be accurate as of the date of the Contract Documents.

1. DIN - Deutsches Institut fur Normung e.V.; www.din.de.
2. IAPMO - International Association of Plumbing and Mechanical Officials; www.iapmo.org.
3. ICC - International Code Council; www.iccsafe.org.
4. ICC-ES - ICC Evaluation Service, LLC; www.icc-es.org.

C. Standards and Regulations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the standards and regulations in the following list. This information is subject to change and is believed to be accurate as of the date of the Contract Documents.
   1. CFR - Code of Federal Regulations; Available from Government Printing Office; www.gpo.gov/fdsys.
   2. DOD - Department of Defense; Military Specifications and Standards; Available from DLA Document Services; www.quicksearch.dla.mil.
   3. DSCC - Defense Supply Center Columbus; (See FS).
   4. FED-STD - Federal Standard; (See FS).
   5. FS - Federal Specification; Available from DLA Document Services; www.quicksearch.dla.mil.
      a. Available from Defense Standardization Program; www.dsp.dla.mil.
      b. Available from General Services Administration; www.gsa.gov.
      c. Available from National Institute of Building Sciences/Whole Building Design Guide; www.wbdg.org.
   6. MILSPEC - Military Specification and Standards; (See DOD).
   7. USAB - United States Access Board; www.access-board.gov.
   8. USATBCB - U.S. Architectural & Transportation Barriers Compliance Board; (See USAB).

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION 014200

SECTION 01 5000
TEMPORARY FACILITIES AND CONTROLS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

B. Related Requirements:
   1. Section 011000 "Summary" for work restrictions and limitations on utility interruptions.

1.03 USE CHARGES

A. General: Installation and removal of and use charges for temporary facilities shall be included in the Contract Sum unless otherwise indicated. Allow other entities engaged in the Project to use temporary services and facilities without cost, including, but not limited to, Engineer, testing agencies, and authorities having jurisdiction.

B. Water and Sewer Service from Existing System: Water from Owner's existing water system is available for use without metering and without payment of use charges. Provide connections and extensions of services as required for construction operations.

C. Electric Power Service from Existing System: Electric power from Owner's existing system is available for use without metering and without payment of use charges. Provide connections and extensions of services as required for construction operations.

1.04 INFORMATIONAL SUBMITTALS

A. Fire-Safety Program: Show compliance with requirements of NFPA 241 and authorities having jurisdiction. Indicate Contractor personnel responsible for management of fire-prevention program.

B. Dust- and HVAC-Control Plan: Submit coordination drawing and narrative that indicates the dust- and HVAC-control measures proposed for use, proposed locations, and proposed time frame for their operation. Include the following:
   1. Locations of dust-control partitions at each phase of work.
   2. HVAC system isolation schematic drawing.
   3. Location of proposed air-filtration system discharge.
   4. Waste-handling procedures.
   5. Other dust-control measures.

1.05 QUALITY ASSURANCE

A. Electric Service: Comply with NECA, NEMA, and UL standards and regulations for temporary electric service. Install service to comply with NFPA 70.

B. Tests and Inspections: Arrange for authorities having jurisdiction to test and inspect each temporary utility before use. Obtain required certifications and permits.

C. Accessible Temporary Egress: Comply with applicable provisions in the United States Access Board's ADA-ABA Accessibility Guidelines and ICC/ANSI A117.1.

1.06 PROJECT CONDITIONS

A. Temporary Use of Permanent Facilities: Engage Installer of each permanent service to assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before Owner's acceptance, regardless of previously assigned responsibilities.

PART 2 PRODUCTS

2.01 MATERIALS

A. Polyethylene Sheet: Reinforced, fire-resistive sheet, 10-mil minimum thickness, with flame-spread rating of 15 or less per ASTM E 84 and passing NFPA 701 Test Method 2.

B. Dust-Control Adhesive-Surface Walk-Off Mats: Provide mats minimum 36 by 60 inches .

C. Insulation: Unfaced mineral-fiber blanket, manufactured from glass, slag wool, or rock wool; with maximum flame-spread and smoke-developed indexes of 25 and 50, respectively.

2.02 TEMPORARY SANITARY FACILITIES

A. Maintain daily in clean and sanitary condition.

2.03 EQUIPMENT

A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.

PART 3 EXECUTION

3.01 TEMPORARY FACILITIES, GENERAL

A. Conservation: Coordinate construction and use of temporary facilities with consideration given to conservation of energy, water, and materials. Coordinate use of temporary utilities to minimize waste.

1. Salvage materials and equipment involved in performance of, but not actually incorporated into, the Work. See other Sections for disposition of salvaged materials that are designated as Owner's property.

3.02 INSTALLATION, GENERAL

A. Locate facilities where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work.

1. Locate facilities to limit site disturbance as specified in Section 011000 "Summary."

B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.03 SUPPORT FACILITIES INSTALLATION

A. General: Comply with the following:

1. Provide construction for temporary offices, shops, and sheds located within construction area or within 30 feet of building lines that is noncombustible according to ASTM E 136. Comply with NFPA 241.
2. Maintain support facilities until Engineer schedules Substantial Completion inspection. Remove before Substantial Completion. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Owner.

B. Parking: On grade parking is available.

3.04 SECURITY AND PROTECTION FACILITIES INSTALLATION

A. Protection of Existing Facilities: Protect existing vegetation, equipment, structures, utilities, and other improvements at Project site and on adjacent properties, except those indicated to be removed or altered. Repair damage to existing facilities.

1. Where access to adjacent properties is required in order to affect protection of existing facilities, obtain written permission from adjacent property owner to access property for that purpose.

B. Temporary Egress: Maintain temporary egress from existing occupied facilities as required by authorities having jurisdiction.

3.05 OPERATION, TERMINATION, AND REMOVAL

A. Supervision: Enforce strict discipline in use of temporary facilities. To minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.

B. Maintenance: Maintain facilities in good operating condition until removal.
   1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on a 24-hour basis where required to achieve indicated results and to avoid possibility of damage.

C. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.
   1. Materials and facilities that constitute temporary facilities are property of Contractor. Owner reserves right to take possession of Project identification signs.
   2. At Substantial Completion, repair, renovate, and clean permanent facilities used during construction period. Comply with final cleaning requirements specified in Section 017700 "Closeout Procedures."

END OF SECTION 015000

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SECTION 01 6000
PRODUCT REQUIREMENTS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and comparable products.

B. Related Requirements:
   1. Section 012500 "Substitution Procedures" for requests for substitutions.
   2. Section 014200 "References" for applicable industry standards for products specified.

1.03 DEFINITIONS

A. Products: Items obtained for incorporating into the Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes the terms "material," "equipment," "system," and terms of similar intent.
   1. Named Products: Items identified by manufacturer's product name, including make or model number or other designation shown or listed in manufacturer's published product literature that is current as of date of the Contract Documents.
   2. New Products: Items that have not previously been incorporated into another project or facility. Products salvaged or recycled from other projects are not considered new products.
   3. Comparable Product: Product that is demonstrated and approved by Engineer through submittal process to have the indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.

B. Basis-of-Design Product Specification: A specification in which a single manufacturer's product is named and accompanied by the words "basis-of-design product," including make or model number or other designation. In addition to the basis-of-design product description, product attributes and characteristics may be listed to establish the significant qualities related to type, function, in-service performance and physical properties, weight, dimension, durability, visual characteristics, and other special features and requirements for purposes of evaluating comparable products of additional manufacturers named in the specification.

C. Subject to Compliance with Requirements: Where the phrase "Subject to compliance with requirements" introduces a product selection procedure in an individual Specification Section, provide products qualified under the specified product procedure. In the event that a named product or product by a named manufacturer does not meet the other requirements of the specifications, select another named product or product from another named manufacturer that does meet the requirements of the specifications. Submit a comparable product request, if applicable.

1.04 ACTION SUBMITTALS

A. Comparable Product Request Submittal: Submit request for consideration of each comparable product. Identify basis-of-design product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.
   1. Include data to indicate compliance with the requirements specified in "Comparable Products" Article.
   2. Engineer's Action: If necessary, Engineer will request additional information or documentation for evaluation within seven days of receipt of a comparable product

request. Engineer will notify Contractor of approval or rejection of proposed comparable product request within 15 days of receipt of request, or seven days of receipt of additional information or documentation, whichever is later.

a. Form of Engineer's Approval of Submittal: As specified in Section 013300 "Submittal Procedures."
b. Use product specified if Engineer does not issue a decision on use of a comparable product request within time allocated.

B. Basis-of-Design Product Specification Submittal: Comply with requirements in Section 013300 "Submittal Procedures." Show compliance with requirements.

1.05 QUALITY ASSURANCE

A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, select product compatible with products previously selected, even if previously selected products were also options.

B. Identification of Products: Except for required labels and operating data, do not attach or imprint manufacturer or product names or trademarks on exposed surfaces of products or equipment that will be exposed to view in occupied spaces or on the exterior.

1. Labels: Locate required product labels and stamps on a concealed surface, or, where required for observation following installation, on a visually accessible surface that is not conspicuous.
2. Equipment Nameplates: Provide a permanent nameplate on each item of service-connected or power-operated equipment. Locate on a visually accessible but inconspicuous surface. Include information essential for operation, including the following:
   a. Name of product and manufacturer.
   b. Model and serial number.
   c. Capacity.
   d. Speed.
   e. Ratings.
3. See individual identification sections in Divisions 23 and 26 for additional identification requirements.

1.06 PRODUCT DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, and handle products using means and methods that will prevent damage, deterioration, and loss, including theft and vandalism. Comply with manufacturer's written instructions.

B. Delivery and Handling:

1. Schedule delivery to minimize long-term storage at Project site and to prevent overcrowding of construction spaces.
2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
3. Deliver products to Project site in an undamaged condition in manufacturer's original sealed container or other packaging system, complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
4. Inspect products on delivery to determine compliance with the Contract Documents and to determine that products are undamaged and properly protected.

C. Storage:

1. Store products to allow for inspection and measurement of quantity or counting of units.
2. Store products that are subject to damage by the elements, under cover in a weathertight enclosure above ground, with ventilation adequate to prevent condensation.
3. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.

4. Protect stored products from damage and liquids from freezing.

1.07 PRODUCT WARRANTIES

A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by the Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of the Contract Documents.
   1. Manufacturer's Warranty: Written warranty furnished by individual manufacturer for a particular product and specifically endorsed by manufacturer to Owner.
   2. Special Warranty: Written warranty required by the Contract Documents to provide specific rights for Owner.

B. Special Warranties: Prepare a written document that contains appropriate terms and identification, ready for execution.
   1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
   2. See other Sections for specific content requirements and particular requirements for submitting special warranties.

C. Submittal Time: Comply with requirements in Section 017700 "Closeout Procedures."

PART 2 PRODUCTS

2.01 PRODUCT SELECTION PROCEDURES

A. General Product Requirements: Provide products that comply with the Contract Documents, are undamaged and, unless otherwise indicated, are new at time of installation.
   1. Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.
   2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types that have been produced and used successfully in similar situations on other projects.
   3. Owner reserves the right to limit selection to products with warranties meeting requirements of the Contract Documents.
   4. Where products are accompanied by the term "as selected," Engineer will make selection.
   5. Descriptive, performance, and reference standard requirements in the Specifications establish salient characteristics of products.
   6. Or Equal: For products specified by name and accompanied by the term "or equal," or "or approved equal," or "or approved," comply with requirements in "Comparable Products" Article to obtain approval for use of an unnamed product.
      a. Submit additional documentation required by Engineer in order to establish equivalency of proposed products. Evaluation of "or equal" product status is by the Engineer, whose determination is final.

B. Product Selection Procedures:
   1. Sole Product: Where Specifications name a single manufacturer and product, provide the named product that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
      a. Sole product may be indicated by the phrase: "Subject to compliance with requirements, provide the following: …"
   2. Sole Manufacturer/Source: Where Specifications name a single manufacturer or source, provide a product by the named manufacturer or source that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered.
      a. Sole manufacturer/source may be indicated by the phrase: "Subject to compliance with requirements, provide products by the following: …"
   3. Limited List of Products: Where Specifications include a list of names of both manufacturers and products, provide one of the products listed that complies with

requirements. Comparable products or substitutions for Contractor's convenience will not be considered unless otherwise indicated.

a. Limited list of products may be indicated by the phrase: "Subject to compliance with requirements, provide one of the following: …"

4. Non-Limited List of Products: Where Specifications include a list of names of both available manufacturers and products, provide one of the products listed, or an unnamed product, which complies with requirements.
   a. Non-limited list of products is indicated by the phrase: "Subject to compliance with requirements, available products that may be incorporated in the Work include, but are not limited to, the following: …"
5. Limited List of Manufacturers: Where Specifications include a list of manufacturers' names, provide a product by one of the manufacturers listed that complies with requirements. Comparable products or substitutions for Contractor's convenience will not be considered unless otherwise indicated.
   a. Limited list of manufacturers is indicated by the phrase: "Subject to compliance with requirements, provide products by one of the following: …"
6. Non-Limited List of Manufacturers: Where Specifications include a list of available manufacturers, provide a product by one of the manufacturers listed, or a product by an unnamed manufacturer, which complies with requirements.
   a. Non-limited list of manufacturers is indicated by the phrase: "Subject to compliance with requirements, available manufacturers whose products may be incorporated in the Work include, but are not limited to, the following: …"
7. Basis-of-Design Product: Where Specifications name a product, or refer to a product indicated on Drawings, and include a list of manufacturers, provide the specified or indicated product or a comparable product by one of the other named manufacturers. Drawings and Specifications indicate sizes, profiles, dimensions, and other characteristics that are based on the product named. Comply with requirements in "Comparable Products" Article for consideration of an unnamed product by one of the other named manufacturers.
   a. For approval of products by unnamed manufacturers, comply with requirements in Section 012500 "Substitution Procedures" for substitutions for convenience.

C. Visual Matching Specification: Where Specifications require "match Engineer's sample," provide a product that complies with requirements and matches Engineer's sample. Engineer's decision will be final on whether a proposed product matches.
   1. If no product available within specified category matches and complies with other specified requirements, comply with requirements in Section 012500 "Substitution Procedures" for proposal of product.

D. Visual Selection Specification: Where Specifications include the phrase "as selected by Engineer from manufacturer's full range" or similar phrase, select a product that complies with requirements. Engineer will select color, gloss, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.02 COMPARABLE PRODUCTS

A. Conditions for Consideration of Comparable Products: Engineer will consider Contractor's request for comparable product when the following conditions are satisfied. If the following conditions are not satisfied, Engineer may return requests without action, except to record noncompliance with these requirements:
   1. Evidence that proposed product does not require revisions to the Contract Documents, is consistent with the Contract Documents, will produce the indicated results, and is compatible with other portions of the Work. Detailed comparison of significant qualities of proposed product with those named in the Specifications. Significant product qualities include attributes such as type, function, in-service performance and physical properties,

weight, dimension, durability, visual characteristics, and other specific features and requirements.

2. Evidence that proposed product provides specified warranty.
3. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.
4. Samples, if requested.

B. Submittal Requirements: Approval by the Engineer of Contractor's request for use of comparable product is not intended to satisfy other submittal requirements. Comply with specified submittal requirements.

PART 3 EXECUTION (NOT USED)

END OF SECTION 016000

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SECTION 01 7300
EXECUTION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes general administrative and procedural requirements governing execution of the Work including, but not limited to, the following:
   1. Construction layout.
   2. Field engineering and surveying.
   3. Installation of the Work.
   4. Cutting and patching.
   5. Coordination of Owner-installed products.
   6. Progress cleaning.
   7. Starting and adjusting.
   8. Protection of installed construction.

B. Related Requirements:
   1. Section 011000 "Summary" for limits on use of Project site.
   2. Section 013300 "Submittal Procedures" for submitting surveys.
   3. Section 024119 "Selective Demolition" for demolition and removal of selected portions of the building.

1.03 DEFINITIONS

A. Cutting: Removal of in-place construction necessary to permit installation or performance of subsequent work.

B. Patching: Fitting and repair work required to restore construction to original conditions after installation of subsequent work.

1.04 PREINSTALLATION MEETINGS

A. Cutting and Patching Conference: Conduct conference at Project site.
   1. Prior to submitting cutting and patching plan, review extent of cutting and patching anticipated and examine procedures for ensuring satisfactory result from cutting and patching work. Require representatives of each entity directly concerned with cutting and patching to attend, including the following:
      a. Contractor's superintendent.
      b. Trade supervisor responsible for cutting operations.
      c. Trade supervisor(s) responsible for patching of each type of substrate.
      d. Mechanical, electrical, and utilities subcontractors' supervisors, to the extent each trade is affecting by cutting and patching operations.
   2. Review areas of potential interference and conflict. Coordinate procedures and resolve potential conflicts before proceeding.

1.05 INFORMATIONAL SUBMITTALS

A. Certificates: Submit certificate signed by land surveyor certifying that location and elevation of improvements comply with requirements.

B. Cutting and Patching Plan: Submit plan describing procedures at least 10 days prior to the time cutting and patching will be performed. Include the following information:
   1. Extent: Describe reason for and extent of each occurrence of cutting and patching.

2. Changes to In-Place Construction: Describe anticipated results. Include changes to structural elements and operating components as well as changes in building appearance and other significant visual elements.
3. Products: List products to be used for patching and firms or entities that will perform patching work.
4. Dates: Indicate when cutting and patching will be performed.
5. Utilities and Mechanical and Electrical Systems: List services and systems that cutting and patching procedures will disturb or affect. List services and systems that will be relocated and those that will be temporarily out of service. Indicate length of time permanent services and systems will be disrupted.
    a. Include description of provisions for temporary services and systems during interruption of permanent services and systems.

1.06 QUALITY ASSURANCE

A. Cutting and Patching: Comply with requirements for and limitations on cutting and patching of construction elements.
    1. Structural Elements: When cutting and patching structural elements, notify Engineer of locations and details of cutting and await directions from Engineer before proceeding. Shore, brace, and support structural elements during cutting and patching. Do not cut and patch structural elements in a manner that could change their load-carrying capacity or increase deflection.
    2. Operational Elements: Do not cut and patch operating elements and related components in a manner that results in reducing their capacity to perform as intended or that results in increased maintenance or decreased operational life or safety. Operational elements include the following:
        a. Primary operational systems and equipment.
        b. Fire separation assemblies.
        c. Air or smoke barriers.
        d. Fire-suppression systems.
        e. Plumbing piping systems.
        f. Mechanical systems piping and ducts.
        g. Control systems.
        h. Communication systems.
        i. Fire-detection and -alarm systems.
        j. Conveying systems.
        k. Electrical wiring systems.
        l. Operating systems of special construction.
    3. Other Construction Elements: Do not cut and patch other construction elements or components in a manner that could change their load-carrying capacity, that results in reducing their capacity to perform as intended, or that results in increased maintenance or decreased operational life or safety. Other construction elements include but are not limited to the following:
        a. Water, moisture, or vapor barriers.
        b. Membranes and flashings.
        c. Exterior curtain-wall construction.
        d. Sprayed fire-resistive material.
        e. Equipment supports.
        f. Piping, ductwork, vessels, and equipment.
        g. Noise- and vibration-control elements and systems.
    4. Visual Elements: Do not cut and patch construction in a manner that results in visual evidence of cutting and patching. Do not cut and patch exposed construction in a manner that would, in Engineer's opinion, reduce the building's aesthetic qualities. Remove and replace construction that has been cut and patched in a visually unsatisfactory manner.

B. Manufacturer's Installation Instructions: Obtain and maintain on-site manufacturer's written recommendations and instructions for installation of products and equipment.

PART 2 PRODUCTS

2.01 MATERIALS

A. General: Comply with requirements specified in other Sections.
   1. For projects requiring compliance with sustainable design and construction practices and procedures, use products for patching that comply with sustainable design requirements.

B. In-Place Materials: Use materials for patching identical to in-place materials. For exposed surfaces, use materials that visually match in-place adjacent surfaces to the fullest extent possible.
   1. If identical materials are unavailable or cannot be used, use materials that, when installed, will provide a match acceptable to Engineer for the visual and functional performance of in-place materials.

PART 3 EXECUTION

3.01 EXAMINATION

A. Existing Conditions: The existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify the existence and location of underground utilities, mechanical and electrical systems, and other construction affecting the Work.
   1. Before construction, verify the location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping; underground electrical services; and other utilities.
   2. Furnish location data for work related to Project that must be performed by public utilities serving Project site.

B. Examination and Acceptance of Conditions: Before proceeding with each component of the Work, examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.
   1. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.
   2. Examine walls, floors, and roofs for suitable conditions where products and systems are to be installed.
   3. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.

C. Written Report: Where a written report listing conditions detrimental to performance of the Work is required by other Sections, include the following:
   1. Description of the Work.
   2. List of detrimental conditions, including substrates.
   3. List of unacceptable installation tolerances.
   4. Recommended corrections.

D. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.02 PREPARATION

A. Existing Utility Information: Furnish information to Owner that is necessary to adjust, move, or relocate existing utility structures, utility poles, lines, services, or other utility appurtenances located in or affected by construction. Coordinate with authorities having jurisdiction.

B. Field Measurements: Take field measurements as required to fit the Work properly. Recheck measurements before installing each product. Where portions of the Work are indicated to fit to other construction, verify dimensions of other construction by field measurements before

fabrication. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

C. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.

D. Review of Contract Documents and Field Conditions: Immediately on discovery of the need for clarification of the Contract Documents caused by differing field conditions outside the control of Contractor, submit a request for information to Engineer according to requirements in Section 013100 "Project Management and Coordination."

3.03 CONSTRUCTION LAYOUT

A. Verification: Before proceeding to lay out the Work, verify layout information shown on Drawings, in relation to the property survey and existing benchmarks. If discrepancies are discovered, notify Engineer promptly.

B. General: Engage a land surveyor to lay out the Work using accepted surveying practices.
   1. Establish limits on use of Project site.
   2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
   3. Inform installers of lines and levels to which they must comply.
   4. Check the location, level and plumb, of every major element as the Work progresses.
   5. Notify Engineer when deviations from required lines and levels exceed allowable tolerances.
   6. Close site surveys with an error of closure equal to or less than the standard established by authorities having jurisdiction.

C. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and rim and invert elevations.

D. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.

E. Record Log: Maintain a log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make the log available for reference by Engineer.

3.04 FIELD ENGINEERING

A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning the Work. Preserve and protect permanent benchmarks and control points during construction operations.
   1. Do not change or relocate existing benchmarks or control points without prior written approval of Engineer. Report lost or destroyed permanent benchmarks or control points promptly. Report the need to relocate permanent benchmarks or control points to Engineer before proceeding.
   2. Replace lost or destroyed permanent benchmarks and control points promptly. Base replacements on the original survey control points.

B. Benchmarks: Establish and maintain a minimum of two permanent benchmarks on Project site, referenced to data established by survey control points. Comply with authorities having jurisdiction for type and size of benchmark.
   1. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.
   2. Where the actual location or elevation of layout points cannot be marked, provide temporary reference points sufficient to locate the Work.

3. Remove temporary reference points when no longer needed. Restore marked construction to its original condition.

3.05 INSTALLATION

A. General: Locate the Work and components of the Work accurately, in correct alignment and elevation, as indicated.
   1. Make vertical work plumb and make horizontal work level.
   2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
   3. Conceal pipes, ducts, and wiring in finished areas unless otherwise indicated.

B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.

C. Install products at the time and under conditions that will ensure the best possible results. Maintain conditions required for product performance until Substantial Completion.

D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.

E. Sequence the Work and allow adequate clearances to accommodate movement of construction items on site and placement in permanent locations.

F. Tools and Equipment: Where possible, select tools or equipment that minimize production of excessive noise levels.

G. Templates: Obtain and distribute to the parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other portions of the Work to confirm that adequate provisions are made for locating and installing products to comply with indicated requirements.

H. Attachment: Provide blocking and attachment plates and anchors and fasteners of adequate size and number to securely anchor each component in place, accurately located and aligned with other portions of the Work. Where size and type of attachments are not indicated, verify size and type required for load conditions.
   1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Engineer.
   2. Allow for building movement, including thermal expansion and contraction.
   3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

I. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for the best visual effect. Fit exposed connections together to form hairline joints.

J. Repair or remove and replace damaged, defective, or nonconforming Work.
   1. Comply with Section 017700 "Closeout Procedures" for repairing or removing and replacing defective Work.

3.06 CUTTING AND PATCHING

A. Cutting and Patching, General: Employ skilled workers to perform cutting and patching. Proceed with cutting and patching at the earliest feasible time, and complete without delay.
   1. Cut in-place construction to provide for installation of other components or performance of other construction, and subsequently patch as required to restore surfaces to their original condition.

B. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during installation or cutting and patching operations, by methods and with materials so as not to void existing warranties.

C. Temporary Support: Provide temporary support of work to be cut.

D. Protection: Protect in-place construction during cutting and patching to prevent damage. Provide protection from adverse weather conditions for portions of Project that might be exposed during cutting and patching operations.

E. Adjacent Occupied Areas: Where interference with use of adjoining areas or interruption of free passage to adjoining areas is unavoidable, coordinate cutting and patching according to requirements in Section 011000 "Summary."

F. Existing Utility Services and Mechanical/Electrical Systems: Where existing services/systems are required to be removed, relocated, or abandoned, bypass such services/systems before cutting to minimize interruption to occupied areas.

G. Cutting: Cut in-place construction by sawing, drilling, breaking, chipping, grinding, and similar operations, including excavation, using methods least likely to damage elements retained or adjoining construction. If possible, review proposed procedures with original Installer; comply with original Installer's written recommendations.
   1. In general, use hand or small power tools designed for sawing and grinding, not hammering and chopping. Cut holes and slots neatly to minimum size required, and with minimum disturbance of adjacent surfaces. Temporarily cover openings when not in use.
   2. Finished Surfaces: Cut or drill from the exposed or finished side into concealed surfaces.
   3. Concrete : Cut using a cutting machine, such as an abrasive saw or a diamond-core drill.
   4. Mechanical and Electrical Services: Cut off pipe or conduit in walls or partitions to be removed. Cap, valve, or plug and seal remaining portion of pipe or conduit to prevent entrance of moisture or other foreign matter after cutting.
   5. Proceed with patching after construction operations requiring cutting are complete.

H. Patching: Patch construction by filling, repairing, refinishing, closing up, and similar operations following performance of other work. Patch with durable seams that are as invisible as practicable. Provide materials and comply with installation requirements specified in other Sections, where applicable.
   1. Inspection: Where feasible, test and inspect patched areas after completion to demonstrate physical integrity of installation.
   2. Exposed Finishes: Restore exposed finishes of patched areas and extend finish restoration into retained adjoining construction in a manner that will minimize evidence of patching and refinishing.
      a. Clean piping, conduit, and similar features before applying paint or other finishing materials.
      b. Restore damaged pipe covering to its original condition.
   3. Floors and Walls: Where walls or partitions that are removed extend one finished area into another, patch and repair floor and wall surfaces in the new space. Provide an even surface of uniform finish, color, texture, and appearance. Remove in-place floor and wall coverings and replace with new materials, if necessary, to achieve uniform color and appearance.
      a. Where patching occurs in a painted surface, prepare substrate and apply primer and intermediate paint coats appropriate for substrate over the patch, and apply final paint coat over entire unbroken surface containing the patch. Provide additional coats until patch blends with adjacent surfaces.
   4. Ceilings: Patch, repair, or rehang in-place ceilings as necessary to provide an even-plane surface of uniform appearance.
   5. Exterior Building Enclosure: Patch components in a manner that restores enclosure to a weathertight condition and ensures thermal and moisture integrity of building enclosure.

I. Cleaning: Clean areas and spaces where cutting and patching are performed. Remove paint, mortar, oils, putty, and similar materials from adjacent finished surfaces.

3.07 PROGRESS CLEANING

A. General: Clean Project site and work areas daily, including common areas. Enforce requirements strictly. Dispose of materials lawfully.
   1. Comply with requirements in NFPA 241 for removal of combustible waste materials and debris.
   2. Do not hold waste materials more than seven days during normal weather or three days if the temperature is expected to rise above 80 deg F .
   3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.
      a. Use containers intended for holding waste materials of type to be stored.
   4. Coordinate progress cleaning for joint-use areas where Contractor and other contractors are working concurrently.

B. Site: Maintain Project site free of waste materials and debris.

C. Work Areas: Clean areas where work is in progress to the level of cleanliness necessary for proper execution of the Work.
   1. Remove liquid spills promptly.
   2. Where dust would impair proper execution of the Work, broom-clean or vacuum the entire work area, as appropriate.

D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials that are not hazardous to health or property and that will not damage exposed surfaces.

E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.

F. Exposed Surfaces in Finished Areas: Clean exposed surfaces and protect as necessary to ensure freedom from damage and deterioration at time of Substantial Completion.

G. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers or into waterways. Comply with waste disposal requirements in Section 015000 "Temporary Facilities and Controls."

H. During handling and installation, clean and protect construction in progress and adjoining materials already in place. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.

I. Clean and provide maintenance on completed construction as frequently as necessary through the remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.

J. Limiting Exposures: Supervise construction operations to ensure that no part of the construction, completed or in progress, is subject to harmful, dangerous, damaging, or otherwise deleterious exposure during the construction period.

3.08 STARTING AND ADJUSTING

A. Coordinate startup and adjusting of equipment and operating components with requirements in Section 019113 "General Commissioning Requirements."

B. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.

C. Adjust equipment for proper operation. Adjust operating components for proper operation without binding.

D. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

E. Manufacturer's Field Service: Comply with qualification requirements in Section 014000 "Quality Requirements."

3.09 PROTECTION OF INSTALLED CONSTRUCTION

A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.

B. Protection of Existing Items: Provide protection and ensure that existing items to remain undisturbed by construction are maintained in condition that existed at commencement of the Work.

C. Comply with manufacturer's written instructions for temperature and relative humidity.

END OF SECTION 017300

SECTION 01 7419
CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for the following:
   1. Salvaging nonhazardous demolition and construction waste.
   2. Disposing of nonhazardous demolition and construction waste.

1.03 DEFINITIONS

A. Construction Waste: Building, structure, and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.

B. Demolition Waste: Building, structure, and site improvement materials resulting from demolition operations.

C. Disposal: Removal of demolition or construction waste and subsequent salvage, sale, recycling, or deposit in landfill, incinerator acceptable to authorities having jurisdiction, or designated spoil areas on Owner’s property.

D. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.

E. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.

F. Salvage and Reuse: Recovery of demolition or construction waste and subsequent incorporation into the Work.

1.04 MATERIALS OWNERSHIP

A. Unless otherwise indicated, demolition and construction waste becomes property of Contractor.

B. Historic items, relics, antiques, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, and other items of interest or value to Owner that may be uncovered during demolition remain the property of Owner.
   1. Carefully salvage in a manner to prevent damage and promptly return to Owner.

1.05 QUALITY ASSURANCE

A. Regulatory Requirements: Comply with transportation and disposal regulations of authorities having jurisdiction.

PART 2 PRODUCTS

PART 3 EXECUTION

3.01 SALVAGING DEMOLITION WASTE

A. Comply with requirements in Section 024119 "Selective Demolition" for salvaging demolition waste.

B. Salvaged Items for Reuse in the Work: Salvage items for reuse and handle as follows:
   1. Clean salvaged items.
   2. Pack or crate items after cleaning. Identify contents of containers with label indicating elements, date of removal, quantity, and location where removed.
   3. Store items in a secure area until installation.
   4. Protect items from damage during transport and storage.

5. Install salvaged items to comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make items functional for use indicated.

C. Salvaged Items for Owner's Use: Salvage items for Owner's use and handle as follows:
   1. Clean salvaged items.
   2. Pack or crate items after cleaning. Identify contents of containers with label indicating elements, date of removal, quantity, and location where removed.
   3. Store items in a secure area until delivery to Owner.
   4. Transport items to Owner's storage area on-site designated by Owner.
   5. Protect items from damage during transport and storage.

D. Equipment: Drain tanks, piping, and fixtures. Seal openings with caps or plugs. Protect equipment from exposure to weather.

E. Lighting Fixtures: Separate lamps by type and protect from breakage.

F. Electrical Devices: Separate switches, receptacles, switchgear, transformers, meters, panelboards, circuit breakers, and other devices by type.

3.02 DISPOSAL OF WASTE

A. General: Except for items or materials to be salvaged or recycled, remove waste materials from Project site and legally dispose of them in a landfill or incinerator acceptable to authorities having jurisdiction.
   1. Except as otherwise specified, do not allow waste materials that are to be disposed of accumulate on-site.
   2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.

B. General: Except for items or materials to be salvaged or recycled, remove waste materials and legally dispose of at designated spoil areas on Owner’s property.

C. Burning: Do not burn waste materials.

END OF SECTION 017419

SECTION 01 7700
CLOSEOUT PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for contract closeout, including, but not limited to, the following:
   1. Substantial Completion procedures.
   2. Final completion procedures.
   3. Warranties.
   4. Final cleaning.
   5. Repair of the Work.

B. Related Requirements:
   1. Section 013233 "Photographic Documentation" for submitting final completion construction photographic documentation.
   2. Section 017823 "Operation and Maintenance Data" for additional operation and maintenance manual requirements.
   3. Section 017839 "Project Record Documents" for submitting Record Drawings, Record Specifications, and Record Product Data.
   4. Section 017900 "Demonstration and Training" for requirements to train the Owner's maintenance personnel to adjust, operate, and maintain products, equipment, and systems.

1.03 ACTION SUBMITTALS

A. Product Data: For each type of cleaning agent.

B. Contractor's List of Incomplete Items: Initial submittal at Substantial Completion.

C. Certified List of Incomplete Items: Final submittal at final completion.

1.04 CLOSEOUT SUBMITTALS

A. Certificates of Release: From authorities having jurisdiction.

B. Certificate of Insurance: For continuing coverage.

C. Field Report: For pest control inspection.

1.05 MAINTENANCE MATERIAL SUBMITTALS

A. Schedule of Maintenance Material Items: For maintenance material submittal items specified in other Sections.

1.06 SUBSTANTIAL COMPLETION PROCEDURES

A. Contractor's List of Incomplete Items: Prepare and submit a list of items to be completed and corrected (Contractor's punch list), indicating the value of each item on the list and reasons why the Work is incomplete.

B. Submittals Prior to Substantial Completion: Complete the following a minimum of days prior to requesting inspection for determining date of Substantial Completion. List items below that are incomplete at time of request.
   1. Certificates of Release: Obtain and submit releases from authorities having jurisdiction permitting Owner unrestricted use of the Work and access to services and utilities. Include occupancy permits, operating certificates, and similar releases.

2. Submit closeout submittals specified in other Division 01 Sections, including project record documents, operation and maintenance manuals, damage or settlement surveys, property surveys, and similar final record information.
3. Submit closeout submittals specified in individual Sections, including specific warranties, workmanship bonds, maintenance service agreements, final certifications, and similar documents.
4. Submit maintenance material submittals specified in individual Sections, including tools, spare parts, extra materials, and similar items, and deliver to location designated by Engineer. Label with manufacturer's name and model number.
    a. Schedule of Maintenance Material Items: Prepare and submit schedule of maintenance material submittal items, including name and quantity of each item and name and number of related Specification Section. Obtain Engineer's signature for receipt of submittals.
5. Submit testing, adjusting, and balancing records.
6. Submit sustainable design submittals not previously submitted.
7. Submit changeover information related to Owner's occupancy, use, operation, and maintenance.

C. Procedures Prior to Substantial Completion: Complete the following a minimum of 10 days prior to requesting inspection for determining date of Substantial Completion. List items below that are incomplete at time of request.
1. Advise Owner of pending insurance changeover requirements.
2. Make final changeover of permanent locks and deliver keys to Owner. Advise Owner's personnel of changeover in security provisions.
3. Complete startup and testing of systems and equipment.
4. Perform preventive maintenance on equipment used prior to Substantial Completion.
5. Instruct Owner's personnel in operation, adjustment, and maintenance of products, equipment, and systems. Submit demonstration and training video recordings specified in Section 017900 "Demonstration and Training."
6. Advise Owner of changeover in utility services.
7. Participate with Owner in conducting inspection and walkthrough with local emergency responders.
8. Terminate and remove temporary facilities from Project site, along with mockups, construction tools, and similar elements.
9. Complete final cleaning requirements.
10. Touch up paint and otherwise repair and restore marred exposed finishes to eliminate visual defects.

D. Inspection: Submit a written request for inspection to determine Substantial Completion a minimum of 10 days prior to date the Work will be completed and ready for final inspection and tests. On receipt of request, Engineer either proceed with inspection or notify Contractor of unfulfilled requirements. Owner will prepare the Certificate of Substantial Completion after inspection or will notify Contractor of items, either on Contractor's list or additional items identified by Engineer, that must be completed or corrected before certificate will be issued.
1. Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.
2. Results of completed inspection will form the basis of requirements for final completion.

1.07 FINAL COMPLETION PROCEDURES

A. Submittals Prior to Final Completion: Before requesting final inspection for determining final completion, complete the following:
1. Submit a final Application for Payment according to Section 012900 "Payment Procedures."
2. Certified List of Incomplete Items: Submit certified copy of Engineer's Substantial Completion inspection list of items to be completed or corrected (punch list), endorsed and

dated by Engineer. Certified copy of the list shall state that each item has been completed or otherwise resolved for acceptance.

3. Certificate of Insurance: Submit evidence of final, continuing insurance coverage complying with insurance requirements.
4. Submit pest-control final inspection report.
5. Submit final completion photographic documentation.

B. Inspection: Submit a written request for final inspection to determine acceptance a minimum of 10 days prior to date the work will be completed and ready for final inspection and tests. On receipt of request, Engineer will either proceed with inspection or notify Contractor of unfulfilled requirements. Engineer will prepare a final Certificate for Payment after inspection or will notify Contractor of construction that must be completed or corrected before certificate will be issued.

1. Request reinspection when the Work identified in previous inspections as incomplete is completed or corrected.

1.08 LIST OF INCOMPLETE ITEMS (PUNCH LIST)

A. Organization of List: Include name and identification of each space and area affected by construction operations for incomplete items and items needing correction including, if necessary, areas disturbed by Contractor that are outside the limits of construction.

1. Organize list of spaces in sequential order, starting with exterior areas first.
2. Organize items applying to each space by major element, including categories for ceiling, individual walls, floors, equipment, and building systems.
3. Include the following information at the top of each page:
   a. Project name.
   b. Date.
   c. Name of Engineer.
   d. Name of Contractor.
   e. Page number.
4. Submit list of incomplete items in the following format:
   a. MS Excel electronic file. Engineer will return annotated file.
   b. PDF electronic file. Engineer will return annotated file.
   c. Web-based project software upload. Utilize software feature for creating and updating list of incomplete items (punch list).

1.09 SUBMITTAL OF PROJECT WARRANTIES

A. Time of Submittal: Submit written warranties on request of Engineer for designated portions of the Work where warranties are indicated to commence on dates other than date of Substantial Completion, or when delay in submittal of warranties might limit Owner's rights under warranty.

B. Partial Occupancy: Submit properly executed warranties within 15 days of completion of designated portions of the Work that are completed and occupied or used by Owner during construction period by separate agreement with Contractor.

C. Organize warranty documents into an orderly sequence based on the table of contents of Project Manual.

D. Warranty Electronic File: Provide warranties and bonds in PDF format. Assemble complete warranty and bond submittal package into a single electronic PDF file with bookmarks enabling navigation to each item. Provide bookmarked table of contents at beginning of document.

1. Submit by uploading to web-based project software site .

E. Warranties in Paper Form to Owner:

1. Bind warranties and bonds in heavy-duty, three-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2-by-11-inch paper.
2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify the product or installation. Provide a typed description of the product or

installation, including the name of the product and the name, address, and telephone number of Installer.

3. Identify each binder on the front and spine with the typed or printed title "WARRANTIES," Project name, and name of Contractor.

F. Provide additional copies of each warranty to include in operation and maintenance manuals.

PART 2 PRODUCTS

2.01 MATERIALS

A. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

1. Use cleaning products that comply with Green Seal's GS-37, or if GS-37 is not applicable, use products that comply with the California Code of Regulations maximum allowable VOC levels.

PART 3 EXECUTION

3.01 FINAL CLEANING

A. General: Perform final cleaning. Conduct cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.

1. Complete the following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a designated portion of Project:
    a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.
    b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
    c. Rake grounds that are not planted, mulched, or paved to a smooth, even-textured surface.
    d. Remove tools, construction equipment, machinery, and surplus material from Project site.
    e. Remove snow and ice to provide safe access to building.
    f. Clean exposed exterior and interior hard-surfaced finishes to a dirt-free condition, free of stains, films, and similar foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to their original condition.
    g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
    h. Sweep concrete floors broom clean in unoccupied spaces.
    i. Vacuum carpet and similar soft surfaces, removing debris and excess nap; clean according to manufacturer's recommendations if visible soil or stains remain.
    j. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and other noticeable, vision-obscuring materials. Polish mirrors and glass, taking care not to scratch surfaces.
    k. Remove labels that are not permanent.
    l. Wipe surfaces of mechanical and electrical equipment and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.
    m. Clean plumbing fixtures to a sanitary condition, free of stains, including stains resulting from water exposure.

n. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
o. Clean ducts, blowers, and coils if units were operated without filters during construction or that display contamination with particulate matter on inspection.
p. Clean HVAC system in compliance with NADCA ACR. Provide written report on completion of cleaning.
q. Clean luminaires, lamps, globes, and reflectors to function with full efficiency.
r. Leave Project clean and ready for occupancy.

C. Pest Control: Comply with pest control requirements in Section 015000 "Temporary Facilities and Controls." Prepare written report.

D. Construction Waste Disposal: Comply with waste disposal requirements in Section 015000 "Temporary Facilities and Controls."

3.02 REPAIR OF THE WORK

A. Complete repair and restoration operations before requesting inspection for determination of Substantial Completion.

B. Repair, or remove and replace, defective construction. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment. Where damaged or worn items cannot be repaired or restored, provide replacements. Remove and replace operating components that cannot be repaired. Restore damaged construction and permanent facilities used during construction to specified condition.
1. Remove and replace chipped, scratched, and broken glass, reflective surfaces, and other damaged transparent materials.
2. Touch up and otherwise repair and restore marred or exposed finishes and surfaces. Replace finishes and surfaces that that already show evidence of repair or restoration.
a. Do not paint over "UL" and other required labels and identification, including mechanical and electrical nameplates. Remove paint applied to required labels and identification.
3. Replace parts subject to operating conditions during construction that may impede operation or reduce longevity.
4. Replace burned-out bulbs, bulbs noticeably dimmed by hours of use, and defective and noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.

END OF SECTION 017700

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SECTION 01 7823
OPERATION AND MAINTENANCE DATA

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for preparing operation and maintenance manuals, including the following:
   1. Operation and maintenance documentation directory manuals.
   2. Emergency manuals.
   3. Systems and equipment operation manuals.
   4. Systems and equipment maintenance manuals.
   5. Product maintenance manuals.

B. Related Requirements:
   1. Section 013300 "Submittal Procedures" for submitting copies of submittals for operation and maintenance manuals.
   2. Section 019113 "General Commissioning Requirements" for verification and compilation of data into operation and maintenance manuals.

1.03 DEFINITIONS

A. System: An organized collection of parts, equipment, or subsystems united by regular interaction.

B. Subsystem: A portion of a system with characteristics similar to a system.

1.04 CLOSEOUT SUBMITTALS

A. Submit operation and maintenance manuals indicated. Provide content for each manual as specified in individual Specification Sections, and as reviewed and approved at the time of Section submittals. Submit reviewed manual content formatted and organized as required by this Section.
   1. Engineer and Commissioning Authority will comment on whether content of operation and maintenance submittals is acceptable.
   2. Where applicable, clarify and update reviewed manual content to correspond to revisions and field conditions.

B. Format: Submit operation and maintenance manuals in the following format:
   1. Submit by uploading to web-based project software site. Enable reviewer comments on draft submittals.

C. Initial Manual Submittal: Submit draft copy of each manual at least 30 days before commencing demonstration and training. Engineer, Owner and Commissioning Authority will comment on whether general scope and content of manual are acceptable.

D. Final Manual Submittal: Submit each manual in final form prior to requesting inspection for Substantial Completion and at least 15 days before commencing demonstration and training. Engineer and Commissioning Authority will return copy with comments.
   1. Correct or revise each manual to comply with Engineer's and Commissioning Authority's comments. Submit copies of each corrected manual within 15 days of receipt of Engineer's and Commissioning Authority's comments and prior to commencing demonstration and training.

E. Comply with Section 017700 "Closeout Procedures" for schedule for submitting operation and maintenance documentation.

1.05 FORMAT OF OPERATION AND MAINTENANCE MANUALS

A. Manuals, Electronic Files: Submit manuals in the form of a multiple file composite electronic PDF file for each manual type required.
   1. Electronic Files: Use electronic files prepared by manufacturer where available. Where scanning of paper documents is required, configure scanned file for minimum readable file size.
   2. File Names and Bookmarks: Bookmark individual documents based on file names. Name document files to correspond to system, subsystem, and equipment names used in manual directory and table of contents. Group documents for each system and subsystem into individual composite bookmarked files, then create composite manual, so that resulting bookmarks reflect the system, subsystem, and equipment names in a readily navigated file tree. Configure electronic manual to display bookmark panel on opening file.

1.06 REQUIREMENTS FOR EMERGENCY, OPERATION, AND MAINTENANCE MANUALS

A. Organization of Manuals: Unless otherwise indicated, organize each manual into a separate section for each system and subsystem, and a separate section for each piece of equipment not part of a system. Each manual shall contain the following materials, in the order listed:
   1. Title page.
   2. Table of contents.
   3. Manual contents.

B. Title Page: Include the following information:
   1. Subject matter included in manual.
   2. Name and address of Project.
   3. Name and address of Owner.
   4. Date of submittal.
   5. Name and contact information for Contractor.
   6. Name and contact information for Construction Manager.
   7. Name and contact information for Engineer.
   8. Name and contact information for Commissioning Authority.
   9. Cross-reference to related systems in other operation and maintenance manuals.

C. Table of Contents: List each product included in manual, identified by product name, indexed to the content of the volume, and cross-referenced to Specification Section number in Project Manual.
   1. If operation or maintenance documentation requires more than one volume to accommodate data, include comprehensive table of contents for all volumes in each volume of the set.

D. Manual Contents: Organize into sets of manageable size. Arrange contents alphabetically by system, subsystem, and equipment. When necessary due to size, assemble instructions for subsystems, equipment, and components of one system into a single/individual document.

E. Identification: In the documentation directory and in each operation and maintenance manual, identify each system, subsystem, and piece of equipment with same designation used in the Contract Documents. If no designation exists, assign a designation according to ASHRAE Guideline 4, "Preparation of Operating and Maintenance Documentation for Building Systems."

1.07 OPERATION AND MAINTENANCE DOCUMENTATION DIRECTORY MANUAL

A. Operation and Maintenance Documentation Directory: Prepare a separate manual that provides an organized reference to emergency, operation, and maintenance manuals. List items and their location to facilitate ready access to desired information. Include the following:
   1. List of Systems and Subsystems: List systems alphabetically. Include references to operation and maintenance manuals that contain information about each system.
   2. List of Equipment: List equipment for each system, organized alphabetically by system. For pieces of equipment not part of system, list alphabetically in separate list.

3. Tables of Contents: Include a table of contents for each emergency, operation, and maintenance manual.

1.08 EMERGENCY MANUALS

A. Emergency Manual: Assemble a complete set of emergency information indicating procedures for use by emergency personnel and by Owner's operating personnel for types of emergencies indicated.

B. Content: Organize manual into a separate section for each of the following:
   1. Type of emergency.
   2. Emergency instructions.
   3. Emergency procedures.

C. Type of Emergency: Where applicable for each type of emergency indicated below, include instructions and procedures for each system, subsystem, piece of equipment, and component:
   1. System, subsystem, or equipment failure.

D. Emergency Instructions: Describe and explain warnings, trouble indications, error messages, and similar codes and signals. Include responsibilities of Owner's operating personnel for notification of Installer, supplier, and manufacturer to maintain warranties.

E. Emergency Procedures: Include the following, as applicable:
   1. Instructions on stopping.

1.09 SYSTEMS AND EQUIPMENT OPERATION MANUALS

A. Systems and Equipment Operation Manual: Assemble a complete set of data indicating operation of each system, subsystem, and piece of equipment not part of a system. Include information required for daily operation and management, operating standards, and routine and special operating procedures.
   1. Engage a factory-authorized service representative to assemble and prepare information for each system, subsystem, and piece of equipment not part of a system.
   2. Prepare a separate manual for each system and subsystem, in the form of an instructional manual for use by Owner's operating personnel.

B. Content: In addition to requirements in this Section, include operation data required in individual Specification Sections and the following information:
   1. System, subsystem, and equipment descriptions. Use designations for systems and equipment indicated on Contract Documents.
   2. Performance and design criteria if Contractor has delegated design responsibility.
   3. Operating standards.
   4. Operating procedures.
   5. Operating logs.
   6. Wiring diagrams.
   7. Control diagrams.
   8. Piped system diagrams.

C. Descriptions: Include the following:
   1. Product name and model number. Use designations for products indicated on Contract Documents.
   2. Manufacturer's name.
   3. Equipment identification with serial number of each component.
   4. Equipment function.
   5. Operating characteristics.
   6. Limiting conditions.
   7. Performance curves.
   8. Engineering data and tests.
   9. Complete nomenclature and number of replacement parts.

D. Operating Procedures: Include the following, as applicable:
   1. Startup procedures.
   2. Equipment or system break-in procedures.
   3. Routine and normal operating instructions.
   4. Regulation and control procedures.
   5. Instructions on stopping.
   6. Normal shutdown instructions.
   7. Seasonal and weekend operating instructions.
   8. Required sequences for electric or electronic systems.
   9. Special operating instructions and procedures.

E. Systems and Equipment Controls: Describe the sequence of operation, and diagram controls as installed.

F. Piped Systems: Diagram piping as installed, and identify color coding where required for identification.

1.10 SYSTEMS AND EQUIPMENT MAINTENANCE MANUALS

A. Systems and Equipment Maintenance Manuals: Assemble a complete set of data indicating maintenance of each system, subsystem, and piece of equipment not part of a system. Include manufacturers' maintenance documentation, preventive maintenance procedures and frequency, repair procedures, wiring and systems diagrams, lists of spare parts, and warranty information.
   1. Engage a factory-authorized service representative to assemble and prepare information for each system, subsystem, and piece of equipment not part of a system.
   2. Prepare a separate manual for each system and subsystem, in the form of an instructional manual for use by Owner's operating personnel.

B. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranties and bonds as described below.

C. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual and drawing or schedule designation or identifier where applicable.

D. Manufacturers' Maintenance Documentation: Include the following information for each component part or piece of equipment:
   1. Standard maintenance instructions and bulletins; include only sheets pertinent to product or component installed. Mark each sheet to identify each product or component incorporated into the Work. If data include more than one item in a tabular format, identify each item using appropriate references from the Contract Documents. Identify data applicable to the Work and delete references to information not applicable.
   2. Prepare supplementary text if manufacturers' standard printed data are not available and where the information is necessary for proper operation and maintenance of equipment or systems.
   3. Drawings, diagrams, and instructions required for maintenance, including disassembly and component removal, replacement, and assembly.
   4. Identification and nomenclature of parts and components.
   5. List of items recommended to be stocked as spare parts.

E. Maintenance Procedures: Include the following information and items that detail essential maintenance procedures:
   1. Test and inspection instructions.

2. Troubleshooting guide.
3. Precautions against improper maintenance.
4. Disassembly; component removal, repair, and replacement; and reassembly instructions.
5. Aligning, adjusting, and checking instructions.
6. Demonstration and training video recording, if available.

F. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
1. Scheduled Maintenance and Service: Tabulate actions for daily, weekly, monthly, quarterly, semiannual, and annual frequencies.
2. Maintenance and Service Record: Include manufacturers' forms for recording maintenance.

G. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.

H. Maintenance Service Contracts: Include copies of maintenance agreements with name and telephone number of service agent.

I. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.
1. Include procedures to follow and required notifications for warranty claims.

J. Drawings: Prepare drawings supplementing manufacturers' printed data to illustrate the relationship of component parts of equipment and systems and to illustrate control sequence and flow diagrams. Coordinate these drawings with information contained in record Drawings to ensure correct illustration of completed installation.
1. Do not use original project record documents as part of maintenance manuals.

1.11 PRODUCT MAINTENANCE MANUALS

A. Product Maintenance Manual: Assemble a complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into the Work.

B. Content: Organize manual into a separate section for each product, material, and finish. Include source information, product information, maintenance procedures, repair materials and sources, and warranties and bonds, as described below.

C. Source Information: List each product included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual and drawing or schedule designation or identifier where applicable.

D. Product Information: Include the following, as applicable:
1. Product name and model number.
2. Manufacturer's name.
3. Color, pattern, and texture.
4. Material and chemical composition.
5. Reordering information for specially manufactured products.

E. Maintenance Procedures: Include manufacturer's written recommendations and the following:
1. Inspection procedures.
2. Types of cleaning agents to be used and methods of cleaning.
3. List of cleaning agents and methods of cleaning detrimental to product.
4. Schedule for routine cleaning and maintenance.
5. Repair instructions.

F. Repair Materials and Sources: Include lists of materials and local sources of materials and related services.

G. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that would affect validity of warranties or bonds.
   1. Include procedures to follow and required notifications for warranty claims.

1.12 P2 PRODUCTS (NOT USED)

1.13 P3 EXECUTION (NOT USED)

END OF SECTION 017823

SECTION 01 7839
PROJECT RECORD DOCUMENTS

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract including other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for project record documents, including the following:
   1. Record Drawings.
   2. Record Specifications.
   3. Record Product Data.
   4. Miscellaneous record submittals.

B. Related Requirements:
   1. Section 017300 "Execution" for final property survey.
   2. Section 017700 "Closeout Procedures" for general closeout procedures.
   3. Section 017823 "Operation and Maintenance Data" for operation and maintenance manual requirements.

1.03 CLOSEOUT SUBMITTALS

A. Record Drawings: Comply with the following:
   1. Number of Copies: Submit copies of record drawings as follows:
      a. Initial Submittal:
         1) Submit one paper-copy set(s) of marked-up record prints to owner.
         2) Submit PDF electronic files of scanned record prints (to engineer and owner) and one of file prints (to owner).
         3) Engineer will indicate whether general scope of changes, additional information recorded, and quality of drafting are acceptable.
      b. Final Submittal:
         1) Submit PDF electronic files of scanned record prints (to engineer and owner) and three set(s) of prints (to owner).

B. Record Specifications: Submit annotated PDF electronic files of Project's Specifications, including addenda and contract modifications.

C. Record Product Data: Submit annotated PDF electronic files and directories of each submittal.
   1. Where record Product Data are required as part of operation and maintenance manuals, submit duplicate marked-up Product Data as a component of manual.

D. Miscellaneous Record Submittals: See other Specification Sections for miscellaneous record-keeping requirements and submittals in connection with various construction activities. Submit annotated PDF electronic files and directories of each submittal.

1.04 RECORD DRAWINGS

A. Record Prints: Maintain one set of marked-up paper copies of the Contract Drawings and Shop Drawings, incorporating new and revised drawings as modifications are issued.
   1. Preparation: Mark record prints to show the actual installation where installation varies from that shown originally. Require individual or entity who obtained record data, whether individual or entity is Installer, subcontractor, or similar entity, to provide information for preparation of corresponding marked-up record prints.
      a. Give particular attention to information on concealed elements that would be difficult to identify or measure and record later.
      b. Accurately record information in an acceptable drawing technique.

c. Record data as soon as possible after obtaining it.
d. Record and check the markup before enclosing concealed installations.
e. Cross-reference record prints to corresponding photographic documentation.

2. Content: Types of items requiring marking include, but are not limited to, the following:
   a. Dimensional changes to Drawings.
   b. Revisions to details shown on Drawings.
   c. Locations and depths of underground utilities.
   d. Revisions to routing of piping and conduits.
   e. Revisions to electrical circuitry.
   f. Actual equipment locations.
   g. Duct size and routing.
   h. Locations of concealed internal utilities.
   i. Changes made by Change Order or Work Change Directive.
   j. Changes made following Engineer's written orders.
   k. Details not on the original Contract Drawings.
   l. Field records for variable and concealed conditions.
   m. Record information on the Work that is shown only schematically.
3. Mark the Contract Drawings and Shop Drawings completely and accurately. Use personnel proficient at recording graphic information in production of marked-up record prints.
4. Mark record sets with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of the Work at same location.
5. Mark important additional information that was either shown schematically or omitted from original Drawings.
6. Note Construction Change Directive numbers, alternate numbers, Change Order numbers, and similar identification, where applicable.

B. Record Digital Data Files: Immediately before inspection for Certificate of Substantial Completion, review marked-up record prints with Engineer. When authorized, prepare a full set of corrected digital data files of the Contract Drawings, as follows:
   1. Format: Annotated PDF electronic file with comment function enabled.
   2. Incorporate changes and additional information previously marked on record prints. Delete, redraw, and add details and notations where applicable.
   3. Refer instances of uncertainty to Engineer for resolution.
   4. Engineer will furnish Contractor with one set of digital data files of the Contract Drawings for use in recording information.
      a. See Section 013100 "Project Management and Coordination" for requirements related to use of Engineer's digital data files.
      b. Engineer will provide data file layer information. Record markups in separate layers.

C. Format: Identify and date each record Drawing; include the designation "PROJECT RECORD DRAWING" in a prominent location.
   1. Record Prints: Organize record prints into manageable sets. Bind each set with durable paper cover sheets. Include identification on cover sheets.
   2. Format: Annotated PDF electronic file with comment function enabled.
   3. Record Digital Data Files: Organize digital data information into separate electronic files that correspond to each sheet of the Contract Drawings. Name each file with the sheet identification. Include identification in each digital data file.
   4. Identification: As follows:
      a. Project name.
      b. Date.
      c. Designation "PROJECT RECORD DRAWINGS."
      d. Name of Engineer.
      e. Name of Contractor.

1.05 RECORD SPECIFICATIONS

A. Preparation: Mark Specifications to indicate the actual product installation where installation varies from that indicated in Specifications, addenda, and contract modifications.
   1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
   2. Mark copy with the proprietary name and model number of products, materials, and equipment furnished, including substitutions and product options selected.
   3. Record the name of manufacturer, supplier, Installer, and other information necessary to provide a record of selections made.
   4. For each principal product, indicate whether record Product Data has been submitted in operation and maintenance manuals instead of submitted as record Product Data.
   5. Note related Change Orders , record Product Data, and record Drawings where applicable.

B. Format: Submit record Specifications as annotated PDF electronic file.

1.06 RECORD PRODUCT DATA

A. Recording: Maintain one copy of each submittal during the construction period for project record document purposes. Post changes and revisions to project record documents as they occur; do not wait until end of Project.

B. Preparation: Mark Product Data to indicate the actual product installation where installation varies substantially from that indicated in Product Data submittal.
   1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.
   2. Include significant changes in the product delivered to Project site and changes in manufacturer's written instructions for installation.
   3. Note related Change Orders, record Specifications, and record Drawings where applicable.

C. Format: Submit record Product Data as annotated PDF electronic file.
   1. Include record Product Data directory organized by Specification Section number and title, electronically linked to each item of record Product Data.

1.07 MISCELLANEOUS RECORD SUBMITTALS

A. Assemble miscellaneous records required by other Specification Sections for miscellaneous record keeping and submittal in connection with actual performance of the Work. Bind or file miscellaneous records and identify each, ready for continued use and reference.

B. Format: Submit miscellaneous record submittals as PDF electronic file.
   1. Include miscellaneous record submittals directory organized by Specification Section number and title, electronically linked to each item of miscellaneous record submittals.

1.08 MAINTENANCE OF RECORD DOCUMENTS

A. Maintenance of Record Documents: Store record documents in the field office apart from the Contract Documents used for construction. Do not use project record documents for construction purposes. Maintain record documents in good order and in a clean, dry, legible condition, protected from deterioration and loss. Provide access to project record documents for Engineer's reference during normal working hours.

PART 2 PRODUCTS

PART 3 EXECUTION

END OF SECTION 017839

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SECTION 01 7900
DEMONSTRATION AND TRAINING

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including ther Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes administrative and procedural requirements for instructing Owner's personnel, including the following:
   1. Minimium of two hours (2) of instruction in operation and maintenance of project systems, subsystems, and equipment.

1.03 INFORMATIONAL SUBMITTALS

A. Attendance Record: For each training module, submit list of participants and length of instruction time.

1.04 CLOSEOUT SUBMITTALS

1.05 QUALITY ASSURANCE

A. Instructor Qualifications: A factory-authorized service representative, complying with requirements in Section 014000 "Quality Requirements," experienced in operation and maintenance procedures and training.

1.06 COORDINATION

A. Coordinate instruction schedule with Owner's operations. Adjust schedule as required to minimize disrupting Owner's operations and to ensure availability of Owner's personnel.

B. Coordinate instructors, including providing notification of dates, times, length of instruction time, and course content.

C. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data have been reviewed and approved by Engineer.

1.07 INSTRUCTION PROGRAM

A. Program Structure: Develop an instruction program that includes individual training modules for each system and for equipment not part of a system, as required by individual Specification Sections.

B. Training Modules: Develop a learning objective and teaching outline for each module. Include a description of specific skills and knowledge that participant is expected to master. For each module, include instruction for the following as applicable to the system, equipment, or component:
   1. Basis of System Design, Operational Requirements, and Criteria: Include the following:
      a. System, subsystem, and equipment descriptions.
      b. Performance and design criteria if Contractor is delegated design responsibility.
      c. Operating standards.
      d. Regulatory requirements.
      e. Equipment function.
      f. Operating characteristics.
      g. Limiting conditions.
      h. Performance curves.
   2. Documentation: Review the following items in detail:
      a. Emergency manuals.
      b. Systems and equipment operation manuals.

c. Systems and equipment maintenance manuals.
d. Product maintenance manuals.
e. Project Record Documents.
f. Identification systems.
g. Warranties and bonds.
h. Maintenance service agreements and similar continuing commitments.

3. Operations: Include the following, as applicable:
   a. Startup procedures.
   b. Equipment or system break-in procedures.
   c. Routine and normal operating instructions.
   d. Regulation and control procedures.
   e. Control sequences.
   f. Safety procedures.
   g. Instructions on stopping.
   h. Normal shutdown instructions.
   i. Operating procedures for emergencies.
   j. Operating procedures for system, subsystem, or equipment failure.
   k. Seasonal and weekend operating instructions.
   l. Required sequences for electric or electronic systems.
   m. Special operating instructions and procedures.
4. Adjustments: Include the following:
   a. Alignments.
   b. Checking adjustments.
   c. Noise and vibration adjustments.
   d. Economy and efficiency adjustments.
5. Troubleshooting: Include the following:
   a. Diagnostic instructions.
   b. Test and inspection procedures.
6. Maintenance: Include the following:
   a. Inspection procedures.
   b. Types of cleaning agents to be used and methods of cleaning.
   c. List of cleaning agents and methods of cleaning detrimental to product.
   d. Procedures for routine cleaning.
   e. Procedures for preventive maintenance.
   f. Procedures for routine maintenance.
   g. Instruction on use of special tools.
7. Repairs: Include the following:
   a. Diagnosis instructions.
   b. Repair instructions.
   c. Disassembly; component removal, repair, and replacement; and reassembly instructions.
   d. Instructions for identifying parts and components.
   e. Review of spare parts needed for operation and maintenance.

1.08 PREPARATION

A. Assemble educational materials necessary for instruction, including documentation and training module. Assemble training modules into a training manual organized in coordination with requirements in Section 017823 "Operation and Maintenance Data."

B. Set up instructional equipment at instruction location.

1.09 INSTRUCTION

A. Facilitator: Engage a qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Owner for number of participants, instruction times, and location.

B. Engage qualified instructors to instruct Owner's personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.
   1. Owner will furnish an instructor to describe Owner's operational philosophy.
   2. Owner will furnish Contractor with names and positions of participants.

C. Scheduling: Provide instruction at mutually agreed-on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.
   1. Schedule training with Owner with at least seven days' advance notice.

D. Training Location and Reference Material: Conduct training on-site in the completed and fully operational facility using the actual equipment in-place. Conduct training using final operation and maintenance data submittals.

E. Cleanup: Collect used and leftover educational materials and remove from Project site . Remove instructional equipment. Restore systems and equipment to condition existing before initial training use.

PART 2 PRODUCTS

PART 3 EXECUTION

END OF SECTION 017900

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SECTION 01 9113
GENERAL COMMISSIONING REQUIREMENTS

PART 1 GENERAL

1.01 SUMMARY

A. SCOPE
1. Commissioning requirements common to all Sections
2. Systems and equipment functional performance testing
3. Validation of proper and thorough installation of systems and equipment
4. Equipment performance verification
5. Documentation of tests, procedures, and observations
6. Review of Owner Training agenda
7. Building automation system requirements for post occupancy data collection and remote monitoring systems

B. GENERAL DESCRIPTION
1. Commissioning is the process of verifying and validating that all building systems are installed and perform interactively according to the design intent; that systems are efficient and cost effective to operate and meet the Owner's operational needs; that the installation is adequately documented; and that Operators are adequately trained. It serves as a tool to minimize post-occupancy operational problems. It establishes testing and communication protocols in an effort to advance the building systems from installation to full dynamic operation and optimization.
2. Commissioning Authority shall work with the Contractor and the Design Engineer to direct and oversee the Commissioning process and perform Functional Performance Testing.
3. The Commissioning Plan outlines the commissioning process.
4. The commissioning process does not take away from or reduce the responsibility of the system designers or installing contractors to provide a finished and fully functioning product.
5. This Section and other Sections of the specifications detail the Contractor's responsibilities relative to the Commissioning process. It expands on the Commissioning Plan, which covers the roles and responsibilities of all Parties. It also indicates the details of the Functional Performance Testing in which the Contractor must participate.

C. COMMISSIONING SEQUENCE AND SCHEDULE
1. Where applicable, in order to expedite the close out of the facility, various systems can be in various stages of the commissioning process. CxA and Contractor shall cooperate to schedule the Commissioning tasks to minimize the duration of Commissioning activities.
2. Commissioning will be categorized into Phases as indicated below. Note that per schedule, different systems and/or areas may be in different phases at any given time given that the Commissioning and testing process will be integrated into the construction process:
    a. Construction Phase: This is the period of time where the systems are installed, much of the commissioning documentation is developed, the systems are started, and the majority of the contractor required training is performed. On any given system or area, the Construction Phase will end when the CxA determines that the prerequisite start-up requirements have been met and the system is ready for FPT (Functional Performance Testing). This period shall also be used to flush-out the systems and allow the CxA to evaluate BAS trending to optimize system performance.
        1) Pre-Start Up checklists completed by the installing contractor.
        2) Start-up checklists, tests and OEM procedures completed by the installing contractor.
        3) HVAC TAB completed and final report is available for review prior to beginning FPT.

b. Acceptance Phase: This is the period of time where the systems will be functionally tested and will operate through a Transition period.
   1) Equipment and Systems functional performance testing
   2) Integrated systems functional performance testing
   3) Trend data setup and archiving verification for remote monitoring systems
c. Warranty Phase: Includes the early occupancy of the building and can continue through the Warranty Period and at least into the opposite season from when it was initially tested.
   1) Deferred Functional Testing: FPTs that are performed later during the warranty phase, after substantial completion , due to partial occupancy, equipment, seasonal requirements, design or other site conditions that disallow the test from being performed.
   2) Continued coordination of integration of trend data into remote monitoring systems
d. Commissioning Kick Off and Coordination Meetings
   1) CxA shall schedule and conduct a Commissioning Kick Off meeting near the beginning of construction. The following should be discussed at this meeting:
      (a) CxA will present
         (1) The Commissioning Documents
         (2) Requirements of Commissioning
         (3) Responsibilities of the construction parties
         (4) Management protocols
         (5) Required submittals
         (6) Schedule
      (b) Cx coordination meetings will be organized and led by the CxA as required to ensure the successful completion of the project.
e. Phasing Plan
   1) If contractor intends to start, run, or occupy portions of systems in phases, contractor shall submit a plan for phasing in areas/portions of systems that will be connected subsequent to the initial portions. Specifically address:
      (a) Pipe and Duct Cleaning: Indicate the configurations and protocols for isolating subsequent regions and then protecting the preceding regions when the subsequent region is cleaned, flushed and connected.
      (b) Pipe disinfection: Indicate the plan for disinfecting each region of potable water that requires disinfection. Indicate how the preceding regions of the system will be protected when connecting subsequent regions.
      (c) System Modifications: Indicate the protocols for making subsequent changes to the systems of pipe and duct when the systems have already been cleaned, flushed, pressure tested, disinfected, certified, etc.
      (d) HVAC Equipment: Indicate the protocols for operating the HVAC equipment in a construction environment. The plan shall, at a minimum, include; filtering, operation, maintenance, and Control System responsibilities.

1.02 ADMINISTRATIVE REQUIREMENTS

A. RELATED WORK AND DOCUMENTS
   1. Commissioning Plan: The Commissioning Plan is a part of the Contract Documents and outlines many of responsibilities, procedures and tasks throughout the Commissioning process. It also describes the Functional Performance Tests that will be performed during the Acceptance Phase. The Contractor must have an understanding of commissioning process and the Contractor requirements within the plan.
   2. Individual Specification Sections (Divisions): Individual sections stipulate installation, start-up, warranty, operation and maintenance documentation, and training requirements for the system or device specified in the Section.

a. Division 23 - Heating, Ventilating, and Air-Conditioning (HVAC)
b. Division 26 - Electrical

3. Reference Standards
   a. ASHRAE Guideline 202-2024, “Commissioning Process for Buildings and Systems”
   b. ASHRAE Guideline 1.1-2025, "Application of the Commissioning Process to New HVAC&R Systems"
   c. ASHRAE Guideline 1.4-2019, “Preparing Systems Manuals for Facilities”
   d. BCA - Building Commissioning Association
   e. NEBB - Procedural Standards for Building Systems Commissioning
   f. NETA-ATS, Testing Standards
   g. National Electric Code (NEC)
   h. American Society for Testing and Materials (ASTM)
   i. Electronics Industry Association/Telecommunications Industry Association (EIA/TIA)
   j. Illuminating Engineering Society (IES)
   k. Institute of Electrical and Electronics Engineers (IEEE)
   l. International Electrical Testing Association (NETA)
   m. National Electrical Manufacturers Associates (NEMA)
   n. National Fire Protection Association (NFPA)
   o. Underwriters Laboratory, Inc. (UL)

B. DEFINITIONS AND ABBREVIATIONS
   1. Acceptance Phase: This is the phase of the project when the facility and its systems and equipment are inspected, tested, verified, and documented; and when most of the Functional Performance Testing and formal training occurs. This will generally occur after the Construction Phase is complete (start-up and checks have been accomplished). The Acceptance Phase typically begins with Substantial Completion and ends with Functional Completion.
   2. Observation Log: This is a list that is maintained on a Web-Based Project Software (such as smartsheet) that includes all Observation Items that relate to Commissioning activities and site observations requiring contractor action or response.
   3. Building Automation System (BAS): The computer-based heating, ventilation and air-conditioning (HVAC) control or automation system
   4. Commissioning Authority (CA or CxA): The Party retained by the Owner who will oversee the Commissioning process as well as develop and stipulate many of the Commissioning requirements. They will also manage the Commissioning process, and ensure and validate that systems and equipment are designed, installed and tested to meet the Owner’s requirements.
   5. Commissioning Coordinator (CxC): This refers to the Individual within each of the various Parties that is designated the point of contact for that Party relative to Commissioning activities.
   6. Commissioning Portal: This is an internet hub for the collaboration on Commissioning information. This portal will act as a hub for posting electronic information.
   7. Commissioning Team (CxT): The group of Parties involved in the commissioning process for any given system. The Commissioning Team will include a core group involved with all systems. This core group will typically include the CxA, [the Construction Manager’s Commissioning Coordinator (CM-CxC)], [the Owner’s Commissioning Coordinator (O/O-CxC)] and the General Contractor’s Commissioning Coordinator (GC-CxC). On any given system, the Commissioning Team will also include the Commissioning Coordinator for the Contractor(s) responsible for the system or equipment.
   8. Contractor: As used herein, ‘Contractor’ is a general reference to the installing Party and can therefore refer to the General Contractor, subcontractors, or vendors as inferred by its usage.
   9. Construction Manager (CM): The Party retained by the Owner to represent the Owner and make decisions on the Owner’s behalf throughout the design and construction process.

10. Construction Phase: Phase of the project during which the facility is constructed and/or systems and equipment are installed and started. Contractor and subcontractors complete the installation, complete start-up documentation, submit operation and maintenance information, establish trends, and perform any other applicable requirements to get systems started. Contractor and Vendors may also conduct equipment specific training. The Construction Phase will generally end upon completed start-up and test, adjust and balance of systems and equipment.
11. Deficiency: A condition in the installation or function of a component, piece of equipment or system that is not in compliance with the Contract Documents (that is, does not perform properly or is not complying with the design intent).
12. Engineer of Record - Design Team Engineer lead-design entity
13. Factory Authorized Representative: An individual fully trained on the equipment and certified by the manufacturer to perform the respective task.
14. Factory Testing: Testing of equipment off-site at the manufacturer's facility. The testing may be witnessed by the members of the project team.
15. Functional Performance Testing (FPT): The detailed and thorough testing of building systems and their interactions with building components and other building systems.
16. IAQ: Indoor Air Quality
17. Operation and Maintenance (O&M) Documentation: This refers to Contractor-developed documentation designed to address the needs of facilities personnel and customized for the context of the specific facility and installation. The foundation of O&M Documentation is manufacturer's literature (including 'O&M Manuals', parts lists, troubleshooting guides, etc.) as well as Contractor-developed instructions for start-up and shut-down, sequences, and other installation-specific information. O&M Documentation content is a subset of the Facility Manual, so it is common for only one or the other to be specified. All documentation shall be submitted to Owner in electronic format.
18. Transition Period: Time period after FPT completed to operate systems to purge the building and stabilize equipment operations. Time is also used by the CxA to test system performance.
19. Opposite Season: The season opposite that when the majority of the testing occurs.
20. Pre-Start Up: Preliminary testing accomplished during a scheduled system outage to verify system functionality prior to placing the system/equipment into preliminary service.
21. Start-Up: Refers to the quality control process whereby the Contractor verifies the proper installation of a device or piece of equipment, executes the manufacturer's starting procedures, completes the Start-Up Checklist, energizes the device, verifies that it is in proper working order and ready for dynamic testing, including Start-Up Tests.
22. TAB: Can refer to the test, adjust, and balance process or the Testing, Adjusting, and Balancing Contractor.
23. Trending: Monitoring and recording a history of parameters typically using the BAS.
24. Warranty Phase: Includes the early occupancy of the building and can continue through the Warranty Period and at least into the opposite season from when it was initially tested.

C. COORDINATION MANAGEMENT PROTOCOL
1. Coordination responsibilities and management protocols relative to Commissioning are initially defined below but will be refined and documented in the Commissioning Plan. Contractor shall have input in the protocols and all parties will commit to scheduled obligations. The CxA will record and distribute.
    a. Submittals and Shop Drawings: CM shall distribute the submittal log to the CxA. CxA shall review the submittal log and communicate which submittals need to be forwarded.
    b. CxA Review Comments for Shop Drawings: An email reply is sent directly to the CM, Engineer of Record, and Owner by the CxA. The Owner and Engineer of Record will consider and incorporate at their discretion.

c. Deficiencies Identified by the Commissioning Authority: When the CxA identifies a deficiency the CxA shall make a good faith assessment of responsible parties. Those parties, as well as the Owner and CM shall be notified of the perceived deficiency. This communication is FOR INFORMATION ONLY and is not a direction to resolve the deficiency or to take any action. Contractor may elect to accept responsibility and resolve the deficiency. If the contractor contests either the deficiency or responsibility for that deficiency, Contractor shall respond to that deficiency indicating disagreement. If responsibility is not agreed to via the Commissioning dialogue, CM shall issue a work directive or RFI via the normal contractual channels to resolve the issue.
d. Requests for Meetings: In general, request by the contractor for a meeting with the CxA shall be routed through Owner and CM who will then determine the validity. Note that every attempt should be made to deal with Commissioning issues at regularly scheduled Commissioning Meetings.
e. Control Sequence Modifications: CxA shall review the sequences during the design and submittal phases and address any known concerns prior to the submittal approval. Any recommendations for control sequence modification must be presented to the Owner and Engineer of Record for approval. Any and all changes approved by the Owner and Engineer of Record must be thoroughly documented in the record documents.
f. Notification of Completion Milestones: Contractor shall notify the CxA, Owner and CM at least one week prior to any anticipated Commissioning activity or Commissioning milestone (such as FPT). The Owner or CM (as applicable) shall then coordinate the scheduling of the activity between all required parties. Notification shall be via email.
g. Observation Log: The observation log is maintained on a Web-Based Project Software (such as smartsheet) which tracks the Commissioning related action items. The log is accessible at all times to the Cx team members.
h. Start-Up Checklist and Test Documents: Installing contractors will provide an electronic copy of in-house start-up checklists and test documents to the Owner, CxA, and CM for review, comment and acceptance. The Contractor then performs the reviewed and approved Start-Up procedures, completes the documentation and signs, and submits it. CxA subsequently spot checks the procedures and documentation during the FPT. They are then included in the Commissioning Record.
i. Functional Performance Test Documents: FPTs are witnessed and documented by the CxA but performed by the contractor. They are developed during the construction phase generally after completed submittals. CxA drafts and forwards the FPT procedures to the CM to be subsequently distributed to the subcontractors for review by the CM. Contractors review and comment on the procedures. Throughout the Commissioning process, CxA maintains a current record of the testing procedures and keeps the documentation up to date and accessible for all to access the current progress. CxA normally distributes hard copies of the testing procedures at completion of any significant stage of Commissioning.

2. Coordination Between Testing Parties
   a. Factory Start-Ups: For many systems and equipment, Factory Start-Ups are specified. These Factory Start-Ups will be reviewed and checked during FPT. All costs associated with the Factory Start-Ups are included with the bid unless otherwise noted. Contractor shall make notification of when Factory Start-Ups are occurring and coordinate these with witnessing Parties. The CxA and CxT members may witness Factory Start-Ups at their discretion. Aspects of FPT accomplished during the Factory Start-Ups may be accomplished and approved by the CxA if they meet the intent of the FPT.
   b. Independent Testing Agencies: For systems where contractor’s independent testing agencies are specified, the cost of this testing is included with the bid unless

otherwise noted. Much of the testing performed by these independent agencies will cover aspects required in the Start-Up Procedures and FPTs.

c. Contractor and testing agencies shall coordinate with the CxA so that the CxA can witness the testing and approve the applicable aspects of the FPTs. The Contractor should not start up equipment or systems without CxA approval.
d. The CxA may independently spot-check work of the testing agencies if the tests were not witnessed. However, it is not the intent for the CxA to re-accomplish testing by others that is specified in the construction specifications.
e. Contractor is responsible for coordinating the efforts of testing agency with that of the Cx process. Documentation shall be contiguous and seamless and duplication will be avoided. Testing agencies shall complete the documentation of the Cx process as required.

D. CONTRACTOR RESPONSIBILITIES
1. Construction Phase: The following delineates the commissioning-related responsibilities of the Contractor (and their subcontractors) during the Construction Phase.
a. Include Commissioning requirements in price and plan for work.
b. Designate a CxC from each major subcontractor with activities related to commissioning. These CxCs are to be the primary contacts for Commissioning activities.
c. Attend Construction Phase Commissioning Kick Off Meeting. The CxC and Project Manager from each major subcontractor shall attend at a minimum.
d. The CxC shall attend all Commissioning progress meetings unless otherwise agreed to by the CxA.
e. Remedy any deficiencies identified throughout construction.
f. Submit Start-Up Procedures along with the manufacturer's application, installation and start-up information to the CxA for review prior to implementation.
g. TAB shall submit sample balancing forms and TAB Plan for review and approval prior to starting work.
h. Contractor shall incorporate the Commissioning process into the construction schedule outlining generic Commissioning tasks with precedents or prerequisites to each task. These tasks will apply to many systems and the Contractor shall incorporate as such. Examples of enumerated tasks include:
1) Contractor preparation of the Training Plan
2) Testing Agency activities
3) Contractor documentation of pipe pressure testing, flushing, and cleaning of applicable systems
4) Documentation of the Start-Up Procedures for equipment and systems
5) TAB of applicable system
6) Training Events
7) Preparation of the O&M Manual content
8) FPT and Acceptance
9) Observation Period and System Optimization
10) Occupant or other Regulatory Agency testing or approval process
i. Coordinate the work of subcontractors, vendors, manufacturers, and Testing Agencies provided with the bid, and ensure that they are informed of and are adhering to the requirements of the Commissioning process specified throughout the contract documents. Particular reference is made to providing the required O&M Documentation; submittal of training materials and documentation of that training; collaboration with the overall start-up and testing process; developing comprehensive integrated procedures for scheduling and task notification and documenting them in a common format; and electronic delivery requirements if applicable.
j. Provide assistance to the CxA in preparation for the specific FPT procedures. Contractors, subcontractors and vendors shall review the FPTs to ensure feasibility,

safety and equipment protection and provide necessary written alarm limits to be used during the tests. Damage caused to equipment performed in accordance with the approved procedures that is the result of malfunctioning equipment or contract deficiencies, shall be the responsibility of the Contractor.

k. Thoroughly complete and inspect installation of systems and equipment as detailed throughout Contract Documents, as required by reference or industry standards, and as specifically indicated elsewhere this section. The Contractor (and subcontractors) shall record, in the form of photographs, compliance to and/or deviation from IAQ standards.

l. Contractor shall notify the CxA at least 7 days in advance of any tests, start-ups, or training. CxA shall witness selected tests and start-ups. Notification shall be accompanied by a schedule showing the coordinated start date and task duration and all open prerequisites

m. Start-up, TAB of systems and equipment prior to verification and FPT by the CxA. Start-up procedures shall be in accordance with Contract Documents, reference or industry standards, and Commissioning specs. Provide skilled technicians who are qualified to do the work required. Provide factory trained/authorized technicians where required by the contract documents and stated in the applicable technical section. Generally start-up and testing shall proceed from device checkout, to component checkout, to system checkout, to inter-system checkout.

n. Record start-up and testing procedures on start-up forms or checklists and certify that the systems and equipment have been started and/or tested in accordance with the requirements specified above. Each task or item shall be indicated with the party actually performing the task or procedure.

o. Demonstrate the operation of all systems as specified.

p. Certify that systems have been installed and are operating per Contract Documents and OEM prior to FPT and acceptance.

q. Maintain an updated set of Record Documentation as required by the Contract Documents.

r. Conduct and document Equipment and Systems Training events as required by this Section and by applicable sections of the Specifications pertaining to each piece of equipment or system and general training requirements.

s. Operate systems under direction of the CxA during FPT’s and other acceptance testing.

t. Coordinate trend point naming convention with CxA to ensure that systems are properly setup for data storage and remote monitoring capabilities.

2. Acceptance Phase: The following delineates the commissioning-related responsibilities of the Contractor (and their subcontractors) during the Acceptance Phase.

   a. Will work in conjunction with CxA in FPT and will generally include the following:
      1) Operate and Manipulate systems and equipment to facilitate testing (as dictated in this section, relevant technical sections and the Commissioning Plan).
      2) Operate and Manipulate BAS and other control systems to facilitate FPT (as dictated in this section, relevant technical sections and the Commissioning Plan).

   b. Correct any work not in accordance with Contract Documents.

   c. Maintain record documentation and update and resubmit it after Functional Completion.

   d. Compensate the Owner for additional CxA fees and expenses incurred to retest equipment and systems following testing failures.

   e. Monitor systems, equipment and areas throughout the Transition Period. Log and diagnose all alarms during this period. Maintain trends and logs of all critical parameters. Forward the logs and trends on a weekly basis throughout all Transition Periods.

f. Setup trends for archiving as required by CxA for specified data collection and remote monitoring capabilities.

3. Warranty Phase: The following delineates the commissioning-related responsibilities of the Contractor (and their subcontractors) during the Warranty Phase.
   a. Provide warranty service
   b. Conduct BAS Sequence Training
   c. Respond to and document Warranty issues
   d. Participate as required in the opposite season testing
   e. Correct any deficiencies identified throughout the Warranty Phase
   f. Update record documentation to reflect any changes made throughout the Warranty Phase and resubmit final Record Drawings and data records at the close of the Warranty period.
   g. Assist CxA with any additional point naming or trend archiving requests to assist with on-going remote monitoring.

E. OBSERVATION LOG
1. CxA shall maintain an Observation Log (required information, identified deficiencies, work required, etc.) that relates to Commissioning. Each item shall be tracked with the initiator, the parties responsible, due date, the date of closure, and a description of the resolution. Each item shall be categorized for sorting and tracking and for documentation on applicable forms.
2. CxA will disseminate this list as appropriate to keep all parties informed.
3. All parties indicated as responsible for an action item shall respond. The preferred response method is via e-mail. Response with a plan of action (either repair or plan to resolve) is expected within 48 hours.

1.03 SUBMITTAL

A. The CxA will provide appropriate contractors with a specific request for the type of submittal documentation the CxA requires to facilitate the commissioning work. These requests will be integrated into the normal submittal process and protocol of the construction team. At minimum, the request will include the manufacturer and model number, the manufacturer's printed installation and detailed start-up procedures, full sequences of operation, O&M data, performance data, any performance test procedures, control drawings and details of owner contracted tests. In addition, the installation and checkout materials that are actually shipped inside the equipment and the actual field checkout sheet forms to be used by the factory or field technicians shall be submitted to the Commissioning authority. All documentation requested by the CxA will be included by the Subs in their O&M manual contributions.

B. The Commissioning authority will review and provide comment on submittals related to the commissioned equipment for conformance to the Contract Documents as it relates to the commissioning process, to the functional performance of the equipment and adequacy for developing test procedures. This review is intended primarily to aid in the development of functional testing procedures and only secondarily to verify compliance with equipment specifications. The Commissioning authority will notify the CM, Owner Representative, or Engineer of Record as requested, of items missing or areas that are not in conformance with Contract Documents and which require resubmission.

C. The CxA may request additional design narrative from the Engineer of Record and Controls Contractor, depending on the completeness of the design intent documentation and sequences provided with the Specifications.

D. These submittals to the CxA do not constitute compliance for O&M manual documentation. The O&M manuals are the responsibility of the Contractor, though the CxA will review and approve them.

E. Contractor's responsibility for deviations in submittals from requirements of the Contract Documents is not relieved by the Commissioning Authority's review.

PART 2 PRODUCTS

2.01 INSTRUMENTATION

A. All test instruments described in this section shall be acceptable for any portion of the commissioning process herein described. All instruments shall conform to the standards specified in the most recent edition of "NEBB Procedural Standards for Testing, Adjusting, and Balancing of Environmental Systems" in regard to accuracy and calibration status. Current calibration certificates must be available to the CxA if requested.

B. Test instrument accuracy and resolution must match or exceed that of the system component being verified or calibrated.

C. Test instruments must be used within guidelines as recommended by instrument manufacturer. All measuring methods must be appropriate to the instrument application and measurements must be repeatable under equivalent conditions.

D. Standard Testing Instrumentation: Standard instrumentation normally used for performance assessment and diagnosis will be provided by Cx Authority. These include:
   1. Electronic Manometer (for Air and Flow Hood)
   2. Electronic Manometer (for Water)
   3. Temperature Instruments
   4. Humidity Instruments
   5. CO2 Instrument
   6. Sound Meter
   7. Electronic Multimeter
   8. Tachometer
   9. Ultrasonic Flow Meter
   10. Others as required

PART 3 EXECUTION

3.01 PRE-START UP

A. PREREQUISITES
   1. All equipment, components, and devices applicable to the Pre-Start Up must be installed, and the Pre-Start Up must be documented and approved. This includes installation, identification labeling, insulation, and all other requirements for placing systems into dynamic operation.

B. COMMON ELEMENTS
   1. Required submittal documentation shall be present and located convenient to testing area. Validate that all required documentation has been submitted and is per the contract requirements.
   2. Contractor shall provide the completed Pre-Start Up Procedures at the time of testing. CxA shall review the Pre-Start Up Procedure documentation and spot-check at the beginning of Start Up.
   3. Contractor shall demonstrate that access is sufficient to perform required maintenance.
   4. System and equipment configurations shall be compared against the contract documents.

3.02 START-UP

A. PREREQUISITES
   1. All equipment, components, and devices applicable to the FPT must be started, and the Start-Up must be documented and approved. This includes completion of Start-Up Procedures, pressure testing (of equipment, duct and piping), flushing/cleaning, identification labeling, insulation, and all other requirements for placing systems into dynamic operation.
   2. Unless specifically agreed to by the Owner and CxA, all support systems shall be complete prior to FPT. For instance, a chiller will require that:

a. The electrical system serving it is completed and tested.
b. The hydronic systems serving it have been pressure tested, flushed, and functional performance tested.
c. Balancing has been completed.
d. The control systems have been started and calibrated.
e. The CxA shall determine the optimal sequence of testing.

B. COMMON ELEMENTS
1. Required submittal documentation shall be present and located convenient to testing area. Validate that all required documentation has been submitted and is per the contract requirements.
2. Contractor shall provide the completed Start-Up Procedures at the time of testing. CxA shall review the Start-Up Procedure documentation and spot-check at the beginning of FPT.
3. Contractor shall demonstrate that access is sufficient to perform required maintenance.
4. BAS trends shall have been established as required in the documents. These shall generally be reviewed prior to or during FPT.
5. Capacities and adjusted/balanced conditions as applicable shall be subject to review.
6. Sequencing Verification: For applicable systems and equipment, all modes of operation shall be verified for proper sequencing.
7. System and equipment configurations shall be compared against the contract documents.
8. Verify Modes (such as heating and cooling) are coordinated and do not overlap or ‘fight’.
9. All adjusted, balanced, controlled systems shall be assessed to determine the optimal setting for the system as applicable. The optimal settings should be determined to establish reliable, efficient, safe and stable operation.
10. BAS or Local Panel Dynamic Graphics: The graphic displays for all components, systems, and areas required to be represented by a graphic shall be checked for adequacy and accuracy. Furthermore, when setpoints or other parameters are required to be adjustable, CxA shall verify that they can be adjusted directly from the graphic screen.

C. START-UP PROCEDURE
1. Purpose: The Commissioning process requires that the normal quality control processes involved with preparing systems and equipment for operation are performed to a high standard of care and are thoroughly documented. The required commissioning-related Start-Up Procedures involve nothing additional over that which would be done for a proper installation. These procedures shall be performed on all installed systems and equipment and no sampling strategy is used for the start-up process. The Commissioning process requires all Parties to collaborate to establish the optimal standard of care for starting systems and equipment. After the procedures are established, the Contractor performs them and documents them with the Start-up Procedures that are developed by the Contractor.
2. Start-Up Procedures: The content of these Start-Up Procedures shall provide the minimally acceptable content in accordance with the OEM field quality control requirements. Generic refers to the fact that the protocols may be created before the shop drawings are finalized. These procedures and protocols will normally be common across different manufacturers.
3. Content of Start-Up Procedures: Start-Up Procedures shall generally include the following for each item of equipment or system (as applicable):
a. Project-specific designation, location and service.
b. Indication of the Party performing and documenting the Start-Up Procedure.
c. Clear explanation of the inspection, test, measurement, and outcome with a Pass/Fail indication and a record of measure parameters.
d. A Start-up Checklist item indicating that proper maintenance clearances have been maintained.

4. Recording and Documentation of Factory Start-Up: Manufacturer's start-up protocols shall be executed and forms shall be completed by a qualified/authorized technician.
5. Recording and Documentation of non-Factory Start-Up: The start-up tests and checklists shall be completed by a qualified technician.
6. Commissioning Authority Review: CxA will review and spot-check procedures during FPT.
7. Documentation Completion: The individual executing the start-up must complete the start-up and pre-functional documentation for any given equipment and acknowledge acceptability with the indication of who did the associated task.
8. Sampling and Final Submission: All (100% of) systems are started and documented per the approved procedures and NO sampling strategy is used. Completed Start-up and pre-functional checklists for all pieces of equipment associated with independent systems shall be submitted to CxA prior to any associated FPT. Any outstanding item shall be clearly indicated and an associated Action Item must be entered to track resolution.
9. Owner Access: Contractor shall allow access by Owner representatives to inspect the equipment and ensure its proper operation.

3.03 TEST, ADJUST, AND BALANCE

A. TEST, ADJUST, AND BALANCE
   1. CxA shall review TAB reports.
   2. The CxA will select up to 10% of the readings from the Balancing Reports and spot check them, as part of the time allocations for the various systems. If subsequent failures are found, the TAB contractor will be required to justify noted failures or rebalance and re-document the system. The maximum failure rate for this sample is 10%. The readings selected by the CxA may include water flow readings, amp readings, and water pressure drop readings through coils, heat exchangers, and other hydronic elements. For all readings a deviation of more than 10% between the verification reading and reported data shall be considered as failing the FPT.

3.04 FUNCTIONAL PERFORMANCE TESTING

A. FUNCTIONAL PERFORMANCE TESTING
   1. Objectives and Scope
      a. The objective of FPT is to demonstrate that each system is operating according to the documented design intent and Contract Documents. Functional testing facilitates bringing the systems from a state of substantial completion to full dynamic operation. Additionally, during the testing process, areas of deficient performance are identified and corrected, thus improving the function and operation of the systems.
      b. Each system shall be operated through all modes of operation (seasonal, occupied, unoccupied, warm-up, cool-down, normal and emergency power, fire alarm, part- and full-load) where there is a specified system response. Verifying each sequence in the sequences of operation is required. Proper responses to such modes and conditions as power failure, freeze condition, no flow, equipment failure, etc. shall also be tested.
   2. Development of Test Procedures
      a. CxA shall develop specific test procedures and forms to verify and document proper operation of each piece of equipment and system. Prior to execution, the CxA shall provide a copy of the test procedures to the Contractor who shall review the tests for feasibility, safety, equipment and warranty protection, and scope. The CxA will also submit the tests to the Engineer of Record for review.
      b. Contractor shall review the FPTs in detail and submit edits and comments to the CxA for possible incorporation.
      c. The purpose of any given specific FPT is to verify and document compliance with the stated criteria of acceptance, modes of operation and performance.
   3. Scheduling: After Contractors' notification that systems are ready for testing and submittal and review of all the required submittals has occurred, CM shall schedule the testing. To

the extent practical, tests shall be scheduled to allow efficient and contiguous testing of inter-related systems and equipment.

4. Phasing: Non-interdependent segments of the project testing can be phased. Phasing of this project will be determined as the project progresses.
5. Participation: CxA will direct, witness and document FPTs performed by the contractor after Start-Up Procedure documentation of systems and equipment has been reviewed and accepted. CxA will orchestrate the execution of the FPTs unless otherwise specified. Contractor shall perform the FPTs as described in section 3.4 with manipulation of the systems or equipment, provision of supporting equipment or materials (lifts, ladders, specialty test equipment, safety equipment), and on-the-spot remediation of minor identified deficiencies whenever possible.
    a. Required participating Parties shall be indicated in the individual FPT. Typically, multiple Parties are required for any given test, yet participation for any given Party is only required for the respective portion of the test for which the Party is responsible.
    b. Frequently, on multiple samples where a given party does not directly perform the test, the participation of that party will only be required for an initial quantity of systems/equipment. Whenever practical and at the discretion of the CxA with the contractor's full approval, the CxA will continue with the remaining portion of the sample without assistance from the Contractor(s) not directly performing the test. However, the Contractor is allowed to be present for any or all FPTs conducted.
    c. The required parties shall be available on-site throughout the testing of any given system for which they are required participants. Therefore, time for which they are not directly involved can be spent performing other work (typically addressing identified punch list items or failed tests).
    d. No party involved with the project is prohibited from participation in or witnessing of any tests. Any Contractor may elect to witness all tests on their systems even if their involvement is not directly required (for instance, BAS Vendor involvement is sometimes required on the first few of a sample and not on the entire sample).
    e. CxA will endeavor to coordinate effectively with the individual Contractors throughout FPT and minimize their required involvement.
6. Completeness: All systems must be completed and ready for FPT. All start up, factory authorized field testing, independent testing agency tests, and TAB procedures must be complete and the control systems must be tested and started for the respective system or component.
7. Test Documentation: CxA will witness and document the tests. CxA will record all test results on the forms developed for the testing. CxA will 'Pass' or 'Fail' the testing and record the date and time of the test. Deficiencies shall clearly be indicated when the test is failed. When all related testing is completed successfully, CxA shall recommend acceptance of the system or component. In the case of specialized testing, witness (at CxA's discretion) and review the testing reports prepared by the Contractor.

B. FPT ACCEPTANCE
1. The Acceptance Criteria shall be as follows unless specifically indicated within applicable individual specification sections or test procedures. CxA may exercise professional judgment to relax requirements and pass tests and recommend approval by the Owner and Engineer when appropriate.
    a. Accuracy/repeatability on sensing devices will be as specified for the device. CxA and TAB will use calibrated gages for independent validation and use judgment in passing or failing the devices. In many cases, the coordination of multiple related sensors is more important than absolute accuracy.
    b. HVAC sequence-related criteria will be as explicitly specified in the documents and as interpreted by the CxA.

C. FPT DEFICIENCIES

1. Non-Conformance: Non-conformance deficiencies identified during FPT shall be resolved as follows:
    a. The CxA will record the results of the functional test. All deficiencies or non-conformance issues shall be noted as Observation Log Items and reported to the Owner and CM.
    b. Corrections of identified minor deficiencies may be made during the tests at the discretion of the CxA. In such cases the deficiency will be noted on the FPT documents.
    c. Every effort will be made by the CxA to expedite the testing process and minimize unnecessary delays, while not compromising the integrity of the procedures.
    d. As tests progress and deficiencies are identified, the CxA will discuss the issue with the executing Contractor.
        1) When there is no dispute on the deficiency and the Contractor accepts responsibility to correct it:
            (a) The CxA shall document the deficiency along with the Contractor's response and intentions, and they go on to another test or sequence. A copy/email of the deficiency shall be generated and provided to the Contractor and CxA. The Contractor corrects the deficiency, completes the Action Item response certifying that the issue is resolved and/or the equipment is ready to be retested, and sends it back to the CxA.
            (b) The CxA reschedules the test and the test is repeated.
            (c) The CxA has an allowance of one re-test, cost for additional re-tests greater than this allowance will be charged to the Contractor.
        2) If there is a dispute about a deficiency, regarding whether it is a deficiency and/or who is responsible:
            (a) The deficiency shall be documented as an Observation Log Item with the Contractor's response and the Owner and CM will be notified. The CM will track this issue under the construction contract dispute resolution provisions.
            (b) Final interpretive authority is with the Engineer of Record. Final acceptance authority is with the Owner and CM.
            (c) The CxA documents the resolution to the Observation Log Item.
            (d) Once the interpretation and resolution have been decided, the appropriate party corrects the deficiency, and responds to the Action Item indicating completion. The CxA reschedules the test and the test is repeated until satisfactory performance is achieved. CxA then closes the Action Item.
2. Max Failure Limit and Sample Percentages: A Maximum Failure Limit is indicated along with the Sampling Percentages. The Max Failure Limit indicates the maximum percentage of the tested devices that may have any test that fails before an entirely new sample must be tested. This is based on the concept that if many failures occur, it is a result of inadequate start-up by the Contractor. When the maximum number of failures is reached, testing on that sample will be terminated and re-testing will be scheduled.
    a. If no Max Failure Limit is indicated, all tested samples must pass (Max Failure Limit 0%).
    b. Where sample tests involve multiple systems (i.e., checking strainers on different hydronic systems) the Maximum Failure Limit will apply per system.
    c. The responsible Contractors shall reimburse the Owner for the CxA's cost of that sample test, and redo the start-up and TAB for the applicable devices/systems.
    d. All work necessitated by sample failures shall be at no cost to the Owner.
3. Failure Due to Manufacturer's Defects: If 10% of identical pieces of equipment fail to perform to the Contract Documents (mechanically or substantively) due to manufacturing defect, all identical units may be considered unacceptable by the CM. (For the purposes of defining 'identical equipment' for this Section, size or capacity alone does not constitute

a difference.) In case of failure due to manufacturer's defects, the Contractor shall provide the Owner with the following:

a. Manufacturer's response in writing as to the cause of the failure and proposed resolution.
b. Manufacturer shall implement their proposed resolution on a representative sample of the product.
c. The CM will determine whether a replacement of all identical units or a repair is acceptable.
d. Upon acceptance, the manufacturer shall replace or repair all identical items at their expense and shall extend the warranty accordingly (if the original equipment warranty had begun).
e. Manufacturer shall pay the costs of all retesting necessitated by the failure.

3.05 CLOSEOUT

A. CX REPORT
1. A final summary report (about four to six pages, not including backup documentation) by the CxA will be provided to the CM, focusing on evaluating commissioning process issues and identifying areas where the process could be improved. All acquired documentation, logs, minutes, reports, deficiency lists, communications, findings, unresolved issues, etc., will be compiled in appendices and provided with the summary report. Pre-Start Up verification, Start Up checklists, TAB, functional tests and monitoring reports will not be part of the final report, but will be stored in the Commissioning Record in the O&M manuals.

B. ACCEPTANCE
1. The CxA notes each satisfactorily demonstrated function on the test form. Formal approval of the functional test is made later after review by the CxA and by the CM, if necessary. The CxA recommends acceptance of each test to the CM using a standard form. The CM gives final approval on each test using the same form, providing a signed copy to the CxA and the Contractor.

C. TRAINING
1. The CM shall be responsible for training coordination and scheduling and ultimately for ensuring that training is completed.
2. The CxA shall be responsible for overseeing and approving the content and adequacy of the training of Owner personnel for commissioned equipment.

3.06 OPERATION AND MAINTENANCE

A. O&M AND SYSTEM MANUAL
1. Prior to substantial completion, the CxA shall review the O&M manuals, documentation and redline as-builds for systems that were commissioned to verify compliance with the Specifications. The CxA will communicate deficiencies in the manuals to the CM or Engineer of Record, as requested. Upon a successful review of the corrections, the CxA recommends approval and acceptance of these sections of the O&M manuals to the CM or Engineer of Record. The CxA also reviews each equipment warranty and verifies that all requirements to keep the warranty valid are clearly stated. This work does not supersede the ENGINEER OF RECORD's review of the O&M manuals according to the Engineer of Record's contract.

B. WARRANTY REVIEW
1. During the warranty period, seasonal testing (tests delayed until weather conditions are closer to the system's design) shall be completed as part of this contract. The CxA shall coordinate this activity. Tests will be executed, documented and deficiencies corrected by the appropriate Subs, with facilities staff and the CxA witnessing. Any final adjustments to the O&M manuals and as-builds due to the testing will be made.

3.07 REQUIREMENTS FOR REMOTE MONITORING

A. Trend Log Object Naming Conventions
   1. The trend log object name shall follow either a client, controls contractor, or engineer supplied naming convention.
   2. If the naming convention is supplied by the controls contractor, an electronic spreadsheet copy shall be provided to the client and Engineer outlining the naming convention for approval prior to implementation.
   3. The trend log object name shall also include the equipment and building name for which it belongs to. Trend point names such as “Dev10001-AI10” are not acceptable.

B. Trend Log Intervals
   1. Binary Objects (BO & BI) - value trended as Change Of Value (COV).
   2. Analog Objects (AO & AI) - value trended on 15-minute intervals.
   3. Any setpoint (AV) that is dynamic (i.e.: is reset by control logic) - value trended on 15-minute intervals.
   4. Any setpoint (AV) that is static (fixed value that can be overridden through BAS user interface) - value trended as COV.
   5. Binary or Multi-State Mode Points (BV, MO, & MV) - value trended as COV.

C. Trend Archiving
   1. BAS controllers shall be configured and have the capacity to store a minimum of three (3) days of point trend data for all objects described above for each controller. Building controllers shall be provided to support Trend Log Objects for all control objects in controllers that do not support Trend Log Objects. The network and trending configuration shall be designed to collect required trend data without negatively affecting system performance.
   2. Enterprise Server - Trend Log data shall be collected daily from BAS controllers and stored for a minimum of 12 months on the server.

D. Owner Coordination
   1. Controls contractor to work with Owner and Engineer of Record to make available on-going scheduled extracts/exports from automation system trends or databases to Engineer for remote monitoring in a common, compatible format.

END OF SECTION

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SECTION 02 4119
SELECTIVE DEMOLITION

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. Demolition and removal of selected mechanical equipment elements.

B. Related Requirements:
   1. Section 011000 "Summary" for restrictions on use of the premises, Owner-occupancy requirements, and phasing requirements.
   2. Section 017300 "Execution" for cutting and patching procedures.
   3. Section 013516 "Alteration Project Procedures" for general protection and work procedures for alteration projects.

1.03 DEFINITIONS

A. Remove: Detach items from existing construction and dispose of them off-site unless indicated to be salvaged or reinstalled.

B. Remove and Salvage: Detach items from existing construction, in a manner to prevent damage, and deliver to Owner ready for reuse.

C. Remove and Reinstall: Detach items from existing construction, in a manner to prevent damage, prepare for reuse, and reinstall where indicated.

D. Existing to Remain: Leave existing items that are not to be removed and that are not otherwise indicated to be salvaged or reinstalled.

E. Dismantle: To remove by disassembling or detaching an item from a surface, using gentle methods and equipment to prevent damage to the item and surfaces; disposing of items unless indicated to be salvaged or reinstalled.

1.04 MATERIALS OWNERSHIP

A. Unless otherwise indicated, demolition waste becomes property of Contractor.

B. Historic items, relics, antiques, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, and other items of interest or value to Owner that may be uncovered during demolition remain the property of Owner.
   1. Carefully salvage in a manner to prevent damage and promptly return to Owner.

1.05 PREINSTALLATION MEETINGS

A. Predemolition Conference: Conduct conference at Project site.
   1. Inspect and discuss condition of construction to be selectively demolished.
   2. Review structural load limitations of existing structure.
   3. Review and finalize selective demolition schedule and verify availability of materials, demolition personnel, equipment, and facilities needed to make progress and avoid delays.
   4. Review requirements of work performed by other trades that rely on substrates exposed by selective demolition operations.
   5. Review areas where existing construction is to remain and requires protection.

1.06 INFORMATIONAL SUBMITTALS

A. Schedule of Selective Demolition Activities: Indicate the following:

1. Detailed sequence of selective demolition and removal work, with starting and ending dates for each activity. Ensure Owner's building manager's and other tenants' on-site operations are uninterrupted.
2. Interruption of utility services. Indicate how long utility services will be interrupted.
3. Coordination for shutoff, capping, and continuation of utility services.
4. Use of elevator and stairs.
5. Coordination of Owner's continuing occupancy of portions of existing building and of Owner's partial occupancy of completed Work.

B. Predemolition Photographs or Video: Show existing conditions of adjoining construction, including finish surfaces, that might be misconstrued as damage caused by demolition operations. Comply with Section 013233 "Photographic Documentation." Submit before Work begins.

1.07 CLOSEOUT SUBMITTALS

1.08 FIELD CONDITIONS

A. Owner will occupy portions of building immediately adjacent to selective demolition area. Conduct selective demolition so Owner's operations will not be disrupted.

B. Notify Engineer of discrepancies between existing conditions and Drawings before proceeding with selective demolition.

C. Storage or sale of removed items or materials on-site is not permitted.

D. Utility Service: Maintain existing utilities indicated to remain in service and protect them against damage during selective demolition operations.
1. Maintain fire-protection facilities in service during selective demolition operations.

1.09 COORDINATION

A. Arrange selective demolition schedule so as not to interfere with Owner's operations.

PART 2 PRODUCTS

2.01 PERFORMANCE REQUIREMENTS

A. Regulatory Requirements: Comply with governing EPA notification regulations before beginning selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.

B. Standards: Comply with ASSE A10.6 and NFPA 241.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that utilities have been disconnected and capped before starting selective demolition operations.

B. Review Project Record Documents of existing construction or other existing condition and hazardous material information provided by Owner. Owner does not guarantee that existing conditions are same as those indicated in Project Record Documents.

C. Survey of Existing Conditions: Record existing conditions by use of preconstruction photographs or video.
1. Comply with requirements specified in Section 013233 "Photographic Documentation."
2. Inventory and record the condition of items to be removed and salvaged. Provide photographs or video of conditions that might be misconstrued as damage caused by salvage operations.
3. Before selective demolition or removal of existing building elements that will be reproduced or duplicated in final Work, make permanent record of measurements, materials, and construction details required to make exact reproduction.

3.02 PREPARATION

3.03 UTILITY SERVICES AND MECHANICAL/ELECTRICAL SYSTEMS

A. Existing Services/Systems to Remain: Maintain services/systems indicated to remain and protect them against damage.

B. Existing Services/Systems to Be Removed, Relocated, or Abandoned: Locate, identify, disconnect, and seal or cap off utility services and mechanical/electrical systems serving areas to be selectively demolished.
   1. Arrange to shut off utilities with utility companies.
   2. If services/systems are required to be removed, relocated, or abandoned, provide temporary services/systems that bypass area of selective demolition and that maintain continuity of services/systems to other parts of building.
   3. Disconnect, demolish, and remove mechanical, plumbing, and HVAC systems, equipment, and components indicated on Drawings to be removed.
      a. Piping to Be Removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.
      b. Equipment to Be Removed: Disconnect and cap services and remove equipment.
      c. Equipment to Be Removed and Reinstalled: Disconnect and cap services and remove, clean, and store equipment; when appropriate, reinstall, reconnect, and make equipment operational.
      d. Ducts to Be Removed: Remove portion of ducts indicated to be removed and plug remaining ducts with same or compatible ductwork material.

3.04 PROTECTION

A. Temporary Protection: Provide temporary barricades and other protection required to prevent injury to people and damage to adjacent buildings and facilities to remain.
   1. Provide protection to ensure safe passage of people around selective demolition area and to and from occupied portions of building.
   2. Provide temporary weather protection, during interval between selective demolition of existing construction on exterior surfaces and new construction, to prevent water leakage and damage to structure and interior areas.
   3. Protect walls, ceilings, floors, and other existing finish work that are to remain or that are exposed during selective demolition operations.
   4. Cover and protect furniture, furnishings, and equipment that have not been removed.
   5. Comply with requirements for temporary enclosures, dust control, heating, and cooling specified in Section 015000 "Temporary Facilities and Controls."

B. Remove temporary barricades and protections where hazards no longer exist.

3.05 SELECTIVE DEMOLITION, GENERAL

A. General: Demolish and remove existing construction only to the extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations and as follows:
   1. Proceed with selective demolition systematically, from higher to lower level. Complete selective demolition operations above each floor or tier before disturbing supporting members on the next lower level.
   2. Neatly cut openings and holes plumb, square, and true to dimensions required. Use cutting methods least likely to damage construction to remain or adjoining construction. Use hand tools or small power tools designed for sawing or grinding, not hammering and chopping. Temporarily cover openings to remain.
   3. Cut or drill from the exposed or finished side into concealed surfaces to avoid marring existing finished surfaces.
   4. Do not use cutting torches until work area is cleared of flammable materials. At concealed spaces, such as duct and pipe interiors, verify condition and contents of hidden space

before starting flame-cutting operations. Maintain portable fire-suppression devices during flame-cutting operations.

5. Maintain fire watch during and for at least 1 hours after flame-cutting operations.
6. Maintain adequate ventilation when using cutting torches.
7. Remove decayed, vermin-infested, or otherwise dangerous or unsuitable materials and promptly dispose of off-site.
8. Remove structural framing members and lower to ground by method suitable to avoid free fall and to prevent ground impact or dust generation.
9. Locate selective demolition equipment and remove debris and materials so as not to impose excessive loads on supporting walls, floors, or framing.
10. Dispose of demolished items and materials promptly. Comply with requirements in Section 017419 "Construction Waste Management and Disposal."

B. Site Access and Temporary Controls: Conduct selective demolition and debris-removal operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

C. Removed and Salvaged Items:
1. Transport items to Owner's storage area designated by Owner.

D. Removed and Reinstalled Items:
1. Clean and repair items to functional condition adequate for intended reuse.
2. Pack or crate items after cleaning and repairing. Identify contents of containers.
3. Protect items from damage during transport and storage.
4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use indicated.

3.06 SELECTIVE DEMOLITION PROCEDURES FOR SPECIFIC MATERIALS

A. Concrete: Demolish in small sections. Using electric power-driven saw (gas power not permitted indoors), cut concrete to a depth of at least 3/4 inch at junctures with construction to remain. Dislodge concrete from reinforcement at perimeter of areas being demolished, cut reinforcement, and then remove remainder of concrete. Neatly trim openings to dimensions indicated.

B. Concrete: Demolish in sections. Cut concrete full depth at junctures with construction to remain and at regular intervals using power-driven saw, and then remove concrete between saw cuts.

C. Masonry: Demolish in small sections. Cut masonry at junctures with construction to remain, using power-driven saw, and then remove masonry between saw cuts.

D. Concrete Slabs-on-Grade: Saw-cut perimeter of area to be demolished, and then break up and remove.

3.07 DISPOSAL OF DEMOLISHED MATERIALS

A. Remove demolition waste materials from Project site and dispose of them in an EPA-approved construction and demolition waste landfill acceptable to authorities having jurisdiction.
1. Do not allow demolished materials to accumulate on-site.
2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
3. Remove debris from elevated portions of building by chute, hoist, or other device that will convey debris to grade level in a controlled descent.
4. Comply with requirements specified in Section 017419 "Construction Waste Management and Disposal."

B. Burning: Do not burn demolished materials.

3.08 CLEANING

A. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

END OF SECTION 024119

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SECTION 03 3000
CAST-IN-PLACE CONCRETE

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes cast-in-place concrete, including formwork, reinforcement, concrete materials, mixture design, placement procedures, and finishes, for the following:
   1. Equipment/housekeeping pads.

1.03 DEFINITIONS

A. Cementitious Materials: Portland cement alone or in combination with one or more of the following: blended hydraulic cement, fly ash and other pozzolans, ground granulated blast-furnace slag, and silica fume; subject to compliance with requirements.

1.04 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Design Mixtures: For each concrete mixture. Submit alternate design mixtures when characteristics of materials, Project conditions, weather, test results, or other circumstances warrant adjustments.
   1. Indicate amounts of mixing water to be withheld for later addition at Project site.
   2. Submit substantiating data for each concrete mix design contemplated for use to the Engineer no less than four weeks prior to first concrete placement. Data for each mix shall include the following:
      a. Mix identification number (unique for each mix submitted).
      b. Statement of intended mix use.
      c. Mixture proportions.
      d. Water/cementitious materials ratio.
      e. Wet and dry unit weight.
      f. Total air content.
      g. Design slump and allowable range after additions of all admixtures.
      h. Compressive strength tests.
   3. Shrinkage testing per ASTM C 157.

C. Steel Reinforcement Shop Drawings: Placing drawings that detail fabrication, bending, and placement. Include bar sizes, lengths, material, grade, bar schedules, stirrup spacing, bent bar diagrams, bar arrangement, splices and laps, mechanical connections, tie spacing, hoop spacing, and supports for concrete reinforcement.
   1. Show all reinforcing, top and bottom profile of concrete element, supports below, including beams, columns and walls, grade beams, concrete walls, joists, etc. framing into element.
   2. Provide one continuous elevation at 1/4" (1:48) scale for all beams, joists or walls in a continuous line. Show pockets and openings in shear walls, structural slabs, beams, elevations of top of beams, walls, columns, sections through beams, pilasters, columns, and placing sequence of reinforcing for items with more than one reinforcing layer.
   3. Show locations of approved construction joints, locations of pour strips, splices of reinforcing, type of splice used and splice location. Identify all ASTM A706 and epoxy coated reinforcing locations.

D. Construction Joint Layout: Indicate proposed construction joints required to construct the structure.
   1. Location of construction joints is subject to approval of Owner.

E. Samples: For waterstops, vapor retarder.

1.05 INFORMATIONAL SUBMITTALS

A. Material Certificates: For each of the following, signed by manufacturers:
   1. Cementitious materials.
   2. Admixtures.
   3. Form materials and form-release agents.
   4. Steel reinforcement and accessories.
   5. Waterstops.
   6. Curing compounds.
   7. Bonding agents.
   8. Vapor retarders.
   9. Repair materials.

B. Material Test Reports: For the following, from a qualified testing agency, indicating compliance with requirements:
   1. Aggregates: Submit test reports indicating that aggregates are not potentially reactive based on the ASTM C295 or ASTM 1260 testing limits set forth in section 5.1 of “Guide Specification for Concrete Subject to Alkali-Silica Reactions” (2007 Portland Cement Association). Alternatively, submit ASTM C1567 test reports indicating that the combination of mix ingredients reduces the expansion due to Alkali aggregate reactivity such that the mix complies with section 5.2 of “Guide Specification for Concrete Subject to Alkali-Silica Reactions” (2007 Portland Cement Association). All tests for submitted reports shall have been performed within one year of the submittal date.

C. Minutes of preinstallation conference.

D. Placement Notification: Submit notification at least 24 hours in advance of placement.

E. Proposed location of saw cut joints not indicated on Drawings.

F. Curing compound data demonstrating specified moisture loss performance.

G. Evaporative retarder product and application data.

1.06 QUALITY ASSURANCE

A. Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94/C 94M requirements for production facilities and equipment.

B. Testing Agency Qualifications: An independent agency, acceptable to authorities having jurisdiction, qualified according to ASTM C 1077 and ASTM E 329 for testing indicated.
   1. Personnel conducting field tests shall be qualified as ACI Concrete Field Testing Technician, Grade 1, according to ACI CP-1 or an equivalent certification program.
   2. Personnel performing laboratory tests shall be ACI-certified Concrete Strength Testing Technician and Concrete Laboratory Testing Technician - Grade I. Testing Agency laboratory supervisor shall be an ACI-certified Concrete Laboratory Testing Technician - Grade II.
   3. Personnel inspecting concrete reinforcing steel have current certification as an ACI Concrete Construction Inspector or have experience in concrete construction acceptable to the Owner.

C. Source Limitations: Obtain each type or class of cementitious material of the same brand from the same manufacturer's plant, obtain aggregate from single source, and obtain admixtures from single source from single manufacturer.

D. Welding Qualifications: Qualify procedures and personnel according to AWS D1.4/D 1.4M, "Structural Welding Code - Reinforcing Steel."

E. Formwork: Contractor shall be responsible for design and engineering of formwork. Design of formwork and preparation of formwork drawings shall be performed under supervision of a qualified engineer registered in the state of the project.

F. ACI Publications: Comply with the following unless modified by requirements in the Contract Documents:
   1. ACI 301, "Specifications for Structural Concrete," Sections 1 through 5
   2. ACI 117, "Specifications for Tolerances for Concrete Construction and Materials."

G. Preinstallation Conference: Conduct conference at Project site.
   1. Before submitting design mixtures, review concrete design mixture and examine procedures for ensuring quality of concrete materials. Require representatives of each entity directly concerned with cast-in-place concrete to attend, including the following:
      a. Contractor's superintendent.
      b. Ready-mix concrete manufacturer.
      c. Concrete subcontractor.
      d. Special concrete finish subcontractor.
      e. Owner’s Testing/Inspection Agency.
   2. Review as applicable to Project special inspection and testing and inspecting agency procedures for field quality control, concrete finishes and finishing, cold- and hot-weather concreting procedures, curing procedures, construction contraction and isolation joints, and joint-filler strips, semirigid joint fillers, forms and form removal limitations, shoring and reshoring procedures, vapor-retarder installation, anchor rod and anchorage device installation tolerances, steel reinforcement installation, floor and slab flatness and levelness measurement, concrete repair procedures, and concrete protection.
   3. Minutes of the meeting shall be recorded by Contractor and distributed to all parties within five days. Provide one copy to Owner’s representative and Engineer.

H. Record of Work: Maintain a record listing time and date of all structural concrete placement. Such record shall be kept until completion of Project and shall be available to Engineer for examination at any time.

I. Pre-Placement Inspection: Formwork installation, reinforcing steel placement and installation of all items to be embedded or cast into concrete shall be verified by Contractor prior to placement.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Steel Reinforcement: Deliver, store, and handle steel reinforcement to prevent bending and damage. Avoid damaging coatings on steel reinforcement if present.

B. Waterstops: Store waterstops under cover to protect from moisture, sunlight, dirt, oil, and other contaminants.

PART 2 PRODUCTS

2.01 FORM-FACING MATERIALS

A. Smooth-Formed Finished Concrete: Form-facing panels that will provide continuous, true, and smooth concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.
   1. Plywood, metal, or other approved panel materials.
   2. Exterior-grade plywood panels, suitable for concrete forms, complying with DOC PS 1, and as follows:
      a. High-density overlay, Class 1 or better.
      b. Medium-density overlay, Class 1 or better; mill-release agent treated and edge sealed.
      c. Structural 1, B-B or better; mill oiled and edge sealed.
      d. B-B (Concrete Form), Class 1 or better; mill oiled and edge sealed.

B. Board-Formed Finished Concrete: Elastomeric form liners for texturing architectural concrete.

1. Product: Fitzgerald Formliners, pattern 16020: Rough sawn plank
2. Accessories: Provide fasteners, sealants, rustication and backup strips, form release agents and sealers as schedule or required.

C. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.

D. Chamfer Strips: Wood, metal, PVC, or rubber strips, 3/4 by 3/4 inch (19 by 19 mm), minimum.

E. Rustication Strips: Wood, metal, PVC, or rubber strips, kerfed for ease of form removal.

F. Form-Release Agent: Commercially formulated form-release agent that will not bond with, stain, or adversely affect concrete surfaces and will not impair subsequent treatments of concrete surfaces.
1. Formulate form-release agent with rust inhibitor for steel form-facing materials.

G. Form Ties: Factory-fabricated, removable or snap-off metal or glass-fiber-reinforced plastic form ties designed to resist lateral pressure of fresh concrete on forms and to prevent spalling of concrete on removal.
1. Furnish units that will leave no corrodible metal closer than 1 1/2" inch (38 mm) to the plane of exposed concrete surface.
2. Furnish ties that, when removed, will leave holes no larger than 1 inch (25 mm) in diameter in concrete surface.
3. Furnish ties with integral water-barrier plates to walls indicated to receive dampproofing or waterproofing.

2.02 STEEL REINFORCEMENT

A. Recycled Content of Steel Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 25 percent.

B. Reinforcing Bars: ASTM A 615/A 615M, Grade 60 (Grade 420), deformed.

C. Low-Alloy-Steel Reinforcing Bars: ASTM A 706/A 706M, deformed, where welding of reinforcement or field bending is noted on Drawings.

D. Epoxy-Coated Reinforcing Bars: ASTM A 615/A 615M, Grade 60 (Grade 420) ASTM A 706/A 706M for bars that may be field bent, deformed bars, ASTM A 775/A 775M or ASTM A 934/A 934M for bars that are prefabricated, epoxy coated, with less than 2 percent damaged coating in each 12-inch (300-mm) bar length.

E. Plain-Steel Wire: ASTM A 82/A 82M, as drawn.

F. Epoxy-Coated Wire: ASTM A 884/A 884M, Class A, Type 1 coated, as-drawn, plain steel wire, with less than 2 percent damaged coating in each 12-inch (300-mm) wire length.

G. Plain-Steel Welded Wire Reinforcement: ASTM A 185/A 185M, plain, fabricated from as-drawn steel wire into flat sheets.

H. Epoxy-Coated Welded Wire Reinforcement: ASTM A 884/A 884M, Class A coated, Type 1, plain steel.

2.03 REINFORCEMENT ACCESSORIES

A. Epoxy Repair Coating: Liquid, two-part, epoxy repair coating; compatible with epoxy coating on reinforcement and complying with ASTM A 775/A 775M.

B. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded wire reinforcement in place. Manufacture bar supports from steel wire, plastic, or precast concrete according to CRSI's "Manual of Standard Practice," of greater compressive strength than concrete and as follows:
1. For concrete surfaces exposed to view where legs of wire bar supports contact forms, use CRSI Class 1 plastic-protected steel wire or CRSI Class 2 stainless-steel bar supports.
2. For epoxy-coated reinforcement, use epoxy-coated or other dielectric-polymer-coated wire bar supports.

2.04 CONCRETE MATERIALS

A. Cementitious Material: Use the following cementitious materials, of the same type, brand, and source, throughout Project:
   1. Portland Cement: ASTM C 150, Type I/II, gray. Alternate cementitious materials when proposed to control alkali-silica reactions and tested as part of a representative concrete mix in accordance with ASTM C1567, may be used subject to approval. Supplement with the following:
      a. Fly Ash: ASTM C 618, Class F or C.

B. Normal-Weight Aggregates: ASTM C 33, Class 3S coarse aggregate or better, graded. Provide aggregates from a single source with documented service record data of at least 10 years' satisfactory service in similar applications and service conditions using similar aggregates and cementitious materials. All coarse and fine aggregate shall be tested per ASTM C 295 or ASTM C 1293 in accordance with section 5.1 of “Guide Specification for Concrete Subject to Alkali-Silica Reactions” (2007 Portland Cement Association).
   1. Maximum Coarse-Aggregate Size: As indicated on Drawings.
   2. Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.

C. Lightweight Aggregate: ASTM C 330, with size as indicated on Drawings.

D. Water: ASTM C 94/C 94M and potable.

2.05 ADMIXTURES

A. Air-Entraining Admixture: ASTM C 260.

B. Chemical Admixtures: Provide admixtures certified by manufacturer to be compatible with other admixtures and that will not contribute water-soluble chloride ions exceeding those permitted in hardened concrete. Do not use calcium chloride or admixtures containing calcium chloride.
   1. Water-Reducing Admixture: ASTM C 494/C 494M, Type A.
   2. Retarding Admixture: ASTM C 494/C 494M, Type B.
   3. Water-Reducing and Retarding Admixture: ASTM C 494/C 494M, Type D.
   4. High-Range, Water-Reducing Admixture: ASTM C 494/C 494M, Type F.
   5. High-Range, Water-Reducing and Retarding Admixture: ASTM C 494/C 494M, Type G.
   6. Plasticizing and Retarding Admixture: ASTM C 1017/C 1017M, Type II.
   7. Mid-Range Water Reducing Admixture: ASTM C 494/C 494M, Type A.

C. Color Pigment: ASTM C 979, synthetic mineral-oxide pigments or colored water-reducing admixtures; color stable, free of carbon black, nonfading, and resistant to lime and other alkalis.
   1. Manufacturers: available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
      a. ChemMasters.
      b. Davis Colors.
      c. Dayton Superior Corporation.
      d. Hoover Color Corporation.
      e. Lambert Corporation.
      f. QC Construction Products.
      g. Rockwood Pigments NA, Inc.
      h. Scofield, L. M. Company.
      i. Solomon Colors, Inc.
   2. Color: “Mesa Buff” 5447 by Davis Colors.

2.06 WATERSTOPS

A. Self-Expanding Butyl Strip Waterstops: Manufactured rectangular or trapezoidal strip, butyl rubber with sodium bentonite or other hydrophilic polymers, for adhesive bonding to concrete, 3/4 by 1 inch (19 by 25 mm).

1. Products: Subject to compliance with requirements, provide one of the following:
   a. Carlisle Coatings & Waterproofing, Inc.; MiraSTOP.
   b. CETCO; Volclay Waterstop-RX.
   c. Concrete Sealants Inc.; Conseal CS-231.
   d. Greenstreak; Swellstop.
   e. Henry Company, Sealants Division; Hydro-Flex.
   f. JP Specialties, Inc.; Earth Shield Type 20.

2.07 VAPOR RETARDERS

A. Sheet Vapor Retarder: ASTM E 1745, Class A. Include manufacturer's recommended adhesive or pressure-sensitive tape.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Carlisle Coatings & Waterproofing, Inc.; Blackline 400.
      b. Fortifiber Building Systems Group; Moistop Ultra 15.
      c. Grace Construction Products, W. R. Grace & Co.; Florprufe 120.
      d. Insulation Solutions, Inc.; Viper VaporCheck 16.
      e. Meadows, W. R., Inc.; Perminator 15 mil.
      f. Raven Industries Inc.; Vapor Block 15.
      g. Stego Industries, LLC; Stego Wrap 15 mil Class A.

B. Granular Fill: Clean mixture of crushed stone or crushed or uncrushed gravel; ASTM D 448, Size 57, with 100 percent passing a 1-1/2-inch (37.5-mm) sieve and 0 to 5 percent passing a No. 8 (2.36-mm) sieve.

C. Fine-Graded Granular Material: Clean mixture of crushed stone, crushed gravel, and manufactured or natural sand; ASTM D 448, Size 10, with 100 percent passing a 3/8-inch (9.5-mm) sieve, 10 to 30 percent passing a No. 100 (0.15-mm) sieve, and at least 5 percent passing No. 200 (0.075-mm) sieve; complying with deleterious substance limits of ASTM C 33 for fine aggregates.

2.08 CURING MATERIALS

A. Evaporation Retarder: Waterborne, monomolecular film forming, manufactured for application to fresh concrete.

B. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. (305 g/sq. m) when dry.

C. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

D. Water: Potable.

E. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C 309, Type 1, Class B, dissipating.

F. Clear, Solvent-Borne, Membrane-Forming Curing and Sealing Compound: ASTM C 1315, Type 1, Class A.

G. Clear, Waterborne, Membrane-Forming Curing and Sealing Compound: ASTM C 1315, Type 1, Class A.

2.09 RELATED MATERIALS

A. Expansion- and Isolation-Joint-Filler Strips: ASTM D 1751, asphalt-saturated cellulosic fiber

B. Bonding Agent: ASTM C 1059/C 1059M, Type II, non-redispersible, acrylic emulsion or styrene butadiene.

C. Reglets: Fabricate reglets of not less than 0.022-inch- (0.55-mm-) thick, galvanized-steel sheet. Temporarily fill or cover face opening of reglet to prevent intrusion of concrete or debris.

2.10 REPAIR MATERIALS

A. Repair Underlayment: Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 1/8 inch (3.2 mm) and that can be feathered at edges to match adjacent floor elevations.
   1. Cement Binder: ASTM C 150, portland cement or hydraulic or blended hydraulic cement as defined in ASTM C 219.
   2. Primer: Product of underlayment manufacturer recommended for substrate, conditions, and application.
   3. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch (3.2 to 6 mm) or coarse sand as recommended by underlayment manufacturer.
   4. Compressive Strength: Not less than 5000 psi (29 MPa) at 28 days when tested according to ASTM C 109/C 109M.

B. Repair Overlayment: Cement-based, polymer-modified, self-leveling product that can be applied in thicknesses from 1/4 inch (6.4 mm) and that can be filled in over a scarified surface to match adjacent floor elevations.
   1. Cement Binder: ASTM C 150, portland cement or hydraulic or blended hydraulic cement as defined in ASTM C 219.
   2. Primer: Product of topping manufacturer recommended for substrate, conditions, and application.
   3. Aggregate: Well-graded, washed gravel, 1/8 to 1/4 inch (3.2 to 6 mm) or coarse sand as recommended by topping manufacturer.
   4. Compressive Strength: Not less than 5000 psi (34.5 MPa) at 28 days when tested according to ASTM C 109/C 109M.

2.11 CONCRETE MIXTURES, GENERAL

A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, according to ACI 301.
   1. Use a qualified testing agency for preparing and reporting proposed mixture designs based on laboratory trial mixtures.

B. Cementitious Materials: Limit percentage, by weight, of cementitious materials other than portland cement in concrete as follows:
   1. Fly Ash: 25 percent.

C. Limit water-soluble, chloride-ion content in hardened concrete to 0.15 for reinforced concrete exposed to chlorides in service, and 0.30 for other reinforced concrete percent by weight of cement.

D. Admixtures: Use admixtures according to manufacturer's written instructions.
   1. Use water-reducing, high-range water-reducing, or plasticizing admixture in concrete, as required, for placement and workability.
   2. Use water-reducing and retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.
   3. Use water-reducing admixture in pumped concrete and concrete with a water-cementitious materials ratio below 0.50.

2.12 CONCRETE MIXTURES FOR BUILDING ELEMENTS

A. Proportion structural normal-weight concrete mixture as noted on Drawings, unless aggregates are “potentially reactive” with alkalis based on the ASTM C 295 or ASTM C 1260 or ASTM C 1293 testing limits of Section 5.1 of “Guide Specification of Concrete Subject to Alkali-Silica Reactions” (2007 Portland Cement Association). When aggregates are “potentially reactive”, compliance with Section 5.2 of “Guide Specification for Concrete Subject to Alkali-Silica Reactions” (2007 Portland Cement Association) must be established through ASTM C 1567 testing for proposed alternate concrete mixture. Submit test reports in accordance with Part I of this Specification.

2.13 FABRICATING REINFORCEMENT

A. Fabricate steel reinforcement according to CRSI's "Manual of Standard Practice."

2.14 CONCRETE MIXING

A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C 94/C 94M and ASTM C 1116/C 1116M, and furnish batch ticket information.

PART 3 EXECUTION

3.01 FORMWORK

A. Design, erect, shore, brace, and maintain formwork, according to ACI 301, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until structure can support such loads.

B. Construct formwork so concrete members and structures are of size, shape, alignment, elevation, and position indicated, within tolerance limits of ACI 117. Concrete adjacent to elevator hoistway shall be installed to tolerances required by elevator manufacturer.

C. Limit concrete surface irregularities, designated by ACI 347 as abrupt or gradual, as follows:
   1. Class A, 1/8 inch (3.2 mm) for smooth-formed finished surfaces.
   2. Class B, 1/4 inch (6 mm) for rough-formed finished surfaces.
   3. The permissible irregularity is a cumulative value due to all sources including layout, plumbness, member size, formwork offsets, joints, and member levelness. The permissible irregularity shall also apply between adjacent concrete surfaces on opposite sides of construction joint.

D. Construct forms tight enough to prevent loss of concrete mortar.

E. Fabricate forms for easy removal without hammering or prying against concrete surfaces. Provide crush or wrecking plates where stripping may damage cast concrete surfaces. Provide top forms for inclined surfaces steeper than 1.5 horizontal to 1 vertical.
   1. Install keyways, reglets, recesses, and the like, for easy removal.
   2. Do not use rust-stained steel form-facing material.

F. Set edge forms, bulkheads, and intermediate screed strips for slabs to achieve required elevations and slopes in finished concrete surfaces. Provide and secure units to support screed strips; use strike-off templates or compacting-type screeds.

G. Provide temporary openings for cleanouts and inspection ports where interior area of formwork is inaccessible. Close openings with panels tightly fitted to forms and securely braced to prevent loss of concrete mortar. Locate temporary openings in forms at inconspicuous locations.

H. Chamfer exterior corners and edges of permanently exposed concrete.

I. Form openings, chases, offsets, sinkages, keyways, reglets, blocking, screeds, and bulkheads required in the Work. Determine sizes and locations from trades providing such items.

J. Clean forms and adjacent surfaces to receive concrete. Remove chips, wood, sawdust, dirt, and other debris just before placing concrete.

K. Retighten forms and bracing before placing concrete, as required, to prevent mortar leaks and maintain proper alignment.

L. Coat contact surfaces of forms with form-release agent, according to manufacturer's written instructions, before placing reinforcement.

M. All formwork surfaces that support concrete exposed to view must be accepted by Owner prior to concrete placement.

3.02 EMBEDDED ITEMS

A. Place and secure anchorage devices and other embedded items required for adjoining work that is attached to or supported by cast-in-place concrete. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
   1. Install anchor rods, accurately located, to elevations required and complying with tolerances in Section 7.5 of AISC 303 "Code of Standard Practice for Steel Buildings and Bridges."
      a. Tolerance of Embedded Items: Comply with ACI 117.
         1) Anchor Rods:
            (a) Plumbness: Within +/- 1/16 inch (2 mm) over anchor rod projection.
         2) Embedded Plates and Weldments:
            (a) Location: +/- 1 inch (25 mm) vertical, +/- 1 inch (25 mm) [2 inches (51 mm)] horizontal.
            (b) Plumb and Alignment: 1/4 inch in 12 inches (1:48).
   2. Install reglets to receive waterproofing and to receive through-wall flashings in outer face of concrete frame at exterior walls, where flashing is shown at lintels, shelf angles, and other conditions.

3.03 REMOVING AND REUSING FORMS

A. General: Formwork for sides of beams, walls, columns, and similar parts of the Work that does not support weight of concrete may be removed after cumulatively curing at not less than 50 deg F (10 deg C) for 24 hours after placing concrete. Concrete has to be hard enough to not be damaged by form-removal operations and curing and protection operations need to be maintained.
   1. Leave formwork for beam soffits, joists, slabs, and other structural elements that supports weight of concrete in place until concrete has achieved at least 70 percent of its 28-day design compressive strength.
   2. Remove forms only if shores have been arranged to permit removal of forms without loosening or disturbing shores.
   3. Leave formwork and shoring in place a minimum of 15 days after concrete placement unless reshoring is used.

B. Clean and repair surfaces of forms to be reused in the Work. Split, frayed, delaminated, or otherwise damaged form-facing material will not be acceptable for exposed surfaces. Apply new form-release agent.

C. When forms are reused, clean surfaces, remove fins and laitance, and tighten to close joints. Align and secure joints to avoid offsets. Do not use patched forms for exposed concrete surfaces unless approved by Engineer.

3.04 SHORES AND RESHORES

A. Comply with ACI 318 (ACI 318M) and ACI 301 for design, installation, and removal of shoring and reshoring.
   1. Do not remove shoring or reshoring until measurement of slab tolerances is complete.

B. Plan sequence of removal of shores and reshore to avoid damage to concrete. Locate and provide adequate reshoring to support construction without excessive stress or deflection.

C. Reshoring:
   1. If formwork and shoring are removed before concrete is 15 days old, reshoring shall remain in place a minimum of 15 days after placement irrespective of concrete strength.
   2. For multi-story construction, reshoring shall remain in place a minimum of 15 days after placing of concrete deck above irrespective of concrete strength.

3.05 VAPOR RETARDERS

A. Sheet Vapor Retarders: Place, protect, and repair sheet vapor retarder according to ASTM E 1643 and manufacturer's written instructions.
   1. Lap joints 6 inches (150 mm) and seal with manufacturer's recommended tape.

B. Bituminous Vapor Retarders: Place, protect, and repair bituminous vapor retarder according to manufacturer's written instructions.

C. Granular Course: Cover vapor retarder with granular fill, moisten, and compact with mechanical equipment to elevation tolerances of plus 0 inch (0 mm) or minus 3/4 inch (19 mm).
   1. Place and compact a 1/2-inch- (13-mm-) thick layer of fine-graded granular material over granular fill.

3.06 STEEL REINFORCEMENT

A. General: Comply with CRSI's "Manual of Standard Practice" for placing reinforcement.
   1. Do not cut or puncture vapor retarder. Repair damage and reseal vapor retarder before placing concrete.

B. Clean reinforcement of loose rust and mill scale, earth, ice, and other foreign materials that would reduce bond to concrete.

C. Accurately position, support, and secure reinforcement against displacement. Locate and support reinforcement with bar supports to maintain minimum concrete cover. Do not tack weld crossing reinforcing bars.
   1. Weld reinforcing bars according to AWS D1.4/D 1.4M, where indicated.

D. Size, length, number and placement of supports shall be sufficient as to maintain reinforcing position within specified tolerances during construction traffic and concrete placement.

E. On vertical formwork, use approved bar chairs or spacers as required to maintain concrete cover and bar position. Do not staple or use any other metallic fastener to secure bolsters, chairs, etc. to formwork for concrete surfaces exposed to exterior.

F. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

G. Install welded wire reinforcement in longest practicable lengths on bar supports spaced to minimize sagging. Lap edges and ends of adjoining sheets at least one mesh spacing. Offset laps of adjoining sheet widths to prevent continuous laps in either direction. Lace overlaps with wire.

H. Epoxy-Coated Reinforcement: Repair cut and damaged epoxy coatings with epoxy repair coating according to ASTM D 3963/D 3963M. Use epoxy-coated steel wire ties to fasten epoxy-coated steel reinforcement.

3.07 JOINTS

A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.

B. Construction Joints: Install so strength and appearance of concrete are not impaired, at locations indicated or as approved by Engineer.
   1. Place joints perpendicular to main reinforcement. Continue reinforcement across construction joints unless otherwise indicated.
   2. Form keyed joints as indicated. Embed keys at least 1-1/2 inches (38 mm) into concrete.
   3. Locate horizontal joints in walls and columns at underside of floors, slabs, beams, and girders and at the top of footings or floor slabs.
   4. Space vertical joints in walls as indicated. Locate joints near corners, and in concealed locations where possible.

C. Slabs Exposed to View: Locate control joints where shown on the Contract Drawings. If not shown, locate topping slab control joints at column centerlines, over girders and at intervals not more than 10 feet (3 m) feet each way.

3.08 WATERSTOPS

A. Flexible Waterstops: Install in construction joints and at other joints indicated to form a continuous diaphragm. Install in longest lengths practicable. Support and protect exposed waterstops during progress of the Work. Field fabricate joints in waterstops according to manufacturer's written instructions.

B. Self-Expanding Strip Waterstops: Install in construction joints and at other locations indicated, according to manufacturer's written instructions, adhesive bonding, mechanically fastening, and firmly pressing into place. Install in longest lengths practicable.

3.09 CONCRETE PLACEMENT

A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections have been performed.

B. Before test sampling and placing concrete, water may be added at Project site, subject to limitations of ACI 301.
   1. Do not add water to concrete after adding high-range water-reducing admixtures to mixture.

C. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete will be placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.
   1. Deposit concrete in horizontal layers of depth to not exceed formwork design pressures and in a manner to avoid inclined construction joints.
   2. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.
   3. Do not use vibrators to transport concrete inside forms. Insert and withdraw vibrators vertically at uniformly spaced locations to rapidly penetrate placed layer and at least 6 inches (150 mm) into preceding layer. Do not insert vibrators into lower layers of concrete that have begun to lose plasticity. At each insertion, limit duration of vibration to time necessary to consolidate concrete and complete embedment of reinforcement and other embedded items without causing mixture constituents to segregate.

D. Deposit and consolidate concrete for floors and slabs in a continuous operation, within limits of construction joints, until placement of a panel or section is complete.
   1. Consolidate concrete during placement operations so concrete is thoroughly worked around reinforcement and other embedded items and into corners.
   2. Maintain reinforcement in position on chairs during concrete placement.
   3. Screed slab surfaces with a straightedge and strike off to correct elevations.
      a. Slabs supported by metal deck shall be gaged to provide the specified slab thickness over beams.
   4. Slope surfaces uniformly to drains where required.
   5. Begin initial floating using bull floats or darbies to form a uniform and open-textured surface plane, before excess bleedwater appears on the surface. Do not further disturb slab surfaces before starting finishing operations.

E. Cold-Weather Placement: Comply with ACI 306.1 and as follows. Protect concrete work from physical damage or reduced strength that could be caused by frost, freezing actions, or low temperatures.
   1. When average high and low temperature is expected to fall below 40 deg F (4.4 deg C) for three successive days, maintain delivered concrete mixture temperature within the temperature range required by ACI 301.
   2. Do not use frozen materials or materials containing ice or snow. Do not place concrete on frozen subgrade or on subgrade containing frozen materials.
   3. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.

F. Hot-Weather Placement: Comply with ACI 301 and as follows:
   1. Maintain concrete temperature below 90 deg F (32 deg C) at time of placement. Chilled mixing water or chopped ice may be used to control temperature, provided water equivalent of ice is calculated to total amount of mixing water. Using liquid nitrogen to cool concrete is Contractor's option.

2. Fog-spray forms, steel reinforcement, and subgrade just before placing concrete. Keep subgrade uniformly moist without standing water, soft spots, or dry areas.

3.10 FINISHING FORMED SURFACES

A. Rough-Formed Finish: As-cast concrete texture imparted by form-facing material with tie holes and defects repaired and patched. Remove fins and other projections that exceed specified limits on formed-surface irregularities.
   1. Apply to concrete surfaces not exposed to public view.

B. Smooth-Formed Finish: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Repair and patch tie holes and defects. Remove fins and other projections that exceed specified limits on formed-surface irregularities.
   1. Apply to concrete surfaces exposed to public view.

C. Related Unformed Surfaces: At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a texture matching adjacent formed surfaces. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces unless otherwise indicated.

3.11 FINISHING SLABS

A. General: Comply with ACI 302.1R recommendations for screeding, restraightening, and finishing operations for concrete surfaces. Do not wet concrete surfaces.

B. Float Finish: Consolidate surface with power-driven floats or by hand floating if area is small or inaccessible to power driven floats. Restraighten, cut down high spots, and fill low spots. Repeat float passes and restraightening until surface is left with a uniform, smooth, granular texture.
   1. Apply float finish to surfaces to be covered with fluid-applied or sheet waterproofing

C. Broom Finish: Apply a broom finish to exterior concrete platforms, steps, ramps, and elsewhere as indicated.
   1. Immediately after float finishing, slightly roughen trafficked surface by brooming with fiber-bristle broom perpendicular to main traffic route. Coordinate required final finish with Engineer before application.

3.12 MISCELLANEOUS CONCRETE ITEMS

A. Filling In: Fill in holes and openings left in concrete structures after work of other trades is in place unless otherwise indicated. Mix, place, and cure concrete, as specified, to blend with in-place construction. Provide other miscellaneous concrete filling indicated or required to complete the Work.

B. Curbs: Provide monolithic finish to interior curbs by stripping forms while concrete is still green and by steel-troweling surfaces to a hard, dense finish with corners, intersections, and terminations slightly rounded.

C. Equipment Bases and Foundations: Provide machine and equipment bases and foundations as shown on Drawings. Set anchor bolts for machines and equipment at correct elevations, complying with diagrams or templates from manufacturer furnishing machines and equipment.

D. Steel Pan Stairs: Provide concrete fill for steel pan stair treads, landings, and associated items. Cast-in inserts and accessories as shown on Drawings. Screed, tamp, and trowel finish concrete surfaces.

3.13 CONCRETE PROTECTING AND CURING

A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 301 for hot-weather protection during curing.

B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h (1 kg/sq. m x h) before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.

C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for the remainder of the curing period.

D. Unformed Surfaces: Begin curing immediately after finishing concrete. Cure unformed surfaces, including floors and slabs, concrete floor toppings, and other surfaces.

E. Cure concrete according to ACI 308.1, by one or a combination of the following methods:
   1. Moisture Curing: Keep surfaces continuously moist for not less than seven days with the following materials:
      a. Water.
      b. Continuous water-fog spray.
      c. Absorptive cover, water saturated, and kept continuously wet. Cover concrete surfaces and edges with 12-inch (300-mm) lap over adjacent absorptive covers.
   2. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches (300 mm), and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period using cover material and waterproof tape. Moisture-retaining-cover shall be inspected each day by Contractor. Any areas which do not show condensation on underside of cover or any slab areas which are not wet shall be immediately rewetted and cover replaced to prevent moisture loss.
      a. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive floor coverings.
      b. Moisture cure or use moisture-retaining covers to cure concrete surfaces to receive penetrating liquid floor treatments.
      c. Cure concrete surfaces to receive floor coverings with either a moisture-retaining cover or a curing compound that the manufacturer certifies will not interfere with bonding of floor covering used on Project.
   3. Curing Compound: Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.
      a. Removal: After curing period has elapsed, remove curing compound without damaging concrete surfaces by method recommended by curing compound manufacturer.
   4. Curing and Sealing Compound: Apply uniformly to floors and slabs indicated in a continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Repeat process 24 hours later and apply a second coat. Maintain continuity of coating and repair damage during curing period.

3.14 CONCRETE SURFACE REPAIRS

A. Defective Concrete: Repair and patch defective areas when approved by Engineer. Remove and replace concrete that cannot be repaired and patched to Engineer's approval.

B. Patching Mortar: Mix dry-pack patching mortar, consisting of one part portland cement to two and one-half parts fine aggregate passing a No. 16 (1.18-mm) sieve, using only enough water for handling and placing.

C. Repairing Formed Surfaces: Surface defects include color and texture irregularities, cracks, spalls, air bubbles, honeycombs, rock pockets, fins and other projections on the surface, and stains and other discolorations that cannot be removed by cleaning and that are unacceptable to Engineer. Allow Engineer and Structural Engineer to observe concrete surfaces upon removal of forms and prior to repair of surface defects. Defects in structural concrete shall be brought to the attention of the Engineer and Structural Engineer.
   1. Immediately after form removal, cut out honeycombs, rock pockets, and voids more than 1/2 inch (13 mm) in any dimension to solid concrete. Limit cut depth to 3/4 inch (19 mm). Make edges of cuts perpendicular to concrete surface. Clean, dampen with water, and brush-coat holes and voids with bonding agent. Fill and compact with patching mortar before bonding agent has dried. Fill form-tie voids with patching mortar or cone plugs secured in place with bonding agent.
   2. Repair defects on surfaces exposed to view by blending white portland cement and standard portland cement so that, when dry, patching mortar will match surrounding color. Patch a test area at inconspicuous locations to verify mixture and color match before proceeding with patching. Compact mortar in place and strike off slightly higher than surrounding surface.
   3. Repair defects on concealed formed surfaces that affect concrete's durability and structural performance as determined by Engineer.

D. Repairing Unformed Surfaces: Test unformed surfaces, such as floors and slabs, for finish and verify surface tolerances specified for each surface. Correct low and high areas. Test surfaces sloped to drain for trueness of slope and smoothness; use a sloped template. Submit proposed repair to Engineer for review prior to commencement of work.
   1. Repair finished surfaces containing defects that are unacceptable to Engineer. Surface defects include spalls, popouts, honeycombs, rock pockets, crazing and cracks in excess of 0.01 inch (0.25 mm) wide or that penetrate to reinforcement or completely through unreinforced sections regardless of width, and other objectionable conditions.
   2. After concrete has cured at least 14 days, correct high areas by grinding.
   3. Correct localized low areas during or immediately after completing surface finishing operations by cutting out low areas and replacing with patching mortar. Finish repaired areas to blend into adjacent concrete.
   4. Correct other low areas scheduled to receive floor coverings with a repair underlayment. Prepare, mix, and apply repair underlayment and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface. Feather edges to match adjacent floor elevations.
   5. Correct other low areas scheduled to remain exposed with a repair topping. Cut out low areas to ensure a minimum repair topping depth of 1/4 inch (6 mm) to match adjacent floor elevations. Prepare, mix, and apply repair topping and primer according to manufacturer's written instructions to produce a smooth, uniform, plane, and level surface.
   6. Repair defective areas, except random cracks and single holes 1 inch (25 mm) or less in diameter, by cutting out and replacing with fresh concrete. Remove defective areas with clean, square cuts and expose steel reinforcement with at least a 3/4-inch (19-mm) clearance all around. Dampen concrete surfaces in contact with patching concrete and apply bonding agent. Mix patching concrete of same materials and mixture as original concrete except without coarse aggregate. Place, compact, and finish to blend with adjacent finished concrete. Cure in same manner as adjacent concrete.
   7. Repair random cracks and single holes 1 inch (25 mm) or less in diameter with patching mortar. Groove top of cracks and cut out holes to sound concrete and clean off dust, dirt, and loose particles. Dampen cleaned concrete surfaces and apply bonding agent. Place patching mortar before bonding agent has dried. Compact patching mortar and finish to match adjacent concrete. Keep patched area continuously moist for at least 72 hours.

E. Perform structural repairs of concrete, subject to Engineer's approval, using epoxy adhesive and patching mortar.

F. Repair materials and installation not specified above may be used, subject to Engineer's approval.

3.15 FIELD QUALITY CONTROL

A. Testing and Inspection: As indicated on Drawings.

END OF SECTION

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SECTION 23 0010
HVAC (HEATING, VENTILATION AND AIR CONDITIONING) GENERAL PROVISIONS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section. Consult them for further instructions and be governed by the requirements thereunder.

B. Related Work is specified elsewhere in Divisions 26 and when issued by the Owner, is hereby incorporated.

1.02 GENERAL

A. Work Included
   1. Furnish all labor, materials, equipment and related items, and perform all operations required to complete work within the intent of the Drawings and Specifications, whether or not specifically mentioned, and to deliver complete and fully operational HVAC systems subject to the conditions of the Contract. For this reason, the Contractor shall visit the premises and site before submitting their bid and familiarize themselves with the areas in which work is to be done.
   2. Provide HVAC, plumbing, and electrical details not mentioned or shown which are necessary for the successful operation of all systems. Clean, sterilize, flush and fill all systems per requirements to make them operational; including labor and materials for final fill of water, refrigerant, oils, grease, gases, antifreeze and brine.
   3. Set all sleeves and cut and patch all miscellaneous holes necessary for the convenient, orderly and proper installation of the work. Required holes through masonry and concrete construction with an area less than thirty five (35) square inches (20 inch diameter and less) shall be considered miscellaneous holes.
   4. Any work installed without regard to the work of other crafts which must, in the opinion of the Owner or Engineer, be moved to permit the installation of other work, shall be moved and replaced as a part of this work at no additional charge.
   5. Rough-in for and connect, as shown on the drawings, all equipment furnished by the
   6. Prove satisfactory operation of all equipment and controls to the Owner, and/or Engineer upon request.

B. Work Not Included
   1. Certain labor and materials may be furnished and/or installed under other divisions of these specifications. Coordinate with other trades and arrange the work to make the parts fit together. The following items are to be accomplished under other divisions of these specifications.
      a. Temporary Heat: See "Temporary Heat" Paragraph in this Specification Section and Division 01.
      b. Roof Curbs: See "Roof Curbs" Paragraph in this Specification Section.
      c. Concrete: See "Concrete" Paragraph in this Specification Section.
      d. Electrical Equipment and Wiring: See "Electrical Equipment and Wiring for Mechanical Division" Paragraph in this Specification Section.
      e. Temporary Water and Toilet: See Division 01.

C. Equipment Furnished by Owner
   1. The Owner will award contracts, which will commence concurrently with this contract. Specifically this work will include:
   2. Equipment Installation: Refer to appropriate drawings for equipment furnished by the Owner.

3. Rough-in service pipes to locations as required by Engineer and mechanical drawings and equipment shop drawings. Provide service valves on all pipes except waste and vent pipes, plug or cap these. Final connections to equipment shall be made by Contractor.

1.03 QUALITY ASSURANCE

A. Qualifications of Contractor: All materials and equipment shall be new and all work shall be executed with the maximum speed consistent with current accepted trade practices. Furnish materials and equipment promptly after authorization to proceed, and proceed with work in progress with the other contractors on the project. Perform all work included in the contract in a manner that will not cause interferences or delays to, or interfere with, the progress of other contractors.

B. At all times when work is not in progress, keep all open ends of pipe, ductwork, fittings, equipment and fixtures securely closed and protected.

C. All welding shall comply with the requirements and recommendations of the American Welding Society and all applicable codes.
   1. Weld metal shall not project creating an obstruction.
   2. Chip or grind out all weld metal before re-welding.
   3. Caulking / preening of welds is not allowed.
   4. Welder shall be certified to work on service/utility type indicated.

1.04 CODES AND REFERENCED STANDARDS

A. Comply with specified Codes and Standards. If conflict exists between Codes or Standards and drawings, project specifications manual or addenda requirements, the most stringent requirement shall apply.

B. Conform to the installation rules and regulations of the standards listed including all subsequently published amendments thereto issued prior to the date of the bidding documents.

C. Conform to the requirements of all local, state and federal agencies which have authority over the project. Include all items of labor and material required to meet such requirements regardless of the failure to specify in the project manual or indicated on the drawings each individual item.

D. All equipment, apparatus and systems shall be rated, tested, fabricated and installed with the applicable industry standards.

E. The applicable portions of the following standards form a part of this project manual to the same force and effect as if repeated herein.
   1. American National Standards Institute (ANSI)
   2. International Mechanical Code (IMC)
   3. International Plumbing Code (IPC)
   4. American National Standards Institute (ANSI)
   5. American Gas Association, Inc. (AGA)
   6. American Society for Testing Materials (ASTM)
   7. American Society of Heating, Refrigeration, and Air Conditioning Engineers (ASHRAE)
   8. American Society of Mechanical Engineers (ASME)
   9. American Water Works Association (AWWA)
   10. National Electrical Code (NEC)
   11. National Electric Manufacturers Association (NEMA)
   12. National Fire Protection Association (NFPA)
   13. Sheet Metal and Air Conditioning Contractors National Association, Inc. (SMACNA)
   14. Underwriters Laboratories, Inc. (UL)
   15. Environmental Protection Agency (EPA)

F. Conform to the International Building, Fire, and Mechanical Codes, 2024 edition.

G. Conform to the International Energy Conservation Code, 2024 edition.

H. Conform to applicable sections of NFPA 13 and NFPA 24.

I. Conform to the National Electrical Code, 2023 Edition.

J. Conform to the requirements of the Americans with Disabilities Acts (ADA) and American National Standards Institute (ANSI) Standard 117.

K. Conform to Colorado Department of Public Health and Environment, Health Facilities and Emergency Medical Services Division, "Standards for Hospitals and Health Facilities".

L. All work shall be furnished and installed in complete accordance with Code seismic requirements.

M. In case of difference between these specifications, codes, laws, industry standards, and/or utility company regulations the most stringent requirement shall govern.

1.05 EXAMINATION OF BIDDING DOCUMENTS

A. Each bidder shall examine the bidding documents carefully, and not later than seven days prior to the date of receipt of bids, shall make written request to the Engineer for interpretation or correction of any discrepancies, ambiguity, inconsistency, or error therein which he may discover. Any interpretation or correction will be issued as an addendum by the Engineer. Only a written interpretation or correction by addendum shall be binding. No bidder shall rely upon interpretations or corrections given by any other method. If discrepancies, ambiguity, inconsistency, or error are not covered by addendum or written directive, Contractor shall include in their bid, labor materials and methods of construction resulting in higher cost. After award of contract, no allowance or extra compensation will be made on behalf of the Contractor due to his failure to make the written requests as described above.

B. In order to become familiar with the scope of the work involved, visit the existing site, before submitting bid, and carefully examine the existing conditions in order to have full knowledge and understanding of the conditions and restrictions affecting the performance of the work required. Include in bid all work which is reasonably inferred by the contract drawings and specifications, whether specifically shown or not, as a result of existing conditions, construction, irregularities and interferences which may affect work and is necessary for fully functioning system. No additional compensations will be considered for misunderstanding the conditions to be met.

C. The person submitting the request will be responsible for its prompt delivery. Failure to so request clarification of any inadequacy, omission, or conflict will not relieve the Contractor of responsibility. The signing of the Contract will be considered as implicitly denoting that the Contractor has a thorough comprehension of full intent and scope of the working drawings and specifications.

1.06 BIDS AND SUBSTITUTIONS

A. Materials, equipment or services listed by several identifying names are intended to be bidder’s choice, and any of the listed names may be bid without soliciting prior acceptance. In all cases where more than one name is given in the specifications, the first named manufacturer’s material, equipment or services shall be the basis of design which has been contemplated in coordination and production of the Contract Documents. Where equipment is noted in the drawings or on schedules in the drawings, the manufacturer and model number listed in the schedule shall be considered the basis of design and shall supercede listing order in the specification. Any changes, including spatial requirements, and costs required to accommodate materials or equipment other than the basis of design shall be the responsibility of the Contractor bidding other than basis of design equipment.

B. If the Contractor wishes to submit a substitute to the named manufacturer(s) for any equipment, they shall submit in writing on Engineer’s Substitution Request form, prior to bid, stating the manufacturer's name, model number, and detailed product data. In all cases, if the substitute manufacturer is used, the Contractor shall bear all additional costs including, but not limited to,

responsibility of coordination with all other trades, any changes incurred in plumbing, electrical, mechanical, general, etc., which result from equipment or material substitution.

C. Where materials or equipment are specified by name, the proposed material or equipment must be identical to the specified material or equipment in all characteristics of quality, function and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture. Work performed or constructed with unapproved materials/equipment is at Contractor's risk, and any required correction of Work incorporating materials/equipment shall be at Contractor's sole cost and expense.

D. Performance Specification of Products:
   1. Where any product is specified only by requirement to meet an industry standard or regulating body standard such as UL, AGA, AWWA, ANSI, etc. and the item proposed carries approval of that body, no prior acceptance by the Engineer is needed.
   2. When any product or service is specified by requirement to meet a performance standard or is specified by a generic specification, (no manufacturers name listed) no prior acceptance by the Engineer is needed except as specifically called for in these specifications.

E. Approval by the Engineer of equipment other than the specified does not relieve Contractor of this Responsibility.

F. Acceptance
   1. In all instances, Contractor shall assume full responsibility for proof of equality of the stature to the equipment hereinafter specified. All data and information necessary for proof of equality, function and space requirements shall be prepared and accompany the submittal of the Substitution Request to the Engineer.
   2. Specified material and equipment shall be considered the basis of design, and while not specifically mentioned, characteristics such as material types, gauges, weights, appearance and space requirements of the basis of design materials and equipment must be met by any proposed substitutions.
   3. Action for substitutions specified herein will be given only after the receipt of formal Substitution Request accompanied by complete data showing performance over entire range, physical dimensions and material construction all specifically marked for the individual item in accordance with requirements for Submittals of Product Data.
   4. Where the substitution requires any changes in piping, electrical wiring, clear space for service requirement, venting, ducting, submit Coordination Drawings with the Substitution Request indicating changes required and conclusively coordinating changes required for the HVAC Division as well as changes required for all other Divisions. Contractor submitting the Substitution Request shall:
      a. Be responsible to coordinate all Divisions and make all changes required to accommodate the Substitution.
      b. Pay for all changes required of both the HVAC Division and all other Divisions to accommodate the Substitution.
   5. Approval of the Substitution Request by the Engineer does not relieve the Contractor of the above responsibilities. See General Conditions for method of notification of acceptance.

G. In the event the substituted material or equipment does not perform to meet the design intent, fit or meet quality standards, the Contractor shall provide the specified material or equipment and bear all costs to replace the substitute item(s).

1.07 PERMITS, FEES AND NOTICES

A. Apply for and pay for all permits, fees, licenses and inspections for this Division of work.

1. Do not include the cost of any "Plant Investment Fee" or "System Development Charge" for sewer and/or water charged by the City. This will be arranged for and paid for by the Owner.
2. Do not include the cost of any "Gas Application Fee" charged by the Utility Company. This will be arranged and paid for by the Owner.

B. Notify proper authorities when work is ready for inspections required by applicable codes, rules and regulations, allowing sufficient time for inspections to be made without hindering progress of the work. Furnish to the Owner copies of inspection certificates of acceptance.

1.08 PLANS AND SPECIFICATIONS

A. The intention of the plans and specifications is to provide all piping, fixtures, and equipment. Contractor shall furnish all material and equipment and shall perform all labor to achieve this intent, whether or not such material or equipment is indicated herein. Wherever the term "provide" is used, it shall mean "furnish and install".

B. All plans and specifications including electrical and mechanical plans shall be examined by Contractor prior to submission of quotations to determine systems interface and conditions which could cause interference or deviations in equipment locations and routing. Errors or discrepancies on plans or in specifications shall be reported to the Engineer in writing and written instructions obtained for the discrepancy prior to submittal of bid to the Owner.

C. All changes from the plans necessary to make the work conform to buildings as constructed and to fit work of other trades, or to conform to rules of all governing authorities and regulations, shall be met by each Contractor without extra cost to the Owner.

D. Routing of piping and location of equipment and other devices are shown on plans in a diagrammatic manner for general guidance. Plans shall not be scaled for dimensions. Take all dimensions from certified equipment drawings, and from the site inspection itself before fabricating any work. This Contractor shall coordinate his work with other Contractors and shall provide necessary deviations in routing as far as 10 feet from those shown to provide systems as specified or implied, without interference and pursuant to these requirements at no additional cost to the Owner or Engineer.

E. Manufacturer's drawings and instructions shall be followed in all cases where the makers of devices and equipment furnish directions covering point not shown on the drawings or described in the specifications. Install all equipment in accordance with manufacturer’s recommendations, unless approval is given in writing by the Engineer for deviation.

F. Layout and installation of HVAC work shall be coordinated with the overall construction schedule of various trades to prevent delay in completion of the project. Complete drawings and specifications for the entire job shall be maintained and updated at the job site. Coordinate the HVAC work with and be responsible to the General Contractor for satisfactory progress of the work. Coordinate HVAC work with all other trades on the project without additional cost to the Owner.

G. Priority of interpretation of discrepancies in Contract Documents shall be complimentary from specifications to drawings. Where discrepancies occur between various specifications, drawings and specifications or codes and standards, the most demanding requirement shall take precedence, except where written interpretation from the Engineer indicates otherwise.

H. All work and materials covered by drawings and specifications shall be subject to review at any time by representatives of the Engineer and Owner. If the Engineer or Owner’s agent finds any materials or installation that does not conform to these drawings and specifications, Contractor shall remove the material from the premises and correct the installation to the satisfaction of the agent.

I. In acceptance or rejection of installed mechanical systems, no allowance will be made for lack of skill on the part of the installers.

1.09 SUBMITTALS

A. General
1. Submit under provisions of Division 01 and the requirements below.
2. The purpose of shop drawing submittals by the Contractor is to demonstrate to the Engineer that the Contractor understands the design concept. Contract shall demonstrate their understanding by indicating which equipment and material they intend to furnish and install, and shall detail the fabrication and installation methods they intends to use. Contractor further agrees that if deviations, discrepancies, or conflicts between shop drawing submittals and contract documents are discovered either prior to or after shop drawings and specifications shall control and shall be followed.
3. Review of Submittals is rendered as a service only and shall not be considered as a guarantee of measurements or of building conditions; nor shall it be construed as relieving the Contractor of basic responsibilities under their Contract. Engineer will check submittals only for conformance with the design concept of the project. Review shall not be construed as:
   a. Permitting any departure from the contract requirements.
   b. Relieving the Contractor of the responsibility for any error in details, dimensions or otherwise that may exist in the Shop Drawings.
   c. Contractor agrees that shop drawing submittals processed by the Engineer are not change orders.
4. Submittal Schedule: Generate a Schedule of anticipated initial submittals for this Division and provide to General Contractor. Include in schedule specified durations time for review, submittal revision, resubmittal, and subsequent review as indicated in Division 01. Structure submittal schedule to allow for construction and delivery long lead and critical path items based on the overall project construction schedule. Place orders for all equipment in time to prevent any delay in construction schedule or completion of project. If any materials or equipment are not ordered in time, additional charges made by equipment manufacturers to complete their equipment in time to meet construction schedule, together with any special handling charges, shall be borne by the Contractor.

B. Shop Drawings
1. Proposed Product List: At the onset of Project's submittal phase, this Contractor shall submit a complete list of all material and equipment they propose to use in the installation. This list shall include all products specified in this Division and indicate manufacturer, catalog numbers and other identifying information. Engineer reserves the right to not review individual Product Data and Shop Drawing submittals until the Proposed Product List has been submitted and reviewed.
2. After review of the Proposed Product List by the Engineer, the Contractor shall submit Product Data and Shop Drawings including descriptive literature of the equipment to be provided under this contract. Drawings shall state capacities, sizes, etc., of all equipment and shall be certified.
3. The shop drawings shall be reviewed by the Contractor and stamped and signed certifying they have reviewed and found them to be 100% complete and accurate, prior to submission. Mark project name and location on each shop drawing, catalog cut and/or specification sheet along with Contractors signed stamp. Engineer reserves the right to reject shop drawings submitted without project name, location or stamp of Contractor's review.
4. Clearly mark each shop drawing, catalog cut and/or specification sheet to indicate those products and features which are intended to be furnished. Use highlights, arrows, underlines, circles and strikethroughs to identify exact features, options, capacities, characteristics and dimensions of equipment to be provided. Strike through information that does not apply to the intended product, options not intended to be included, and any manufacturer's disclaimers such as "Dimensions Subject to Change Without Notice."

Options not specifically struck will be understood to be included. Engineer reserves the right to reject shop drawings not adequately marked in the above manner.

5. Review and approval of shop drawings shall not relieve the Contractor from the responsibility of furnishing equipment and materials of proper dimension, size, quantity, quality and all performance characteristics to efficiently perform the requirements and intent of the contract documents. Review and approval does not relieve the Contractor from responsibility for errors on the shop drawings. If the shop drawings deviate from the contract documents, Contractor shall advise the Engineer of such deviations in writing accompanying the shop drawings submittal, including the reasons for the deviations. Coordinate all required changes with the other trades affected. If the changes are occasioned by the Contractor, they shall pay any costs involved.

C. Coordination Drawings
   1. This Contractor shall provide the support role in development of a combined set of coordination electronic drawing(s) to conclusively coordinate spatial arrangements of their materials and equipment with all other trades, and to define sequencing and coordination of installations for efficient flow of the Work. Coordination drawings shall be minimum scale of 1/4” = 1/0” along with a detailed 3-D model showing locations, dimensions and height of installation of all major pieces of equipment, ductwork and piping provided under their respective contracts. Drawings shall include the following:
      a. Bottom of duct height and size
      b. Piping elevations and size
      c. Power and Data Conduit elevations and size
      d. Hanger support locations (Ductwork / Piping / Conduit / Cable Tray)
      e. Lighting devices
      f. Building structure background
      g. Proposed locations for access panels (Ductwork/Piping/Conduit/Cable Tray)
      h. Indicate the locations of all equipment and materials, including clearances required for servicing and maintaining equipment.
      i. Indicate movement and positioning of large equipment into the building during construction.
   2. Order of space preference throughout the building shall be:
      a. Recessed light fixtures
      b. Ductwork
      c. Soil, Waste, vent and storm piping
      d. Domestic water piping
      e. Sprinkler piping
      f. Electrical conduit
      g. Exception: Plumbing lines below or behind plumbing fixtures shall have precedence over all other work. Electrical conduit above or below switchgear, panel boards and control panels shall have precedence over all other work. Do not install any fluid conveying piping over electrical or elevator equipment
   3. Prepare and submit the coordination drawings including detailed three dimensional model for review. By approving and submitting Shop Drawings, Product Data, Samples and similar submittals, the Contractor represents that the Contractor has determined and verified materials, field measurements and field construction criteria related thereto, or will do so, and has checked and coordinated the information contained within such submittals with the requirements of the Work and of the Contract Documents.
   4. The Contractor shall perform no portion of the Work requiring submittal and review of Shop Drawings, Product Data, Samples or similar submittals until the respective submittals has been approved by the Engineer.
   5. Where Contractor has failed to provide proper space for equipment and required clearances (as required by local AHJ, as related to code requirements, as noted or shown on plans or as noted on submittals) Contractor shall relocate the equipment as directed by

Engineer. Contractor shall be held responsible for any and all changes resulting from such relocations and shall be held responsible for any and all changes resulting from such relocations and shall bear any and all increase costs to Contractor as well as costs to other trades in making said revisions.

D. Engineer's Review of Submittals:
   1. The Contractor shall not be relieved of responsibility for deviations from requirements of the Contract Documents by the Engineer's approval of Shop Drawings, Product Data, Samples or similar submittals unless the Contractor has specifically informed the Engineer in writing of such deviation at the time of submittal and the Engineer has given written approval of the specific deviation. The Contractor shall not be relieved of responsibility for errors or omissions in Shop Drawings Product Data, Samples or similar submittals by the Engineer's approval thereof.
   2. Shop Drawings, Product Data, Samples and similar submittals are not Contract Documents. The purpose of their submittal is to demonstrate for those portions of the Work for which submittals are required the way the Contractor proposes to conform to the information given and the design concept expressed in the Contract Documents.
   3. The Engineer will review and approve or take other appropriate action upon contractor's submittals such as Shop Drawings, Product Data and Samples, but only for the limited purpose of checking for conformance with information given and the design concept expressed in the Contract Documents. The Engineer's action will be taken with such reasonable promptness as to cause no delay in the Work of the Contractor or in the activities of the other Contractors, the Owner, General Contractor, or the Owner's Representative, while allowing sufficient time in the Engineer's professional judgment to permit adequate review. Review of such submittals is not conducted for the purpose of determining the accuracy and completeness of other details such as dimensions and quantities, or for substantiating instructions for installation of performance of equipment or systems, all of which remain the responsibility of the Contractor as required by the Contract Documents. The Engineer's review of the Contractor's submittals shall not relieve the Contractor of the obligations per the Contract Documents.
   4. Processing Time: Allow time for submittal review, including time for resubmittals, as follows. Time for review shall commence on Engineer's receipt of submittal. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing, including resubmittals.
      a. Initial Review: Allow (10) consecutive working days from when submittal is recieved by design team for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required. Engineer will advise Contractor when a submittal being processed must be delayed for coordination. Additional days will be required for the review of any shop drawings and other submittals requiring review by the design team if received in quantity equal to or less than fifty (50) sheets during five (5) consecutive working days. For each sheet or other item in excess of over fifty (50) sheets received in five (5) consecutive working days, additional time will be required for review time. The Engineer will advise the Contractor of additional time required.
      b. Intermediate Review: If intermediate submittal is necessary, process it in same manner as initial submittal.
      c. Resubmittal Review: Allow (10) consecutive working days from when submittal is recieved by design team for review of each resubmittal.
      d. Concurrent Consultant Review: Where concurrent review of submittals by Owner, or other parties is required, allow twenty (20) working days for initial review of each submittal.
      e. No extension of the Contract Time will be authorized because of:
         1) Failure to comply with approved Submittal Schedule.
         2) Failure to transmit submittals enough in advance of the Work to permit processing.

5. Resubmittals: Make resubmittals in same form and number of copies as initial submittal.
   a. Note date and content of previous submittal.
   b. Note date and content of revision in label or title block and clearly indicate extent of revision.
   c. Resubmit submittals until they are marked with approval notation from Engineer's action stamp.
   d. Resubmittals shall include complete submittal information regardless if certain items within the previous submittal did not require a resubmittal. Partial resubmittals are not acceptable. The resubmittal shall serve a complete consolidated submittal of all items being provided.
   e. Engineer review of shop drawings include up to (1) resubmittal of the initial submittal. Engineer reserves the right to bill for additional review time at current hourly rates for any additonal resubmittals. Contractor shall be responsible for any additional expenses due to additional review / resubmittal efforts by the Engineer.

1.10 EXISTING UTILITIES

A. The plans indicate the location, type and sizes of various utilities within the site where known. These utilities are indicated as accurately as possible. If utilities are encountered during construction, which are not shown on the drawings, ask for instructions from the Engineer. Any relocation or remodeling required will then be directed by change order. Assume all responsibility for protection of all utilities, shown or not, and repair any damage caused by this construction at no extra charge to the Owner.

B. Investigate with proper authorities for all existing water taps, etc. and make arrangements to pay for all removal charges in original bid.

C. Owners of all underground utilities shall be notified prior to excavation so that they can locate and mark underground utilities.

1.11 MECHANICAL INSTALLATIONS

A. Coordinate mechanical equipment and materials installation with other building components.

B. Verify all dimensions by field measurements.

C. Arrange for chases, slots, and openings in other building components to allow for mechanical installations.

D. Coordinate the installation of required supporting devices and sleeves to be set in poured in place concrete and other structural components, as they are constructed.

E. Sequence, coordinate, and integrate installations of mechanical materials and equipment for efficient flow of the Work. Give particular attention to large equipment requiring positioning prior to closing-in the building.

F. Coordinate the cutting and patching of building components to accommodate the installation of mechanical equipment and materials.

G. Where mounting heights are not detailed or dimensioned, install mechanical services and overhead equipment to provide the maximum headroom possible, and in accordance with minimum required clearances as specified in codes and regulations.

H. The word “concealed” as used in this specification refers to such spaces as pipe and duct chases, pipe and duct trenches, above plastered ceilings, in walls and buried where pipe and/or duct is inaccessible when building is complete. “Exposed” is intended to be within equipment rooms, unfinished areas, above “push up” ceilings, accessible pipe and duct tunnels.

I. The term “furnish” means supply and deliver to Project, unless otherwise defined in greater detail. The term “install” is used to describe operations at Project, from inspecting and unloading, to completion in place, ready for intended use. The term “provide” means furnish and install, complete and ready for intended use, unless otherwise defined in greater detail.

1.12 RELATED REQUIREMENTS

PART 2 - PRODUCTS

2.01 MATERIALS AND EQUIPMENT

A. Use only new materials, of the best quality of their respective kinds.

B. All materials, in general, shall be Underwriter's Laboratories listed and labeled.

C. All airside equipment provided with filtration shall be sized based on a dirty filter allowance/pressure drop not a clean/new filter allowance. Filter allowance shall be based on filter manufacturer's recommendation for dirty filter conditions.

2.02 EQUIPMENT MANUFACTURER

A. Equipment in the following categories shall be of one manufacturer or available through one manufacturer for each category to facilitate ease of maintenance for the Owner.
   1. Access Doors
   2. Motors (open drip-proof squirrel cage)
   3. HVAC Fixture Trim
   4. Pressure Gauges
   5. Starters, including Variable Frequency Drives
   6. Strainers
   7. Temperature Controls
   8. Thermometers
   9. Valves

PART 3 - EXECUTION

3.01 ENGINEERING BY CONTRACTOR

A. The construction of this building requires the contractor to design several systems or subsystems. All such design shall be the completed responsibility of the contractor. Hangers, supports and seismic restraint systems and design details are hereby delegated to the contractor to design, detail, and submit with seal of Registered Professional Engineer for review by Engineer.

3.02 SAFETY AND MAINTENANCE OF WORK AREAS

A. The Engineer has no contractual responsibility in connection with job site safety measures or precautions as related to means, methods, techniques, sequences or procedures. Contractor shall be solely and completely responsible for conditions of the job site, including safety of all persons and property during performance of the work. This requirement will apply continuously and not be limited to normal working hours. The Engineer's observations of the Contractor's performance are not intended to include review of the adequacy of the Contractor's safety measures in, on, or near the construction site. Contractor shall be responsible for providing all such safety measures and shall consult with the Local, State or Federal Safety Inspector for interpretation whenever in doubt as to whether safe conditions do or do not exist; or whether he is or is not in compliance with Safety Regulations.

B. During the project, this Contractor shall maintain his work area in an organized manner, shall not allow debris to accumulate, and shall store equipment, tools and supplies in a manner which shall not cause interference with the activities of others engaged on the project

3.03 HANDLING AND STORAGE OF MATERIALS AND EQUIPMENT

A. Provide for proper storage of all materials and equipment and assume responsibility for losses due to any cause. All storage shall be within the contact limit lines of the building site. Cover and store all equipment and materials out of elements; any rusted or weather damaged item shall not be used.

B. It is recognized that space at the project for storage of materials and products is limited. Coordinate the deliveries of HVAC systems materials and products with the scheduling and

sequencing of the work so that storage requirements at the project are minimized. In general, do not deliver individual items of equipment to the project substantially ahead of the time of installation. All materials shall be covered prior to installation and protected until final acceptance.

3.04 COORDINATION AND INSTALLATION OF THE WORK:

A. COORDINATION AMONGST TRADES:

1. This Contractor shall provide a qualified person during the construction phase of the shop drawing development that has the skills to convert the diagrammatic project drawings into composite assembly drawings in the CAD format determined by the Mechanical Contractor, for integration of systems into the combined master per floor coordination documents being assembled by the Mechanical Contractor. These coordination / fabrication drawings will conform to the performance criteria identified in the project documents and will fit into the 3 Dimensional allotted space for all systems. Where the piping dimensions, sizes and general arrangements of elements will require to be adjusted due to coordination of new or existing building structure, and other trade's materials and or equipment space needs, qualified person will provide proposed recommendations of sizes, elevation changes, offsets, etc. for inclusion in a combined drawing(s) for the coordination meetings that will be reviewed by the Engineer, Owner and other trades to provide an overall acceptance of coordination per floor.
2. The layout shown on the drawings is necessarily diagrammatic but shall be followed as closely as other work will permit. Changes from these drawings required to make this work conform to the building construction shall be made only with prior written approval of the Engineer. All proposed changes shall be shown on shop drawings. All measurements shall be verified by actual observation and all work shall fit in place meeting the approval of the Engineer.
3. Provide openings required in new and existing construction that may be necessary. All patching and repairing shall be done by workmen competent in the trade required, at the expense of this Contractor. Arrange work so that minimum cutting will be required. All rubbish and excess materials involved in such cutting shall be promptly removed from the site and disposed of. Cutting through the floor or roof systems or load bearing walls shall be done only with the prior written approval of the Engineer so as to avoid damaging the structural system.

B. SEQUENCING, SCHEDULING:

1. Confer with the other contractors regarding the location and size of pipes, equipment, ducts, openings and special architectural treatments in order that there may be no interferences between the installation or the progress of the work of any contractor on the project.
2. In the case of interconnection of the work of two or more contractors, verify at the site or on shop drawings all dimensions relating to such work. All errors due to the failure to so verify any such dimensions shall be promptly rectified.
3. Access panels, in walls or ceilings, required (i.e., automatic or manual damper, fire or smoke damper, coil or control instrument mounted in a duct or pipe) shall be provided by the respective contractor and installed by workman competent in the trade required at the expense of this Contractor. Access panels are not required in areas where the ceiling system is lay-in tile; however, sufficient space must be available in and through the ceiling system to allow maintenance and adjustment of dampers, and cleaning of coils as necessary, or a suitable access panel shall be provided for that purpose. Access panels shall be approximately 24 inches by 24 inches wherever possible and shall be provided with flush trim and an allen-key operated camlock fastener. Submit manufacturer's product data to Engineer for approval.
4. Items of equipment may be specified in the singular however, provide and install the number of items of equipment as required for a complete system.

5. Provide excavating, pumping, backfilling and compacting required, as shown on the drawings.
6. Equipment and devices which have factory prime coat or final surface finish shall be replaced, repaired or refinished if defective or damaged during installation.
7. Arrange all work so a minimum period of interruption or outages will occur in the temporary or permanent transfer of services as required for all electrical revisions. Not less than two (2) weeks notification to the Using Agency/Owner shall be required before approval will be granted for any disruption of services. The outage request shall include the extent of the work to be done, length of outage time required and the time at which the outage is to begin. No allowance will be made for extra payment as a result of scheduling "overtime" work necessary to perform before or after normal or regular working hours to accomplish the work intended.
8. Submit a "Sequence of Work Schedule" in respect to all temporary and permanent utility and service cutovers after final determination. This schedule shall be submitted for approval to the Engineer. The submittal shall designate priority order, service or utility affected, date of cutover, and time of day to start and finish.

C. MANUFACTURER'S INSTRUCTIONS
1. Except where more stringent requirements are indicated, comply with the product manufacturer's instructions and recommendations.
2. Consult with manufacturer's technical representatives, who are recognized as technical experts, for specific instructions on special project conditions.
3. All items which are a source of noise generation and/or mechanical vibration shall be installed with proper attenuation provisions including absorbers, isolators, or mufflers as required to prevent objectionable noises and vibrations.
4. If a conflict exists, notify the Engineer in writing and obtain instruction before proceeding with the work in question.

D. HOISTING AND MOVEMENT OF EQUIPMENT
1. This Contractor shall be responsible for hoisting of all materials and equipment furnished or installed under this Section of the Specifications, in accordance with all city, state and federal rules and regulations.
2. Wherever possible, arrange for the movement and positioning of equipment so that enclosing partitions, walls and roofs will not be delayed or need to be removed.
3. Otherwise, advise Contractor of opening requirements to be maintained for the subsequent entry of equipment.
4. Coordinate the movement of heavy items with shoring and bracing so that the building structure will not be overloaded during the movement and installation.
5. Where mechanical products to be installed on the existing roof are too heavy to be hand-carried, do not transport across the existing roof deck; position by crane or other device so as to avoid overloading the roof deck.

E. CONCRETE
1. All poured in place concrete required for equipment furnished under this Division shall be furnished & installed by this Division in accordance with requirements specified under the Architectural Divisions of these Specifications.
2. Concrete pads shall be minimum 4 inch high and anchored to the floor with dowels.

F. CLEARANCES
1. Install piping and equipment:
   a. Straight and true.
   b. Aligned with other work.
   c. Close to walls and overhead structure (allowing for insulation).
   d. Concealed, where possible, in occupied spaces.
   e. Out-of-the-way with maximum passageway and headroom remaining in each space.

f. Conceal all piping in finished areas of the building except where otherwise noted on the drawings.

2. Except as otherwise indicated, arrange mechanical services and overhead equipment with a minimum of:
   a. 7’0” headroom in storage spaces.
   b. 8’6” headroom in other spaces.
3. Do not obstruct windows, doors or other openings.
4. Give the right-of way to piping systems required to slope for drainage (over other service lines and ductwork).
5. Offsets, transitions and changes in direction in pipes and ducts shall be made as required to maintain proper head room and pitch of sloping pipes whether or not indicated on the drawings. Furnish and install all traps, air vents, sanitary vents, etc., as required to affect these offsets, transitions and changes in direction.

G. ACCESS AND ACCESS DOORS
1. Install all work to permit removal (without damage to other parts) of coils, heat exchanger bundles, boiler tubes, fan shafts and wheels, filters, belt guards, sheaves and drives, and all other parts which might require periodic replacement or maintenance. Arrange pipes, ducts, and equipment to permit ready access to valves, traps, starters, motors, control components and to clear the openings of doors and of access panels. Furnish access panels for all mechanical equipment and valves requiring access in concealed locations for installation by contractor.
2. Provide shop drawings to the Engineer showing locations of all access panels.
3. Furnish access doors of type suitable to Engineer and provide to General Contractor to construct into the building. Access doors should be provided in all locations where access is required.
4. Provide painted, steel (unless noted otherwise) access doors with key lock suitable for the surface in which they are installed and satisfactory to the Engineer.
   a. Plaster finish walls and ceilings: Recessed style.
   b. Drywall walls and ceilings: Flush panel.
   c. Remodel Applications: Flanged flush panel.
   d. Corrosive environments, including but not limited to, restrooms, locker rooms, pool equipment rooms, and natatoriums: Panel and frame shall be aluminum or stainless steel. All associated hardware and fasteners shall be stainless steel.
   e. Panels in fire and/or smoke rated assemblies shall be listed for the application and carry the appropriate rating for the assembly in which they are installed.

3.05 PROTECTION OF WORK AND PROPERTY

A. Where there are existing facilities, be responsible for the protection thereof, whether or not such facility is to be removed or relocated. Moving or removing any facility must be done so as not to cause interruption of the work of Owner's operation.

B. Close all pipe openings with caps or plugs during installation. Cover all fixtures and equipment and protect against injury. At the final completion, clean all work and deliver in an unblemished condition, or refinish and repaint at the discretion of the Engineer.

C. Any equipment, duct or piping systems found to have been damaged or contaminated above “MILL” or “SHOP” conditions shall be replaced or cleaned to the Engineer’s satisfaction.

D. Initial fill of traps: Provide initial water seal fill for all waste p-traps, condensate traps, or similar traps.

3.06 PROTECTION OF POTABLE WATER SYSTEMS

A. All temporary water connections shall be made with an approved back flow preventer.

B. All hose bibbs shall have, as a minimum, a vacuum breaker to prevent back flow.

C. Direct connections to hydronic systems shall only be made through a reduced pressure back flow preventer.

3.07 TEMPORARY HEAT, VENTILATION, AND AIR-CONDITIONING

A. Temporary heat, ventilation, and air-conditioning (HVAC), if required for the construction activities, shall be furnished by the General Contractor. Use of the permanent HVAC system shall not be allowed without written authorization from the Engineer. In the event the permanent HVAC systems are desired to be used prior to substantial completion and final cleaning, the General Contractor shall meet all requirements of the Engineer's written authorization and shall pay all utility costs for operation of the HVAC systems until acceptance by the Owner.

3.08 ELECTRICAL EQUIPMENT AND WIRING FOR HVAC DIVISION

A. Unless otherwise indicated, all motors and controls shall be furnished, set in place and wired in accordance with the following schedule. (MD is Mechanical (HVAC) Division - ED is Electrical Division).

Type	Furnished Under	Set in Place by	Wired & Connected by
Equipment Motors and Thermal overloads, resistance heaters (C)	MD	MD	ED
Motor Controllers; magnetic starters, reduced voltage starters and overload relays.	MD	ED(A)	ED
Disconnect switches, fused or unfused, H.P. rated switches, thermal overload switches and fuses, manual operating switches.	ED(A)	ED(A)	ED
Push-button stations, pilot lights, multi-speed switches, float switches, thermostats, control relays, time clocks, control transformers, control panels, motor valves, damper motors, solenoid valves, EP and PE switches and interlocks	MD	MD(B)	MD(B)
120 volt power for Building Automation System (BAS) panels, fire protection and boiler controls	ED	ED	ED

B. Notes:
1. If furnished as part of factory wired equipment, wiring and connections only by ED.
2. If any of these devices carry the full load current to any motor or resistive element(s) they shall be connected by ED. Control devices carrying full load current furnished by MD and wired by ED shall be located at the device being controlled, unless shown on drawings or mutual agreement is made between the contractors with no change in the contract price.
3. Wiring from alarm contacts to alarm system by ED; all control function wiring by MD. Duct detectors furnished by ED, set in place by MD.

C. GENERAL NOTE:
1. The above list does not attempt to include all components. All items necessary for a complete system shall be included in the base contract.
2. Connections to all controls directly attached to ducts, piping and mechanical equipment shall be made with flexible connectors.

3.09 FLUSHING, CLEANING & STERILIZING

A. Piping material shall be cleaned, free of debris, prior to shipment to site. Pipes ends shall be temporarily sealed to prevent contamination. Remove once welding occurs.

B. Intent: It is the intent of this specification to require that all work, including the inside of equipment, be left in a clean condition with all dust, grease, and construction debris removed.
   1. Piping and connection equipment to be left free of sediments, core sand, grease, etc.
   2. Clean all exposed surfaces of piping, ducts and hangers, etc., sufficiently to receive paint. Vacuum ducts as required for debris removal.
   3. Air systems shall not be operated without filters. Replace the filters or clean permanent type filters just prior to substantial completion. All air systems shall be furnished with one additional set of filters for owner replacement.
   4. Remove and clean all screens, interceptors, strainers, etc., in piping systems just prior to substantial completion.
   5. Clean and wipe dry all plumbing fixtures, exposed valves, faucets, and piping, etc. that are exposed just prior to substantial completion. Clean all equipment and fixtures per manufacturer's specifications to avoid scratching finished surfaces. Leave all plumbing fixtures ready to use.
   6. Clean interior and exterior of all air handling equipment of all construction debris. Clean exterior of all exposed ductwork just prior to substantial completion.
   7. Thoroughly clean all equipment room floors after completion of equipment, pipe and duct cleaning. A condition of final acceptance will be the cleanliness of all exposed systems, equipment, and equipment rooms.

C. Before final connections are made in the piping systems, blow out all piping with air and then wash out with cleaning compounds. Then flush the system to remove of all foreign materials. Furnish all temporary connections, valves, etc, required for this purpose. Clean the boiler and chiller by the same procedure.

D. Clean the boiler by the same procedure.

E. After flushing, sterilize the domestic water system in accordance with Domestic Water Piping section.

3.10 TESTING

A. Test all heating water and chilled water piping at 150 psig hydrostatic pressure before connecting to unit.

B. Test all refrigeration piping under 150 psig pressure using oil pumped, dry nitrogen and tapping of joints if there is any loss of pressure, soap each joint to find leaks. Charge with 10 psig refrigerant and test with halide torch or electronic leak detector. Evacuate using vacuum pump to 500 microns and purge twice with oil pumped, dry nitrogen.

C. All tests must be done to the satisfaction of the local authorities having jurisdiction, before covering.

D. All hydrostatic tests to be held for a minimum of six hours without loss of pressure. Air tests to be held for a minimum of two hours without loss of pressure.

E. Furnish all instruments required for testing.

3.11 EQUIPMENT START UP AND PLACING IN OPERATION

A. Clean all ducts, pipes, equipment, controls etc., of plaster and other foreign debris.

B. Before final acceptance, clean or replace all strainers, oil or grease all bearings and clean out all drains. Clean and recoat all permanent filters, replace throwaway type filters with new filters.

C. The systems shall be put into operation.
   1. The Contractor shall verify that all controls are set to meet operating conditions specified.
      a. Example: Boiler operating and limit controls set where specified.
   2. The contractor shall verify that all pieces of equipment are operable and that all sequences of control are being met.
   3. The contractor to adjust settings through 1st year as required by MECHANICAL ENGINEER.

D. All packaged HVAC equipment shall be started by the manufacturer or under the manufacturer's supervision. Start-up data shall be recorded in logs. Copies of start-up logs shall be forwarded to the Engineer and included in Operation and Maintenance manuals.

3.12 CLEANING

A. Immediately prior to inspection for Substantial Completion, remove any remaining waste materials and rubbish from mechanical and electrical rooms. Remove protective coating, barriers and other protective devices, temporary work, and surplus materials.

B. Leave mechanical rooms and similar unfinished spaces "Broom Clean." Dust equipment, ducts, pipes, and other mechanical and electrical work in mechanical rooms and similar unfinished spaces. Remove construction dirt and debris from interior of equipment, panels, disconnects, ductwork, etc. thoroughly in accordance with manufacturer's instructions.

C. Upon completion of the contract all remaining materials and rubbish shall be removed from the building and premises and the work areas shall be left clean and free from stains, mortar, paint spots, etc.

D. Upon completion of the work, put systems into service maintaining responsibility for the equipment during all testing operations including the lubricating and turning on and off of such apparatus

3.13 CLOSEOUT SUBMITTALS

A. OPERATION AND MAINTENANCE INSTRUCTIONS

1. Operating and Maintenance Manuals shall be delivered to the Owner's authorized representative and the Owner instructed as to their use and the equipment involved.
2. Thirty (30) days Prior to Substantial Completion, submit operating and maintenance manuals for equipment to Engineer for approval. Assemble manuals in accordance with specfication 01 7823 - Operation and Maintenance Data.
3. Provide comprehensive table of contents indicating the contents of all volumes in each set. Organize and bookmark documents in an orderly sequence, based on the table of contents of the project manual. Within each section, organize alphabetically by system or subsystem. Include a section in the directory for each of the following: list of documents, list of systems, and list of equipment.
4. Include copy of each submittal, including Engineer’s review comments, and subsequent resubmittals for record purposes, indicating the actual product installed. Include significant changes in the product delivered to project site and changes in manufacturer's written instructions for installation.
5. Provide organized warranty section in the document/manual (Contractor’s option to provide separate document/manual). Organize the warranty documents into an orderly sequence in separate section, based on the table of contents of the project manual. A copy of each warranty shall also be provided in the respective equipment section.
6. Provide organized equipment failure emergency section in the document/manal (Contractor’s option to provide separate document/manual). Organize into separate sections for each type of emergency and include all necessary steps for safe equipment shut-down and containment. Also, include all necessary contractor and vendor contact information including 24-hour call numbers.
7. Provide comprehensive contact list in the document/manual including contractor and subcontractor's names, addresses, telephone and contact person for Owner's use.
8. Include equipment start up logs for all equipment required to be checked out and intitally started under manufacturer’s supervision.
9. Provide manufacturer's complete data sheets including assembly drawings, spare parts lists, wiring diagrams, mechanical diagrams, installation diagrams, and instructions. Identify equipment in manual and cross-reference by including serial number of actual components.

10. In addition to Manufacturer's maintenance documentation, provide separate schedules for preventative and routine maintenance and service with standard time allotment. Include service and lubrication requirements and list of required lubricants for equipment, application methods and frequencies.
11. Provide Preventative Maintenance (PM) schedule for checking all items such as belt drives, safety controls, motor running load amps, oil, refrigerant and grease charges. Include cleaning schedule of all strainers, traps, coils, tubes, etc.
12. Include record of spare parts provided including signature of Owner's representative showing acceptance of parts. Provide list of additional items recommended to be stocked as spare parts, with parts identified and cross-referenced to manufacturer's maintenance documentation and including local sources of maintenance materials and related services.
13. Provide test reports from manufacturer start-up indicating systems to be operating properly, including TAB reports, thermographics surveys, etc.
14. Provide a PDF copy of documents for review.
15. Provide electronic copy of final operation and maintenance manuals in full color pdf format on CD/DVD or USB drive (per owner's preference). All files shall clearly indicate specific information related to actual products installed. File structure and naming shall be “user-friendly” and organized in a manner such that it may be intuitively navigated by users.
16. Approval will not be given for final payment until the tests, balancing and operating instruction portions have been completed.

B. AS-BUILT DRAWINGS
1. During the process of the work, maintain an accurate record of the installation of the mechanical systems.
2. At the completion of the Work, Engineer shall furnish Contractor with a complete set of the latest revised record drawings in reproducible form, and Contractor shall indicate thereon all as-built changes and such additional details necessary or appropriate to provide a complete reference document for use by Engineer. If variations and details cannot be shown clearly thereon, the Contractor shall prepare supplemental drawings adequate to impart the information. These additional drawings shall also be in reproducible form. The foregoing drawings collectively shall constitute the “As-Built” drawings for the Work.
3. All indications on “As-Built” drawings shall be executed in a legible manner at Contractor's cost, using methods and legend presentations compatible with the overall scheme of the record Drawings with respect to scale, drawing sheet sizes and sequential indexing. All changes shall be marked clearly in red and clouded.
4. Engineer shall review Contractor's “As-Built” drawings and notify Contractor of observed discrepancies or deviations. Contractor shall promptly correct and resubmit revised drawings for Engineer review. Completed “As-Built” drawings shall be delivered to Owner through Engineer.

C. DEMONSTRATION AND TRAINING
1. Refer to Division 01 for additional requirements regarding Demonstration, Training, and O&M Manuals submittals.
2. Prepare and submit operating instructions, as required by Specification Sections, and instruct Owner's personnel in use and maintenance of operating equipment. Explain use operation of each system to the Owner’s designated representative(s). Explain use of the O&M manuals in operating and maintaining the systems.
3. After installation is complete, schedule to meet and instruct Owner's designated personnel in use, operation, care, and cleaning of equipment. Demonstrate that each system operates properly. Instructions shall be given only by qualified personnel, thoroughly familiar with use and maintenance of equipment.
4. Notify the Engineer in writing a minimum of 72 hours in advance of any scheduled Owner equipment training. Following all training, provide record of Owner training including the type of training, who conducted the training, and list of attendees. Include copy in each set of O&M manuals.

5. Owner training sessions shall be video recorded to DVD media or USB drive (per owner's preference). DVDs / USB Drives shall be clearly labeled indicating content of equipment covered by training, and copies provided as follows: one (1) copy to the Owner, one (1) copy to the Engineer, and one (1) copy shall be included with the O&M manuals.

D. WARRANTY
1. Guarantee all work including labor, material and equipment for this project for a period of one year from date of acceptance by Using Agency/Owner. Repair or replace any work found defective without cost to the Owner during the warranty period.
2. All materials and equipment shall be new unless otherwise specified.
3. Guarantee all workmanship, materials and equipment and replace any found defective without cost to the Owner, for one year after final acceptance, as defined in General Conditions.
4. Each warranty for longer than the one year described above (that comes with equipment used on the job) shall be passed on to the owner with dates of start and end of the warranty.

END OF SECTION 23 0010

SECTION 23 0116
DEMOLITION OF HVAC SYSTEMS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. Demolition and removal of selected portions of the HVAC systems including
      a. Ductwork
      b. Piping
      c. Equipment
   2. Abandoning in-place below-grade construction.
   3. Disconnecting, capping or sealing, and abandoning in-place site utilities.
   4. Salvage of existing items for reuse or recycling by Owner.

B. Related Sections:
   1. Section 01 1000 - Summary for use of the premises and phasing requirements.
   2. Section 01 3200 - Construction Progress Documentation for preconstruction photographs taken before building demolition.
   3. Section 02 4119 - Selective Demolition for partial demolition of buildings, structures, and site improvements.

1.03 DEFINITIONS

A. Abandon In-Place: Disconnect existing component from service and label as abandoned, but do not remove. Distribution systems and networks such as piping and ductwork that are abandoned in place shall be labeled abandoned at intervals specified in Division 23 Section "Identification for HVAC".

B. Remove: Detach items from existing construction and legally dispose of them off-site unless indicated to be removed and salvaged.

C. Remove and Salvage: Carefully detach from existing construction, in a manner to prevent damage, and deliver to Owner ready for reuse. Include fasteners or brackets needed for reattachment elsewhere.

D. Remove and Reinstall: Detach items from existing construction, clean, prepare for reuse, and reinstall where indicated.

E. Existing to Remain: Existing items of construction that are not to be permanently removed and that are not otherwise indicated to be removed, removed and salvaged, or removed and reinstalled.

1.04 MATERIALS OWNERSHIP

A. Unless otherwise indicated in this section or in the drawings, demolition waste becomes property of Contractor.

B. Historic items, relics, antiques, and similar objects including, but not limited to, cornerstones and their contents, commemorative plaques and tablets, and other items of interest or value to Owner that may be uncovered during demolition remain the property of Owner.
   1. Carefully salvage in a manner to prevent damage and promptly return to Owner.

1.05 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified refrigerant recovery technician.

B. Proposed Protection Measures: Submit informational report, including Drawings, that indicates the measures proposed for protecting individuals and property, for infection control measures, for dust control, and for noise control. Indicate proposed locations and construction of barriers.

C. Schedule of Building Demolition Activities: Indicate the following:
   1. Detailed sequence of demolition work, with starting and ending dates for each activity.
   2. Temporary interruption of utility services.
   3. Shutoff and capping or re-routing of utility services.

D. Inventory: Submit a list of items to be removed and salvaged and deliver to Owner prior to start of demolition.

E. Predemolition Photographs or Video: Show existing conditions of adjoining construction and site improvements, including finish surfaces, that might be misconstrued as damage caused by demolition operations. Comply with Section 01 3233 - Photographic Documentation. Submit before the Work begins.

F. Statement of Refrigerant Recovery: Signed by refrigerant recovery technician responsible for recovering refrigerant, stating that all refrigerant that was present was recovered and that recovery was performed according to EPA regulations. Include name and address of technician and date refrigerant was recovered.

G. Warranties: Documentation indicating that existing warranties are still in effect after completion of selective demolition.

1.06 CLOSEOUT SUBMITTALS

A. Inventory: Submit a list of items that have been removed and salvaged.

1.07 QUALITY ASSURANCE

A. Refrigerant Recovery Technician Qualifications: Certified by EPA-approved certification program.

B. Regulatory Requirements: Comply with governing EPA notification regulations before beginning demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.

C. Standards: Comply with ANSI/ASSE A10.6 and NNFPA 241.

D. Predemolition Conference: Conduct conference at Project Site.
   1. Inspect and discuss condition of construction to be demolished.
   2. Review and finalize building demolition schedule and verify availability of demolition personnel, equipment, and facilities needed to make progress and avoid delays.
   3. Review and finalize protection requirements.
   4. Review procedures for noise and infection control measuers.
   5. Review procedures for protection of adjacent buildings.
   6. Review items to be salvaged and returned to Owner.

1.08 FIELD CONDITIONS

A. Owner will occupy portions of building immediately adjacent to demolition area. Conduct demolition so Owner's operations will not be disrupted.

B. Conditions existing at time of inspection for bidding purposes will be maintained by Owner as far as practical.

C. Notify Engineer of descrepancies between existing conditions and Drawings before proceeding with HVAC systems demolition.

D. Hazardous Materials: It is not expected that hazardous materials will be encountered in the Work.
   1. Hazardous materials will be removed by Owner before start of the Work.

2. If materials suspected of containing hazardous materials are encountered, do not disturb; immediately notify Engineer and Owner. Hazardous materials will be removed by Owner under a separate contract.

E. Storage or sale of removed items or materials on-site is not permitted.

F. Utility Service: Maintain existing utilities indicated to remain in service and protect them against damage during selective demolition operations.
1. Maintain fire-protection facilities in service during selective demolition operations.

1.09 COORDINATION

A. Arrange demolition schedule so as not to interfere with Owner's on-site operations.

1.10 WARRANTY

A. Existing Warranties: Remove, replace, patch, and repair materials and surfaces cut or damaged during selective demolition, by methods and with materials so as not to void existing warranties. Notify warrantor before proceeding. Existing warranties include the following:

B. Notify warrantor on completion of selective demolition, and obtain documentation verifying that existing system has been inspected and warranty remains in effect. Submit documentation at Project closeout.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 EXAMINATION

A. Verify that utilities have been disconnected and capped before starting demolition operations.

B. Review Project Record Documents of existing construction provided by Owner. Owner does not guarantee that existing conditions are same as those indicated in Project Record Documents.

C. Inventory and record the condition of items to be removed and salvaged. Provide photographs or video of conditions that might be misconstrued as damage caused by salvage operations. Comply with Section 01 3233 - Photographic Documentation.

D. When unanticipated mechanical, electrical, or structural elements that conflict with intended function or design are encountered, investigate and measure the nature and extent of conflict. Promptly submit a written report to Engineer.

3.02 PREPARATION

A. Refrigerant: Remove refrigerant from mechanical equipment according to 40 CFR 82 and regulations of authorities having jurisdiction before starting demolition.

B. Existing Services/Systems: Locate, identify, disconnect, and seal or cap off indicated systems serving areas to be demolished.
1. Arrange shut off of systems/servince with Owner.
2. If systems/services are required to be removed, relocated, or abandoned that will affect adjacent occupied/operational portions of the building(s), then provide temporary utilities that bypass buildings and structures to be demolished and that maintain continuity of service to other buildings and structures.
3. Cut off pipe or conduit a minimum of 24 inches below grade. Cap, valve, or plug and seal remaining portion of pipe or conduit after bypassing according to requirements of authorities having jurisdiction.

C. Existing Utilities: See plumbing and electrical Sections for shutting off, disconnecting, removing, and sealing or capping utilities. Do not start demolition work until utility disconnecting and sealing have been completed and verified in writing.

D. Salvaged Items: Comply with the following:
1. Clean salvaged items of dirt and demolition debris.

2. Pack or crate items after cleaning. Identify contents of containers.
3. Store items in a secure area until delivery to Owner.
4. Transport items to storage area designated by Owner.
5. Protect items from damage during transport and storage.

3.03 UTILITY SERVICES AND MECHANICAL/ELECTRICAL SYSTEMS

A. Existing Services/Systems to Remain: Maintain services/systems indicated to remain and protect them against damage.
   1. Comply with requirements for existing services/systems interruptions specified in Section 01 1000 - Summary.

B. Existing Services/Systems to Be Removed, Relocated, or Abandoned: Locate, identify, disconnect, and seal or cap off indicated utility services and mechanical/electrical systems serving areas to be selectively demolished.
   1. Owner will arrange to shut off indicated services/systems when requested by Contractor.
   2. If services/systems are required to be removed, relocated, or abandoned, provide temporary services/systems that bypass area of selective demolition and that maintain continuity of services/systems to other parts of building.
   3. Disconnect, demolish, and remove fire-suppression systems, plumbing, and HVAC systems, equipment, and components indicated to be removed.
      a. Piping to Be Removed: Remove portion of piping indicated to be removed and cap or plug remaining piping with same or compatible piping material.
      b. Piping to Be Abandoned in Place: Drain piping and cap or plug piping with same or compatible piping material. Label entire extents of system abandoned in place in accordance with Section 23 0553 - Identification for HVAC.
      c. Equipment to Be Removed: Disconnect and cap services and remove equipment.
      d. Equipment to Be Removed and Reinstalled: Disconnect and cap services and remove, clean, and store equipment; when appropriate, reinstall, reconnect, and make equipment operational.
      e. Equipment to Be Removed and Salvaged: Disconnect and cap services and remove equipment and deliver to Owner.
      f. Ducts to Be Removed: Remove portion of ducts indicated to be removed and plug remaining ducts with same or compatible ductwork material.
      g. Ducts to Be Abandoned in Place: Cap or plug ducts with same or compatible ductwork material. Label entire extents of system abandoned in place in accordance with Section 23 0553 - Identification for HVAC.

C. Refrigerant: Remove refrigerant from mechanical equipment to be selectively demolished according to 40 CFR 82 and regulations of authorities having jurisdiction.

3.04 PROTECTION

A. Existing Facilities: Protect adjacent walkways, loading docks, building entries, and other building facilities during demolition operations. Maintain exit access from existing buildings.

B. Existing Utilities: Maintain utility services to remain and protect from damage during demolition operations.
   1. Do not interrupt existing utilities serving operating facilities unless authorized in writing by Owner.
   2. Provide temporary services during interruptions to existing utilities, as acceptable to Owner.
      a. Provide at least 10 days notice to occupants of affected buildings if shutdown of service is required during changeover.

C. Temporary Protection: Erect temporary barriers to maintain pressure differentials necessary to meet the requirements of the Infection Control plan as required by authorities having jurisdiction and as indicated. Comply with requirements in Section 01 5000 - Temporary Facilities and Controls.

1. Erect and maintain dustproof partitions and temporary enclosures to limit dust, noise, and dirt migration to occupied portions of adjacent buildings.
2. Provide HEPA filtered exhaust of the demolition area to maintain airflow direction across the temporary barrier from clean portions of the building to areas of demolition and construction activity.

D. Remove temporary barriers and protections where hazards no longer exist. Where open excavations or other hazardous conditions remain, leave temporary barriers and protections in place.

3.05 DEMOLITION, GENERAL

A. General: Demolish indicated systems completely. Use methods required to complete the Work within limitations of governing regulations and as follows:
1. Do not use cutting torches until work area is cleared of flammable materials. Maintain portable fire-suppression devices during flame-cutting operations.
2. Maintain fire watch during and for at least 8 hours, or duration required by Owner and Fire Marshal, after flame cutting operations.
3. Maintain adequate ventilation when using cutting torches.
4. Locate building demolition equipment and remove debris and materials so as not to impose excessive loads on supporting walls, floors, or framing.

B. Site Access and Temporary Controls: Conduct building demolition and debris-removal operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.
1. Do not close or obstruct streets, walks, walkways, or other adjacent occupied or used facilities without permission from Owner and authorities having jurisdiction. Provide alternate routes around closed or obstructed traffic ways if required by authorities having jurisdiction.
2. Use water mist and other suitable methods to limit spread of dust and dirt. Comply with governing environmental-protection regulations. Do not use water when it may damage adjacent construction or create hazardous or objectionable conditions, such as ice, flooding, and pollution.

C. Explosives: Use of explosives is not permitted.

3.06 DEMOLITION BY MECHANICAL MEANS

A. Proceed with demolition of HVAC systems systematically, from lower to higher level. Complete HVAC systems demolition operations above each floor or tier before disturbing supporting members on the next lower level.

B. Remove debris from elevated portions of the building by chute, hoist, or other device that will convey debris to grade level in a controlled descent.
1. Remove structural framing members and lower to ground by method suitable to minimize ground impact and dust generation.

C. Below-Grade Construction: Abandon foundation walls and other below-grade construction. Cut below-grade construction flush with grade.

D. Below-Grade Construction: Demolish HVAC construction in foundation walls and other below-grade construction that are within footprint of new construction and extending 5 feet outside footprint indicated for new construction. Abandon below-grade construction outside this area.

3.07 REPAIRS

A. Promptly repair damage to adjacent buildings caused by demolition operations.

B. Repair fireproofing where hangers and supports have been removed.

C. Patch holes and openings in walls, floors, ceilings and structure created by removal of piping and ductwork.

3.08 DISPOSAL OF DEMOLISHED MATERIALS

A. General: Except for items or materials indicated to be recycled, reused, salvaged, reinstalled or otherwise indicated to remain Owner's property, remove demolished materials from Project site and legally dispose of them in an EPA-approved landfill.
   1. Do not allow demolished materials to accumulate on-site.
   2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.

3.09 CLEANING

A. Clean adjacent structures and improvements of dust, dirt, and debris caused by selective demolition operations. Return adjacent areas to condition existing before selective demolition operations began.

3.10 SELECTIVE DEMOLITION SCHEDULE

A. Existing Items to be Removed: All ductwork, piping and equipment as indicated in drawings.

B. Existing Items to be Removed and Salvaged: Items of monetary value to the Owner as follows:
   1. Mechanical equipment of salvage value, including unit heaters, control valves, pumps, boilers, chillers, etc.

C. Existing systems to be Abandoned In-Place: Specifically noted on drawings. Typically limited to distribution ductwork and piping located in portions of the building where removal is not economically viable. Label systems as abandoned at intervals specified in Division 23 Section "Identification for HVAC".

D. Existing items to Remove and Reinstall: Specifically noted in drawings.

E. Existing sytsems/services and items to Remain: All system not specifically noted as removed, demolished, or abandoned in place shall be considered existing to remain.

END OF SECTION 23 0116

SECTION 23 0513
COMMON MOTOR REQUIREMENTS FOR HVAC EQUIPMENT

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. This section includes:
   1. Requirements for general-purpose, horizontal, small and medium, squirrel-cage induction motors for use on ac power systems up to 600 V and installed at equipment manufacturer's factory or shipped separately by equipment manufacturer for field installation.

B. Related Sections:
   1. Section 23 0010 - HVAC (Heating, Ventilation and Air Conditioning) General Provisions.
   2. Section 26 0583 - Wiring Connections: Electrical characteristics and wiring connections.

1.03 REFERENCE STANDARDS

A. ABMA STD 9 - Load Ratings and Fatigue Life for Ball Bearings; American Bearing Manufacturer's Association, Inc.; 2015

B. IEEE 112 - IEEE Standard Test Procedure for Polyphase Induction Motors and Generators; Institute of Electrical and Electronic Engineers; 2017.

C. NEMA MG 00001 - Motors and Generators; National Electronic Manufacturers Association; 2024.

D. NFPA 70 - National Electrical Code; National Fire Protection Association; 2023.

1.04 SUBMITTALS

A. Product Data: Provide wiring diagrams with electrical characteristics and connection requirements.

B. Test Reports: Indicate test results verifying nominal efficiency and power factor for three phase motors larger than 1/2 horsepower.

C. Manufacturer's Installation Instructions: Indicate setting, mechanical connections, lubrication and wiring instructions.

D. Operation Data: Include instructions for safe operating procedures.

E. Maintenance Data: Include assembly drawings, bearing data including replacement sizes, and lubrication instructions.

1.05 QUALITY ASSURANCE

A. Conform to NFPA 70.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Protect motors stored on site from weather, moisture, and construction debris by maintaining factory covers and suitable weather-proof covering. For extended outdoor storage, remove motors from equipment and store separately.

1.07 WARRANTY

A. Provide five year manufactruer warranty for motors larger than 20 horsepower.

1.08 COORDINATION

A. Coordinate features of motors, installed units, and accessory devices to be compatible with the following:
   1. Motor controllers.

2. Torque, speed, and horsepower requirements of the load.
3. Ratings and characteristics of supply circuit and required control sequence.
4. Ambient and environmental conditions of installation location.

PART 2 - PRODUCTS

2.01 GENERAL CONSTRUCTION AND REQUIREMENTS

A. Comply with NEMA MG 00001, Design B for centrifugal fans blowers, pumps and compressors.

B. Electrical Service:
   1. Motors smaller than 1/2 HP: 115 volts, single phase, 60 Hz.
   2. Motors 1/2 horsepower and larger: 208 or 480 volts, three phase, 60 Hz.

C. Motors 3/4 horsepower and larger, or motors used in outdoor application (regardless of horsepower) shall be three phase, 60 Hertz.
   1. Motors configured in duplex applications shall have an individual eletrical circuit to each motor.
   2. Motor starter coils and control circuits shall be powered by 120 volt control transformers located within the motor starter enclosure. Refer to Division 26 Section "Enclosed Controllers".

D. Construction:
   1. Open drip-proof type except where specifically noted otherwise.
   2. Motors shall be of cast iron construction.
   3. Design for continuous operation in 104 deg F (40 deg C) environment.
   4. Design for temperature rise in accordance with NEMA MG 00001 limits for insulation class, service factor, and motor enclosure type.

E. Duty: Continuous duty at ambient temperature of 86°F and at altitude of 7,500 feet above sea level.

F. Maximum load factor: Select motors with maximum ratio of brake horsepower (BHP) to nameplate horsepower not exceeding 0.8.

G. Capacity and Torque Characteristics: Sufficient to start, accelerate, and operate connected loads at designated speeds, at installed altitude and envrionment, with indicated operating sequence, and without exceeding nameplate ratings or considering service factor.

H. All motors furnished shall be the manufacturer's premium efficiency design. The nominal efficiency and the minimum guaranteed efficiency shall be stamped on the nameplate of the motor. All efficiency testing and labeling shall be done in accordance with the NEMA MG 00001 standard.

I. All three phase motors shall be "Inverter Approved" or "VFD Approved" and stamped as such on the nameplate of the motor.

J. Explosion-proof motors shall be UL approved and labeled for hazard classification, with over temperature protection.

K. Visible nameplates must be stainless steel, indicating motor horsepower, voltage, phase, cyclecs, RPM, full load amps, locked rotor amps, frame size, manufacturer's name and model number, service factor, power factor, and efficiency.

L. Wiring Terminations:
   1. Motor leads shall be stranded copper and shall be permanently identified and brought out into the motor terminal box through a neoprene gasket. Provide conduit opening suitable for locknut type connector. Include UL listed ground lug in terminal enclosure.
   2. For fractional horsepower motors where connection is made directly, provide conduit opening suitable for locknut type fitting in end frame.

M. All motors that are used with VFDs shall have grounding brushes on the shaft of the motor. The terminal strip provided shall be tied to the building ground. Acceptable products: AEGIS SGR.

N. All equipment selections/pefermance shall be based on a maximum motor speed of 60 Hz. This shall apply to all fan and pump motors. Providing motors in in an "overspeed" state shall not be acceptable.

2.02 POLYPHASE MOTORS

A. Description: NEMA MG 00001, Design B, medium induction motor.

B. Multispeed Motors: Separate winding for each speed.

C. Rotor: Random-wound, squirrel cage.

D. Starting Torque: Between 1 and 1-1/2 times full load torque.

E. Starting Current: Six times full load current.

F. Power Output, Locked Rotor Torque, Breakdown or Pull Out Torque: NEMA Design B characteristics.

G. Design, Construction, Testing, and Performance: Conform to NEMA MG 00001 for Design B motors.

H. Bearings: Grease lubricated, anti-friction ball bearings with housings equipped with provision for relubrication, rated for minimum ABMA STD 9, L-10 life for 100,000 hours. Calculate bearing load with NEMA minimum V-belt pulley with belt center line at end of NEMA standard shaft extension. Stamp bearing sizes on nameplate.

I. Sound Power Levels: To NEMA MG 00001.

J. Temperature Rise: Class F.

K. Insulation: Class F or better.

L. Testing Procedure: In accordance with IIEEE 112. Load test motors to determine free from electrical or mechanical defects in compliance with performance data.

M. Motors shall be 1800 rpm unless indicated otherwise. Enclosures shall be totally enclosed unless noted otherwise. Motors rated two horsepower and larger shall be fan cooled, premium efficiency design.

N. Code Letter Designation:
   1. Motors 15 HP and Larger: NEMA starting Code F or Code G.
   2. Motors Smaller than 15 HP: Manufacturer's standard starting characteristic.

O. Motor Frames: NEMA Standard T-Frames. Stator frame and end brackets shall be a minimum of grade 25 cast iron construction.

P. Other Frame designation (e.g "T" Frame) motor plate rating shall be selected such that the brake horsepower of the equipment to be driven, through the full range of operating conditions, not including the service factor in the motor rating, does not exceed 80 percent of the name plate rating.

Q. "U" frame motor nameplate rating shall not be less thatn the brake horsepower of equipment to be driven, through the full range of operating conditions, not including service factor in the motor rating.

R. Thermistor System (Motor Frame Sizes 254T and Larger): Three PC thermistors imbedded in motor windings and epoxy encapsulated solid state control relay for wiring intor motor stator.

S. Weatherproof Epoxy Sealed Motors: Epoxy seal windings using vacuum and pressure with rotor and stator surfaces protected with epoxy enamel; bearings double shielded with waterproof non-washing grease.

T. Nominal Efficiency: As scheduled at full load and rated voltage when tested in accordance with IEEE 112

U. Nominal Power Factor: As scheduled at full load and rated voltage when tested in accordance with IEEE 112.

2.03 POLYPHASE MOTORS WITH ADDITIONAL REQUIREMENTS

A. Motors Used with Reduced-Voltage and Multispeed Controllers: Match wiring connection requirements for controller with required motor leads. Provide terminals in motor terminal box, suited to control method.

B. Motors Used with Variable Frequency Controllers (VFDs): Ratings, characteristics, and features coordinated with and approved by controller manufacturer.
   1. Windings: Copper magnet wire with moisture-resistant insulation varnish, designed and tested to resist transient spikes, high frequencies, and short time rise pulses produced by pulse-width modulated inverters.
   2. Premium-Efficient Motors: Class B temperature rise; Class F insulation.
   3. Inverter-Duty Motors: Class F temperature rise; Class H insulation.
   4. Thermal Protection: Comply with NEMA MG 00001 requirements for thermally protected motors.

2.04 SINGLE-PHASE MOTORS

A. Motors larger than 1/20 hp and less than 5 HP shall be one of the following, to suit starting torque and requirements of specific motor application:
   1. Permanent-split capacitor (PSC).
   2. Electrically commutated motor (ECM).
   3. Capacitor start, capacitor run (CS/CR).

B. Multispeed Motors: Variable-torque, ECM.

C. Split phase motors shall have the following characteristics:
   1. Starting torque: Less than 150 percent of full load torque.
   2. Starting current: Up to seven times full load current.
   3. Breakdown torque: Approximately 200 percent of full load torque.
   4. Motors: Capacitor in series with starting winding; provide capacitor-start/capacitor-run motors with two capacitors in parallel with run capacitor remaining in circuit at operating speeds.

D. Capacitor start motors shall have the following characteristics:
   1. Starting torque: Three times full load torque.
   2. Starting current: Less than five times full load current.
   3. Pull-up torque: Up to 350 percent of full load torque.
   4. Breakdown torque: Approximately 250 percent of full load torque.
   5. Motors: Capacitor in series with starting winding; provide capacitor-start/capacitor-run motors with two capacitors in parallel with run capacitor remaining in circuit at operating speeds.

E. Drip-proof Enclosure: Class A (50 degress C temperature rise) insulation, NEMA Service Factor.

F. Enclosed Motors: Class A (50 degrees C temperature rise) insulation, 1.0 Service Factor.

G. Bearings: Prelubricated, antifriction ball bearings or sleeve bearings suitable for radial and thrust loading.

H. Motors 1/20 HP and Smaller: Shaded-pole type.

I. Thermal Protection: Internal protection to automatically open power supply circuit to motor when winding temperature exceeds a safe value calibrated to temperature rating of motor

insulation. Thermal-protection device shall automatically reset when motor temperature returns to normal range.

PART 3 - EXECUTION

3.01 APPLICATION

A. Single phase motors for shaft mounted fans and blowers: Permanent split capacitor type.

B. Three phase motors for exterior mounted fans.

C. Motors located in exterior locations: Totally closed type.

3.02 INSTALLATION

A. Install in accordance with manufacturer's instructions.

B. Install securely on firm foundation. Mount ball bearing motors with shaft in any position.

C. Check line voltage and phase and ensure agreement with nameplate.

3.03 MOTOR DYNAMIC BALANCING

A. Perform dynamic balancing and test motors for vibration after manufacture. Self excite vibration velocity of motor shall not exceed 0.157/0.06 inches per second at the bearing caps.

3.04 START UP

A. Inspect, test and start each motor and align with a dial indicator to 0.002"

3.05 SCHEDULE

A. NEMA Open Motor Service Factors.
   1. 1/6 - 1/3 HP: 1800 rpm: 1.35
   2. 1/2 HP: 1800 rpm: 1.25
   3. 3/4 HP: 1800 rpm: 1.25
   4. 1 HP: 1800 rpm: 1.15
   5. 1.5 - 150 HP: 1800 rpm: 1.15

END OF SECTION 23 0513

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SECTION 23 0514
VARIABLE FREQUENCY DRIVES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Specifications throughout all Divisions of the Project Manual are directly applicable to this Section, and this Section is directly applicable to them.

1.02 SUMMARY

A. This section includes:
   1. Perform all Work required to provide and install a complete variable frequency motor drive (VFD) consisting of a pulse width modulated (PWM) inverter designed for use on a standard NEMA Design B induction motor.
   2. The drive manufacturer shall supply the drive and all necessary controls as specified.

B. Related Sections
   1. Section 23 0920.01 - Control Systems.
   2. Section 26 0583 - Wiring Connections.

1.03 REFERENCE STANDARDS

A. The latest published edition of a reference shall be applicable to this Project unless identified by a specific edition date.

B. All reference amendments adopted prior to the effective date of this Contract shall be applicable to this Project.

C. All materials, installation and workmanship shall comply with the applicable requirements and standards addressed within the following references:
   1. IEEE 519, Guide for Harmonic Content and Control.
   2. ANSI/UL Standard UL508C, Underwriter’s Laboratories.
   3. NEMA IS 10033, AC Adjustable Speed Drives, National Electrical Manufacturer’s Association (NEMA).
   4. IEC 16800 Parts 1 and 2.

1.04 QUALITY ASSURANCE

A. Company specializing in manufacturing the products specified in this Section with minimum three (3) years experience.

B. VFD and options shall be UL listed as a complete assembly. VFD’s that require the customer to supply external fuses are not acceptable.

C. VFD and options shall be tested to ANSI/UL Standard 508 and listed by a nationally recognized testing agency such as UL or ETL.

D. VFD and options shall comply with applicable requirements of the latest Standards of ANSI/UL, IEEE, and the NEC.

1.05 SUBMITTALS

A. Product Data:
   1. Submit product data for components and accessories.
   2. All VFD’s for this Project shall be supplied by one manufacturer.
   3. Submit Shop Drawings indicating outline dimensions, enclosure construction, lifting and supporting points, electrical one-line diagram, equipment electrical ratings, noise levels (including driven equipment) and total harmonic distortion (voltage and current).
   4. Manufacturer shall provide terminal block to terminal block wiring diagrams coordinated with the Owner to provide a complete and functional operating system. Furnish detailed

Drawings showing construction, dimensions, wiring diagrams, and installation procedures for Engineer's approval.

B. Operation and Maintenance Data:
   1. Submit manufacturer's written installation instructions.
   2. Submit training outline.
   3. Furnish harmonic analysis verifying compliance with specified distortion levels.
   4. Furnish a list of recommended spare parts.

1.06 DELIVERY, STORAGE AND HANDLING

A. Deliver, store, protect and handle products to the Project Site under provisions of Division 01.

B. Accept products on Site in factory-fabricated protective container with factory installed shipping skids and lifting lugs. Inspect for damage.

C. Store in clean dry place and protect from weather and construction traffic. Handle carefully to avoid damage to components, enclosures and finish.

1.07 EXTRA MATERIALS

A. Submit two insulated-handle tools designed for pulling fuses in accordance with ANSI/IEEE C37.46.

B. Refer to Section 262813 for fuse requirements.

1.08 WARRANTY

A. VFD shall be unconditionally warranted by the manufacturer for two (2) years from the date of Substantial Completion, not to exceed 30 months from date of shipment.

B. Warranty shall include all parts, labor, shipping, field service or technician time, labor or travel expenses, and verbal or written correspondence with the VFD manufacturer or VFD manufacturer's representatives. Include correspondence which might be incidental to the proper installation and operation of the equipment.

PART 2 PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer's product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.

B. Equipment Categories: Equipment types in this section are considered equipment categories that be furnished by one manufacturer or available through one manufacturer to provide consistency within the overall project for equipment configurations, manufacturing techniques, and to facilitate ease of maintenance for the Owner.

2.02 GENERAL

A. All materials shall meet or exceed all applicable referenced standards, federal, state and local requirements, and conform to codes and ordinances of authorities having jurisdiction.

B. Furnish complete VFD controllers that convert incoming fixed frequency three-phase AC power into a variable frequency and voltage for controlling the speed of three-phase AC motors. The motor current shall closely approximate a sine wave. Motor voltage shall be varied with

frequency to maintain desired motor magnetization current suitable for centrifugal pump and fan control and to eliminate the need for motor derating.

C. Contractor shall check equipment schedules on the Contract Drawings to determine if the VFD "bypass" switch option is required to allow the motor to run if the controller malfunctions. If the "bypass" switch option is not required, articles and statements in this section do not apply, and the Contractor shall provide one spare VFD for each equivalent horsepower motor or equipment power load indicated on the Contract Drawings.

D. VFD manufacturer shall verify compatibility of motor furnished on equipment. One controller shall control the speed of one motor with the exception being a wall fan technology system.

E. VFD shall convert 3 phase, 60 Hz utility power to adjustable voltage and frequency, 3 phase AC power for stepless motor speed control from 10 percent to 100 percent of the motor's 60 Hz speed. Input voltage characteristics are 480 volts, 3 phase, 60 Hz.

F. VFD shall include a converter section. The converter section shall convert fixed frequency and voltage AC utility power to a variable DC voltage. VFD's that use silicon controlled rectifiers in the converter bridge shall also include 5 percent reactors. Isolation transformers are not acceptable in lieu of line reactors.

G. VFD shall include an inverter section. The inverter section shall invert the variable DC voltage into a PWM wave form; adjustable voltage and frequency output for stepless motor speed control.

H. Individual or simultaneous operation of VFD's shall not add more than 5 percent total harmonic voltage distortion and no more than 5 percent total harmonic current distortion (per IEEE 516-2021) to the normal bus.
   1. VFD manufacturer shall perform harmonic analysis based on the electrical one-line diagram.
   2. The VFD manufacturer shall provide calculations specific to this installation, showing total harmonic voltage distortion is less than 5 percent.
   3. Input line filters shall be sized and provided as required by the VFD manufacturer to ensure compliance with IEEE Standard 519. All VFD's shall include a minimum of 5 percent impedance reactors, no exceptions.

I. VFD shall include a coordinated AC transient protection system consisting of 4-120 joule rated MOV's (phase to phase and phase to ground), a capacitor clamp, and 5 percent impedance reactors.

J. Alternate Harmonics Specification:
   1. Input line reactors and DC Bus filtered chokes (factory installed and wired in the drive enclosure) shall be provided to allow reliable operation on a typical commercial power distribution system and to minimize harmonics reflected onto the input line.
      a. Shall not interfere with computer and other electronic systems in the building.
      b. If not inherently protected, provide a suitable isolation transformer.
      c. The system shall not produce spikes on the incoming line.
   2. Any inverter that generates sufficient electrical line noise to interfere with operation of sensitive building equipment shall be field modified or replaced by the inverter supplier at no additional cost to the Owner.

K. EMI / RFI filters. All VFD's shall include EMI/RFI filters. The onboard filters shall allow the VFD assembly to be CE Marked and the VFD shall meet product Standard EN 61800-3 for the First Environment restricted level.

L. Low voltage logic and 115V control circuits shall be electrically isolated from the power circuits. Signal circuit common shall be grounded.

M. VFD shall include a power ride-through feature to allow continuous operation up to a three-cycle line loss.

N. Two independently adjustable accel and decel ramps with 1 to 1800 seconds adjustable time ramps. Extended time periods are also acceptable.

O. VFD shall have full function output current limit adjustable from 10 to 100 percent. At the factory with compatible motor, provide at least three lock-out ranges (50 rpm maximum each), two of which can be used to correct any run test problems.

P. Components shall be pretested and complete VFD shall have full burn-in under full load for a minimum of 12 hours. Provide at least three lockout ranges (50 rpm maximum), two of which can be used to correct run test problems.

Q. Ambient noise generated by the VFD shall be limited to an amount equal to the system noise level as designated by the latest ASHRAE noise level guidelines for such equipment at each octave band. Noise level criteria at different octave bands and mid-frequencies shall be furnished with the submittal data.

R. VFD shall include a motor flux optimization circuit that will automatically reduce applied motor voltage to the motor to optimize energy consumption and audible motor noise.

2.03 MANUFACTURERS

A. Danfoss Graham.

B. ABB.

C. Yaskawa Electric.

2.04 ENCLOSURE

A. VFD shall be enclosed in a UL Listed Type 12 enclosure. Enclosure shall be UL listed as a plenum rated VFD. The VFD tolerated voltage window shall allow operation from a line of +30 percent nominal, and -35 percent nominal voltage as a minimum.

B. Environmental operating conditions: 0 to 40 degrees C continuous. VFD's that can operate at 40 degrees C intermittently (during a 24-hour period) are not acceptable and must be oversized. Altitude from 0 to 7500 feet above sea level, less than 95 percent humidity, non-condensing. VFD's without these ratings are not acceptable.

C. The following operator controls shall be located on the front of the enclosure:
   1. Bypass Hand-Off-Auto.
   2. Drive mode selector.
   3. Bypass mode selector.
   4. Bypass fault reset.
   5. Provide the following indicating lights (LED type). In addition, provide test mode or push to test feature:
      a. Power-on (ready).
      b. Run enable (safeties) open.
      c. Drive mode select damper opening.
      d. Bypass mode selected.
      e. Drive running.
      f. Bypass running.
      g. Drive fault.
      h. Bypass fault.
      i. Bypass H-O-A mode.
      j. Automatic transfer to bypass selected.
      k. Safety open.
      l. Damper opening.
      m. Damper end-switch made.
   6. Provide the following relay (form C) outputs from the bypass:
      a. System started.
      b. System running.

c. Bypass overttide enabled.
d. Drive Fault.
e. Bypass fault (motor overload or underload-broken belt).
f. Bypass H-O-A position.

D. Digital inputs for the system shall accept 24V or 115VAC (selectable).

E. Customer Interlock Terminal Strip: Provide a separate terminal strip for connection of freeze, fire, smoke contacts, and external start command. All external safety interlocks shall remain fully functional whether the system is in Hand, Auto, or Bypass modes (not functional in fireman's override 2). The remote start/stop contact shall operate in VFD mode.

F. Dedicated digital input that will transfer motor from VFD mode to bypass mode upon dry contact closure for fireman's override. Two modes of operation are required:
1. The first mode forces the motor to bypass operation and overrides both the VFD and bypass H-O-A switches and forces the motor to operate across the line (test mode). The system will only respond to the digital inputs and motor protections.
2. The second mode operates as the first, but will also defeat the overload and singe-phase protection for bypass and ignore all keypad and digital inputs t the system 9run until destruction).

G. Include a "run permissive circuit" that will provide a normally open contact whenever a run command is provided (local or remote start command in VFD or bypass mode). The VFD system (VFD or bypass) shall not operate the motor until it receives a dry contact closure from a damper or valve end-switch. When the VFD system safety interlock (fire detector, freezestat, high static pressure switch, etc) opens, the motor shall coast to a stop and the run permissive contact shall open, closing the damper or valve.

H. Include Class 20 or 30 (selectable) electronic motor overload protection.

2.05 BYPASS

A. Furnish where indicated on the Drawings, a complete factory wired and tested bypass system consisting of an output contactor and bypass contactor. Overload protection shall be provided in both drive and bypass modes.

B. Bypass to be furnished, built, and mounted by the VFD manufacturer.

C. Provide an internal switch to select manual or automatic bypass.

D. Provide an adjustable current sensing circuit for the bypass to provide loss of load indication (broken belt) when in the bypass mode.

E. Door interlocked, padlockable disconnect that will disconnect all input power from the drive and all internally mounted options.

F. Fused VFD only disconnect (service switch). Fast acting fuses exclusive to the VFD – fast acting fuses allow the VFD to disconnect from the line prior to clearing upstream branch circuit protection, maintaining bypass capability. Bypass designs, which have no such fuses, or that incorporate fuses common to both the VFD and the bypass will not be accepted. The following contactor bypass schemes are not acceptable.
1. Door interlocked main input disconnect switch.
2. Power on light.
3. "Drive-off-bypass" manual mode selector switch.

G. The bypass shall incorporate an internally sourced power supply and shall not require an external power source.

2.06 DISPLAY / KEYPAD

A. All applicable operating values shall be capable of being displayed in engineering (user) units. A minimum of three (3) operating values from the list below shall be capable of being displayed

at all times. The display shall be in complete English words (alpha-numeric codes are not acceptable):
1. Output frequency.
2. Motor speed (RPM, percent, or engineering units).
3. Motor current.
4. Calculated motor torque.
5. Calculated motor power (kW).
6. DC bus voltage.
7. Output voltage.

B. Keypad shall include a backlit LCD display. The display shall be in complete English words for programming and fault diagnostics (alpha-numeric codes are not acceptable). The keypad shall use the following assistants:
1. Start-up assistants.
2. Parameter assistants.
3. Maintenance assistant.
4. Troubleshooting assistant.

C. VFD shall have the same customer interface, including digital display, and keypad, regardless of horsepower rating. Keypad shall be removable, capable of remote mounting and shall allow for uploading and downloading of parameter settings as an aid for start-up of multiple VFD's.

D. Keypad shall include Hand-Off-Auto selections and manual speed control. The drive shall incorporate "bumpless transfer" of speed reference when switching between "Hand" and "Auto" modes. There shall be fault reset and "Help" buttons on the keypad. The Help button shall include "on-line" assistance for programming and troubleshooting.

E. Provide a built-in time clock with battery back-up in the VFD keypad. The time clock shall be used to date and time stamp faults and record operating parameters at the time of fault. If the battery fails, the VFD shall automatically revert to hours of operation since initial power up. The time clock shall also be programmable to control start/stop functions, constant speeds, PID parameter sets and output relays. The VFD shall have a digital input that allows an override to the time clock (when in the off mode) for a programmable time frame. There shall be four (4) separate, independent timer functions that have both weekday and weekend settings.

2.07 SERIAL COMMUNICATION

A. The VFD shall have an EIA-485 port as standard. The standard protocols shall be Modbus, Johnson Controls N2, Siemens Building Technologies FLN, and BACnet. [Optional protocols for LonWorks, Profibus, EtherNet, BACnet IP, and DeviceNet shall be available.] Each individual drive shall have the protocol in the base VFD. The use of third party gateways and multiplexers is not acceptable. All protocols shall be "certified" by the governing authority (i.e. BTL Listing for BACnet). Use of noncertified protocols is not allowed.

B. VFD shall be provided with protocol information specific to the selected BAS Provider and shall be pre-configured at the factory to provide automatic communications without the need for field programming.

C. VFD shall continue to provide serial communications regardless of how inverter is being controlled ("manual" mode via keypad, "automatic" mode via BAS, or "stopped" mode via either keypad or automatic BAS start/stop signal).

D. Serial communications capabilities shall include, but not be limited to:
1. Run/stop control speed set adjustment.
2. Proportional/integral or PID control adjustments.
3. Current limit.
4. Accel/decel time adjustments.

E. VFD shall have the capability of allowing the BAS to monitor the following feedback signals:
1. Process variable.

2. Output speed/frequency.
3. Current.
4. Torque.
5. Power (kW).
6. Operating hours.
7. Kilowatt hours (kWh).
8. Relay outputs.
9. Diagnostic warning and fault information.

F. VFD shall allow the BAS to control the drive's digital and analog outputs and monitor all drive digital and analog inputs via the serial interface.

G. VFD shall be capable of providing the BAS with status signals for bypass operation and external safety trips via serial interface.

2.08 SYSTEM OPERATION

A. Selector switch in the "off" position: controller run circuit shall be open and the system shall not operate.

B. Selector switch in the "manual" position: motor speed shall be controlled by the manual speed potentiometer.

C. Selector switch in the "auto" position: operation shall be via input 0 to 10 VDC or 4-20 mA signal with strategy output speed proportional to the input signal. If required into the controls strategy, VFD manufacturer shall furnish a pressure transducer mounted in the drive enclosure to convert a 3 to 15 psi pressure signal to a 0 to 10 VDC signal or 4-20 mA signal.

2.09 HARMONIC MITIGATION

A. The engineer has completed a preliminary harmonic analysis showing that all drives below 15 HP shall be 6 pulse drives with swinging chokes and all drives 25 HP and above shall be Ultra Low Harmonic designs. Drives with passive harmonic filters, 12 pulse, and 18 pulse drives shall not be acceptable. Compliance to IEEE 519 does not relieve the vendor from the requirement to provide active front end drives, as specified above.

PART 3 EXECUTION

3.01 PREPARATION

A. Verify that surfaces are ready to receive Work.

B. Verify that field measurements are as shown on Shop Drawings and as instructed by manufacturer.

C. Verify that required utilities are available, in the proper location, and ready for use.

3.02 HARMONIC ANALYSIS

A. Perform a harmonic analysis study to identify the effects of nonlinear loads and their associated harmonic contributions on the voltages and currents throughout the electrical system. Analyze designated operating scenarios, including recommendations for VFD input filtering to limit TDD and THD(V) at each VFD to specified levels.

B. Prepare a harmonic analysis study and report complying with IEEE 399 and NETA ATS (Acceptance Testing Specification).

3.03 INSTALLATION

A. Installation shall meet or exceed all applicable federal, state and local requirements, referenced standards and conform to codes and ordinances of authorities having jurisdiction.

B. Coordinate layout and installation of VFDs with other construction including conduit, piping, equipment, and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

C. Wall-Mounting Controllers: Install VFDs on walls with tops at uniform height and with disconnect operating handles not higher than 79 inches (2000 mm) above finished floor unless otherwise indicated, and by bolting units to wall or mounting on lightweight structural-steel channels bolted to wall. For controllers not on walls, provide freestanding racks complying with Section 26 0529 - Hangers and Supports for Electrical Systems

D. Floor-Mounting Controllers: Install VFDs on 4-inch (100-mm) nominal thickness concrete base. Comply with requirements for concrete base specified in
   1. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on 18-inch (450-mm) centers around the full perimeter of concrete base.
   2. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.
   3. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.
   4. Install anchor bolts to elevations required for proper attachment to supported equipment.

E. All installation shall be in accordance with manufacturer’s published recommendations.

F. Power wiring shall be completed by the Electrical Contractor in accordance with the recommendations of the VFD manufacturer as outlined in the installation manual.

G. Temporary Lifting Provisions: Remove temporary lifting eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.

H. Install fuses in each fusible-switch VFD

I. Install fuses in control circuits if not factory installed. Comply with requirements in Section 26 2813 - Fuses.

J. Install heaters in thermal-overload relays. Select heaters based on actual nameplate full-load amperes after motors have been installed.

3.04 IDENTIFICATION

A. Identify VFDs, components, and control wiring. Comply with requirements for identification specified in Section 26 0553 - Identification for Electrical Systems.
   1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.
   2. Label each VFD with engraved nameplate.
   3. Label each enclosure-mounted control and pilot device.

3.05 CONTROL WIRE INSTALLATION

A. Install wiring between VFDs and remote devices and facility's central-control system.

B. Bundle, train, and support wiring in enclosures.

C. Connect selector switches and other automatic control devices where applicable.
   1. Connect selector switches with control circuit in both manual and automatic positions for safety-type control devices such as low- and high-pressure cutouts, high-temperature cutouts, and motor overload protectors

3.06 TESTING

A. Manufacturer shall provide a factory trained technician to inspect, test and start-up the VFD and associated equipment and place the VFD into operation.

B. A harmonic test verifying the distortion level shall be included as part of Start-up and forwarded to the Owner. Any additional equipment, installation and equipment floor space required to meet the distortion level as set forth in the Specification, shall be borne by the VFD manufacturer.

3.07 ADJUSTING

A. Program microprocessors for required operational sequences, status indications, alarms, event recording, and display features. Clear events memory after final acceptance testing and prior to Substantial Completion

B. Set field-adjustable switches, auxiliary relays, time-delay relays, timers, and overload-relay pickup and trip ranges.

C. Adjust the trip settings of MCPs and thermal-magnetic circuit breakers with adjustable, instantaneous trip elements. Initially adjust to six times the motor nameplate full-load amperes and attempt to start motors several times, allowing for motor cool-down between starts. If tripping occurs on motor inrush, adjust settings in increments until motors start without tripping. Do not exceed eight times the motor full-load amperes (or 11 times for NEMA Premium Efficient motors if required). Where these maximum settings do not allow starting of a motor, notify the Cleveland Clinic before increasing settings.

D. Set the taps on reduced-voltage autotransformer controllers.

E. Set field-adjustable circuit-breaker trip ranges.

F. Set field-adjustable pressure switcheS.

END OF SECTION 23 05 13

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SECTION 23 0516
EXPANSION FITTINGS AND LOOPS FOR HVAC

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
1. Flexible, ball-joint, packed expansion joints and slip-joint packed expansion joints require periodic maintenance. Make sure the client is aware you are specifying.
2. Flexible, ball-joint, packed expansion joints.
3. Slip-joints are not suitable for refrigerant piping.
4. Slip-joint packed expansion joints.
5. Expansion compensators are only for pipe less than 4" in small spaces and only absorb axial movement.
6. Expansion-compensator packless expansion joints.
7. Rubber, Expansion-compensator packless expansion joints.
8. Metal-bellows packless expansion joints.
9. Only use grooved-joint if you are allowing "Victaulic" grooved mechanical couplings elsewhere in the project.
10. Grooved-joint expansion joints.
11. Pipe loops and swing connections.
12. Alignment guides and anchors.

B. Related Sections include the following:
1. Division 23 Section "Hydronic Piping"
2. Division 23 Section "Hangers and Supports for HVAC"
3. Division 23 Section "Vibration and Seismic Controls for HVAC"

1.03 PERFORMANCE REQUIREMENTS

A. Compatibility: Products shall be suitable for piping service fluids, materials, working pressures, and temperatures.

B. Capability: Products to absorb 200 percent of maximum axial movement between anchors.

1.04 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

B. Paragraph below is defined in Section 01 3300 - Submittal Procedures as a "Delegated-Design Submittal." Retain if Work of this Section is required to withstand specific design loads and design responsibilities have been delegated to Contractor or if structural data are required as another way to verify compliance with performance requirements. Professional engineer qualifications are specified in Section 01 4000 - Quality Requirements.

C. Delegated-Design Submittal: For each anchor and alignment guide indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
1. Design Calculations: Calculate requirements for thermal expansion of piping systems and for selecting and designing expansion joints, loops, and swing connections.
2. Anchor Details: Detail fabrication of each anchor indicated. Show dimensions and methods of assembly and attachment to building structure.
3. Alignment Guide Details: Detail field assembly and attachment to building structure.
4. Schedule: Indicate type, manufacturer's number, size, material, pressure rating, end connections, and location for each expansion joint.

5. General Note: After Preconstruction coordination and per the delegated design aspect of the above specification, the contractor shall engage appropriate design professional to select piping expansion compensation method for all hot piping and submit as a shop drawing.

1.05 INFORMATIONAL SUBMITTALS

A. Welding certificates.

B. Product Certificates: For each type of expansion joint, from manufacturer.

1.06 CLOSEOUT SUBMITTALS

A. Maintenance Data: For expansion joints to include in maintenance manuals.

1.07 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel according to the following:
   1. AWS D1.1/D1.1M, "Structural Welding Code - Steel."
   2. ASME Boiler and Pressure Vessel Code: Section IX.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer's product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.

B. Equipment Categories: Equipment types in this section are considered equipment categories that be furnished by one manufacturer or available through one manufacturer to provide consistency within the overall project for equipment configurations, manufacturing techniques, and to facilitate ease of maintenance for the Owner.

2.02 PACKLESS EXPANSION JOINTS

A. Metal, Expansion-Compensator Packless Expansion Joints:
   1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
      a. Flexicraft Industries.
      b. Flex-Weld, Inc.
      c. Hyspan Precision Products, Inc.
      d. Metraflex, Inc.
   2. Minimum Pressure Rating: 175 psig nless otherwise indicated.
   3. Configuration for Copper Tubing: Two-ply, phosphor-bronze bellows with copper pipe ends.
      a. End Connections for Copper Tubing NPS 2 and Smaller: Solder joint threaded.
      b. End Connections for Copper Tubing NPS 2-1/2 to NPS 4: Threaded.
   4. Configuration for Steel Piping: Two-ply, stainless-steel bellows; steel-pipe end connections; and carbon-steel shroud.
      a. End Connections for Steel Pipe NPS 2 and Smaller: Threaded.
      b. End Connections for Steel Pipe NPS 2-1/2 to NPS 4: Flanged.

B. Rubber, Expansion-Compensator Packless Expansion Joints:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
   a. Flex-Hose Co., Inc.
   b. Flexicraft Industries.
   c. Mason Industries, Inc.; Mercer Rubber Co.
2. Material: Twin reinforced-rubber spheres with external restraining cables.
3. Minimum Pressure Rating: 150 psig at 170 deg F unless otherwise indicated.

C. Flexible-Hose Packless Expansion Joints:
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
   a. Flex-Hose Co., Inc.
   b. Flexicraft Industries.
   c. Metraflex, Inc.
2. Description: Manufactured assembly with inlet and outlet elbow fittings and two flexible-metal-hose legs joined by long-radius, 180-degree return bend or center section of flexible hose.
3. Flexible Hose: Corrugated-metal inner hoses and braided outer sheaths.
4. Expansion Joints for Copper Tubing NPS 2 and Smaller: Copper-alloy fittings with solder-joint end connections.
   a. Bronze hoses and single-braid bronze sheaths with 450 psig at 70 deg F and 340 psig at 450 deg F ratings.
5. Expansion Joints for Copper Tubing NPS 2-1/2 to NPS 4: Copper-alloy fittings with solder joint or flanged end connections.
   a. Stainless-steel hoses and single-braid, stainless-steel sheaths with 300 psig at 70 deg F and 225 psig at 450 deg F ratings.
6. Expansion Joints for Steel Piping NPS 2 and Smaller: Carbon-steel fittings with threaded end connections.
   a. Stainless-steel hoses and single-braid, stainless-steel sheaths with 450 psig at 70 deg F and 325 psig at 600 deg F ratings.
7. Expansion Joints for Steel Piping NPS 2-1/2 to NPS 6: Carbon-steel fittings with flangedend connections.
   a. Stainless-steel hoses and single-braid, stainless-steel sheaths with 200 psig at 70 deg Fand 145 psig at 600 deg F ratings.
8. Expansion Joints for Steel Piping NPS 8 to NPS 12: Carbon-steel fittings with flangedend connections.
   a. Stainless-steel hoses and double-braid, stainless-steel sheaths with 165 psig at 70 deg F and 120 psig at 600 deg F ratings.
9. Expansion Joints for Steel Piping NPS 14 and Larger: Carbon-steel fittings with flanged end connections.
   a. Stainless-steel hoses and double-braid, stainless-steel sheaths with 165 psig at 70 deg F and 120 psig at 600 deg F ratings.

D. Metal-Bellows Packless Expansion Joints:
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
   a. Flex-Hose Co., Inc.
   b. Flexicraft Industries.
   c. Hyspan Precision Products, Inc.
   d. Metraflex, Inc.
2. Standards: ASTM F 1120 and EJMA's "Standards of the Expansion Joint Manufacturers Association, Inc."
3. Type: Circular, corrugated bellows with external tie rods.
4. Minimum Pressure Rating: 175 psig unless otherwise indicated.

5. Single joint if pipe smaller than 2". Single with base if bigger than 2". Double if pipe will move both directions or is larger than 8".
6. Configuration:
   a. Axial movement only for NPS 2 and Smaller: Single Joint.
   b. Axial movement only for NPS 2-1/2 thru NPS 6: Single Joint with base.
   c. Axial movement only for NPS 8 and Larger: Double joint with base.
   d. Axial and Transverse movement for all pipe sizes: Double joint with base.
7. Expansion Joints for Copper Tubing: Multi-ply phosphor-bronze bellows, copper pipe ends, and brass shrouds.
   a. End Connections for Copper Tubing NPS 2 and Smaller: Solder joint or threaded.
   b. End Connections for Copper Tubing NPS 2-1/2 to NPS 4: Solder joint or flanged.
8. Expansion Joints for Steel Piping: Multi-ply stainless-steel bellows, steel pipe ends, and carbon-steel shroud.
   a. End Connections for Steel Pipe NPS 2 and Smaller: Threaded.
   b. End Connections for Steel Pipe NPS 2-1/2 and Larger: Flanged.

E. Rubber Packless Expansion Joints:
1. Manufacturers: Subject to compliance with requirements, provide products by one fo the following:
   a. Amber/Booth Company, Inc.; a div. of Vibration Isolation Products of Texas, Inc.
   b. Flex-Hose Co., Inc.
   c. Fexicraft Industries.
   d. Mason Industries, Inc.; Mercer Rubber Co.
   e. Metraflex, Inc.
2. Standards: ASTM F 1123 and FSA's "Technical Handbook: Non-Metallic Expansion Joints and Flexible Pipe Connectors."
3. Material: Fabric-reinforced rubber complying with FSA-NMEJ-703.
4. Spherical Type: Multiple spheres. Arch type is not acceptable.
5. Minimum Pressure Rating for NPS 1-1/2 to NPS 4: 150 psig at 220 deg F.
6. Minimum Pressure Rating for NPS 5 and NPS 6: 140 psig at 200 deg F.
7. Minimum Pressure Rating for NPS 8 to NPS 12: 140 psig at 180 deg F.
8. Materials for Fluids Containing Acids, Alkalies, or Chemicals: EPDM.
9. Materials for Fluids Containing Gas, Hydrocarbons, or Oil: Buna-N.
10. Material for Water: EPDM.
11. End Connections: Full-faced, integral steel flanges with steel retaining rings.

2.03 ALIGNMENT GUIDES AND ANCHORS

A. Alignment Guides:
1. Alignment Guides should be placed on both sides of expansion joints, indicate locations on plans. Consult manufacturer rep for locations. Coordinate with structural for installation on steel and/or concrete.
2. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
   a. Advanced Thermal Systems, Inc.
   b. Flex-Hose Co., Inc.
   c. Flexicraft Industries.
   d. Flex-Weld, Inc.
   e. Hyspan Precision Products, Inc.
   f. Metraflex, Inc.
3. Indicate alignment-guide length and maximum spider travel on Drawings.
4. Description: Steel, factory-fabricated alignment guide, with bolted two-section outer cylinder and base for attaching to structure; with two-section guiding spider for bolting to pipe.

B. Anchor Materials:
   1. Steel Shapes and Plates: ASTM A36/A36M M.
   2. Bolts and Nuts: ASME B18.10 or ASTM A183, steel hex head.
   3. Washers: ASTM F844, steel, plain, flat washers.
   4. Mechanical Fasteners: Insert-wedge-type stud with expansion plug anchor for use in hardened portland cement concrete, with tension and shear capacities appropriate for application.
      a. Stainless-steel studs are available. Use if in corrosive environment such as pools, water and wastewater treatment plans.
      b. Stud: Threaded, zinc-coated carbon steel.
      c. Expansion Plug: Zinc-coated steel.
      d. Washer and Nut: Zinc-coated steel.
   5. Chemical Fasteners: Insert-type-stud, bonding-system anchor for use with hardened portland cement concrete, with tension and shear capacities appropriate for application.
      a. Bonding Material: ASTM C881/C881M, Type IV, Grade 3, two-component epoxy resin suitable for surface temperature of hardened concrete where fastener is to be installed.
      b. Stainless-steel studs are available. Use if in corrosive environment such as pools, water and wastewater treatment plans.
      c. Stud: ASTM A307, zinc-coated carbon steel with continuous thread on stud unless otherwise indicated.
      d. Washer and Nut: Zinc-coated steel.

PART 3 - EXECUTION

3.01 EXPANSION-JOINT INSTALLATION

A. Install expansion joints of sizes matching sizes of piping in which they are installed.

B. Install packed-type expansion joints with packing suitable for fluid service.

C. Install metal-bellows expansion joints according to EJMA's "Standards of the Expansion Joint Manufacturers Association, Inc."

D. Install rubber packless expansion joints according to FSA-NMEJ-702.

E. Install grooved-joint expansion joints to grooved-end steel piping

3.02 PIPE LOOP AND SWING CONNECTION INSTALLATION

A. See the 2024 ASHRAE HANDBOOK - "HVAC Systems and Equipment," Chapter 12 which states that cold springing is not recommended for most HVAC piping. If retaining first paragraph below, indicate dimensions of loops and swing connections and locations of guides and anchors on Drawings. Consult ASHRAE Handbook for sizing of pipe loops, L bends, Z bends, U bends, and anchor and guide location.

B. Install pipe loops as shown on drawings and diagrams.

C. Connect risers and branch connections to mains with at least five pipe fittings including tee in main.

D. Connect risers and branch connections to terminal units with at least four pipe fittings including tee in riser.

E. Connect mains and branch connections to terminal units with at least four pipe fittings including tee in main.

3.03 ALIGNMENT-GUIDE AND ANCHOR INSTALLATION

A. Install alignment guides to guide expansion and to avoid end-loading and torsional stress.

B. Indicate locations and number of guides on Drawings.

C. Install two guide(s) on each side of pipe expansion fittings and loops. Install guides nearest to expansion joint not more than four pipe diameters from expansion joint.

D. Coordinate first paragraph below with structural Drawings if welding is included in structural work.

E. Attach guides to pipe and secure guides to building structure.

F. Install anchors at locations to prevent stresses from exceeding those permitted by ASME B31.9 and to prevent transfer of loading and stresses to connected equipment.

G. Anchor Attachments:
   1. Coordinate first subparagraph below with structural Drawings if welding is included in structural work.
   2. Anchor Attachment to Steel Pipe: Attach by welding. Comply with ASME B31.9 and ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."
   3. Anchor Attachment to Copper Tubing: Attach with pipe hangers. Use MSS SP-69, Type 24, U-bolts bolted to anchor.

H. Coordinate first paragraph below with structural Drawings if welding is included in structural work.

I. Fabricate and install steel anchors by welding steel shapes, plates, and bars. Comply with ASME B31.9 and AWS D1.1/D1.1M 1/D1.1M.
   1. Anchor Attachment to Steel Structural Members: Attach by welding.
   2. Anchor Attachment to Concrete Structural Members: Attach by fasteners. Follow fastener manufacturer's written instructions.

J. Use grout to form flat bearing surfaces for guides and anchors attached to concrete.

END OF SECTION 23 0516

SECTION 23 0517
SLEEVES AND SLEEVE SEALS FOR HVAC PIPING

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. Sleeves.
   2. Stack-sleeve fittings.
   3. Grout.

1.03 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer's product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.

B. Equipment Categories: Equipment types in this section are considered equipment categories that be furnished by one manufacturer or available through one manufacturer to provide consistency within the overall project for equipment configurations, manufacturing techniques, and to facilitate ease of maintenance for the Owner.

2.02 SLEEVES

A. Galvanized-Steel Wall Pipes: ASTM A53/A53M, Schedule 40, with plain ends and welded steel collar; zinc coated.

B. Galvanized-Steel-Pipe Sleeves: ASTM A53/A53M, Type E, Grade B, Schedule 40, zinc coated, with plain ends.

C. Galvanized-Steel-Sheet Sleeves: 0.0239-inchminimum thickness; round tube closed with welded longitudinal joint.

2.03 STACK-SLEEVE FITTINGS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
   1. Smith, Jay R. Mfg. Co.
   2. Zurn Specification Drainage Operation; Zurn Plumbing Products Group.
   3. Wade
   4. Josam

B. Description: Manufactured, cast-iron sleeve with integral clamping flange. Include clamping ring, bolts, and nuts for membrane flashing. Provide underdeck clamp with clamping ring and set screws for securement of fitting to slab.

2.04 GROUT

A. Standard: ASTM C1107/C1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

B. Characteristics: Nonshrink; recommended for interior and exterior applications.

C. Design Mix: 5000-psi, 28-day compressive strength.

D. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.01 SLEEVE INSTALLATION

A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.

B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inchannular clear space between piping and concrete slabs and walls.
   1. Sleeves are not required for core-drilled holes.

C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.
   1. Cut sleeves to length for mounting flush with both surfaces.
      a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inchesabove finished floor level.
   2. Using grout, seal the space outside of sleeves in slabs and walls without sleeve-seal system.

D. Install sleeves for pipes passing through interior partitions.
   1. Cut sleeves to length for mounting flush with both surfaces.
   2. Install sleeves that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or outer dimension of pipe insulation for insulated pipe. Coordinate sleeve sizes with requirements of Section 23 0712 - HVAC Insulation.
   3. Increase sleeve size as required to provide additional annular space for project specific seismic conditions. Coordinate sleeve sizes with requirements of Section 23 0548 - Vibration and Seismic Controls for HVAC.
   4. Seal annular space between sleeve and piping or piping insulation; use joint sealants appropriate for size, depth, and location of joint.

E. Fire-Barrier Penetrations: Maintain indicated fire rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with firestop materials.

3.02 STACK-SLEEVE-FITTING INSTALLATION

A. Install stack-sleeve fittings in new slabs as slabs are constructed.
   1. Install fittings that are large enough to provide 1/4-inch annular clear space between sleeve and pipe or pipe insulation. Coordinate sleeve sizes with requirements of Section 23 0712 - HVAC Insulation.
   2. Increase sleeve size as required to provide additional annular space for project specific seismic conditions. Coordinate sleeve sizes with requirements of Section 23 0548 - Vibration and Seismic Controls for HVAC.
   3. Secure flashing between clamping flanges for pipes penetrating floors with membrane waterproofing.
   4. Install section of cast-iron soil pipe to extend sleeve to 2 inches above finished floor level.
   5. Extend cast-iron sleeve fittings below floor slab as required to secure clamping ring if ring is specified.
   6. Using grout, seal the space around outside of stack-sleeve fittings.

B. Fire-Barrier Penetrations: Maintain indicated fire rating of floors at pipe penetrations. Seal pipe penetrations with firestop materials.

3.03 SLEEVE AND SLEEVE-SEAL SCHEDULE

A. Use sleeves and sleeve seals for the following piping-penetration applications:
   1. Concrete Slabs above Grade:
      a. Piping, All sizes: Stack-sleeve fittings.
   2. Interior Partitions:
      a. Piping Smaller Than NPS 6: Galvanized-steel-pipe sleeves.
      b. Piping NPS 6and Larger: Galvanized-steel-sheet sleeves.

END OF SECTION 23 0517

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SECTION 23 0521
THERMOMETERS AND PRESSURE GAUGES

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. Liquid-in-glass thermometers.
   2. Light-activated thermometers.
   3. Thermowells.
   4. Dial-type pressure gauges.
   5. Gauge attachments.
   6. Test plugs.
   7. Test-plug kits.

1.03 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

1.04 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For meters and gauges to include in operation and maintenance manuals.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer's product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.

B. Equipment Categories: Equipment types in this section are considered equipment categories that be furnished by one manufacturer or available through one manufacturer to provide consistency within the overall project for equipment configurations, manufacturing techniques, and to facilitate ease of maintenance for the Owner.

2.02 METERS AND GAUGES MATERIALS

A. Comply with requirements in "Schedules" Article for applications of meters and gauges for specific services.

2.03 LIQUID-IN-GLASS THERMOMETERS

A. Metal-Case, Industrial-Style, Liquid-in-Glass Thermometers:
   1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
      a. Flo Fab Inc.
      b. Miljoco Corporation.
      c. Palmer Wahl Instrumentation Group.

d. Tel-Tru Manufacturing Company.
e. Trerice, H. O. Co.
f. Weiss Instruments, Inc.
g. Winters Instruments - U.S.

2. Standard: ASME B40.200.
3. Case: Precision die cast aluminum; 9-inch (229-mm) nominal size.
4. Case Form: Adjustable angle.
5. Tube: Glass with magnifying lens front, filled with non-toxic, blue organic liquid. Red color not permitted due to similarity with red reflecting mercury.
6. Tube Background: Nonreflective aluminum with white finish and permanently etched scale markings graduated in deg F (deg C), V-shaped backing for optimum readability.
7. Lens: Acrylic to 300 deg F, glass above 300 deg F.
8. Stem: Aluminum, tapered to fit standard industrial thermowells and of length to suit installation.
   a. Design for Air-Duct Installation: With ventilated shroud.
   b. Design for Thermowell Installation: Bare stem.
9. Connection: 1-1/4 inches (32 mm), with ASME B1.1 screw threads.
10. Accuracy: Plus or minus 1 percent of scale range or one scale division, to a maximum of 1.5 percent of scale range.

2.04 LIGHT-ACTIVATED THERMOMETERS

A. Direct-Mounted, Light-Activated Thermometers: Not permitted on steam systems.
   1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
      a. Flo Fab Inc.
      b. Miljoco Corporation.
      c. Trerice, H. O. Co.
      d. Weiss Instruments, Inc.
   2. Case: High impact ABS black plastic.
   3. Case Form: Adjustable angle.
   4. Sensor: Glass passivated thermistor.
   5. Recalibration: Internal potentiometer.
   6. Display: LCD, black character, 0.50” high with gray background.
   7. Illuminance: Bi-directional solar collectors, 35 lux.
   8. Refresh rate: 10 seconds.
   9. Scales: Deg F and deg C with user adjustable switch to select which scale range to display.
   10. Case Form: Adjustable angle.
   11. Connector: 1-1/4 inches, with ASME B1.1 screw threads.
   12. Stem: Aluminum, tapered to fit standard industrial thermowells and of length to suit installation.
      a. Design for Air-Duct Installation: With ventilated shroud.
      b. Design for Thermowell Installation: Bare stem.
   13. Accuracy: Plus or minus 1 deg or 1% of reading.

2.05 THERMOWELLS

A. Thermowells:
   1. Standard: ASME B40.200.
   2. Description: Pressure-tight, socket-type fitting made for insertion into piping tee fitting.
   3. Material for Use with Copper Tubing: CNR or CUNI.
   4. Material for Use with Steel Piping: CRES or CSA.
   5. Type: Stepped shank unless straight or tapered shank is indicated.
   6. External Threads: NPS 3/4, (DN 15, DN 20, or NPS 25,) ASME B1.20.1 pipe threads.

7. Internal Threads: ASME B1.1 screw threads, 1-1/4” thread to mate with the 1-1/4” male thread for liquid-in-glass and light-activated type thermometers.
8. Bore: Diameter required to match thermometer bulb or stem.
9. Insertion Length: Length required to match thermometer bulb or stem.
10. Lagging Extension: Include on thermowells for insulated piping and tubing.

B. Heat-Transfer Medium: Mixture of graphite and glycerin.

2.06 PRESSURE GAUGES

A. Direct-Mounted, Metal-Case, Dial-Type Pressure Gauges:
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
a. Ashcroft Inc.
b. Flo Fab Inc.
c. Miljoco Corporation.
d. Trerice, H. O. Co.
e. Watts Regulator Co.; a div. of Watts Water Technologies, Inc.
f. Weiss Instruments, Inc.
2. Standard: ASME B40.100.
3. Case: Stainless steel, sealed open-front; 4-1/2 inch nominal diameter.
4. Pressure-Element Assembly: Phosphor bronze bourdon tube, brass socket.
5. Pressure Connection: Brass, with NPS 1/4 (DN 8) (DN 15), ASME B1.20.1 pipe threads and bottom-outlet type unless back-outlet type is indicated.
6. Movement: Mechanical, with link to pressure element and connection to pointer.
7. Dial: Nonreflective aluminum white finished with permanently etched black scale markings graduated in psi (kPa).
8. Pointer: Aluminum, black, adjustable
9. Lens: Acrylic.
10. Ring: Stainless steel.
11. Accuracy: Grade A, plus or minus 1 percent of full scale.

2.07 GAUGE ATTACHMENTS

A. Pressure snubbers: For dampening of pressure pulsations to prolong gauge life. ASME B40.100, brass; with NPS 1/4(DN 8 or DN 15), ASME B1.20.1 pipe threads and porous-metal-type surge-dampening device consisting of sintered stainless steel with no moving parts. Include extension for use on insulated piping.

B. Valves: Required on all pressure gauge installations. Brass ball or stainless-steel needle, with NPS 1/4 or NPS 1/2, ASME B1.20.1 pipe threads.

2.08 TEST PLUGS

A. Manufacturers: Subject to compliance with requirements, provide products by one the following:
1. Flow Design, Inc.
2. Miljoco Corporation.
3. National Meter, Inc.
4. Peterson Equipment Co., Inc.
5. Sisco Manufacturing Company, Inc.
6. Trerice, H. O. Co.
7. Watts Regulator Co.; a div. of Watts Water Technologies, Inc.

B. Description: Test-station fitting made for insertion into piping tee fitting.

C. Body: Brass or stainless steel with core inserts and gasketed and threaded cap. Include extended stem on units to be installed in insulated piping.

D. Thread Size: NPS 1/4 (DN 8) or NPS 1/2 (DN 15), ASME B1.20.1 pipe thread.

E. Minimum Pressure and Temperature Rating: 500 psig at 200 deg F (3450 kPa at 93 deg C).

F. Core Inserts: Chlorosulfonated polyethylene synthetic and EPDM self-sealing rubber.

2.09 TEST-PLUG KITS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
   1. Flow Design, Inc.
   2. Miljoco Corporation.
   3. National Meter, Inc.
   4. Peterson Equipment Co., Inc.
   5. Sisco Manufacturing Company, Inc.
   6. Trerice, H. O. Co.
   7. Watts Regulator Co.; a div. of Watts Water Technologies, Inc.

B. Furnish one test-plug kit(s) containing two thermometer(s), one pressure gauge and adapter, and carrying case. Thermometer sensing elements, pressure gauge, and adapter probes shall be of diameter to fit test plugs and of length to project into piping.

C. Low-Range Thermometer: Small, bimetallic insertion type with diameter dial and tapered-end sensing element. Dial range shall be at least 25 to 125 deg F.

D. High-Range Thermometer: Small, bimetallic insertion type with diameter dial and tapered-end sensing element. Dial range shall be at least 0 to 220 deg F.

E. Pressure Gauge: Small, Bourdon-tube insertion type with 2- to 3-inch- diameter dial and probe. Dial range shall be at least 0 to 200 psig.

F. Carrying Case: Metal or plastic, with formed instrument padding.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Install thermowells with socket extending a minimum of 2 inches (51 mm) into fluid or to center of pipe, whichever is smaller, and in vertical position in piping tees. Account for thickness of insulation when determining thermowell and associated thermometer lengths.

B. Install thermowells of sizes required to match thermometer connectors. Include bushings if required to match sizes.

C. Install thermowells with extension on insulated piping.

D. Fill thermowells with heat-transfer medium.

E. Install direct-mounted thermometers in thermowells and adjust vertical and tilted positions.

F. Install duct-thermometer mounting brackets in walls of ducts. Attach to duct with screws.

G. Install direct-mounted pressure gauges in piping tees with pressure gauge located on pipe at the most readable position.

H. Install valve and snubber in piping for each pressure gauge for fluids (except steam).

I. Install valve and syphon fitting in piping for each pressure gauge for steam.

J. Install test plugs in piping tees.

K. Install permanent indicators on walls or brackets in accessible and readable positions.

L. Install connection fittings in accessible locations for attachment to portable indicators.

M. In addition to locations shown on drawings, install pressure gauges at the following locations:
   1. Inlet and outlet of each strainer (in hydronic systems only). Provide valving and piping to allow pressure sampling at each strainer with a single pressure gauge.

N. Install thermometers at locations indicated on drawings.

3.02 CONNECTIONS

A. Install meters and gauges adjacent to machines and equipment to allow service and maintenance of meters, gauges, machines, and equipment.

3.03 ADJUSTING

A. After installation, calibrate meters according to manufacturer's written instructions.

B. Adjust faces of meters and gauges to proper angle for best visibility.

3.04 THERMOMETER TYPE AND SCALE-RANGE SCHEDULE

A. Provide liquid-in-glass thermometer of scale range based on system type as follows. Contractor option to provide light-activated thermometer for all systems except steam and condensate:

1. Chilled-Water Piping: 0 to 120 deg F (Minus 20 to plus 50 deg C).

3.05 PRESSURE-GAUGE SCALE-RANGE SCHEDULE

A. Chilled-Water Piping: 0 to 100 psi.

END OF SECTION 23 0521

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SECTION 23 0529
HANGERS AND SUPPORTS FOR HVAC

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

B. Related Sections:
   1. Division 23 Section - Vibration and Seismic Requirements For HVAC
   2. Division 23 Section - Hydronic Piping

1.02 WORK INCLUDES

A. Base Bid: Hangers and supports of piping and equipment are considered delegated design by Contractor. Design, Furnish and Install hanger and support systems for Piping and Equipment as part of the Base Bid.

1.03 SUMMARY

A. Section Includes:
   1. Metal pipe hangers and supports.
   2. Trapeze pipe hangers.
   3. Metal framing systems.
   4. Thermal-hanger shield inserts.
   5. Fastener systems.
   6. Pipe positioning systems.
   7. Steel Equipment supports.

1.04 PERFORMANCE REQUIREMENTS

A. Structural support and anchor design detailing is the responsibility of the Contractor unless otherwise specified or indicated on drawings. Support and anchor design calculations shall be provided to the Engineer upon request.

B. Structural Performance: Hangers and supports for piping and equipment shall withstand the effects of gravity loads and stresses within limits and under conditions indicated according to ASCE/SEI 7.
   1. Design supports for multiple pipes, including pipe stands, capable of supporting combined weight of supported systems, system contents, and test water.
   2. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.
   3. Design seismic-restraint hangers and supports for piping and equipment and obtain approval from authorities having jurisdiction.

1.05 ACTION SUBMITTALS

A. Shop Drawings: Submit, for approval, complete calculations and details for field fabricated support installations (trapeze hangers, metal framing systems, and equipment supports); designed by a Registered Structural Engineer in the State in which the project is located. Submittal shall include Engineer’s Seal affixed thereto.

B. Show fabrication and installation details and include calculations for the following; include Product Data for components.

1.06 GENERAL REQUIREMENTS

A. Hangers, structural supports, and attachments shall include all labor, supplemental steel, or other smaterial necessary for support of pipe furnished by Contractor. Factory fabricated items shall be complete with nuts, bolts, washers, etc. Where type and size are not indicated, proper

selection shall be determined by Contractor. Refer to Noise & Vibration Isolation section of this specification for special support and hanger requirements.

B. Complete site visit for existing conditions for type of structure. Do NOT hang/support any single point loads greater than 500 lbs. from steel beams/purlins; and NOT greater than 250 lbs. from concrete slab or steel bar joists; unless specifically noted otherwise on the structural drawings. Weights shall not exceed framing member allowance. Where this Contractor violates allowances, any and all resultant damage and costs to correct, as determined by the Engineer, will be borne by this Contractor.

C. Supports include equipment anchoring to floor slab, brackets to wall, hangers to equipment supported from structure above, etc. Refer to drawings for equipment and locations.

D. Supplemental steel spanning structural framing elements may be steel angles or channels specifically sized for equipment/pipe weight. Supports shall be sized with a minimum safety factor of 5 to 1, or as required by code; most stringent requirement shall apply. Contractor may provide supplemental supports as manufactured by Unistrut, B-Line systems, or approved equal.

E. Hangers and supports shall conform to recommendations of Standard Practice No. 58 and 69 of Manufacturer's Standardization Society of the Valve and Fitting Industry (MSS).

F. All exterior supports, hangers, attachments, etc. shall be galvanized or stainless steel; including all accessories, fasteners, etc.

1.07 REFERENCES

A. American Concrete Institute, ACI:
   1. ACI 301: Specifications for Structural Concrete for Buildings.
   2. ACI 304: Recommended Practice for Measuring, Mixing, Transporting and Placing Concrete.
   3. ACI 347: Recommended Practice for Concrete Formwork.

B. American Society of Testing and Materials, ASTM:
   1. ASTM A82: Cold Drawn Steel Wire for Concrete Reinforcement.
   2. ASTM A185: Welded Steel Wire Fabric For Concrete Reinforcement.
   3. ASTM C33: Concrete Aggregates.
   4. ASTM C150: Portland Cement.
   5. ASTM C171: Sheet Materials for Curing Concrete.
   6. ASTM C260: Air-Entraining Admixtures for Concrete.
   7. ASTM C309: Liquid Membrane Forming Compounds for Curing Concrete.
   8. ASTM C404: Aggregates for Masonry Grout.

C. American National Standards Institute, ANSI:
   1. ANSI/ASME B31.9: Building Services Piping.

D. Concrete Reinforcing Steel Institute, CRSI:
   1. CRSI Manual of Standard Practice.

E. Manufacturers Standardization Society of the Valve and Fittings Industry, MSS:
   1. MSS SP-58: Pipe Hangers and Supports - Materials, Design and Manufacture.
   2. MSS SP-69: Pipe Hangers and Supports - Selection and Application.

F. National Electrical Manufacturers Association, NEMA:
   1. NEMA ML 1: Metal Framing (Continuous Slot Metal Channel Systems).

G. Sheet Metal & Air Conditioning Contractor's National Association, Inc., SMACNA:
   1. Duct Hangers: SMACNA Duct Manuals.

1.08 REGULATORY REQUIREMENTS

A. International Building, Mechanical Codes, 2024 version.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer's product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.

B. Equipment Categories: Equipment types in this section are considered equipment categories that be furnished by one manufacturer or available through one manufacturer to provide consistency within the overall project for equipment configurations, manufacturing techniques, and to facilitate ease of maintenance for the Owner.

2.02 ACCEPTABLE MANUFACTURERS

A. Hangers and Supports:
   1. Mason Industries: www.mason-industries.com.
   2. Nibco: www.nibco.com.
   3. PHD Mfg, Inc: www.phd-mfg.com.
   4. Bang-It
   5. Wood-Knocker
   6. Drop-In Anchors

B. Saddles and Shields:
   1. Mason Industries: www.mason-industries.com
   2. Pipe Shields, Inc: www.pipeshieldsinc.com.
   3. PHD, Mfg., Inc.: www.phd-mfg.com.

2.03 HORIZONTAL PIPING HANGERS AND SUPPORTS

A. Single Pipe Support: Provide factory fabricated horizontal piping hangers, in compliance with ANSI/MSS SP-58 and ANSI/MSS SP-69.
   1. Type: Adjustable Steel Clevis; Type 1
   2. Use only one type by one manufacturer for each type of piping service.
   3. Select hangers and supports sized to exactly fit pipe size for bare piping; and to exactly fit around pipe insulation with saddle and shield for insulated piping.
   4. Provide rubber or neoprene lined pipe ring isolators and copper plated hangers and supports for copper piping systems.
   5. Size in accordance with MSS and manufacturer's recommendations, most stringent shall apply; minimum safety factor of 5.
   6. Based on Anvil Fig. 260 for insulated pipe of any type, non-insulated ferrous alloy pipe and non-insulated plastic pipe; Anvil Fig CT-65 copper plated carbon steel light duty adjustable clevis hanger for non-insulated copper piping NPS 1/2 thru 4.

B. Trapeze Pipe Supports: Allowed for parallel runs of piping.
   1. Shop- or Field-fabricated: Provide MSS SP-69, Type 59 pipe support assembly made from structural carbon-steel shapes with MSS SP-58 carbon-steel hanger rods, nuts, saddles, and U-bolts.
   2. Factory-fabricated: Provide channel strut system, B-line B22 or heavier as required for load and/or span lengths. Provide steel blocking of varying thickness welded to channel

under each sloped pipe as required to obtain proper pitch. Strut clamps, straps, and rollers shall be used to maintain proper alignment:
a. Clamps and straps: B-Line series B2000 or B2400.
b. Rollers: B-Line figures B218, B219, B379, B479 or B3126

C. Pipe hangers and supports on insulated piping shall be installed outside the insulation and shall include insulation supports/inserts between the pipe and hanger/shield.
1. Exception: Seismic restraint hangers and supports shall be installed in contact with the pipe. Insulation shall be carried over the hanger or support and sealed.

2.04 VERTICAL PIPING CLAMPS

A. Provide factory fabricated vertical piping clamps, in compliance with ANSI/MSS SP-58 and ANSI/MSS SP-69.
1. Use only one type by one manufacturer for each type of piping service.
2. Select vertical piping clamps to be sized to exactly fit pipe type and size of bare pipe.
3. Provide rubber or neoprene line pipe ring and copper plated clamps for copper piping systems.
4. Vertical risers shall be supported independently of horizontal branch hangers.
5. Type:
a. At floor levels: MSS Type 8; two bolt riser clamp.
1) Ferrous alloy and plastic pipe: Anvil Fig. 261 or Fig. 600/599; steel pipe/socket clamp.
2) Copper pipe: Anvil Fig. CT-121; copper pipe riser clamp; carbon steel with copper finish.
b. Intermediate support between floor levels when floor to floor level exceeds maximum vertical distance allowable between supports: MSS Type 42; four bolt riser clamp; required in addition to riser clamps at floor levels. Provide with shear lugs welded to pipe, weldless eye nuts and hanger rods.

2.05 ATTACHMENTS

A. Hanger Rod Attachments: Provide factory fabricated hanger rod attachments, in compliance with ANSI/MSS SP-58 and ANSI/MSS SP-69
1. Use only one type by one manufacturer for each type of piping service.
2. Select size of hanger rod attachments to suit hanger rods; size attachments to suit equipment/pipe hanger/rod and applied load.
3. Proved copper plated hanger rod attachments for copper piping systems.
4. Type: MSS Type 13 Steel Turnbuckles

B. Building Attachments: Provide factory fabricated hanger rod and brace angle attachments, in compliance with ANSI/MSS SP-58 and ANSI/MSS SP-69
1. Use only one type by one manufacturer for each type of piping service.
2. Size attachments to suit equipment/pipe hanger/rod size and applied load.
3. C-Clamps: MSS Type 23.
4. Steel Brackets:
a. MSS Type 31 for light duty.
b. MSS Type 32 for medium duty.
c. MSS Type 33 for heavy duty.

2.06 SADDLES AND SHIELDS

A. Provide saddles or shields under piping hangers and supports, factory fabricated, for all insulated piping.

B. Acceptable Manufacturers:
1. Anvil
2. Cooper/B-Line
3. TOLCO Pipe Shields

4. Pipe Shields, Inc.; a subsidiary of Piping Technology & Products, Inc.
5. Value Engineered Products, Inc

C. Size saddles and shields for exact fit to mate with pipe insulation

D. Types:
1. Protection Saddles: MSS Type 39, Anvil Fig 160-160A; fill interior voids with segments of insulation matching adjoining insulation.
2. Protection Shields. MSS Type 40, Anvil Fig 167; length recommended by manufacturer to prevent crushing insulation, minimum length 12”.

E. Insulation-Insert Material for Cold Piping: Utilize one of the following with vapor barrier:
1. ASTM C 552, Type II cellular glass with 100-psig (688-kPa) minimum compressive strength, density ranging from 6.2 to 9.7 lb/cu. ft, and suitable for minus 450 to plus 800 deg F application.
2. ASTM C 591, Type VI, Grade 1 polyisocyanurate with 125-psig (862-kPa) minimum compressive strength, 6 lb/cu. ft. density, and suitable for minus 70 to plus 300 deg F application.

F. Insulation-Insert Material for Hot Piping: Utilize one of the following, vapor barrier not required:
1. ASTM C 533, Type I water-repellent treated calcium silicate with 100-psig minimum compressive strength, 15 lb/cu. ft. density, and suitable for 80 to 1200 deg F application.
2. ASTM C 552, Type II cellular glass with 100-psig minimum compressive strength, density ranging from 6.2 to 9.7 lb/cu. ft, and suitable for minus 450 to plus 800 deg F application.
3. Permitted on low temperature heating water systems (200 deg F and lower) only, not permitted on steam and condensate: ASTM C 591, Type VI, Grade 1 polyisocyanurate with 125-psig (862-kPa) minimum compressive strength, 6 lb/cu. ft. density, and suitable for minus 70 to plus 300 deg F application.

G. For Trapeze or Clamped Systems: Insert and shield shall cover entire circumference of pipe.

H. For Clevis or Band Hangers: Insert and shield shall cover lower 180 degrees of pipe.

I. Insert Length: Extend 2 inches beyond sheet metal shield for piping operating below ambient air temperature.

2.07 SPRING HANGERS AND SUPPORTS

A. Provide factory fabricated spring hangers and supports, in compliance with ANSI/MSS SP-58 and ANSI/MSS SP-69
1. Use only one type by one manufacturer for each piping service.
2. Select spring hangers and supports to suit pipe size and loading.
3. Types:
a. Spring Cushion Hangers: MSS Type 48
b. Constant Supports: Selected to suit piping system; including auxiliary stops for erection and hydrostatic test, and field load adjustment capability.
1) Horizontal Type: MSS Type 54
2) Vertical Type: MSS Type 55
3) Trapeze Type: MSS Type 56

2.08 FLASHING MATERIALS

A. Provide locations of roof penetrations to Roofing Contractor for Coordination.
1. Roofing Contractor provides flashing material specified in Division 07.
2. Provide flashings for each penetration of HVAC systems through floors, roofs or waterproofing membranes.

2.09 SLEEVES

A. Pipes through floors where no plumbing fixtures are installed: 18 gauge galvanized steel.

B. Pipes through Beams, Walls, Fireproofing, Footings, Potentially Wet floorfs: Schedule 40 steel pipe or 18 gauge galvanized steel.

C. UL Labeled Sleeves: Prefabricated with insulation and fireproofing.

D. Sleeves through Fire and/or Smoke Walls: Comply with NFPA 90A.

E. Size sleeves to allow for expansion movement and to provide for continuous insulation.

F. Prefabricated Wall Penetration Seals: Modular mechanical type; interlocking synthetic rubber links filling annular space between pipe and wall opening.
   1. Description: Bolt and pressure plate fasteners and rubber seal provide water tight seal between pipe and wall.
   2. Seal provides electrical insulation between pipe and wall.
   3. Based on: Thunderline Corporation Link Seal.

2.10 FIRE AND SMOKE PENETRATION SEALANT

A. UL listed; flexible sealant or pre-fabricated metal device.

2.11 MISCELLANEOUS SUPPORT MATERIALS

A. Metal Framing: NEMA Standard ML 1

B. Structural Steel: ANSI/ASTM A 36, carbon-steel plates, shapes and bars; black and galvanized.

C. Hanger Rods: Steel, threaded both ends, threaded one end, or continuously threaded.

D. Cement Grout: Portland cement, ANSI/ASME C150, Type I or Type III, clean uniformly grades, natural sand, ANSI/ASTM C404, Size No. 2; mix ratio of 1.0 part cement to 3.0 parts sand, by volume, with minimum amount of water for placement and hydration, non-shrink, nonmetallic, suitable for interior and exterior applications.
   1. Properties: Non-staining, noncorrosive, and nongaseous.
   2. Design Mix: 5000-psi, 28-day compressive strength.

E. Heavy Duty Steel Trapezes:
   1. Fabricate from steel shapes selected for applied loads.
   2. Weld steel in accordance with AWS standards.
   3. Pipe Guides: Factory fabricated cast semi-steel or heavy fabricated steel, including bolted 2-section outer cylinder and base with 2-section guiding spider bolted tight to pipe.
      a. Size guide and spiders to clear pipe, insulation and cylinder.
      b. Guides Length: Recommended by manufacturer to allow required travel.

2.12 CONCRETE SUPPORTS

A. Formwork: ACI 347.

B. Reinforcement:
   1. Bars: ASTM A615, deformed, Grade 40, sizes 3 - 18.
   2. Steel Wire: ASTM A82, plain cold drawn.
   3. Welded Wire Fabric: ASTM A185; 6 inch x 6 inch, #10 x #10.
   4. Supports: Concrete Reinforcing Steel Institute.
   5. Concrete Materials:
      a. Portland Cement: ASTM C150, Type 1, above or Type 2, below grade.
      b. Aggregate: ASTM C33, meeting concrete strength requirements.
      c. Water: Clean, free of substances harmful to concrete.
      d. Air-Entraining Admixture: ASTM C260.
      e. Calcium chloride not permitted.
      f. Curing: Burlap cloth cover, 9 oz./sq.yd., AASHTO M182 Class 3 ASTM C171 moisture retaining cover; or ASTM C309, Type 1 membrane forming curing compound.
   6. Mixes:

a. Purpose design mix for approval meeting indicated strength requirements, and submit test data on mix indicating compliance with strength requirements.
b. Use air-entrained concrete at manufacturer's prescribed rate in all exposed concrete including below grade structures such as manholes and vaults.
c. Mix Strength at 28 Days
1) 3000 psi: Minimum 500 lbs, cement/cy; maximum 0.68 w/c/ratio.

2.13 METAL FRAMING SYSTEMS

A. MFMA Manufacturer Metal Framing Systems:
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
a. Cooper B-Line, Inc.
b. Flex-Strut Inc.
c. Thomas & Betts Corporation.
d. Unistrut Corporation; Tyco International, Ltd.
2. Description: Shop- or field-fabricated pipe-support assembly for supporting multiple parallel pipes.
3. Standard: MFMA-4.
4. Channels: Continuous slotted steel channel with inturned lips.
5. Channel Nuts: Formed or stamped steel nuts or other devices designed to fit into channel slot and, when tightened, prevent slipping along channel.
6. Hanger Rods: Continuous-thread rod, nuts, and washer.
7. Coating: Galvanized if required elsewhere in this specification section; otherwise, manufacturer’s standard factory paint coating.

2.14 FASTENER SYSTEMS

A. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

B. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated steel anchors, for use in hardened portland cement concrete; with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

2.15 PIPE POSITIONING SYSTEMS

A. Description: IAPMO PS 42, positioning system of metal brackets, clips, and straps for positioning piping in pipe spaces; for plumbing fixtures in commercial applications.

2.16 EQUIPMENT SUPPORTS

A. Description: Welded, shop- or field-fabricated equipment support made from structural carbon-steel shapes.

PART 3 - EXECUTION

3.01 HANGER AND SUPPORT INSTALLATION

A. General: Comply with MSS SP-69 and MSS SP-89. Install hangers, supports, clamps, and attachments complete with inserts, bolts, rods, nuts, and washers in accordance with manufacturer’s recommendations and as required to properly support piping from the building structure.
1. Install hangers and supports to prevent sway, vibration, and sag.
2. Support piping at drops to equipment to prevent stress at the equipment connections.
3. Locate hanger within three feet of a direction change. Locate hangers within one foot of each horizontal elbow. Provide additional supports at each side of heavy valves and equipment. Valves NPS 3 and larger in horizontal piping shall be supported independently of the adjacent piping.

4. Install hangers with minimum 1/2 inch clear space between finished covering and adjacent work.
5. Use hangers with vertical adjustment range of 1-1/2 inch minimum suited for adjustment after piping is erected.
6. Hanging of pipes from other trades (ie. Ductwork, conduits, ceiling suspension systems, medical equipment structural support systems, etc.) is not acceptable. Hanging and support of piping from other piping is not permitted, even of same trade.
7. Adjust hangers and supports after insulation installation to provide uniform loading and to bring piping to proper levels and elevations.
8. Use bracket type hangers fastened to walls to support piping running adjacent to walls and not supported from structure overhead.
9. Perforated strap iron hangers and wire are not permitted for either temporary or permanent installation.
10. Hangers and supports shall be securely fastened to structural members by beam clamps and clips, concrete inserts, anchors and all required accessories.

B. Non-ferrous Piping: Copper plated and Plastic coated hangers and trapeze may contact non-ferrous piping. Ferrous hangers and supports, including trapeze members, must be isolated from non-ferrous pipe to prevent galvanic corrosion. Provide one of the following for installations mating ferrous hangers and supports with non-ferrous piping:
1. Install split plastic shield, 6 inches long x 1/16 inch thick, 360 degree full circumference coverage with 225 deg F minimum rating between piping and hangers/supports.
2. B-Line Vibra cushion clamps
3. B-Line Iso Pipe rubber tape: Two (2) wraps extended 2 inches each side of hanger or clamp.
4. Black electrical tape or PVC tape: Three (3) wraps extended 2 inches each side of hanger or clamp.
5. Painting of hangers, clamps and trapeze is not considered acceptable substitute to the above.

C. Piping installed below slab on grade: Secure to avoid floating of piping during concrete pour; submit means and methods of securement for Engineer approval. For insulated piping, securements shall not be allowed to pierce the insulation or jacketing under any circumstances.

D. Metal Trapeze Pipe-Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Arrange for grouping of parallel runs of horizontal piping, and support together on field-fabricated trapeze pipe hangers.
1. Pipes of Various Sizes: Support together and space trapezes for smallest pipe size or install intermediate supports for smaller diameter pipes as specified for individual pipe hangers.
2. Field fabricate from ASTM A36/A36M, carbon-steel shapes selected for loads being supported. Weld steel according to AWS D1.1/D1.1M.

E. Metal Framing System Installation: Arrange for grouping of parallel runs of piping, and support together on field-assembled metal framing systems.

F. Thermal-Hanger Shield Installation: Install in pipe hanger or shield for insulated piping.

G. Pipe Positioning-System Installation: Install support devices to make rigid supply and waste piping connections to each plumbing fixture.

H. Install hangers and supports complete with necessary attachments, inserts, bolts, rods, nuts, washers, and other accessories.

I. Equipment Support Installation: Fabricate from welded-structural-steel shapes.

3.02 EQUIPMENT BASES AND SUPPORTS

A. Concrete:

1. Provide for major equipment reinforced concrete bases poured directly on structural floor slab 4 inches thick minimum, extended 4 inches minimum beyond machinery bedplates.
2. Formwork: ACI 347.
   a. Chamfer external corners of exposed concrete work.
   b. Provide openings required by others as required.
   c. Use manufacturer's templates or certified setting diagrams for placement of equipment anchor bolts and other mounting accessories.
   d. Install accessories: Waterstops, expansion joints, and moisture barrier.
   e. Reinforcement: CRSI, Manual of Standard Practice.
   f. Concrete Placement: ACI 301, 304.
   g. Concrete Curing : ACI 301.
   h. Concrete Finishing:
      1) Horizontal Surfaces: Float and trowel to level, smooth, uniform, dense finish; leave struck off surface untroweled; or scratch struck off surfaces.
      2) Curbs: Strip form while concrete is green; steel trowel to hard dense finish with corners, intersections and terminations slightly rounded and coved.
   i. Concrete Grouting: Grout openings and recesses as indicated, around mechanical work using grouting mix, tamped, screened, troweled and cured.
      1) See individual equipment Sections for specification for fine grouting (nonshrink grout) of equipment base plates on foundation bases.
   j. Field Quality Control: Contractor engages testing laboratory to take on site slump or compression tests to ensure compliance with specifications.

B. Install lateral bracing with pipe hangers and supports to prevent swaying.

C. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and to not exceed maximum pipe deflections allowed by ASME B31.9 for building services piping.

D. Trapeze Hangers:
   1. Where piping of various sizes are supported together by trapeze hangers, space hangers to accommodate maximum span permitted for smallest pipe; or install intermediate supports for smaller diameter pipe(s).
   2. Strut clamps, straps, and rollers shall be used to maintain proper alignment. Provide a clamp/strut/roller in each straight run of pipe every 30 feet on center.

E. Vertical Piping Risers: Support vertical piping risers independent of adjacent hangers on horizontal branches.
   1. Copper pipe: Support at the base of the riser and at each floor level; clamps not to exceed ten (10) feet on center. Provide additional supports for floor to floor heights exceeding 10 feet.
   2. Cast iron pipe: Support at the base of the riser and at each floor level; not to exceed fifteen (15) feet on center. Provide additional supports for floor to floor heights exceeding 15 feet.

F. Pipe Movement: Install hangers and supports to allow controlled movement of piping systems
   1. Install hangers and supports to allow controlled movement of piping systems; to permit freedom of movement between piping anchors; and to facilitate action of expansion joints, expansion loops, expansion bends, and similar entities.
      a. Load Distribution: Install hangers and supports so piping live and dead loading and stresses from movement is not transmitted to connected equipment.

G. Polypropylene pipe: Support at maximum 32 inch intervals horizontally and five (5) feet on center vertically with additional support at base of riser and at each floor level.

H. Insulated Piping: Provide thermal-inserts and shields.
   1. Piping Operating above Ambient Air Temperature: Clamps supporting vertical piping may project through insulation. Hangers and supports for horizontal piping may not pass

through insulation with the exception of saddles for roller support, seismic anchors, and seismic guides. Use thermal-hanger shield insert with clamp sized to match OD of insert.

2. Piping Operating below Ambient Air Temperature: Use thermal-hanger shield insert with clamp sized to match OD of insert.
3. Do not exceed pipe stress limits allowed by ASME B31.9 for building services piping.
4. Install MSS SP-58, Type 39, protection saddles for above ambient services only at roller locations. Fill interior voids with insulation that matches adjoining insulation.
5. Install MSS SP-58, Type 40, protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.
6. Shield Dimensions for Pipe: Not less than the following:
   a. NPS 1/4 to NPS 2 (DN 8 to DN 90): 12 inches (305 mm) long and 18 gauge (0.048 inch (1.22 mm) thick).
   b. NPS 2-1/2 to NPS 6: 18 inches long and 16 gauge (0.06 inch thick).
   c. NPS 8 to NPS 14: 24 inches long and 14 gauge (0.075 inch thick).
   d. NPS 16 to NPS 24: 24 inches long and 12 gauge (0.105 inch thick).
7. Pipes NPS 8 (DN 200) and Larger:
   a. Above ambient services: Include wood or reinforced calcium-silicate-insulation inserts of length at least as long as protective shield.
   b. Below ambient services: Include reinforced cellular glass or polyisocyanurate insulation inserts of length at least as long as protective shield.
8. Thermal-Hanger Shields: Install with insulation same thickness as piping insulation.

3.03 PIPE HANGER AND SUPPORT SPACING SCHEDULE

A. Based on piping sizes and material types, install at maximum span distances, with minimum hanger rod sizes as follows. Specification based on individual piping supports and does not apply to trapeze systems:
   1. Horizontal Copper and Steel Piping:
      a. NPS 1/2 to NPS 1-1/4: 6 feet maximum span, 3/8 inch diameter hanger rod.
      b. NPS 1-1/2 to NPS 2: 10 feet maximum span, 3/8 inch diameter hanger rod.
      c. NPS 2-1/2 to NPS 3: 10 feet maximum span, 1/2 inch diameter hanger.
      d. NPS 4 to NPS 6: 10 feet maximum span, 5/8 inch diameter hanger rod.
      e. NPS 8 to NPS 12: 10 feet maximum span, 7/8 inch diameter hanger.
      f. NPS 14 and larger: 20 feet maximum span, 1 inch diameter hanger
   2. Horizontal Cast Iron Piping:
      a. NPS 1-1/2 to NPS 2: 5 feet maximum span, 3/8 inch diameter hanger rod.
      b. NPS 2-1/2 to NPS 3: 5 feet maximum span, 1/2 inch diameter hanger rod.
      c. NPS 4: 5 feet maximum span, 5/8 inch diameter hanger rod.
      d. NPS 6 to NPS 8: 5 feet maximum span, 3/4 inch diameter hanger rod.
      e. NPS 10 to NPS 12: 5 feet maximum span, 7/8 inch diameter hanger rod.
   3. Hangers shall be placed adjacent either side of each piping joint and change in direction in addition to the above spacing criteria.

3.04 BUILDING ATTACHMENTS

A. Space attachments within maximum piping spans indicated in MSS SP-69 and in spacing schedules this section. The most stringent, shortest distances shall apply.

B. Install building attachments within concrete slabs or attach to structural steel.

C. Concrete Structure Attachments:
   1. Install concrete inserts before concrete is placed; fasten inserts to forms and install reinforcing bars through openings at top of inserts.
   2. Install mechanical-expansion anchors in concrete after concrete is placed and completely cured. Install fasteners according to manufacturer's written instructions.

3. Structural attachments shall be approved box type inserts for concrete floors and roof decks.
    a. Continuous inserts may be used for parallel lines of piping and equipment. Furnish spot or strip inserts with Styrofoam or cardboard filler strip to protect the cavity during pour and subsequent construction activities.
    b. Anchor insert attachment to structure shall be installed in accordance with the manufacturer's recommendation for the application and approved by the Structural Engineer.
4. Self-drilling concrete anchors, Phillips Red Head Series, and powder-actuated fasteners, Hilti Fastening Systems or approved equal may be used only with prior approval by the Engineer. If approved:
    a. Install powder-actuated fasteners for use in lightweight concrete or concrete slabs less than 4 inches (100 mm) thick in concrete after concrete is placed and completely cured. Use operators that are licensed by powder-actuated tool manufacturer.
    b. Install fasteners according to powder-actuated tool manufacturer's operating manual.
5. Inserts and anchors shall not be installed in concrete pan construction between joists.

D. Steel Joist Attachments: Connect to top chord of all steel joists for all hangers from steel attached to joists.

E. Install additional building attachments when supporting additional concentrated loads, including valves, flanges, and strainers NPS 2-1/2 and larger and at changes in piping direction, guides, and expansion joints.

F. Anchors:
1. Install anchors at locations preventing stresses from exceeding ANSI/ASME B31; and preventing transfer of loading and stresses to connected equipment.
    a. Install anchors at ends of principal piping runs and at intermediate points in pipe runs between expansion loops and bends.
    b. Preset anchors to accommodate both expansion and contraction of piping.
2. Fabricate and install anchors by welding steel shapes, plates and bars to piping and to structure; in compliance with ANSI/ASME B31 and the American Welding Society (AWS).
3. Anchors for Expansion Compensators: Install anchors in accordance with expansion unit manufacturer's recommendations.
    a. Limit piping movement and forces to maximums recommended by manufacturer for each unit.

3.05 EQUIPMENT SUPPORTS

A. Fabricate structural-steel stands to suspend equipment from structure overhead or to support equipment above floor. Brace and fasten with flanges bolted to structure.
1. Supports: Configured of structural steel members or steel piping and fittings.
2. Pipe Stands: Welded fabrication.
    a. Provide flat plate supports for each leg, bolted to 6 inch high x 12 inch x 12 inch concrete base poured on floor. Alternatively, bolt support to floor and encase flange and anchor bolts in 6 inch high x 12 inch x 12 inch concrete base to be water proofed and sealed after curing.
    b. Attach top of each leg to ceiling or roof structure as indicated or approved.

B. Grouting: Place grout under supports for equipment and make bearing surface smooth.

C. Provide lateral bracing, to prevent swaying, for equipment supports.

D. Provide rigid anchoring for piping immediately after vibration connections to equipment and immediately after vibration isolator at transition from spring isolated piping rack to rigidly hung piping rack.

3.06 METAL FABRICATIONS

A. Cut, drill, and fit miscellaneous metal fabrications for trapeze pipe hangers equipment supports.

B. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.

C. Field Welding: Comply with AWS D1.1/D1.1M procedures for shielded, metal arc welding; appearance and quality of welds; and methods used in correcting welding work; and with the following:
   1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.
   2. Obtain fusion without undercut or overlap.
   3. Remove welding flux immediately.
   4. Finish welds at exposed connections so no roughness shows after finishing and so contours of welded surfaces match adjacent contours.

3.07 FLASHING

A. Flash and counterflash where mechanical equipment passes through the roof, weather or waterproofed walls and floors.
   1. Coordinate with and accommodate specified roofing systems.
   2. Ensure compatibility with number and type of roofing plies.
   3. Vertical Surfaces: Lap flashings minimum 3 inches.
   4. Embed metal flashing flanges in roofing or composition flashing or stripping, by extending flanges minimum 6 inches for embedment.
   5. Pipes Through Outside Walls: Turn flange back into wall and seal.
   6. Curbs:
      a. Provide curbs for mechanical roof installations 12 inches minimum high.
      b. Flash and counterflash with steel, soldered and waterproofed.

3.08 SLEEVES

A. Setting: Set sleeves in advance of concrete work; provide reinforcing around sleeves.

B. Wet Areas: Extend sleeves through potentially wet floors 2 inch above finished floor level.
   1. Caulk sleeves full depth and provide floor plate.
   2. Insulation: Where piping or ductwork passes through floor, ceiling or wall, close off space between pipe or duct and construction with noncombustible mineral wool insulation.
      a. Provide tight fitting metal caps on both sides and caulk.
   3. Install chrome plated escutcheons where piping passes through finished surfaces.
   4. When duct sleeves pass through fire and smoke barriers, pack void space between duct and sleeve with fireproof mineral wool.
      a. Provide sheet metal duct flanges at wall to cover openings.
   5. Prefabricated Wall Penetration Seals: Install and tighten seals in accordance with manufacturer's recommendations.
      a. Ensure watertight pipe penetration.

3.09 FIRE AND SMOKE PENETRATION SEALANT

A. Where supports pass through fire or smoke rated walls, partitions, floors, and ceilings, seal openings with UL rated materials.

3.10 ADJUSTING

A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.

B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches (40 mm).

3.11 PAINTING

A. Touchup: Provide Cleanup and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal.

B. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

C. Exposed Steel Surfaces: Prime coat all exposed steel hangers and supports. Hangers and supports in crawl spaces, pipe shafts, and suspended ceiling spaces are not considered exposed.

3.12 HANGER AND SUPPORT SCHEDULE

A. Specific hanger and support requirements are in Sections specifying piping systems and equipment.

B. Comply with MSS SP-69 for pipe-hanger selections and applications that are not specified in piping system Sections.

C. Use hangers and supports with galvanized metallic coatings for piping and equipment that will not have field-applied finish.

D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.

E. Use carbon-steel metal trapeze pipe hangers and metal framing systems and attachments for general service applications.

F. Use stainless-steel pipe hangers and stainless-steel attachments for hostile environment applications.

G. Use copper-plated pipe hangers and copper or stainless-steel attachments for copper piping and tubing.

H. Use padded hangers for piping that is subject to scratching.

I. Use thermal-hanger shield inserts for insulated piping and tubing.

J. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
   1. Individual Hangers and Supports: Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of noninsulated or insulated, stationary pipes NPS 1/2 to NPS 30
   2. Trapeze support system (MSS Type 59) with piping clamps or U-bolts (MSS Type 24) for support of pipes NPS 1/2 to NPS 30.
   3. Adjustable Pipe Saddle Supports (MSS Type 38): For stanchion-type support for pipes NPS 2-1/2 to NPS 36 with vertical adjustment capability, steel-pipe base stanchion support and cast-iron floor flange.
   4. Single-Pipe Rolls (MSS Type 41): For suspension of pipes NPS 1 to NPS 30 from two rods where longitudinal movement caused by expansion or contraction occurs.
   5. Adjustable Roller Hangers (MSS Type 43): For suspension of pipes NPS 2-1/2 to NPS 24, from single rod where horizontal movement caused by piping expansion or contraction occurs.
   6. Adjustable Pipe Roll and Base Units (MSS Type 46): For support of pipes NPS 2 to NPS 30 where horizontal movement caused by piping expansion and contraction occurs and with vertical and lateral adjustment capability during installation.

K. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
   1. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers NPS 3/4 to NPS 24 (DN 24 to DN 600).
   2. Carbon- or Alloy-Steel Riser Clamps (MSS Type 42): For support of pipe risers NPS 3/4 to NPS 24 if longer ends are required for riser clamps or if additional intermediate supports between floors utilizing eyelets and additional hanger rod are required.

L. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel Turnbuckles (MSS Type 13): For adjustment up to 6 inches (150 mm) for heavy loads.
2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.
3. Malleable-Iron Sockets (MSS Type 16): For attaching hanger rods to various types of building attachments.
4. Steel Weldless Eye Nuts (MSS Type 17): For 120 to 450 deg F piping installations.

M. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel or Malleable Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.
2. Top-Beam C-Clamps (MSS Type 19): For use under roof installations with bar-joist construction, to attach to top flange of structural shape.
3. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.
4. Top-Beam Clamps (MSS Type 25): For top of beams if hanger rod is required tangent to flange edge.
5. Welded-Steel Brackets: For support of pipes from below or for suspending from above by using clip and rod. Use one of the following for indicated loads:
   a. Light (MSS Type 31): 750 lb (340 kg).
   b. Medium (MSS Type 32): 1500 lb.
   c. Heavy (MSS Type 33): 3000 lb.

N. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Steel-Pipe-Covering Protection Saddles (MSS Type 39): To fill interior voids with insulation that matches adjoining insulation.
2. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.
3. Thermal-Hanger Shield Inserts: For supporting insulated pipe.

O. Spring Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:
1. Restraint-Control Devices (MSS Type 47): Where indicated to control piping movement.
2. Spring Cushions (MSS Type 48): For light loads if vertical movement does not exceed 1-1/4 inches (32 mm).
3. Spring-Cushion Roll Hangers (MSS Type 49): For equipping Type 41, roll hanger with springs.
4. Spring Sway Braces (MSS Type 50): To retard sway, shock, vibration, or thermal expansion in piping systems.
5. Variable-Spring Hangers (MSS Type 51): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from hanger.
6. Variable-Spring Base Supports (MSS Type 52): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from base support.
7. Variable-Spring Trapeze Hangers (MSS Type 53): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from trapeze support.
8. Constant Supports: For critical piping stress and if necessary to avoid transfer of stress from one support to another support, critical terminal, or connected equipment. Include auxiliary stops for erection, hydrostatic test, and load-adjustment capability. These supports include the following types:
   a. Horizontal (MSS Type 54): Mounted horizontally.
   b. Vertical (MSS Type 55): Mounted vertically.
   c. Trapeze (MSS Type 56): Two vertical-type supports and one trapeze member.

P. Comply with MSS SP-69 for trapeze pipe-hanger selections and applications that are not specified in piping system Sections.

Q. Comply with MFMA-103 for metal framing system selections and applications that are not specified in piping system Sections.

R. Use powder-actuated fasteners mechanical-expansion anchors instead of building attachments where required in concrete construction.

END OF SECTION 23 0529

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SECTION 23 0533
HEAT TRACING FOR HVAC PIPING

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Self-regulating parallel resistance electric heating cable.

B. Cable outer jacket markings.

C. Connection kits.

D. Accessories.

E. Controls.

1.02 RELATED REQUIREMENTS

A. Section 26 0519 - Electrical Conductors and Cables.

B. Section 26 0526 - Grounding and Bonding for Electrical Systems.

C. Section 26 0533 - Conduit for Electrical Systems.

D. Section 26 0534 - Boxes for Electrical Systems.

E. Section 26 0583 - Wiring Connections.

1.03 REFERENCE STANDARDS

A. IEEE 515.1 - IEEE Standard for the Testing, Design, Installation, and Maintenance of Electrical Resistance Trace Heating for Commercial Applications; 2012.

B. NFPA 70 - National Electrical Code; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.

C. UL (DIR) - Online Certifications Directory; Current Edition.

1.04 ADMINISTRATIVE REQUIREMENTS

A. Sequencing: Ensure that utility connections are achieved in an orderly and expeditious manner.

B. Coordinate the work with other trades to provide ground fault protection for electric heat tracing circuits as required by NFPA 70.

C. Coordinate the work with other trades to provide circuit breaker ratings suitable for installed circuit lengths.

1.05 SUBMITTALS

A. See Section 01 3000 - Administrative Requirements, for submittal procedures.

B. Product Data: Provide data for electric heat tracing.

C. Shop Drawings: Indicate electric heat tracing layout, electrical terminations, thermostats, controls, and branch circuit connections.

1.06 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section with minimum three years of documented experience.

1.07 WARRANTY

A. See Section 01 7800 - Closeout Submittals, for additional warranty requirements.

PART 2 PRODUCTS

2.01 SELF-REGULATING PARALLEL RESISTANCE ELECTRIC HEATING CABLE

A. Manufacturer: Listed as compatible with existing.

B. Provide products listed, classified, and labeled by UL (DIR) or testing firm acceptable to authorities having jurisdiction (AHJ).

C. Factory Rating and Testing: Comply with IEEE 515.1.

D. Heating Element:
   1. Provide pair of parallel No.16 tinned or nickel coated stranded copper bus wires embedded in cross linked conductive polymer core with varying heat output in response to temperature along its length.
   2. Terminations: Waterproof, factory assembled, non-heating leads with connector at one end and water-tight seal at opposite end.
   3. Capable of crossing over itself without overheating.

E. Insulated Jacket: Flame retardant polyolefin.

F. Cable Cover: Provide tinned copper and polyolefin outer jacket with UV inhibitor.

G. Maximum Power-On Operating Temperature: 150 degrees F.

H. Maximum Power-Off Exposure Temperature: 185 degrees F.

I. Electrical Characteristics:
   1. 120 volts, single phase, 60 Hz.

2.02 CABLE OUTER JACKET MARKINGS

A. Name of manufacturer, trademark, or other recognized symbol of identification.

B. Catalog number, reference number, or model.

C. Month and year of manufacture, date coding, applicable serial number, or equivalent.

D. Agency listing or approval.

2.03 CONNECTION KITS

A. Provide power connection, splice/tee, and end seal kits compatible with the heating cable and without requiring cutting of the cable core to expose bus wires.

B. Provide with NEMA 4X rating for prevention of corrosion and water ingress.

2.04 ACCESSORIES

A. Provide Accessories As Indicated or As Required for Complete Installation, Including but Not Limited To:
   1. High temperature, glass filament tape for attachment of heating cable to metal piping.
   2. Cable ties.
   3. Silicone end seals and splice kits.
   4. Installation clips.

2.05 CONTROLS

A. Controls are pre-existing. New heat trace installation shall be incorporated into the existing system and not require additional pipe mounted thermostats, contactors, high-limit safeties, alarms or programmable controller hardware.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that piping and equipment are ready to receive work.

B. Verify field measurements are as indicated on shop drawings.

C. Verify required power is available, in proper location, and ready for use.

3.02 PREPARATION

A. Clean exposed surfaces prior to installation.

B. Prepare surfaces using approved methods as recommended by manufacturer.

3.03 INSTALLATION

A. Install in accordance with manufacturer's written installation instructions.

B. Comply with installation requirements of IEEE 515.1 and NFPA 70, Article 427.

C. Apply heating cable linearly on pipe with fiberglass tape only after piping has successfully completed any required pressure testing.

D. Comply with applicable local building codes and requirements of authorities having jurisdiction.

E. Grounding: Refer to Section 26 0526 - Grounding and Bonding for Electrical Systems.

F. Grounding: Refer to Section 26 0526.

G. Identification:
   1. After thermal insulation installation, apply external pipeline decals to indicate presence of the thermal insulation cladding at intervals not to exceed 20 ft including cladding over each valve or other equipment that may require maintenance.
   2. Refer to Section 23 0553 - Identification for HVAC and 26 0553 - Identification for Electrical Systems.

H. Equipment Wiring: Refer to Section 26 0583 - Wiring Connections

I. Electrical Connections: Refer to Section 26 0519 - Electrical Conductors and Cables

3.04 FIELD QUALITY CONTROL

A. See Section 01 4000 - Quality Requirements, for additional requirements.

B. Perform start-up by factory technician or factory representative as per Owner's requirements.

C. Field Testing and Inspections:
   1. Commission system in accordance with installation and operation manual.
   2. Inspect for sources of water entry and proper sealing.
   3. Inspect weather barrier to confirm that no sharp edges are contacting the trace heating.
   4. Insulation Resistance: Greater than 20 megohms at a test voltage of 2500 VDC for polymer insulated trace heaters.
   5. Test heating cable integrity with megohmmeter at the following intervals:
   6. Measure voltage and current at each unit.
   7. Controls:

3.05 CLOSEOUT ACTIVITIES

A. Demonstrate operation of controls.

3.06 PROTECTION

A. Protect installed products from damage until Date of Substantial Completion.

END OF SECTION

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SECTION 23 0548
VIBRATION AND SEISMIC CONTROLS FOR HVAC

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section

1.02 SUMMARY

A. This Section includes the following:
   1. Supplemental Steel bases.
   2. Vibration isolators.

B. Related Sections:
   1. Division 23 Section "Hangers and Supports for HVAC"
   2. Division 23 Section "Hydronic Piping"

1.03 SUBMITTALS

A. Product Data: Provide schedule of vibration isolator type with location and load on each. Include spring constant, spring horizontal to vertical stiffness, maximum load capacity, color codes, neoprene durometer.

B. Shop Drawings: Indicate inertia bases and locate vibration isolators, with static and dynamic load on each.

C. Manufacturer's Instructions: Indicate installation instructions with special procedures and setting dimensions.

D. Design: All spring isolators shall be completely stable in operation and shall be designed for not less than 30 percent reserve deflection beyond actual operating conditions. Open spring isolators shall be designed such that the horizontal to vertical spring stiffness ratio shall be 0.8 or greater for stability. All elastomeric isolators shall be of neoprene high quality synthetic rubber with anti-ozone and anti-oxidant additives. Steel components shall be cleaned and painted. All nuts, bolts, and washers shall be zinc electroplated. Structural steel bases thoroughly cleaned of welding slag and primed with zinc-chromate or metal etching primer. Isolators for equipment installed outdoors shall be designed to provide adequate restraint due to normal wind conditions and to withstand wind loads of 20 lb/sq. ft. applied to any exposed surface of the isolated equipment.

E. Delegated Design Submittal:
   1. Details: Indicate fabrication and arrangement. Detail attachments of restraints to the restrained items and to the structure. Show attachment locations, methods, and spacings. Identify components, list their strengths, and indicate directions and values of forces transmitted to the structure during seismic events. Indicate association with vibration isolation devices.
   2. Coordinate vibration isolation details with wind-restraint details required for equipment mounted outdoors. Comply with requirements in other Division 23 Sections for equipment mounted outdoors,
   3. Preapproval and Evaluation Documentation: By an agency acceptable to authorities having jurisdiction, showing maximum ratings of restraint items and the basis for approval (tests or calculations).
   4. Coordination Drawings; Show coordination of seismic bracing for Division 23 and 26 equipment with other systems and equipment in the vicinity, including other supports.

F. Welding certificates.

G. Qualification Data: Professional engineer and testing agency

1.04 QUALITY ASSURANCE

A. Testing Agency Qualifications: An Independent agency, with the experience and capability to conduct the testing indicated, that is a nationally recognized testing laboratory (NRTL) as defined by OSHA in 29 CFR 1910, 7, and that is acceptable to authorities having jurisdiction.

B. Welding: Qualify procedures and personnel according to AWS D1.1/D1.1M, “Structural Welding Code - Steel.”

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Subject to compliance with requirements, provide products by one of the following:
   1. Vibration Mountings and Controls - Amber Booth
   2. Kinetics Noise Control, Inc
   3. Mason Industries

2.02 SUPPLEMENTAL STEEL BASES

A. Single Steel I-beams
   1. Dimensions: Beams shall have a minimum section depth of 8% of the longest span between vibration isolator supports; down to a minimum of 6"; of size and shape required for the equipment to be supported.
   2. Loading: Structural steel base shall provide a rigid, distortion free, mounting base for supported equipment, which allows no excessive differential motion between the equipment and structural steel base.
   3. Anchoring: Provide prelocated and drilled anchor bolt holes into all structural steel bases to bolt down equipment.

2.03 VIBRATION ISOLATORS

A. Open Spring Isolators:
   1. Springs: Minimum horizontal stiffness equal to 80% vertical stiffness, with working deflection between 0.3 and 0.6 of maximum deflection. Color code springs for load carrying capacity.
   2. Spring Mounts: Provide with leveling devices, minimum 0.10 inch thick neoprene sound pads, and zinc chromate plated hardware.
   3. For equipment demonstrating lateral dynamic forces greater than 50% of the equipment weight, provide base plate and bolt holes for attachment to the support structure.

B. Housed Spring Isolators with limit stops:
   1. Springs: Minimum horizontal stiffness equal to 80% vertical stiffness, with working deflection between 0.3 and 0.6 of maximum deflection. Color code springs for load carrying capacity.
   2. Spring Mounts: Provide with leveling devices, minimum 0.10 inch thick neoprene sound pads, and zinc chromate plated hardware.
   3. Mounting: Complete with bolt holes in the base plates to provide for attachment to the equipment and support structure.
   4. Maintain a minimum 1/4" clearance around restraining bolts or housing limit stops to prevent interference with spring action.
   5. Housing: Steel enclosed constructed to carry equipment weight and serve as blocking during equipment installation.

C. Spring Hangers:
   1. Springs: Minimum horizontal stiffness equal to 80 percent vertical stiffness, with working deflection between 0.3 and 0.6 of maximum deflection. Color code springs for load carrying capacity.
   2. Neoprene element: Molded with a rod isolation bushing and double deflection neoprene block.

3. Spring diameters and hanger box lower holes; large enough to permit the hanger rod to swing through 30 degree arc before contacting the isolator housing and short circuiting the spring.
4. Housing: Rectangular steel box constructed to carry the equipment weight without deformation.

D. Neoprene Pad Isolators:
1. Neoprene waffle pads.
   a. Hardness: 30 durometer, or durometer required for specified load deflection.
   b. Thickness: Minimum 3/4 inch (20 mm).
   c. Maximum Loading: 50 psi.
   d. Rib Height: Maximum 0.7 times width.
2. Configuration: Single layer, or waffle pads bonded each side of 1/4 inch (6 mm) thick steel plate.
3. For equipment demonstrating lateral dynamic forces greater than 50% of the equipment weight, provide base plate and bolt holes for attachment to the support structure.

2.04 VIBRATION ISOLATION EQUIPMENT BASES

A. Steel Base: Factory-fabricated, welded, structural-steel bases and rails.

B. Design requirements: Lowest possible mounting height with no less than 1-inch clearance above the floor, include equipment anchor bolts and auxiliary motor slide bases or rails.

C. Include supports for suction and discharge elbows for pumps.

D. Structural Steel: Steel shapes, plates, and bars complying with ASTM A36/A36M. Bases shall have shape to accommodate supported equipment.

E. Support Brackets: Factory-welded steel brackets on frame for outrigger isolation mountings and to provide for anchor bolts and equipment support.

F. Fabrication: Fabricate steel templates to hold equipment anchor-bolt cleaves and anchors in place during placement of concrete. Obtain anchor-belt templates from supported equipment manufacturer.

G. Concrete Inertia Base: Factory-fabricated, welded, structural-steel bases and rails ready for placement of cast-in-place concrete.
1. Design Requirements: Lowest possible mounting height with not less than 1-inch (25-mm) clearance above the floor. Include equipment anchor bolts and auxiliary motor slide bases or rails. Retain first subparagraph below if inertia bases are required for pumps.
   a. Include supports for suction and discharge elbows for pumps.
2. Structural Steel: Steel shapes, plates, and bars complying with ASTM A36/A36M. Bases shall have shape to accommodate supported equipment.
3. Support Brackets: Factory-welded steel brackets on frame for outrigger isolation mountings and to provide for anchor bolts and equipment support.
4. Fabrication: Fabricate steel templates to hold equipment anchor-bolt sleeves and anchors in place during placement of concrete. Obtain anchor-bolt templates from supported equipment manufacturer.

PART 3 - EXECUTION

3.01 INSTALLATION

A. General
1. Installation and Quality Control: Except as otherwise indicated, the contractor shall comply with manufacturer's instructions for installation and load application to vibration isolation materials and units. Adjust to ensure that units do not exceed rated operating deflections or bottom out under loading. Remove space blocks and similar devices intended for temporary protection against overloading during installation. Anchor and attach units to structural supports and equipment as required. Adjust leveling devices as

required where leveling uniformly onto isolators. Shim units as required where leveling devices cannot be used to distribute loading properly. Where equipment base frames are specified, install base frames on isolator units so that a minimum of 1/2" clearance below the base will result when the supported equipment has been installed and loaded for operation. (2" clearance for inertia bases). The contractor shall adjust all vibration isolators after the system has achieved full operating weight so that: no vibration isolation element is short circuited; no undue stress is applied to the equipment, connected piping or structure; and no noise is transmitted to any support structure, connected surface or adjacent environment. After equipment installation and operation, it shall be the contractors' responsibility to demonstrate that the isolation system meets all of the criteria specified in this Section and on the Drawings, and has been properly installed.

2. On housed spring isolators, adjust so side stabilizers are clear under normal operating conditions.
3. Prior to making piping connections to equipment with operating weights substantially different from installed weights, block up equipment with temporary shims to final height. When full load is applied, adjust isolators to load to allow shim removal.
4. Provide isolated or integral seismic snubbers for all equipment, piping, and ductwork mounted on isolators. Each inertia base shall have minimum of four seismic snubbers located close to isolators. Snub equipment designated for post-disaster use to 1/8 inch maximum clearance.
5. Support piping connections to equipment mounted on isolators using isolators or resilient hangers for the first three points of support.

3.02 VIBRATION-CONTROL AND SEISMIC-RESTRAINT DEVICE INSTALLATION

A. General
   1. Comply with requirements in Division 07 Section "Roof Accessories" for installation of roof curbs, equipment supports, and roof penetrations.

B. Install cables so they do not bend across edges of adjacent equipment or building structure.

C. Install busing assemblies for anchor bolts for floor-mounted equipment, arranged to provide resilient media between anchor bolt and mounting hole in concrete base.

D. Install bushing assemblies for mounting bolts for wall-mounted equipment, arranged to provide resilient media where equipment or equipment-mounting channels are attached to wall.

E. Attachment to Structure: If specific attachment is not indicated, anchor bracing to structure at flanges of beams, at upper truss chords of bar joists, or at concrete members.

F. Drilled-in Anchors:
   1. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not damage existing reinforcing or embedded items during coring or drilling. Notify the structural engineer if reinforcing steel or other embedded items are encountered during drilling. Locate and avoid pre-stressed tendons, electrical and telecommunications conduit, and gas lines.
   2. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength.
   3. Wedge Anchors: Protect threads from damage during anchor Installation. Heavy-duty sleeve anchors shall be installed with sleeve fully engaged in the structural element to which anchor is to be fastened.
   4. Adhesive Anchors. Clean holes to remove loose material and drilling dust prior to installation of adhesive. Place adhesive in holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive.
   5. Set anchors to manufacturer's recommended torque, using a torque wrench.
   6. Install zinc-coated steel anchors for interior and stainless-steel anchors for exterior applications.

3.03 VIBRATION CONTROL AND SEISMIC-RESTRAINT DEVICE SCHEDULE

A. All major rotating and reciprocating mechanical equipment (5 HP and larger) shall be mounted with spring isolators with a minimum of 2 inch static deflection.

B. All piping connected to rotating and reciprocating shall be mounted with spring isolated hangers within the mechanical loom or 50 feet whichever is greater.

C. Housed spring mounts that accommodate vibration requirements shall be submitted for review prior in acceptance.

3.04 FIELD QUALITY CONTROL

A. Testing Agency; Engage a qualified testing agency to perform tests and inspections.

B. Tests and inspections:
   1. Provide evidence of recent calibration of test equipment by a testing agency acceptable to authorities having jurisdiction.
   2. Schedule test with Owner, through Engineer, before connecting anchorage device to restrained component (unless post connection testing has been approved), and with at least seven days advance notice.
   3. Obtain Engineers's approval before transmitting test loads to structure. Provide temporary load-spreading members.

C. Test at least four of each type and size of installed anchors and fasteners selected by Engineer.

D. Test to 90 percent of rated proof load of device.

E. Measure isolator restraint clearance.

F. Measure isolator deflection.

G. Verify snubber minimum clearances.

3.05 VIBRATION ISOLATION EQUIPMENT BASES INSTALLATION

A. Coordinate the location of embedded connection hardware with supported equipment attachment and mounting points and with requirements for concrete reinforcement and formwork specified in Section 03 3000 - CAST-IN-PLACE CONCRETE and Section 03 3053 "Miscellaneous Cast-in-Place Concrete".

END OF SECTION {CH#10002543}

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SECTION 23 0553
IDENTIFICATION FOR HVAC

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes
   1. Equipment Nameplates.
   2. Pipe Labels.
   3. Valve Tags.
   4. Stencils.

1.03 REFERENCE STANDARDS

A. ANSI/ASME A13.1 - Scheme for the Identification of Piping Systems; The American Society of Mechanical Engineers; 2026.

1.04 SUBMITTALS

A. Product Data: Provide manufacturers catalog literature for each product required.

B. Valve Tag Layout: Provide sample drawing of proposed valve tag with letter sizing and arrangement.

C. Samples: For color, letter style, and graphic representation required for each identification material and device.

D. Valve numbering scheme.

E. Closeout submittals: Valve tag chart included in O&M's and valve tag chart in frame for mounting in location determined by Owner.

1.05 COORDINATION

A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be applied.

B. Coordinate installation of identifying devices with location of access panels and doors.

C. Install identifying devices before installing acoustical ceiling and similar concealment.

D. Coordinate identification methods with facility's standards for consistency with methods in use elsewhere in the existing facility.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Subject to compliance with requirements, provide products by one of the following.
   1. Brady Corporation
   2. Brimar Industries, Inc.
   3. Carlton Industries, LP
   4. Craftmark Identification Systems
   5. Emedco
   6. Kolbi Pipe Marker Co.
   7. LEM Products, Inc.
   8. Marking Services, Inc.
   9. Seton Identification Products.

2.02 IDENTIFICATION APPLICATIONS

A. Provide identifying devices for the following systems including piping and associated equipment:
   1. Chilled Water.

2.03 EQUIPMENT NAMEPLATES

A. Description: Laminated three-layer plastic with engraved letters.
   1. Letter Color: White.
   2. Letter Height: 1/2 inch.
   3. Background Color: Black.
   4. Thickness: 1/8 inch.

B. Label Content: Include equipment's Drawing designation where drawing designation is unique or unique. Where equipment number is not unique in the drawing set (ie multiple fan coils tagged as FCU-1), submit proposed unique designation system for use in identifying each individual piece of equipment.

2.04 TAGS

A. Metal Tags: 0.032-inch minimum thick brass with predrilled or stamped hole for attachment hardware. Letters to be stamped; tag size minimum 1-1/2 inch diameter with smooth edges.

B. Tag Lettering size: Minimum 1/4 inch height for name of units where viewing distance is less than 24 inches, 1/2 inch minimum height for distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of the principal lettering.

C. Valve Tag Fastener: Brass wire link or beaded chain with S-hook.

D. Valve Tag Chart: Typewritten letter size list in anodized aluminum frame.

E. Valve Tag Content: On each valve tag include system designation (ie CW = Cold Water), numerical valve designation, (ie -001), and valve vintage (ie 2014) to designate year of installation.

2.05 PIPE LABELS

A. Comply with ANSI/ASME A13.1.

B. Label all piping systems with factory-fabricated, preprinted, color-coded, with lettering indicating service type and pipe size, and showing flow direction.

C. Construction type and Lettering size shall be based on pipe or pipe covering approximate sizes.
   1. Pipes or Pipe covering 3/4 inch and smaller: Letter size 1/2 inch, permanent phenolic tags or label constructed of flexible, precoiled, semi-rigid plastic, pre-formed to cover full circumference of pipe or pipe covering and to attach to pipe without fasteners or adhesive.
   2. Pipes or Pipe covering 1 inch to 1-1/4 inch Letter size 1/2 inch, label constructed of flexible, precoiled, semi-rigid plastic, pre-formed to cover full circumference of pipe or pipe covering and to attach to pipe without fasteners or adhesive.
   3. Pipe or Pipe covering size 1-1/2 inch to 2-1/4 inch diameter: Letter size 3/4 inch; label constructed of flexible, precoiled, semi-rigid plastic, pre-formed to cover full circumference of pipe or pipe covering and to attach to pipe without fasteners or adhesive.
   4. Pipe or Pipe covering size 2-1/2 inch to 6 inch diameter: Letter size 1-1/4 inch; label constructed of flexible, precoiled, semi-rigid plastic, pre-formed to cover full circumference of pipe or pipe covering and to attach to pipe without fasteners or adhesive.
   5. Pipe or Pipe covering size 6 inch: Letter size 1-1/4 inch; label constructed of flat, semi-rigid plastic, and attached to pipe with heavy duty nylon ties.
   6. Pipe or Pipe covering size 8 inch to 10 inch diameter: Letter size 2-1/2 inch; label constructed of flat, semi-rigid plastic, and attached to pipe with heavy duty nylon ties.
   7. Pipe or Pipe covering size larger than 10 inch diameter: Letter size 3-1/2 inch; label constructed of flat, semi-rigid plastic, and attached to pipe with heavy duty nylon ties.

D. Pipe Label Contents:
   1. Include identification of the the piping service (fluid conveyed) utilizing common English terminology (ie. Chilled Water Supply) in lieu of abbreviations used on Drawings.
   2. Indicate Pipe Size at every label location.
   3. Indicate fluid flow direction with Arrows at every label location.
   4. Insulated piping shall be labeled as "Non-Asbestos".

E. Schedule for banding and labeling of pipe and conduit will conform to ANSI/ANSI A13.1.

2.06 VALVE AND DEVICE IDENTIFICATION CEILING TAGS

A. Description: 3/4" x variable length vinyl tape for use with thermal transfer printer. 3.0-mil film, acrylic-adhesive with polished face. UV, moisture, and abrasion resistant with 5 year warranty.

B. Provide clear plastic printed ceiling identification tags for HVAC valves, controls, and other above ceiling equipment components. In addition, label all above ceiling components at each location in accordance with the Facility's established standards.

C. Color: Ceiling tags to be clear background with black letters.

D. Contents: Label text shall be in accordance with the Facility's established standards.

E. Install ceiling identification tags per the Facility's established standards.

F. All ceiling grid tags shall be installed prior to the ceiling cover inspection.

2.07 STENCILS

A. Pipe Systems Stenciling: Not permitted.

PART 3 - EXECUTION

3.01 IDENTIFICATION APPLICATIONS

A. Provide the following:
   1. Automatic Controls: Tags. Key to control schematic.
   2. Control Panels: Nameplates.
   3. Equipment (i.e Chillers, Pumps, etc.): Nameplates.
   4. Heat Transfer Equipment: Nameplates.
   5. Instrumentation: Tags.
   6. Major Control Components: Nameplates.
   7. Piping: Labels printed with service and direction of flow.
   8. Pumps: Nameplates.
   9. Relays: Tags.
   10. Small-sized Equipment: Tags.
   11. Tanks: Nameplates.
   12. Thermostats: Nameplates.
   13. Valves: Tags.

3.02 PREPARATION

A. Clean piping, and equipment surfaces of substances that could impair the bond of identification devices, including dirt, oil, grase, release agents, and imcompatible primers, paints, and encapsulants.

B. Degrease and clean surfaces to receive adhesive for identification materials.

3.03 EQUIPMENT NAMEPLATE INSTALLATION

A. Locate equipment nameplates where accessible, readily visible, and not subject to physical abrasion by routine equipment maintenance.

B. Install plastic nameplates with corrosive-resistant mechanical fasteners, or adhesive. Apply with sufficient adhesive to ensure permanent adhesion and seal with clear lacquer.

C. Identify pumps, compressors, tanks, water heaters, water treatment devices, and all other pieces of equipment with plastic nameplates.

3.04 PIPE LABEL INSTALLATION

A. Install plastic pipe markers in accordance with manufacturer's instructions. Lettering shall be visible from the floor.

B. Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; equipment rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:
   1. Near each valve and control device.
   2. Near each branch connection, excluding short take-offs for fixtures and terminal untis. Mark each pipe at branch where flow pattern is not obvious.
   3. Near penetrations through walls, floors, ceilings and inaccessble enclosures.
   4. At access doors, manholes, and similar access points that permit view of concealed piping.
   5. Mear major equipment items and other points of originaation and termination.
   6. Spaced at maximum intervals of 20 feet along each run. Reduce intervals to 15 feet in areas of congested piping .

C. Pipe Label Color Schedule:
   1. Chilled Water Piping: White letters on a green background
   2. Refrigerant Piping: Black letters on an orange background

3.05 VALVE TAG INSTALLATION

A. Install tags with corrosion resistant chain on valves and control devices in piping systems, except check valves; valves within factory-fabricated equipment units; shutoff valves; faucets; convenience and hose connections; and similar roughing-in connections of end-use fixtures and units. List tagged valves in a valve schedule.

B. Use tags in addition to labels on piping 3/4 inch diameter and smaller.
   1. Identify service, flow direction, and pressure.
   2. Install in clear view and align with axis of piping.
   3. Locate identification not to exceed 20 feet on straight runs including risers and drops, adjacent to each valve and Tee, at each side of penetration of structure or enclosure, and at each obstruction. Identification of Non-potable CW piping not to exceed 4 feet.

C. Identify valves in main and branch piping with tags.

D. Provide ceiling tacks to locate valves above T-bar type panel ceilings. Locate in corner of panel closest to valve.

END OF SECTION 23 0553

SECTION 23 0591
HVAC SYSTEM STARTUP

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes:
   1. Startup of all installed equipment and systems. Provide all information and assistance for cooperation with the testing, adjusting and balancing services.
   2. All work specified herein shall be complete prior to the start of the balancing procedures.

B. Related Sections:
   1. Division 23 Section - Hydronic Piping
   2. Division 23 Section - Testing-Adjust-Balance

1.03 SUBMITTALS

A. Product Data: Including performance data for temporary circulation pump to be used for flush and cleansing of closed loop hydronic systems.

PART 2 - PRODUCTS (NOT APPLICABLE)

PART 3 - EXECUTION

3.01 START-UP PROCEDURES

A. Bearings:
   1. Inspect for cleanliness. Clean and remove foreign materials.
   2. Verify alignment.
   3. Replace defective bearings and those which run rough and noisy.
   4. Lubricate as recommended by manufacturers.

B. Drives:
   1. Adjust tension in V-belt drives; adjust veripitch sheaves and drives for proper equipment speed.
   2. Adjust drives for alignment of sheaves and V-belts.
   3. Clean, remove foreign materials before starting operation.
   4. Replace fan sheaves as required and identified by the TAB Contractor to achieve design air flows.

C. Motors:
   1. Inspect each motor for amperage comparison to nameplate value.
   2. Correct conditions which produce excessive current flow which exist due to equipment malfunction.

D. Pumps:
   1. Inspect:
      a. Mechanical seals for cleanliness and adjustment before operating pump.
      b. Shaft sleeves for scoring.
      c. Mechanical faces, chambers and seal rings. Replace if defective.
   2. Verify that piping system is free of dirt and scale before circulating liquid through pump.

E. Control Valves:
   1. Inspect both hand and automatic control valves; clean bonnets and stems.
   2. tighten packing glands to assure no leakage, but permit valve stems to operate with galling.
   3. Replace:

a. Packing in valves to retain maximum adjustment after system is judged complete.
b. Packing on all valves which continue to leak.

4. Remove and replace bonnets which leak.
5. After cleaning, coat packing gland threads and valve stems with a surface lubricant.

F. Verify that control valve seats are free from foreign material and are properly positioned for intended service.

G. Tighten flanges after system has been placed in operation.
1. Replace flange gaskets which show sign of leaking after tightening.

H. Inspect screwed joints for leakage. Promptly remake each joint which appears to be faulty; do not wait for rust to form.

I. After system has been placed in operation, clean strainers, dirt pockets, orifices, valve seats and headers in fluid systems to assure being free from foreign materials.

J. Remove rust, scale and foreign materials from equipment and renew surfaces.

K. Install clean air filters. Set and calibrate draft gauges of air filters and other equipment.

L. Inspect each electrical control circuit to assure that operation complies with specifications and requirements to provide desired performance.

M. Inspect each pressure gauge and thermometer for calibration. Replace items which are defaced, broken, or which read incorrectly.

N. Repair damaged insulation.

O. Vent gasses trapped in systems. Verify that liquids are drained from all parts of gas or air systems.

P. Inspect piping for leaks at every joint, and at every screwed, flanged or welded connection, using a proper leak detecting method.

3.02 SYSTEM CLEANING

A. Prior to start-up, new or extensively extended chilled water and heating hot water systems shall be cleaned. Provide system valves, hose connections and access points to flush, fill, and clean new portions of system without connecting to existing systems that remains in service throughout this project. Provide temporary circulation pump of appropriately sized to circulate fluid and cleansing agent through system without using existing pumping systems already in service.

B. After cleaning, the system's water shall not contain a suspended solids count above 25 parts per million of solids 1.2 microns or larger, shall be essentially free of cutting thread oil (below 5 parts per million), and the system's pH shall be 8.0 or above. There are three different procedures for the chemical flushing of systems: 1) Trisodium Phosphate - Caustic, 2) Acrysol A-1 and Dequest 2010, and 3) Drewsperse 4945/Drewsperse 4395.
1. Trisodium Phosphate - Caustic: After the system is filled with water, 3 pounds of trisodium phosphate and 3 pounds of sodium hydroxide shall be added for every 1000 gallons of volume. This solution shall be circulated throughout the system for 12 hours, drained, and rinsed twice. CAUTION: This procedure is not to be used if metals such as, aluminum or galvanized steel are present in the system. The alternate methods 2 and 3 may be used in these systems.
2. Acrysol A-1 and Dequest 2010: After the system is filled with water, add 1 gallon of cleaning concentrate for every 100 gallons of system volume. The cleaning concentrate is prepared as follows:
a. In 100 gallons of water, add the following chemicals and mix until thoroughly dissolved. Add chemicals in the following order and quantity. Use a polyethylene drum.
1) Acrysol Grade A-1 (Rohm and Haas): 41.5 pounds
2) Dequest 2010 (Monsanto): 41.5 pounds

3) Potassium Hydroxide: 12.5 pounds

b. The cleaning solution shall be circulated throughout the system for 24 hours, preferably under light heat (100° - 120°F) if possible, drained, and rinsed twice.

3. Drewsperse 4945/Drewsperse 4395: These are preblended proprietary cleaning solutions. Drewsperse 4945 is a combination of polymers, chelate, and a gluconic acid derivative designed to remove iron and copper oxides from the system. Drewsperse 4395 is a low-foaming, non-ionic surfactant which effectively disperses mud, silt, clay, oil, and other suspended matter from the system, allowing these to be carried out of the system by flushing. After the system is filled with water, add 1 gallon of Drewsperse 4945 and 1 gallon of Drewsperse 4395 for every 1000 gallons of system volume. This solution shall be circulated throughout system for 12 hours at 120°F if possible, drained, and rinsed twice.

C. All automatic valves shall be open when cleaning system to allow circulation through coils. After rinsing, a sample shall be submitted for analysis to determine that the system has been adequately cleaned prior to installing glycol or adding corrosion inhibitors.

END OF SECTION 23 0591

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SECTION 23 0593
TESTING, ADJUSTING, AND BALANCING FOR HVAC

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. Balancing Hydronic Piping Systems:
      a. Variable-flow hydronic systems.
   2. Testing, Adjusting, and Balancing Equipment:
      a. Motors.
      b. Chillers.
   3. Testing, adjusting, and balancing existing systems and equipment.
   4. Sound tests.
   5. Vibration tests.
   6. Control system verification.

1.03 DEFINITIONS

A. AABC: Associated Air Balance Council.

B. BAS: Building automation systems.

C. NEBB: National Environmental Balancing Bureau.

D. TAB: Testing, adjusting, and balancing.

E. TABB: Testing, Adjusting, and Balancing Bureau.

F. TAB Specialist: An independent entity meeting qualifications to perform TAB work.

G. TDH: Total dynamic head.

1.04 PREINSTALLATION MEETINGS

A. TAB Conference: If requested by the Owner, conduct a TAB conference at Project Site after approval of the TAB strategies and procedures plan to develop a mutual understanding of the details. Provide a minimum of 14 days' advance notice of scheduled meeting time and location.
   1. Minimum Agenda Items:
      a. The Contract Documents examination report.
      b. The TAB plan.
      c. Needs for coordination and cooperation of trades and subcontractors.
      d. Proposed procedures for documentation and communication flow.

1.05 ACTION SUBMITTALS

A. LEED Submittals:
   1. Air-Balance Report for Prerequisite IEQ 1: Documentation indicating that work complies with ASHRAE Std 62.1, Section 7.2.2 - "Air Balancing."
   2. TAB Report for Prerequisite EA 2: Documentation indicating that work complies with ASHRAE/IESNA 90.1, Section 6.7.2.3 - "System Balancing."

1.06 INFORMATIONAL SUBMITTALS

A. Coordinate timing of informational submittals duration after Contractor's Notice to Proceed with overall project schedule to not delay delivery of the TAB scope of work

B. Qualification Data: Within 30 days of Contractor's Notice to Proceed, submit documentation that the TAB specialist and this Project's TAB team members meet the qualifications specified in "Quality Assurance" Article.

C. Contract Documents Examination Report: Within 60 days of Contractor's Notice to Proceed, submit the Contract Documents review report as specified in Part 3.

D. Strategies and Procedures Plan: Within 90 days of Contractor's Notice to Proceed, submit TAB strategies and step-by-step procedures as specified in "Preparation" Article.

E. System Readiness Checklists: Within 90 days of Contractor's Notice to Proceed, submit system readiness checklists as specified in "Preparation" Article.

F. Examination Report: Submit a summary report of the examination review required in "Examination" Article.

G. Certified Testing of Existing systems reports.

H. Certified TAB reports.

I. Sample report forms.

J. Instrument calibration reports, to include the following:
   1. Instrument type and make.
   2. Serial number.
   3. Application.
   4. Dates of use.
   5. Dates of calibration.

1.07 QUALITY ASSURANCE

A. TAB Specialists Qualifications: Certified by AABC, NEBB or TABB.
   1. TAB Field Supervisor: Employee of the TAB specialist and certified by AABC, NEBB or TABB.
   2. TAB Technician: Employee of the TAB specialist and certified by AABC, NEBB or TABB as a TAB technician.

B. Instrumentation Type, Quantity, Accuracy, and Calibration: Comply with requirements in ASHRAE Std 111, Section 4, "Instrumentation."

C. ASHRAE Compliance: Applicable requirements in ASHRAE Std 62.1, Section 7.2.2 - "Air Balancing."

D. ASHRAE/IESNA Compliance: Applicable requirements in ASHRAE/IESNA 90.1, Section 6.7.2.3 - "System Balancing."

1.08 FIELD CONDITIONS

A. Full Owner Occupancy: Owner will occupy the site and existing building during entire TAB period. Cooperate with Owner during TAB operations to minimize conflicts with Owner's operations.

B. Partial Owner Occupancy: Owner may occupy completed areas of building before Substantial Completion. Cooperate with Owner during TAB operations to minimize conflicts with Owner's operations.

PART 2 PRODUCTS (NOT APPLICABLE)

PART 3 EXECUTION

3.01 EXAMINATION

A. Examine the Contract Documents to become familiar with Project requirements and to discover conditions in systems designs that may preclude proper TAB of systems and equipment.

B. Examine installed systems for balancing devices, such as test ports, gage cocks, thermometer wells, flow-control devices, balancing valves and fittings, and manual volume dampers. Verify that locations of these balancing devices are applicable for intended purpose and are accessible.

C. Examine the approved submittals for HVAC systems and equipment.

D. Examine design data including HVAC system descriptions, statements of design assumptions for environmental conditions and systems output, and statements of philosophies and assumptions about HVAC system and equipment controls.

E. Examine equipment performance data including pump curves.
   1. Relate performance data to Project conditions and requirements, including system effects that can create undesired or unpredicted conditions that cause reduced capacities in all or part of a system.
   2. Calculate system-effect factors to reduce performance ratings of HVAC equipment when installed under conditions different from the conditions used to rate equipment performance. Compare results with the design data and installed conditions.

F. Examine system and equipment installations and verify that field quality-control testing, cleaning, and adjusting specified in individual Sections have been performed.

G. Examine test reports specified in individual system and equipment Sections.

H. Examine HVAC equipment and verify that bearings are greased, belts are aligned and tight, filters are clean, and equipment with functioning controls is ready for operation.

I. Examine strainers. Verify that startup screens have been replaced by permanent screens with indicated perforations.

J. Examine control valves for proper installation for their intended function of throttling, diverting, or mixing fluid flows.

K. Examine system pumps to ensure absence of entrained air in the suction piping.

L. Examine operating safety interlocks and controls on HVAC equipment.

M. Report deficiencies discovered before and during performance of TAB procedures. Observe and record system reactions to changes in conditions. Record default set points if different from indicated values.

3.02 PREPARATION

A. Prepare a TAB plan that includes the following:
   1. Equipment and systems to be tested.
   2. Strategies and step-by-step procedures for balancing the systems.
   3. Instrumentation to be used.
   4. Sample forms with specific identification for all equipment.

B. Perform system-readiness checks of HVAC systems and equipment to verify system readiness for TAB work. Include, at a minimum, the following:
   1. Hydronics:
      a. Verify leakage and pressure tests on water distribution systems have been satisfactorily completed.
      b. Piping is complete with terminals installed.
      c. Water treatment is complete.
      d. Systems are flushed, filled, and air purged.
      e. Strainers are pulled and cleaned.
      f. Control valves are functioning per the sequence of operation.
      g. Shutoff and balance valves have been verified to be 100 percent open.
      h. Pumps are started and proper rotation is verified.
      i. Pump gage connections are installed directly at pump inlet and outlet flanges or in discharge and suction pipe prior to valves or strainers.
      j. Variable-frequency controllers' startup is complete and safeties are verified.
      k. Suitable access to balancing devices and equipment is provided.

3.03 GENERAL PROCEDURES FOR TESTING AND BALANCING

A. Perform testing and balancing procedures on each system according to the procedures contained in AABC's "National Standards for Total System Balance", which meets or exceeds requirements in ASHRAE Std 111 and SMACNA's "HVAC Systems - Testing, Adjusting, and Balancing" and in this Section.

B. Cut insulation, ducts, pipes, and equipment cabinets for installation of test probes to the minimum extent necessary for TAB procedures.
   1. After testing and balancing, install test ports and duct access doors that comply with requirements in Section 23 3300 - Air Duct Accessories.
   2. Install and join new insulation that matches removed materials. Restore insulation, coverings, vapor barrier, and finish according to 23 0712 - HVAC Insulation.

C. Mark equipment and balancing devices, including damper-control positions, valve position indicators, fan-speed-control levers, and similar controls and devices, with paint or other suitable, permanent identification material to show final settings.

D. Take and report testing and balancing measurements in inch-pound (IP) units.

3.04 GENERAL PROCEDURES FOR HYDRONIC SYSTEMS

A. Prepare test reports for pumps, coils, and heat exchangers. Obtain approved submittals and manufacturer-recommended testing procedures. Crosscheck the summation of required coil and heat exchanger flow rates with pump design flow rate.

B. Prepare schematic diagrams of systems' "as-built" piping layouts.

C. In addition to requirements in "Preparation" Article, prepare hydronic systems for testing and balancing as follows:
   1. Check liquid level in expansion tank.
   2. Check highest vent for adequate pressure.
   3. Check flow-control valves for proper position.
   4. Locate start-stop and disconnect switches, electrical interlocks, and motor starters.
   5. Verify that motor starters are equipped with properly sized thermal protection.
   6. Check that air has been purged from the system.

3.05 PROCEDURES FOR VARIABLE-FLOW HYDRONIC SYSTEMS

A. Balance systems with automatic two- and three-way control valves by setting systems at maximum flow through heat-exchange terminals, and proceed as specified above for hydronic systems.

B. Adjust the variable-flow hydronic system as follows:
   1. Verify that the differential-pressure sensor is located as indicated.
   2. Determine whether there is diversity in the system.

C. For systems with diversity:
   1. Determine diversity factor.
   2. Simulate system diversity by closing required number of control valves, as approved by the design engineer.
   3. Adjust pumps to deliver total design gpm.
      a. Measure total water flow.
         1) Position valves for full flow through coils.
         2) Measure flow by main flow meter, if installed.
         3) If main flow meter is not installed, determine flow by pump TDH or exchanger pressure drop.
      b. Measure pump TDH as follows:
         1) Measure discharge pressure directly at the pump outlet flange or in discharge pipe prior to any valves.

2) Measure inlet pressure directly at the pump inlet flange or in suction pipe prior to any valves or strainers.
3) Convert pressure to head and correct for differences in gage heights.
4) Verify pump impeller size by measuring the TDH with the discharge valve closed. Note the point on manufacturer's pump curve at zero flow and verify that the pump has the intended impeller size.
5) With valves open, read pump TDH. Adjust pump discharge valve until design water flow is achieved.

c. Monitor motor performance during procedures and do not operate motor in an overloaded condition.

4. Adjust flow-measuring devices installed in mains and branches to design water flows.
   a. Measure flow in main and branch pipes.
   b. Adjust main and branch balance valves for design flow.
   c. Re-measure each main and branch after all have been adjusted.
5. Open control valves that were shut. Close a sufficient number of control valves that were previously open to maintain diversity, and balance terminals that were just opened.
6. Prior to verifying final system conditions, determine system differential-pressure set point.
7. If the pump discharge valve was used to set total system flow with variable-frequency controller at 60 Hz, at completion open discharge valve 100 percent and allow variable-frequency controller to control system differential-pressure set point. Record pump data under both conditions.
8. Mark final settings and verify that memory stops have been set.
9. Verify final system conditions as follows:
   a. Re-measure and confirm that total water flow is within design.
   b. Re-measure final pumps' operating data, TDH, volts, amps, and static profile.
   c. Mark final settings.
10. Verify that memory stops have been set.

3.06 PROCEDURES FOR MOTORS

A. Motors 1/2 HP and Larger: Test at final balanced conditions and record the following data:
   1. Manufacturer's name, model number, and serial number.
   2. Motor horsepower rating.
   3. Motor rpm.
   4. Phase and hertz.
   5. Nameplate and measured voltage, each phase.
   6. Nameplate and measured amperage, each phase.
   7. Starter size and thermal-protection-element rating.
   8. Service factor and frame size.

B. Motors Driven by Variable-Frequency Controllers: Test manual bypass of controller to prove proper operation.

3.07 PROCEDURES FOR AIR-COOLED CHILLERS

A. Balance water flow through each evaporator to within specified tolerances of indicated flow with all pumps operating. With only one chiller operating in a multiple chiller installation, do not exceed the flow for the maximum tube velocity recommended by the chiller manufacturer. Measure and record the following data with each chiller operating at design conditions:
   1. Evaporator-water entering and leaving temperatures, pressure drop, and water flow.
   2. Evaporator and condenser refrigerant temperatures and pressures, using instruments furnished by chiller manufacturer.
   3. Power factor if factory-installed instrumentation is furnished for measuring kilowatts.
   4. Kilowatt input if factory-installed instrumentation is furnished for measuring kilowatts.
   5. Capacity: Calculate in tons of cooling.

6. Verify condenser-fan rotation and record fan and motor data including number of fans and entering- and leaving-air temperatures.

3.08 SOUND TESTS

A. After the systems are balanced and construction is Substantially Complete, measure and record sound levels at 5 locations as designated by the Engineer.

B. Instrumentation:
   1. The sound-testing meter shall be a portable, general-purpose testing meter consisting of a microphone, processing unit, and readout.
   2. The sound-testing meter shall be capable of showing fluctuations at minimum and maximum levels, and measuring the equivalent continuous sound pressure level (LEQ).
   3. The sound-testing meter must be capable of using 1/3 octave band filters to measure mid-frequencies from 31.5 Hz to 8000 Hz.
   4. The accuracy of the sound-testing meter shall be plus or minus one decibel.

C. Test Procedures:
   1. Perform test at quietest background noise period. Note cause of unpreventable sound that affects test outcome.
   2. Equipment should be operating at design values.
   3. Calibrate the sound-testing meter prior to taking measurements.
   4. Use a microphone suitable for the type of noise levels measured that is compatible with meter. Provide a windshield for outside measurements.
   5. Record a set of background measurements in dBA and sound pressure levels in the eight un-weighted octave bands 63 Hz to 8000 Hz (NC) with the equipment off. Record sound readings and sound pressure levels again with equipment operating.
   6. Take readings no closer than 36 inches from a wall or from the operating equipment and approximately 60 inches from the floor, with the meter held or mounted on a tripod.
   7. For outdoor measurements, move sound-testing meter slowly and scan area that has the most exposure to noise source being tested. Use A-weighted scale for this type of reading.

D. Reporting:
   1. Report shall record the following:
      a. Location.
      b. System tested.
      c. dBA reading.
      d. Sound pressure level in each octave band with equipment on and off.
   2. Plot sound pressure levels on NC worksheet with equipment on and off.
   3. Plot sound pressure levels on RC worksheet with equipment on and off.

3.09 VIBRATION TESTS

A. After systems are balanced and construction is Substantially Complete, measure and record vibration levels on equipment having motor horsepower equal to or greater than 10.

B. Instrumentation:
   1. Use portable, battery-operated, and microprocessor-controlled vibration meter with or without a built-in printer.
   2. The meter shall automatically identify engineering units, filter bandwidth, amplitude, and frequency scale values.
   3. The meter shall be able to measure machine vibration displacement in mils of deflection, velocity in inches per second, and acceleration in inches per second squared.
   4. Verify calibration date is current for vibration meter before taking readings.

C. Test Procedures:
   1. To ensure accurate readings, verify that accelerometer has a clean, flat surface and is mounted properly.

2. With the unit running, set up vibration meter in a safe, secure location. Connect transducer to meter with proper cables. Hold magnetic tip of transducer on top of the bearing, and measure unit in mils of deflection. Record measurement, then move transducer to the side of the bearing and record in mils of deflection. Record an axial reading in mils of deflection by holding nonmagnetic, pointed transducer tip on end of shaft.
3. Change vibration meter to velocity (inches per second) measurements. Repeat and record above measurements.
4. Record CPM or rpm.
5. Read each bearing on motor, fan, and pump as required. Track and record vibration levels from rotating component through casing to base.

D. Reporting:
1. Report shall record location and the system tested.
2. Include horizontal-vertical-axial measurements for tests.
3. Verify that vibration limits follow Specifications, or, if not specified, follow the General Machinery Vibration Severity Chart or Vibration Acceleration General Severity Chart from the AABC National Standards. Acceptable levels of vibration are normally "smooth" to "good."
4. Include in report General Machinery Vibration Severity Chart, with conditions plotted.

3.10 CONTROLS VERIFICATION

A. In conjunction with system balancing, perform the following:
1. Verify temperature control system is operating within the design limitations.
2. Confirm that the sequences of operation are in compliance with Contract Documents.
3. Verify that controllers are calibrated and function as intended.
4. Verify that controller set points are as indicated.
5. Verify the operation of lockout or interlock systems.
6. Verify the operation of valve and damper actuators.
7. Verify that controlled devices are properly installed and connected to correct controller.
8. Verify that controlled devices travel freely and are in position indicated by controller: open, closed, or modulating.
9. Verify location and installation of sensors to ensure that they sense only intended temperature, humidity, or pressure.

B. Reporting: Include a summary of verifications performed, remaining deficiencies, and variations from indicated conditions.

3.11 PROCEDURES FOR TESTING, ADJUSTING, AND BALANCING EXISTING SYSTEMS

A. Perform preliminary testing of systems that are currently in place and whose capacities are expanded to meet the build out scope. Systems and equipment in this category include:
1. [INPUT SYSTEM TYPES, DESCRIPTION OR TAGS]

B. Perform a preconstruction inspection of existing equipment that is to remain and be reused.
1. Measure and record the operating speed, airflow, and static pressure of each fan.
2. Measure motor voltage and amperage. Compare the values to motor nameplate information.
3. Check the condition of coils.
4. Check the operation of the drain pan and condensate-drain trap.
5. Check bearings and other lubricated parts for proper lubrication.
6. Report on the operating condition of the equipment and the results of the measurements taken. Report deficiencies.

C. Before performing testing and balancing of existing systems, inspect existing equipment that is to remain and be reused to verify that existing equipment has been cleaned and refurbished. Verify the following:
1. New filters are installed.

2. Coils are clean and fins combed.
3. Drain pans are clean.
4. Fans are clean.
5. Bearings and other parts are properly lubricated.
6. Deficiencies noted in the preconstruction report are corrected.

D. Perform testing and balancing of existing systems to the extent that existing systems are affected by the renovation work.
1. Compare the indicated airflow of the renovated work to the measured fan airflows, and determine the new fan speed and the face velocity of filters and coils.
2. Verify that the indicated airflows of the renovated work result in filter and coil face velocities and fan speeds that are within the acceptable limits defined by equipment manufacturer.
3. If calculations increase or decrease the airflow rates and water flow rates by more than 5 percent, make equipment adjustments to achieve the calculated rates. If increase or decrease is 5 percent or less, equipment adjustments are not required.

3.12 TOLERANCES

A. Set HVAC system's airflow rates and water flow rates within the following tolerances:
1. Cooling-Water Flow Rate: Plus or minus 10 percent.

B. Maintaining pressure relationships as designed shall have priority over the tolerances specified above.

3.13 PROGRESS REPORTING

A. Initial Construction-Phase Report: Based on examination of the Contract Documents as specified in "Examination" Article, prepare a report on the adequacy of design for systems balancing devices. Recommend changes and additions to systems balancing devices to facilitate proper performance measuring and balancing. Recommend changes and additions to HVAC systems and general construction to allow access for performance measuring and balancing devices.

B. Status Reports: Prepare weekly progress reports to describe completed procedures, procedures in progress, and scheduled procedures. Include a list of deficiencies and problems found in systems being tested and balanced. Prepare a separate report for each system and each building floor for systems serving multiple floors.

3.14 FINAL REPORT

A. General: Prepare a certified written report; tabulate and divide the report into separate sections for tested systems and balanced systems.
1. Include a certification sheet at the front of the report, signed and sealed by the certified testing and balancing engineer.
2. Include a list of instruments used for procedures, along with proof of calibration.
3. Certify validity and accuracy of field data.

B. Final Report Contents: In addition to certified field-report data, include the following:
1. Pump curves.
2. Manufacturers' test data.
3. Field test reports prepared by system and equipment installers.
4. Other information relative to equipment performance; do not include Shop Drawings and Product Data.

C. General Report Data: In addition to form titles and entries, include the following data:
1. Title page.
2. Name and address of the TAB specialist.
3. Project name.
4. Project location.

5. Engineer's name and address.
6. Contractor's name and address.
7. Report date.
8. Signature of TAB supervisor who certifies the report.
9. Table of Contents with the total number of pages defined for each section of the report. Number each page in the report.
10. Summary of contents including the following:
    a. Indicated versus final performance.
    b. Notable characteristics of systems.
    c. Description of system operation sequence if it varies from the Contract Documents.
11. Nomenclature sheets for each item of equipment.
12. Notes to explain why certain final data in the body of reports vary from indicated values.

D. System Diagrams: Include schematic layouts of hydronic distribution systems. Present each system with single-line diagram and include the following:
1. Water flow rates.
2. Pipe and valve sizes and locations.
3. Balancing stations.
4. Position of balancing devices.

E. Apparatus-Coil Test Reports:
1. Coil Data:
    a. System identification.
    b. Location.
    c. Coil type.
    d. Number of rows.
    e. Fin spacing in fins per inch o.c.
    f. Make and model number.
    g. Face area in sq. ft..
    h. Tube size in NPS.
    i. Tube and fin materials.
    j. Circuiting arrangement.
2. Test Data (Indicated and Actual Values):
    a. Water flow rate in gpm.
    b. Water pressure differential in feet of head or psig.
    c. Entering-water temperature in deg F.
    d. Leaving-water temperature in deg F.
    e. Refrigerant expansion valve and refrigerant types.
    f. Refrigerant suction pressure in psig.
    g. Refrigerant suction temperature in deg F.

F. Pump Test Reports: Calculate impeller size by plotting the shutoff head on pump curves and include the following:
1. Unit Data:
    a. Unit identification.
    b. Location.
    c. Service.
    d. Make and size.
    e. Model number and serial number.
    f. Water flow rate in gpm.
    g. Water pressure differential in feet of head or psig.
    h. Required net positive suction head in feet of head or psig.
    i. Pump rpm.
    j. Impeller diameter in inches.

k. Motor make and frame size.
l. Motor horsepower and rpm.
m. Voltage at each connection.
n. Amperage for each phase.
o. Full-load amperage and service factor.
p. Seal type.

2. Test Data (Indicated and Actual Values):
a. Static head in feet of head or psig.
b. Pump shutoff pressure in feet of head or psig.
c. Actual impeller size in inches.
d. Full-open flow rate in gpm.
e. Full-open pressure in feet of head or psig.
f. Final discharge pressure in feet of head or psig.
g. Final suction pressure in feet of head or psig.
h. Final total pressure in feet of head or psig.
i. Final water flow rate in gpm.
j. Voltage at each connection.
k. Amperage for each phase.

G. Instrument Calibration Reports:
1. Report Data:
a. Instrument type and make.
b. Serial number.
c. Application.
d. Dates of use.
e. Dates of calibration.

3.15 VERIFICATION OF TAB REPORT

A. The TAB specialist's test and balance engineer shall conduct the inspection in the presence of Construction Manager.

B. Construction Manager shall randomly select measurements, documented in the final report, to be rechecked. Rechecking shall be limited to either 10 percent of the total measurements recorded or the extent of measurements that can be accomplished in a normal 8-hour business day.

C. If rechecks yield measurements that differ from the measurements documented in the final report by more than the tolerances allowed, the measurements shall be noted as "FAILED."

D. If the number of "FAILED" measurements is greater than 10 percent of the total measurements checked during the final inspection, the testing and balancing shall be considered incomplete and shall be rejected.

E. If TAB work fails, proceed as follows:
1. TAB specialists shall recheck all measurements and make adjustments. Revise the final report and balancing device settings to include all changes; resubmit the final report and request a second final inspection.
2. If the second final inspection also fails, Owner may contract the services of another TAB specialist to complete TAB work according to the Contract Documents and deduct the cost of the services from the original TAB specialist's final payment.
3. If the second verification also fails, Owner, design professional or Engineer may contact AABC Headquarters regarding the AABC National Performance Guaranty.

F. Prepare test and inspection reports.

3.16 ADDITIONAL TESTS

A. Within 90 days of completing TAB, perform additional TAB to verify that balanced conditions are being maintained throughout and to correct unusual conditions.

B. Seasonal Periods: If initial TAB procedures were not performed during near-peak summer and winter conditions, perform additional TAB during near-peak summer and winter conditions.

END OF SECTION

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SECTION 23 0712
HVAC INSULATION

CEPART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General Conditions and Division 01 Specification Sections, apply to this Section.

B. Specifications throughout all Divisions of the Project Manual are directly applicable to this Section, and this Section is directly applicable to them.

1.02 SUMMARY

A. Section includes insulating the following HVAC duct, piping and associated equipment.
   1. Piping Systems:
      a. Chilled Water Piping

B. Related Sections:
   1. 22 0719 - Plumbing Piping Insulation
   2. 23 2113 - Hydronic Piping
   3. 23 0529 - Hangers and Supports for HVAC

1.03 REFERENCE STANDARDS

A. ASTM B 209 - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate; 2007.

B. ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications; 2024b.

C. ASTM C177 - Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus; 2019.

D. ASTM C195 - Standard Specification for Mineral Fiber Thermal Insulating Cement; 2007 (Reapproved 2019).

E. ASTM C449 - Standard Specification for Mineral Fiber Hydraulic-Setting Thermal Insulating and Finishing Cement; 2007 (Reapproved 2013).

F. ASTM C518 - Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus; 2017.

G. ASTM C534/C534M - Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form; 2023.

H. ASTM C552 - Standard Specification for Cellular Glass Thermal Insulation; 2022.

I. ASTM C585 - Standard Practice for Inner and Outer Diameters of Thermal Insulation for Nominal Sizes of Pipe and Tubing; 2022.

J. ASTM C591 - Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation; 2022.

K. ASTM C795 - Standard Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel; 2008 (Reapproved 2018).

L. ASTM C871 - Standard Test Methods for Chemical Analysis of Thermal Insulation Materials for Leachable Chloride, Fluoride, Silicate, and Sodium Ions; 2018.

M. ASTM C1136 - Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation; 2023.

N. ASTM D1784 - Standard Classification System and Basis for Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds; 2020.

O. ASTM D2842 - Standard Test Method for Water Absorption of Rigid Cellular Plastics; 2019.

P. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2023b.

Q. ASTM E96/E96M - Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials; 2022.

R. ASTM F1249 - Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor; 2020.

S. MIL-DTL-3316 - Adhesives, Fire-Resistant, Thermal Insulation; 2020d.

T. NFPA 90A - Standard for the Installation of Air-Conditioning and Ventilating Systems; 2021.

U. NFPA 90B - Standard for the Installation of Warm Air Heating and Air-Conditioning Systems; 2024.

V. UL 723 - Standard for Test for Surface Burning Characteristics of Building Materials; Current Edition, Including All Revisions.

W. ASTM C177 - Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded Hot Plate Apparatus; 2004.

X. ASTM C518 - Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus; 2004.

Y. ASTM C534/C534M - Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form; 2008.

Z. ASTM C552 - Standard Specification for Cellular Glass Thermal Insulation; 2007.

AA. ASTM C585 - Standard Practice for Inner and Outer Diameters of Rigid Thermal Insulation for Nominal Sizes of Pipe and Tubing (NPS System); 1990 (Reapproved 2004).

BB. ASTM C591 - Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation; 2007.

CC. ASTM D2842 - Standard Test Method for Water Absorption of Rigid Cellular Plastics; 2006.

DD. ASTM E96/E96M - Standard Test Methods for Water Vapor Transmission of Materials; 2005.

EE. NFPA 255 - Standard Method of Test of Surface Burning Characteristics of Building Materials; National Fire Protection Association; 2006.

FF. UL 723 - Standard for Test for Surface Burning Characteristics of Building Materials; Underwriters Laboratories Inc.; 2003.

1.04 ACTION SUBMITTALS

A. Product Data: For each type of product indicated. Include thermal conductivity, water-vapor permeance thickness, and jackets (both factory- and field-applied if any).

B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
   1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.
   2. Detail application of field-applied jackets.
   3. Detail application at linkages of control devices.

C. Samples: For each type of insulation and jacket indicated. Identify each Sample, describing product and intended use. Sample sizes are as follows:
   1. Preformed Pipe Insulation Materials: 12 inches long by NPS 2.
   2. Jacket Material for Pipe: 12 inches long by NPS 2.
   3. Sheet Form Insulation Materials: 12 inches square.
   4. Sheet Jacket Materials: 12 inches square.
   5. Manufacturer's Color Charts: For products where color is specified, show the full range of colors available for each type of finish material.

1.05 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified Installer.

B. Material Test Reports: From a qualified testing agency acceptable to authorities having jurisdiction indicating, interpreting, and certifying test results for compliance of insulation materials, sealers, attachments, cements, and jackets, with requirements indicated. Include dates of tests and test methods employed.

C. Field quality-control reports.

1.06 QUALITY ASSURANCE

A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.

B. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E84, by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.
   1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.
   2. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

C. Mockups: Before installing insulation, build mockups for each type of insulation and finish listed below to demonstrate quality of insulation application and finishes. Build mockups in the location indicated or, if not indicated, as directed by Engineer. Use materials indicated for the completed Work.
   1. Piping Mockups:
      a. One ten foot section of NPS 2 straight pipe.
      b. One each of a 90-degree threaded, welded, and flanged elbow.
      c. One each of a threaded, welded, and flanged tee fitting.
      d. One NPS 2 or smaller valve, and one NPS 2-1/2 or larger valve.
      e. Four support hangers including hanger shield and insert.
      f. One threaded strainer and one flanged strainer with removable portion of insulation.
      g. One threaded reducer and one welded reducer.
      h. One pressure temperature tap.
      i. One mechanical coupling.
   2. Equipment Mockups:
      a. One hydronic pump.
      b. One tank or vessel.
   3. For each mockup, fabricate cutaway sections to allow observation of application details for insulation materials, adhesives, mastics, attachments, and jackets.
   4. Notify Engineer seven days in advance of dates and times when mockups will be constructed.
   5. Obtain Engineer's approval of mockups before starting insulation application.
   6. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Engineer specifically approves such deviations in writing.
   7. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
   8. Demolish and remove mockups when directed.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Packaging: Insulation material containers shall be marked by manufacturer with appropriate ASTM standard designation, type and grade, and maximum use temperature.

B. Deliver material to site in factory fabricated containers with manufacturer's stamp or label, showing fire and smoke hazard ratings of products.

C. Store material in original wrappings and protect from weather and construction traffic.

D. Protect against sun, dirt, water, chemical and mechanical damage.

E. Remove damaged insulation from project site. Do not install.

1.08 FIELD CONDITIONS

A. Maintain ambient conditions required by manufacturers of each product.

B. Maintain temperature before, during, and after installation for minimum of 24 hours.

1.09 COORDINATION

A. Coordinate sizes and locations of supports, hangers, and insulation shields specified in Section 23 0529 - Hangers and Supports for HVAC.

B. Coordinate clearance requirements with duct Installer for duct insulation application. Before preparing piping and equipment Shop Drawings, establish and maintain clearance requirements for installation of insulation and field-applied jackets and finishes and for space required for maintenance.

C. Coordinate installation and testing of heat tracing.

1.10 SCHEDULING

A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.

B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

PART 2 - PRODUCTS

2.01 INSULATION MATERIALS

A. Comply with requirements in Part 3 schedule articles for where insulating materials shall be applied.

B. Products shall not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C871.

D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C795.

E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.

F. Hydrous Calcium Silicate Pipe and Tank insulation
   1. Manufacturer:
      a. Johns Manville Corporation: www.jm.com.
      b. Substitutions: See Section 01 6000 - Product Requirements.
   2. Insulation: ASTM C533; rigid molded, asbestos free, gold color.
      a. 'K' Value: 0.40 at 300 degrees F, when tested in accordance with ASTM C177 or ASTM C518.
      b. Maximum Service Temperature: 1200 degrees F.
      c. Density: 15 lb/cu ft.
   3. Shapes: Preformed pipe sections consisting of flat-, curved-, and grooved-block section.
   4. Insulating Cement: ASTM C449.

G. Flexible Elastomeric Insulation: Closed-cell, sponge- or expanded-rubber materials. Comply with ASTM C 534, Type I for tubular materials, Type II for sheet materials. Use molded tubular material wherever possible.
   1. Products: Subject to compliance with requirements, provide products by one of the following:
      a. Aeroflex USA, Inc. Aeroflex EPDM.
      b. Armacell LLC; AP Armaflex.
      c. Armacell LLC; Arma Tuff (Exterior Ductwork)
      d. Armacell LLC; ArmaFlex Shield (Exterior Piping)
      e. K-Flex USA; Insul-Sheet, K-Flex Gray Duct Liner, and K-Flex LS
   2. Maximum Thermal Conductivity ('K' Value): 0.28 at 75 deg F.
   3. Minimum Service Temperature: -40 deg F.
   4. Maximum Service Temperature: 220 deg F.
   5. Water Vapor Permeability: 0.08 perm-inch
   6. Connection: Waterproof vapor barrier adhesive.
   7. Polyethylene and Polyolefin materials are not acceptable.

H. Exterior Pipe Insulation Systems: Based on Flexible Elastomeric Insulation systems specified above with a factory adhered moisture-resistant and UV-resistant jacket. Based on Armaflex ArmaFlex Shield product specifically designed for exterior pipe insulation application. Use with manufacturer's adhesive and jacket lay up materials.

I. Mineral-Fiber, Pipe and Tank Insulation: Mineral or glass fibers bonded with a thermosetting resin. Semirigid board material with factory-applied ASJ complying with ASTM C 1393, Type II or Type IIIA Category 2, or with properties similar to ASTM C 612, Type IB. Nominal density is 2.5 lb/cu. ft. or more. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Johns Manville: MicroFlex.
      b. Knauf Insulation Inc.: Pipe and Tank Insulation
      c. Owens Corning: Fiberglas Pipe and Tank Insulation.
   2. Maximum Thermal conductivity (K-value): 0.24 at 75 deg F
   3. Maximum service temperature: 850 deg F.
   4. Moisture Vapor Permeability: 0.02 perms, when tested in accordance with ASTM E96/E96M.

2.02 INSULATING CEMENTS

A. Mineral-Fiber Insulating Cement: Comply with ASTM C195.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Insulco, a Division of MFS, Inc.; Triple I.
      b. P.K. Insulation Mfg, Co., Inc,; Super-Stik
      c. Ramco Insulation, Inc.; Super-Stik.

B. Mineral-Fiber, Hydraulic-Setting Insulating and Finishing Cement: Comply with ASTM C449.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Insulco, a Division of MFS, Inc.; SmoothKote.
      b. P.K. Insulation Mfg. Co., Inc.; PK No. 127 and Quik-Cote.
      c. Ramco Insulation, Inc.; Ramcote 1200.

2.03 ADHESIVES

A. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated unless otherwise indicated.

B. Flexible Elastomeric Adhesive: Air dried, contact adhesive, compatible with insulation and compliant with MIL-A-24179A, Type II, Class I.
   1. Products: Subject to compliance with requirements, provide one of the following:

a. Aeroflex USA, Inc.: Aeroseal.
b. Armacell LLC: Armaflex 520 Adhesive.
c. K-Flex USA: R-373 Contact Adhesive.

2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

C. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand Specialty Construction Brands, Incl, a business of H.B. Fuller Company: CP-127
b. Eagles Bridges - Marathon Industries; 225.
c. Mon-Eco Industries, Inc.: 22-25
2. For indoor applications, adhesive shall have a VOC content of 80 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

D. ASJ Adhesive, and FSK Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand Specialty Construction Brands, Incl, a business of H.B. Fuller Company: CP-82
b. Eagles Bridges - Marathon Industries; 225.
c. Mon-Eco Industries, Inc.: 22-25
2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

E. PVC Jacket Adhesive: Compatible with PVC jacket.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Dow Corning Corporation: 739 Dow Silicone.
b. Johns Mansville: Zeston Perma-Weld, CEEL-TITE Solvent Welding Adhesive.
c. P.I.C Plastics, Inc.: Welding Adhesive.
d. Speedline Corporation: Polyco VP Adhesive.
2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

2.04 MASTICS

A. Materials shall be compatible with insulation materials, jackets, and substrates; comply with MIL-PRF-19565C, Type II.
1. For indoor applications, use mastics that have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

B. Vapor-Barrier Mastic: Water based; suitable for indoor use on below ambient services.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Foster Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: 30-03/11-90.

b. Vimasco Corporation: 749.
2. Water-Vapor Permeance: ASTM E96/E96M , Procedure B, 0.013 perm at 43-mildry film thickness.
3. Service Temperature Range: Minus 20 to plus 180 deg F.
4. Solids Content: ASTM D 1644, 58 percent by volume and 70 percent by weight.
5. Color: White.

C. Vapor-Barrier Mastic: Solvent based; suitable for indoor use on below ambient services.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: CP-30
b. Eagle Bridges - Marathon Industries: 501.
c. Mon-Eco Industries, Inc.: 55-10.
2. Water-Vapor Permeance: ASTM F1249, 0.05 at 35-mil dry film thickness.
3. Service Temperature Range: 0 to 180 deg F.
4. Solids Content: ASTM D 1644, 44 percent by volume and 62 percent by weight.
5. Color: White.

D. Vapor-Barrier Mastic: Solvent based; suitable for outdoor use on below ambient services.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: Encacel.
b. Eagle Bridges - Marathon Industries: 570.
c. Mon-Eco Industries, Inc.: 55-70.
2. Water-Vapor Permeance: ASTM F1249 ASTM F 1249, 0.05 perm at 30-mildry film thickness.
3. Service Temperature Range: Minus 50 to plus 220 deg F.
4. Solids Content: ASTM D 1644, 33 percent by volume and 46 percent by weight.
5. Color: White.

E. Breather Mastic: Water based; suitable for indoor and outdoor use on above ambient services.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: CP-10.
b. Eagle Bridges - Marathon Industries: 550.
c. Mon-Eco Industries, Inc.: 55-50.
2. Water-Vapor Permeance: ASTM F1249, 1.8 perms at 0.0625-inch dry film thickness.
3. Service Temperature Range: Minus 20 to plus 180 deg F.
4. Solids Content: 60 percent by volume and 66 percent by weight.
5. Color: White.

2.05 LAGGING ADHESIVES

A. Description: Comply with MIL-DTL-3316 C, Class I, Grade A and shall be compatible with insulation materials, jackets, and substrates.
1. For indoor applications, use lagging adhesives that have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
2. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: CP-50 AHV2
b. Eagle Bridges - Marathon Industries: 130.
c. Mon-Eco Industries, Inc.: 11-30.
d. Vimasco Corporation: 713 and 714
3. Fire-resistant, water-based lagging adhesive and coating for use indoors to adhere fire-resistant lagging cloths over duct insulation.

4. Service Temperature Range: 0 to plus 180 deg F.
5. Color: White.

2.06 SEALANTS

A. FSK and Metal Jacket Flashing Sealants:
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: CP-76.
b. Eagle Bridges - Marathon Industries: 405
c. Mon-Eco Industries, Inc.: 44-05
2. Materials shall be compatible with insulation materials, jackets, and substrates.
3. Fire- and water-resistant, flexible, elastomeric sealant.
4. Service Temperature Range: Minus 40 to plus 250 deg F.
5. Color: Aluminum.
6. For indoor applications, sealants shall have a VOC content of 420 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
7. Sealants shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

B. ASJ Flashing Sealants, and Vinyl and PVC Jacket Flashing Sealants:
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: CP-76
2. Materials shall be compatible with insulation materials, jackets, and substrates.
3. Fire- and water-resistant, flexible, elastomeric sealant.
4. Service Temperature Range: Minus 40 to plus 250 deg F.
5. Color: White.
6. For indoor applications, sealants shall have a VOC content of 420 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
7. Sealants shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

2.07 FACTORY-APPLIED JACKETS

A. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:
1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C1136, Type I.
2. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C1136, Type I.
3. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C1136, Type II.
4. FSP Jacket: Aluminum-foil, fiberglass-reinforced scrim with polyethylene backing; complying with ASTM C1136, Type II.
5. Vinyl Jacket: White vinyl with a permeance of 1.3 permswhen tested according to ASTM E96/E96M, Procedure A, and complying with NFPA 90A and NFPA 90B.

2.08 FIELD-APPLIED FABRIC-REINFORCING MESH

A. Woven Glass-Fiber Fabric for Duct and Equipment Insulation: Approximately 6 oz./sq. yd.with a thread count of 5 strands by 5 strands per sq. in. for covering ducts and equipment.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: Chil-Glas No. 5.

B. Woven Glass-Fiber Fabric for Pipe Insulation: Approximately 2 oz./sq. yd.with a thread count of 10 strands by 10 strands per sq. in. for covering ducts and equipment.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Vimasco Corporation: Elastafab 894.

C. Woven Polyester Fabric: Approximately 1 oz./sq. yd.with a thread count of 10 strands by 10 strands per square inch in a Leno weave, for ducts, piping and equipment.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Foster Brand, Specialty Construction Brands, Inc., a business of H.B. Fuller Company: Mast-A-Fab.
      b. Vimasco Corporation: Elastafab 894.

2.09 FIELD-APPLIED CLOTHS

A. Woven Glass-Fiber Fabric: Comply with MIL-C-20079H, Type I, plain weave, and presized a minimum of 8 oz./sq. yd.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Alpha Associates, Inc.; Alpha-Maritex 84215 and 84203/1185RW, Luben 59.

2.10 FIELD-APPLIED JACKETS

A. Field-applied jackets shall comply with ASTM C 921, Type I, unless otherwise indicated.

B. FSK Jacket: Aluminum-foil-face, fiberglass-reinforced scrim with kraft-paper backing.

C. PVC Jacket: High-impact-resistant, UV-resistant PVC complying with ASTM D1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Johns Mansville; Zeston.
      b. P.I.C Plastics, Inc.; FG Series.
      c. Proto Corporation; LoSmoke.
      d. Speedline Corporation; SmokeSafe.
   2. Adhesive: As recommended by jacket material manufacturer.
   3. Color: Color-code jackets based on system. Color as selected by Engineer.
   4. Factory- fabricated fitting covers to match jacket if available; otherwise, field fabricate:
      a. Shapes: 45- and 90-degree, short- and long radisu elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, and mechanical joints.

D. Metal Jacket:
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Childers Brand Specialty Construction Brands, Inc., a businees of H.B. Fuller Company; Metal Jacketing Systems.
      b. ITW Insulation Systems; Aluminum and Stainless Steel Jacketing.
      c. RPR Products, Inc.; Insul-Mate.
   2. Aluminum Jacket: Comply with ASTM B 209, Alloy 3003, 3005, 3105, or 5005, Temper H-14. Factory cut and rolled to size. Finish and thicknesses are indicated in field applied jacket schedules.
   3. Stainless-Steel Jacket: ASTM A 167 or ASTM A 240. Factory cut and rolled to size. Finish and thicknesses are indicated in field applied jacket schedules.
   4. Moisture barrier for galvanic corrosion protection:
      a. Moisture Barrier for Indoor Applications: 1-mil-thick, heat-bonded polyethylene and kraft paper.
      b. Moisture Barrier for Outdoor Applications: 3-mil-thick, heat-bonded polyethylene and kraft paper.
      c. Metal Jacket Fitting Covers: Factory-fabricated fitting covers of same material, finish, and thickness as jacket for:

1) Elbows: Pre-formed 2-piece or gore, 45- and 90-degree, short- and long-radius elbows.
2) Tee covers.
3) Flange and union covers.
4) End caps.
5) Beveled collars.
6) Valve colvers.
7) Field fabricate fitting covers only if factory-fabricated fitting covers are not available.

E. Self-Adhesive Outdoor Jackets such as Polyguard Products' Alumaguard 60 which can generally be described as a 60-mil-thick, laminated vapor barrier and waterproofing membrane for installation over insulation located aboveground outdoors; consisting of a rubberized bituminous resin on a crosslaminated polyethylene film covered with aluminum-foil facing are not considered equivalent to Metal Jackets specified above.

2.11 TAPES

A. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C1136.
1. Products: Subject to compliance with requirements, provide one of the following:
a. ABI, Ideal Tape Division; 428 AWF ASJ.
b. Compac Corporation; 104 and 105.
c. Venture Tape; 1540 CW Plus, 1542 CW Plus, and 1542 CW Plus/SQ.
2. Width: 3 inches
3. Thickness: 11.5 mils.
4. Adhesion: 90 ounces force/inchin width.
5. Elongation: 2 percent.
6. Tensile Strength: 40 lbf/inchin width.
7. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.

B. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C1136.
1. Products: Subject to compliance with requirements, provide one of the following:
a. ABI, Ideal Tape Division; 491 AWF FSK.
b. Compac Corporation; 110 and 111.
c. Venture Tape; 1525 CW NT, 1528 CW, and 1528 CW/SQ.
2. Width: 3 inches.
3. Thickness: 6.5 mils.
4. Adhesion: 90 ounces force/inchin width.
5. Elongation: 2 percent.
6. Tensile Strength: 40 lbf/inch in width.
7. FSK Tape Disks and Squares: Precut disks or squares of FSK tape.

C. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.
1. Products: Subject to compliance with requirements, provide one of the following:
a. ABI, Ideal Tape Division; 370 White PVC tape.
b. Compac Corporation; 130.
c. Venture Tape; 1506 CW NS.
2. Width: 2 inches.
3. Thickness: 6 mils.
4. Adhesion: 64 ounces force/inchin width.
5. Elongation: 500 percent.
6. Tensile Strength: 18 lbf/inchin width.

D. Aluminum-Foil Tape: Vapor-retarder tape with acrylic adhesive.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. ABI, Ideal Tape Division; 488 AWF.
      b. Compac Corporation; 120.
      c. Venture Tape; 3520 CW.
   2. Width: 2 inches.
   3. Thickness: 3.7 mils.
   4. Adhesion: 100 ounces force/inch in width.
   5. Elongation: 5 percent.
   6. Tensile Strength: 34 lbf/inch in width.

2.12 SECUREMENTS

A. Bands:
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. ITW Insulation Systems; Gerrard Strapping and Seals.
      b. RPR Products, Inc.; Insul-Mate Strapping, Seals, and Springs.
   2. Stainless Steel: ASTM A 167 or ASTM A240/A240M, Type 304 or Type 316; 0.015 inch thick, 1/2 inch wide with wing seal or closed seal.
   3. Aluminum: ASTM B 209, Alloy 3003, 3005, 3105, or 5005; Temper H-14, 0.020 inch thick, 1/2 inch wide with wing seal or closed seal.
   4. Springs: Twin spring set constructed of stainless steel with ends flat and slotted to accept metal bands. Spring size determined by manufacturer for application.

B. Staples: Outward-clinching insulation staples, nominal 3/4 inchwide, stainless steel or Monel.

C. Wire: 0.062-inch soft-annealed, stainless steel.
   1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
      a. C&F Wire.
      b. Childers Products.
      c. RPR Products, Inc.

2.13 CORNER ANGLES

A. PVC Corner Angles: 30 mils thick, minimum 1 by 1 inch, PVC according to ASTM D 1784, Class 16354-C. White or color-coded to match adjacent surface.

B. Aluminum Corner Angles: 0.040 inch thick, minimum 1 by 1 inch, aluminum according to ASTM B 209, Alloy 3003, 3005, 3105, or 5005; Temper H-14.

C. Stainless-Steel Corner Angles: 0.024 inch thick, minimum 1 by 1 inch, stainless steel according to ASTM A 167 or ASTM A240/A240M, Type 304 or Type 316.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine substrates and conditions for compliance with requirements for installation tolerances and other conditions affecting performance of insulation application.
   1. Verify that systems to be insulated have been tested and are free of defects.
   2. Verify that surfaces to be insulated are clean and dry.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 PREPARATION

A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

3.03 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of ducts and fittings.

B. Install insulation materials, vapor barriers or retarders, jackets, and thicknesses required for each item of duct system as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Keep insulation materials dry during application and finishing.

G. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

H. Install insulation with least number of joints practical.

I. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.
   1. Install insulation continuously through hangers and around anchor attachments.
   2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
   3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.

J. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

K. Install insulation with factory-applied jackets as follows:
   1. Draw jacket tight and smooth.
   2. Cover circumferential joints with 3-inch-wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.
   3. Overlap jacket longitudinal seams at least 1-1/2 inches. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 4 inches o.c.
      a. For below ambient services, apply vapor-barrier mastic over staples.
   4. Cover joints and seams with tape, according to insulation material manufacturer's written instructions, to maintain vapor seal.
   5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to duct flanges and fittings.

L. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.

M. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

N. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.

3.04 PENETRATIONS

A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.
   1. Seal penetrations with flashing sealant.

2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside roof flashing at least 2 inchesbelow top of roof flashing.
4. Seal jacket to roof flashing with flashing sealant.

B. Insulation Installation at Underground Exterior Wall Penetrations: Terminate insulation flush with sleeve seal. Seal terminations with flashing sealant.

C. Insulation Installation at Aboveground Exterior Wall Penetrations: Install insulation continuously through wall penetrations.
1. Seal penetrations with flashing sealant.
2. For applications requiring only indoor insulation, terminate insulation inside wall surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.
3. Extend jacket of outdoor insulation outside wall flashing and overlap wall flashing at least 2 inches.
4. Seal jacket to wall flashing with flashing sealant.

D. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.

E. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Terminate insulation at fire damper sleeves for fire-rated wall and partition penetrations. Externally insulate damper sleeves to match adjacent insulation and overlap duct insulation at least 2 inches.
1. Comply with requirements in Section 07 8413 "Penetration Firestopping"irestopping and fire-resistive joint sealers.

F. Insulation Installation at Floor Penetrations:
1. Duct: For penetrations through fire-rated assemblies, terminate insulation at fire damper sleeves and externally insulate damper sleeve beyond floor to match adjacent duct insulation. Overlap damper sleeve and duct insulation at least 2 inches.
2. Seal penetrations through fire-rated assemblies. Comply with requirements in Section 07 8413 "Penetration Firestopping."

3.05 GENERAL PIPE INSULATION INSTALLATION

A. Requirements in this article generally apply to all insulation materials except where more specific requirements are specified in various pipe insulation material installation articles.

B. Exposed Piping: Locate insulation and cover seams in least visible locations.

C. Insulated pipes conveying fluids below ambient temperature: Insulate entire system including fittings, valves, unions, flanges, strainers, flexible connections, pump bodies, and expansion joints.

D. Butt insulation and seal joints and breaks with vapor barrier tape or sealer and 2 inches of foil adhered over joint.

E. Inserts and Shields:
1. Shields: Galvanized steel between pipe hangers or pipe hanger rolls and inserts.
2. Insert location: Between support shield and piping and under the finish jacket.
3. Insert configuration: Minimum 6 inches long, of same thickness and contour as adjoining insulation; may be factory fabricated.
4. Insert material: Hydrous calcium silicate insulation or other heavy density insulating material suitable for the planned temperature range.

F. Continue insulation through walls, sleeves, pipe hangers, and other pipe penetrations. Finish at supports, protrusions, and interruptions.

G. Exposed piping in Mechanical Equipment Rooms or unfinished spaces less than 7 feet above finished floor: Provide with aluminum jacket.

H. Exposed piping in Public Spaces: Provide with PVC jacket and fitting covers.

I. Insulate all exposed piping, including thermostatic mixing valves, below fixtures scheduled for use by the disabled in accordance with ADA.

J. Exterior Applications: Provide vapor barrier jacket. Insulate fittings, joints, and valves with insulation of like material and thickness as adjoining pipe, and finish with glass mesh reinforced vapor barrier cement. Cover with aluminum jacket with seams located on bottom side of horizontal piping.

K. Insulation Installation on Fittings, Valves, Strainers, Flanges, and Unions:
   1. Install insulation over fittings, valves, strainers, flanges, unions, and other specialties with continuous thermal and vapor-retarder integrity unless otherwise indicated.
   2. Insulate pipe elbows using preformed fitting insulation or mitered fittings made from same material and density as adjacent pipe insulation. Each piece shall be butted tightly against adjoining piece and bonded with adhesive. Fill joints, seams, voids, and irregular surfaces with insulating cement finished to a smooth, hard, and uniform contour that is uniform with adjoining pipe insulation.
   3. Insulate tee fittings with preformed fitting insulation or sectional pipe insulation of same material and thickness as used for adjacent pipe. Cut sectional pipe insulation to fit. Butt each section closely to the next and hold in place with tie wire. Bond pieces with adhesive.
   4. Insulate valves using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. For valves, insulate up to and including the bonnets, valve stuffing-box studs, bolts, and nuts. Fill joints, seams, and irregular surfaces with insulating cement.
   5. Insulate strainers using preformed fitting insulation or sectional pipe insulation of same material, density, and thickness as used for adjacent pipe. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker. Fill joints, seams, and irregular surfaces with insulating cement. Insulate strainers so strainer basket flange or plug can be easily removed and replaced without damaging the insulation and jacket. Provide a removable reusable insulation cover. For below-ambient services, provide a design that maintains vapor barrier.
   6. Insulate flanges and unions using a section of oversized preformed pipe insulation. Overlap adjoining pipe insulation by not less than two times the thickness of pipe insulation, or one pipe diameter, whichever is thicker.
   7. Cover segmented insulated surfaces with a layer of finishing cement and coat with a mastic. Install vapor-barrier mastic for below-ambient services and a breather mastic for above-ambient services. Reinforce the mastic with fabric-reinforcing mesh. Trowel the mastic to a smooth and well-shaped contour.
   8. For services not specified to receive a field-applied jacket except for flexible elastomeric and polyolefin, install fitted PVC cover over elbows, tees, strainers, valves, flanges, and unions. Terminate ends with PVC end caps. Tape PVC covers to adjoining insulation facing using PVC tape.
   9. Label the outside insulation jacket of each union with the word "union." Match size and color of pipe labels.

L. Install removable insulation covers at strainers, flanges, unions, pumps and other items requiring service access. Installation shall conform to the following.

1. Make removable flange and union insulation from sectional pipe insulation of same thickness as that on adjoining pipe. Install same insulation jacket as adjoining pipe insulation.
2. When flange and union covers are made from sectional pipe insulation, extend insulation from flanges or union long at least two times the insulation thickness over adjacent pipe insulation on each side of flange or union. Secure flange cover in place with stainless-steel or aluminum bands. Select band material compatible with insulation and jacket.
3. Construct removable valve insulation covers in same manner as for flanges, except divide the two-part section on the vertical center line of valve body.
4. When covers are made from block insulation, make two halves, each consisting of mitered blocks wired to stainless-steel fabric. Secure this wire frame, with its attached insulation, to flanges with tie wire. Extend insulation at least 2 inches over adjacent pipe insulation on each side of valve. Fill space between flange or union cover and pipe insulation with insulating cement. Finish cover assembly with insulating cement applied in two coats. After first coat is dry, apply and trowel second coat to a smooth finish.
5. Unless a PVC jacket is indicated in field-applied jacket schedules, finish exposed surfaces with a metal jacket.

3.06 INSTALLATION OF FLEXIBLE ELASTOMERIC INSULATION

A. Seal longitudinal seams and end joints with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

B. Insulation Installation on Pipe Flanges:
   1. Install pipe insulation to outer diameter of pipe flange.
   2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
   3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with cut sections of sheet insulation of same thickness as pipe insulation.
   4. Secure insulation to flanges and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

C. Insulation Installation on Pipe Fittings and Elbows:
   1. Install mitered sections of pipe insulation.
   2. Secure insulation materials and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

D. Insulation Installation on Valves and Pipe Specialties:
   1. Install preformed valve covers manufactured of same material as pipe insulation when available.
   2. When preformed valve covers are not available, install cut sections of pipe and sheet insulation to valve body. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
   3. Install insulation to flanges as specified for flange insulation application.
   4. Secure insulation to valves and specialties and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

3.07 INSTALLATION OF MINERAL-FIBER INSULATION

A. Insulation Installation on Straight Pipes and Tubes.
   1. Secure each layer of preformed pipe insulation to pipe with wire or bands and tighten bands without deforming insulation materials.
   2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.
   3. For insulation with factory-applied jackets on above-ambient surfaces, secure laps with outward clinched staples at 6 inches o.c.

4. For insulation with factory-applied jackets on below-ambient surfaces, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Flanges:
1. Install preformed pipe insulation to outer diameter of pipe flange.
2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.
3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with mineral-fiber blanket insulation.
4. Install jacket material with manufacturer's recommended adhesive, overlap seams at least 1 inch, and seal joints with flashing sealant.

C. Insulation Installation on Pipe Fittings and Elbows:
1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed insulation elbows and fittings are not available, install mitered sections of pipe insulation, to a thickness equal to adjoining pipe insulation. Secure insulation materials with wire or bands.

D. Insulation Installation on Valves and Pipe Specialties:
1. Install preformed sections of same material as straight segments of pipe insulation when available.
2. When preformed sections are not available, install mitered sections of pipe insulation to valve body.
3. Arrange insulation to permit access to packing and to allow valve operation without disturbing insulation.
4. Install insulation to flanges as specified for flange insulation application

E. Mineral-fiber insulated pipes conveying fluids below ambient temperature:
1. Provide vapor barrier jackets, factory-applied or field-applied. Secure with self-sealing longitudinal laps and butt strips with pressure sensitive adhesive. Secure with outward clinch expanding staples and vapor barrier mastic.
2. Insulate fittings, joints, and valves with molded insulation of like material and thickness as adjacent pipe. Finish with glass cloth and vapor barrier adhesive or PVC fitting covers.

F. Mineral fiber insulated pipes conveying fluids above ambient temperature:
1. Provide standard jackets, with or without vapor barrier, factory-applied or field-applied. Secure with self-sealing longitudinal laps and butt strips with pressure sensitive adhesive. Secure with outward clinch expanding staples.
2. Insulate fittings, joints, and valves with insulation of like material and thickness as adjoining pipe. Finish with glass cloth and adhesive or PVC fitting covers.

3.08 FIELD-APPLIED JACKET INSTALLATION

A. Where glass-cloth jackets are indicated, install directly over bare insulation or insulation with factory-applied jackets.
1. Draw jacket smooth and tight to surface with 2-inch overlap at seams and joints.
2. Embed glass cloth between two 0.062-inch-thick coats of lagging adhesive.
3. Completely encapsulate insulation with coating, leaving no exposed insulation.

B. Where FSK jackets are indicated, install as follows:
1. Draw jacket material smooth and tight.
2. Install lap or joint strips with same material as jacket.
3. Secure jacket to insulation with manufacturer's recommended adhesive.
4. Install jacket with 1-1/2 inch laps at longitudinal seams and 3-inch-wide joint strips at end joints.

5. Seal openings, punctures, and breaks in vapor-retarder jackets and exposed insulation with vapor-barrier mastic.

C. Where PVC jackets are indicated, install with 1-inch overlap at longitudinal seams and end joints; for horizontal applications, install with longitudinal seams along top and bottom of tanks and vessels. Seal with manufacturer's recommended adhesive.
   1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.

D. Where metal jackets are indicated, install with 2-inch overlap at longitudinal seams and end joints. Overlap longitudinal seams arranged to shed water. Seal end joints with weatherproof sealant recommended by insulation manufacturer. Secure jacket with stainless-steel bands 12 inches o.c. and at end joints.

3.09 FINISHES

A. Equipment and Pipe Insulation with ASJ, Glass-Cloth, or Other Paintable Jacket Material: Paint jacket with paint system identified below and as specified in Section 09 9113 "Exterior Painting" and Section 09 9123 "Interior Painting."
   1. Flat Acrylic Finish: Two finish coats over a primer that is compatible with jacket material and finish coat paint. Add fungicidal agent to render fabric mildew proof.
      a. Finish Coat Material: Interior, flat, latex-emulsion size.

B. Flexible Elastomeric Thermal Insulation: After adhesive has fully cured, apply two coats of insulation manufacturer's recommended protective coating.

C. Color: Final color as selected by Engineer. Vary first and second coats to allow visual inspection of the completed Work.

D. Do not field paint aluminum or stainless-steel jackets.

3.10 FIELD QUALITY CONTROL

A. Tests and Inspections:
   1. Inspect field-insulated equipment, randomly selected by Engineer, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location(s) for each type of equipment.
   2. Inspect pipe, fittings, strainers, and valves, randomly selected by Engineer, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to three locations of straight pipe, two locations of threaded fittings, two locations of welded fittings, two locations of threaded strainers, two locations of welded strainers, two locations of threaded valves, and two locations of flanged valves for each pipe service identified in the "Piping Insulation Schedule, General" Article.

B. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.11 PIPING INSULATION SCHEDULE, GENERAL

A. Acceptable preformed pipe and tubular insulation materials and thicknesses are identified for each piping system and pipe size range. If more than one material is listed for a piping system, selection from materials listed is Contractor's option.

B. Equipment shall be insulated to equivalent thicknesses a specified for the associated piping system.

C. Items Not Insulated: Unless otherwise indicated, do not install insulation on the following:
   1. Drainage piping located in crawl spaces.
   2. Underground piping.
   3. Chrome-plated pipes and fittings unless there is a potential for personnel injury.

3.12 INDOOR PIPING INSULATION SCHEDULE

A. Condensate and Equipment Drain Water from cooling coils, all pipe sizes:

1. Flexible Elastomeric:
    a. NPS smaller than 1-1/2 inch: Minimum installed R-Value of 2.0 (Minimum of 3/4 inch thick), whichever is greater
    b. NPS 1-1/2 inch and larger: Minimum installed R-Value of 5.0 (Minimum of 1 inch thick), whichever is greater

B. Chilled water piping (above 40 deg F to 60 de g):
1. NPS smaller than 1-1/2 inch:
    a. Flexible Elastomeric: Minimum installed R-Value of 2.0 (Minimum of 3/4 inch thick), whichever is greater
    b. Mineral-Fiber, Preformed Pipe Insulation, Type I: Minimum installed R-Value of 2.0 (Minimum of 1 inch thick), whichever is greater
2. NPS 1-1/2 inch and larger:
    a. Flexible Elastomeric: Minimum installed R-Value of 5.0 (Minimum of 1 inch thick), whichever is greater
    b. Mineral-Fiber, Preformed Pipe Insulation, Type I: Minimum installed R-Value of 5.0 (Minimum of 1-1/2 inches thick), whichever is greater

C. Refrigerant Piping
1. Suction piping operating below ambient: Minimum installed R-Value of 5.0 (Minimum of 1 inch thick), whichever is greater.
2. For VRF systems insulate per manufacturer's recommendation for actual service conditions.
3. Insulate branch selectors per manunufacturer's recommendation.

3.13 OUTDOOR PIPING INSULATION SCHEDULE

A. Chilled Water Piping: Mineral-Fiber, Preformed Pipe Insulation, Type I: 4 inches thick or Flexible Elastomeric: Two layers 2-inches thick each for a total of 4 inches thick.

B. Suction piping operating below ambient: 2 inches thick.

3.14 INDOOR, FIELD-APPLIED JACKET SCHEDULE

A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.

B. If more than one material is listed, selection from materials listed is Contractor's option.

C. Piping exposed in mechanical rooms and subject to abrasion. All piping, ductwork and equipment below an elevation of 8 feet above finished floor shall be considered subject to abrasion.
1. PVC, white: 20 mils thick

3.15 OUTDOOR, FIELD-APPLIED JACKET SCHEDULE

A. All Outdoors Services - Provide 0.016" stucco embossed aluminum or Type 304 stainless steel jacket in addition to specified factory-applied jacket. Lap seams in a manner to not permit water entry and seal water tight.

END OF SECTION 23 0711

SECTION 23 0716
HVAC EQUIPMENT INSULATION

PART 1 GENERAL

1.01 REFERENCE STANDARDS

A. 40 CFR 59, Subpart D - National Volatile Organic Compound Emission Standards for Architectural Coatings; U.S. Environmental Protection Agency; current edition.

B. ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications; 2024b.

C. ASTM B209/B209M - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate; 2021a.

D. ASTM C195 - Standard Specification for Mineral Fiber Thermal Insulating Cement; 2007 (Reapproved 2019).

E. ASTM C449 - Standard Specification for Mineral Fiber Hydraulic-Setting Thermal Insulating and Finishing Cement; 2007 (Reapproved 2013).

F. ASTM C450 - Standard Practice for Fabrication of Thermal Insulating Fitting Covers for NPS Piping, and Vessel Lagging; 2018 (Reapproved 2026).

G. ASTM C533 - Standard Specification for Calcium Silicate Block and Pipe Thermal Insulation; 2017.

H. ASTM C547 - Standard Specification for Mineral Fiber Pipe Insulation; 2022.

I. ASTM C552 - Standard Specification for Cellular Glass Thermal Insulation; 2022.

J. ASTM C553 - Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications; 2013 (Reapproved 2019).

K. ASTM C578 - Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation; 2019.

L. ASTM C585 - Standard Practice for Inner and Outer Diameters of Thermal Insulation for Nominal Sizes of Pipe and Tubing; 2022.

M. ASTM C591 - Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation; 2022.

N. ASTM C612 - Standard Specification for Mineral Fiber Block and Board Thermal Insulation; 2014 (Reapproved 2019).

O. ASTM C795 - Standard Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel; 2008 (Reapproved 2018).

P. ASTM C871 - Standard Test Methods for Chemical Analysis of Thermal Insulation Materials for Leachable Chloride, Fluoride, Silicate, and Sodium Ions; 2018.

Q. ASTM C1126 - Standard Specification for Faced or Unfaced Rigid Cellular Phenolic Thermal Insulation; 2019.

R. ASTM C1136 - Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation; 2023.

S. ASTM C1290 - Standard Specification for Flexible Fibrous Glass Blanket Insulation Used to Externally Insulate HVAC Ducts; 2016 (Reapproved 2021).

T. ASTM C1427 - Standard Specification for Extruded Preformed Flexible Cellular Polyolefin Thermal Insulation in Sheet and Tubular Form; 2021.

U. ASTM D1784 - Standard Classification System and Basis for Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds; 2020.

V. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2023b.

W. ASTM E96/E96M - Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials; 2022.

X. ASTM F1249 - Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor; 2020.

Y. NFPA 90A - Standard for the Installation of Air-Conditioning and Ventilating Systems; 2021.

Z. NFPA 90B - Standard for the Installation of Warm Air Heating and Air-Conditioning Systems; 2024.

1.02 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.03 SUMMARY

A. Section includes insulating the following HVAC equipment that is not factory insulated:
   1. Chilled-water pumps.
   2. Expansion/compression tanks.
   3. Air separators.
   4. Buffer Tanks.

B. Related Sections:
   1. 23 0712 - HVAC Insulation

1.04 ACTION SUBMITTALS

A. Product Data: For each type of product indicated. Include thermal conductivity, water-vapor permeance thickness, and jackets (both factory- and field-applied if any).

B. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
   1. Detail application of protective shields, saddles, and inserts at hangers for each type of insulation and hanger.
   2. Detail attachment and covering of heat tracing inside insulation.
   3. Detail removable insulation at equipment connections.
   4. Detail application of field-applied jackets.
   5. Detail application at linkages of control devices.
   6. Detail field application for each equipment type.

C. Samples: For each type of insulation and jacket indicated. Identify each Sample, describing product and intended use. Sample sizes are as follows:
   1. Preformed Pipe Insulation Materials: 12 inches long by NPS 2 (DN 50).
   2. Sheet Form Insulation Materials: 12 inches square.
   3. Sheet Jacket Materials: 12 inches square.
   4. Manufacturer's Color Charts: For products where color is specified, show the full range of colors available for each type of finish material.

1.05 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified Installer.

B. Material Test Reports: From a qualified testing agency acceptable to authorities having jurisdiction indicating, interpreting, and certifying test results for compliance of insulation materials, sealers, attachments, cements, and jackets, with requirements indicated. Include dates of tests and test methods employed.

C. Field quality-control reports.

1.06 QUALITY ASSURANCE

A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.

B. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E84, by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.
   1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.
   2. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Packaging: Insulation material containers shall be marked by manufacturer with appropriate ASTM standard designation, type and grade, and maximum use temperature.

1.08 COORDINATION

A. Coordinate sizes and locations of supports, hangers, and insulation shields specified in section 23 0529 - Hangers and Supports for HVAC.

B. Coordinate clearance requirements with equipment Installer for equipment insulation application.

C. Coordinate installation and testing of heat tracing.

1.09 SCHEDULING

A. Schedule insulation application after pressure testing systems and, where required, after installing and testing heat tracing. Insulation application may begin on segments that have satisfactory test results.

B. Complete installation and concealment of plastic materials as rapidly as possible in each area of construction.

PART 2 PRODUCTS

2.01 INSULATION MATERIALS

A. Comply with requirements in "Equipment Insulation Schedule"articles for where insulating materials shall be applied.

B. Products shall not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C871.

D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C795.

E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.

F. Flexible Elastomeric Insulation: Closed-cell, sponge- or expanded-rubber materials. Comply with ASTM C 534, Type I for tubular materials and Type II for sheet materials.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Aeroflex USA, Inc.; Aerocel.
      b. Armacell LLC; AP Armaflex.
      c. K-Flex USA; Insul-Sheet and K-FLEX LS.

G. Mineral-Fiber Board Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C612, Type IA or Type IB. Provide insulation with factory-applied ASJ. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. CertainTeed Corp.; CertaPro Commercial Board.
      b. Fibrex Insulations Inc.; FBX.
      c. Johns Manville; 800 Series Spin-Glas.

d. Knauf Insulation; Insulation Board.
e. Manson Insulation Inc.; AK Board.
f. Owens Corning; Fiberglas 700 Series.

H. Mineral-Fiber, Pipe and Tank Insulation: Mineral or glass fibers bonded with a thermosetting resin. Semirigid board material with factory-applied ASJ complying with ASTM C 1393, Type II or Type IIIA Category 2, or with properties similar to ASTM C612, Type IB. Nominal density is 2.5 lb/cu. ft. (40 kg/cu. m) or more. Thermal conductivity (k-value) at 100 degrees Fahrenheit is 0.29 Btu x in./h x sq. ft. x deg F (0.042 W/m x K) or less. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.
1. Products: Subject to compliance with requirements, provide one of the following:
a. CertainTeed Corp.; CrimpWrap.
b. Johns Manville; MicroFlex.
c. Knauf Insulation; Pipe and Tank Insulation.
d. Manson Insulation Inc.; AK Flex.
e. Owens Corning; Fiberglas Pipe and Tank Insulation.

2.02 INSULATING CEMENTS

A. Mineral-Fiber Insulating Cement: Comply with ASTM C195.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Ramco Insulation, Inc.; Super-Stik.

B. Expanded or Exfoliated Vermiculite Insulating Cement: Comply with ASTM C 196.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Ramco Insulation, Inc.; Thermokote V.

C. Mineral-Fiber, Hydraulic-Setting Insulating and Finishing Cement: Comply with ASTM C449.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Ramco Insulation, Inc.; Ramcote 1200 and Quik-Cote.

2.03 ADHESIVES

A. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated unless otherwise indicated.

B. Flexible Elastomeric and Polyolefin Adhesive: Comply with MIL-A-24179A, Type II, Class I.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Aeroflex USA, Inc.; Aeroseal.
b. Armacell LLC; Armaflex 520 Adhesive.
c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 85-75.
d. K-Flex USA; R-373 Contact Adhesive.
2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

C. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-127.
b. Eagle Bridges - Marathon Industries; 225.
c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 85-60/85-70.
d. Mon-Eco Industries, Inc.; 22-25.
2. For indoor applications, adhesive shall have a VOC content of 80 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

D. ASJ Adhesive, and FSK and PVDC Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-82.
      b. Eagle Bridges - Marathon Industries; 225.
      c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 85-50.
      d. Mon-Eco Industries, Inc.; 22-25.
   2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
   3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

E. PVC Jacket Adhesive: Compatible with PVC jacket.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Dow Corning Corporation; 739, Dow Silicone.
      b. Johns Manville; Zeston Perma-Weld, CEEL-TITE Solvent Welding Adhesive.
      c. P.I.C. Plastics, Inc.; Welding Adhesive.
      d. Speedline Corporation; Polyco VP Adhesive.
   2. For indoor applications, adhesive shall have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
   3. Adhesive shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

2.04 MASTICS

A. Materials shall be compatible with insulation materials, jackets, and substrates; comply with MIL-PRF-19565C, Type II.
   1. For indoor applications, use mastics that have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

B. Vapor-Barrier Mastic: Water based; suitable for indoor and outdoor use on below ambient services.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Foster Brand, Specialty Construction Brands, Inc., a business of H .B. Fuller Company; 30-80/30-90.
      b. Vimasco Corporation; 749.
   2. Water-Vapor Permeance: ASTM E96/E96M, Procedure B, 0.013 perm (0.009 metric perm) at 43-mil (1.09-mm) dry film thickness.
   3. Service Temperature Range: Minus 20 to plus 180 degrees Fahrenheit.
   4. Solids Content: ASTM D 1644, 58 percent by volume and 70 percent by weight.
   5. Color: White.

C. Vapor-Barrier Mastic: Solvent based; suitable for indoor use on below ambient services.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-30.
      b. Eagle Bridges - Marathon Industries; 501.
      c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 30-35.

d. Mon-Eco Industries, Inc.; 55-10.
2. Water-Vapor Permeance: ASTM F1249, 0.05 perm (0.03 metric perm) at 35-mil (0.9-mm) dry film thickness.
3. Service Temperature Range: 0 to 180 degrees Fahrenheit.
4. Solids Content: ASTM D 1644, 44 percent by volume and 62 percent by weight.
5. Color: White.

D. Vapor-Barrier Mastic: Solvent based; suitable for outdoor use on below ambient services.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; Encacel.
b. Eagle Bridges - Marathon Industries; 570.
c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 60-95/60-96.
2. Water-Vapor Permeance: ASTM F1249, 0.05 perm (0.033 metric perm) at 30-mil (0.8-mm) dry film thickness.
3. Service Temperature Range: Minus 50 to plus 220 degrees Fahrenheit.
4. Solids Content: ASTM D 1644, 33 percent by volume and 46 percent by weight.
5. Color: White.

E. Breather Mastic: Water based; suitable for indoor and outdoor use on above ambient services.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-10.
b. Eagle Bridges - Marathon Industries; 550.
c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 46-50.
d. Mon-Eco Industries, Inc.; 55-50.
e. Vimasco Corporation; WC-1/WC-5.
2. Water-Vapor Permeance: ASTM F1249, 1.8 perms (1.2 metric perms) at 0.0625-inch (1.6-mm) dry film thickness.
3. Service Temperature Range: Minus 20 to plus 180 degrees Fahrenheit.
4. Solids Content: 60 percent by volume and 66 percent by weight.
5. Color: White.

2.05 LAGGING ADHESIVES

A. Description: Comply with MIL-A-3316C, Class I, Grade A and shall be compatible with insulation materials, jackets, and substrates.
1. For indoor applications, use lagging adhesives that have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
2. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-50 AHV2.
b. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 30-36.
c. Vimasco Corporation; 713 and 714.
3. Fire-resistant, water-based lagging adhesive and coating for use indoors to adhere fire-resistant lagging cloths over equipment insulation.
4. Service Temperature Range: 0 to plus 180 degrees Fahrenheit.
5. Color: White.

2.06 SEALANTS

A. Joint Sealants:

1. Joint Sealants for Cellular-Glass, Phenolic, and Polyisocyanurate Products: Subject to compliance with requirements, provide one of the following:
   a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-76.
   b. Eagle Bridges - Marathon Industries; 405.
   c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 30-45.
   d. Mon-Eco Industries, Inc.; 44-05.
   e. Pittsburgh Corning Corporation; Pittseal 444.
2. Joint Sealants for Polystyrene Products: Subject to compliance with requirements, provide one of the following:
   a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-70.
   b. Eagle Bridges - Marathon Industries; 405.
   c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 30-45.
   d. Mon-Eco Industries, Inc.; 44-05.
3. Materials shall be compatible with insulation materials, jackets, and substrates.
4. Permanently flexible, elastomeric sealant.
5. Service Temperature Range: Minus 100 to plus 300 degrees Fahrenheit.
6. Color: White or gray.
7. For indoor applications, sealants shall have a VOC content of 420 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
8. Sealants shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

B. FSK and Metal Jacket Flashing Sealants:
1. Products: Subject to compliance with requirements, provide one of the following:
   a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-76.
   b. Eagle Bridges - Marathon Industries; 405.
   c. Foster Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; 95-44.
   d. Mon-Eco Industries, Inc.; 44-05.
2. Materials shall be compatible with insulation materials, jackets, and substrates.
3. Fire- and water-resistant, flexible, elastomeric sealant.
4. Service Temperature Range: Minus 40 to plus 250 degrees Fahrenheit.
5. Color: Aluminum.
6. For indoor applications, sealants shall have a VOC content of 420 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
7. Sealants shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

C. ASJ Flashing Sealants, and Vinyl, PVDC, and PVC Jacket Flashing Sealants:
1. Products: Subject to compliance with requirements, provide one of the following:
   a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; CP-76.
2. Materials shall be compatible with insulation materials, jackets, and substrates.
3. Fire- and water-resistant, flexible, elastomeric sealant.
4. Service Temperature Range: Minus 40 to plus 250 degrees Fahrenheit.
5. Color: White.

6. For indoor applications, sealants shall have a VOC content of 420 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).
7. Sealants shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

2.07 FACTORY-APPLIED JACKETS

A. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:
1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C1136, Type I.
2. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C1136, Type I.
3. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C1136, Type II.
4. FSP Jacket: Aluminum-foil, fiberglass-reinforced scrim with polyethylene backing; complying with ASTM C1136, Type II.
5. Vinyl Jacket: White vinyl with a permeance of 1.3 perms (0.86 metric perm) when tested according to ASTM E96/E96M, Procedure A, and complying with NFPA 90A and NFPA 90B.

2.08 FIELD-APPLIED JACKETS

A. Field-applied jackets shall comply with ASTM C 921, Type I, unless otherwise indicated.

B. FSK Jacket: Aluminum-foil-face, fiberglass-reinforced scrim with kraft-paper backing.

C. PVC Jacket: High-impact-resistant, UV-resistant PVC complying with ASTM D1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.
1. Products: Subject to compliance with requirements, provide one of the following:
a. Johns Manville; Zeston.
b. P.I.C. Plastics, Inc.; FG Series.
c. Proto Corporation; LoSmoke.
d. Speedline Corporation; SmokeSafe.
2. Adhesive: As recommended by jacket material manufacturer.
3. Color: White.
4. Factory-fabricated tank heads and tank side panels.

D. Metal Jacket:
1. Products: Subject to compliance with requirements, provide one of the following:
a. Childers Brand, Specialty Construction Brands, Inc., a business of H. B. Fuller Company; Metal Jacketing Systems.
b. ITW Insulation Systems; Aluminum and Stainless Steel Jacketing.
c. RPR Products, Inc.; Insul-Mate.
2. Aluminum Jacket: Comply with ASTM B209/B209M, Alloy 3003, 3005, 3105, or 5005, Temper H-14.
a. Sheet and roll stock ready for shop or field sizing.
b. Finish and thickness are indicated in field-applied jacket schedules.
c. Moisture Barrier for Indoor Applications: 3-mil- (0.075-mm-) thick, heat-bonded polyethylene and kraft paper.
d. Moisture Barrier for Outdoor Applications: 3-mil- (0.075-mm-) thick, heat-bonded polyethylene and kraft paper.
e. Factory-Fabricated Fitting Covers:
1) Same material, finish, and thickness as jacket.
2) Preformed two-piece or gore, 45- and 90-degree, short- and long-radius elbows.

3) Tee covers.
4) Flange and union covers.
5) End caps.
6) Beveled collars.
7) Valve covers.
8) Field fabricate fitting covers only if factory-fabricated fitting covers are not available.

3. Stainless-Steel Jacket: ASTM A 167 or ASTM A240/A240M.
   a. Sheet and roll stock ready for shop or field sizing.
   b. Material, finish, and thickness are indicated in field-applied jacket schedules.
   c. Moisture Barrier for Indoor Applications: 3-mil- (0.075-mm-) thick, heat-bonded polyethylene and kraft paper.
   d. Moisture Barrier for Outdoor Applications: 3-mil- (0.075-mm-) thick, heat-bonded polyethylene and kraft paper.
   e. Factory-Fabricated Fitting Covers:
      1) Same material, finish, and thickness as jacket.
      2) Preformed two-piece or gore, 45- and 90-degree, short- and long-radius elbows.
      3) Tee covers.
      4) Flange and union covers.
      5) End caps.
      6) Beveled collars.
      7) Valve covers.
      8) Field fabricate fitting covers only if factory-fabricated fitting covers are not available.

2.09 TAPES

A. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C1136.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. ABI, Ideal Tape Division; 428 AWF ASJ.
      b. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0836.
      c. Compac Corporation; 104 and 105.
      d. Venture Tape; 1540 CW Plus, 1542 CW Plus, and 1542 CW Plus/SQ.
   2. Width: 3 inches.
   3. Thickness: 11.5 mils (0.29 mm).
   4. Adhesion: 90 ounces force/inch (1.0 N/mm) in width.
   5. Elongation: 2 percent.
   6. Tensile Strength: 40 lbf/inch (7.2 N/mm) in width.
   7. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.

B. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C1136.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. ABI, Ideal Tape Division; 491 AWF FSK.
      b. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0827.
      c. Compac Corporation; 110 and 111.
      d. Venture Tape; 1525 CW NT, 1528 CW, and 1528 CW/SQ.
   2. Width: 3 inches.
   3. Thickness: 6.5 mils (0.16 mm).
   4. Adhesion: 90 ounces force/inch (1.0 N/mm) in width.
   5. Elongation: 2 percent.
   6. Tensile Strength: 40 lbf/inch (7.2 N/mm) in width.
   7. FSK Tape Disks and Squares: Precut disks or squares of FSK tape.

C. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. ABI, Ideal Tape Division; 370 White PVC tape.
      b. Compac Corporation; 130.
      c. Venture Tape; 1506 CW NS.
   2. Width: 2 inches.
   3. Thickness: 6 mils (0.15 mm).
   4. Adhesion: 64 ounces force/inch (0.7 N/mm) in width.
   5. Elongation: 500 percent.
   6. Tensile Strength: 18 lbf/inch (3.3 N/mm) in width.

D. Aluminum-Foil Tape: Vapor-retarder tape with acrylic adhesive.
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. ABI, Ideal Tape Division; 488 AWF.
      b. Avery Dennison Corporation, Specialty Tapes Division; Fasson 0800.
      c. Compac Corporation; 120.
      d. Venture Tape; 3520 CW.
   2. Width: 2 inches.
   3. Thickness: 3.7 mils (0.093 mm).
   4. Adhesion: 100 ounces force/inch (1.1 N/mm) in width.
   5. Elongation: 5 percent.
   6. Tensile Strength: 34 lbf/inch (6.2 N/mm) in width.

2.10 SECUREMENTS

A. Bands:
   1. Products: Subject to compliance with requirements, provide one of the following:
      a. ITW Insulation Systems; Gerrard Strapping and Seals.
      b. RPR Products, Inc.; Insul-Mate Strapping, Seals, and Springs.
   2. Stainless Steel: ASTM A 167 or ASTM A240/A240M, Type 304 or Type 316; 0.015 inch thick, 3/4 inch wide with wing seal or closed seal.
   3. Aluminum: ASTM B 209 (ASTM B 209M), Alloy 3003, 3005, 3105, or 5005; Temper H-14, 0.020 inch thick, 3/4 inch wide with wing seal or closed seal.
   4. Springs: Twin spring set constructed of stainless steel with ends flat and slotted to accept metal bands. Spring size determined by manufacturer for application.

B. Insulation Pins and Hangers:
   1. Capacitor-Discharge-Weld Pins: Copper- or zinc-coated steel pin, fully annealed for capacitor-discharge welding, 0.135-inch- (3.5-mm-) diameter shank, length to suit depth of insulation indicated.
      a. Products: Subject to compliance with requirements, provide one of the following:
         1) AGM Industries, Inc.; CWP-1.
         2) GEMCO; CD.
         3) Midwest Fasteners, Inc.; CD.
         4) Nelson Stud Welding; TPA, TPC, and TPS.
   2. Cupped-Head, Capacitor-Discharge-Weld Pins: Copper- or zinc-coated steel pin, fully annealed for capacitor-discharge welding, 0.135-inch- (3.5-mm-) diameter shank, length to suit depth of insulation indicated with integral 1-1/2-inch (38-mm) galvanized carbon-steel washer.
      a. Products: Subject to compliance with requirements, provide one of the following:
         1) AGM Industries, Inc.; CHP-1.
         2) GEMCO; Cupped Head Weld Pin.
         3) Midwest Fasteners, Inc.; Cupped Head.
         4) Nelson Stud Welding; CHP.

3. Metal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place.
   a. Products: Subject to compliance with requirements, provide one of the following:
      1) AGM Industries, Inc.; Tactoo Perforated Base Insul-Hangers.
      2) GEMCO; Perforated Base.
      3) Midwest Fasteners, Inc.; Spindle.
   b. Baseplate: Perforated, galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
   c. Spindle: Copper- or zinc-coated, low-carbon steel, Aluminum, or Stainless steel, fully annealed, 0.106-inch- (2.6-mm-) diameter shank, length to suit depth of insulation indicated.
   d. Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.
4. Nonmetal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate fastened to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place.
   a. Products: Subject to compliance with requirements, provide one of the following:
      1) GEMCO; Nylon Hangers.
      2) Midwest Fasteners, Inc.; Nylon Insulation Hangers.
   b. Baseplate: Perforated, nylon sheet, 0.030 inch thick by 1-1/2 inches in diameter.
   c. Spindle: Nylon, 0.106-inch- (2.6-mm-) diameter shank, length to suit depth of insulation indicated, up to 2-1/2 inches.
   d. Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.
5. Self-Sticking-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place.
   a. Products: Subject to compliance with requirements, provide one of the following:
      1) AGM Industries, Inc.; Tactoo Self-Adhering Insul-Hangers, Series.
      2) GEMCO; Peel & Press.
      3) Midwest Fasteners, Inc.; Self Stick.
   b. Baseplate: Galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.
   c. Spindle: Copper- or zinc-coated, low-carbon steel, fully annealed, 0.106-inch- (2.6-mm-) diameter shank, length to suit depth of insulation indicated.
   d. Adhesive-backed base with a peel-off protective cover.
6. Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch- (0.41-mm-) thick, stainless-steel sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.
   a. Products: Subject to compliance with requirements, provide one of the following:
      1) AGM Industries, Inc.; RC-150.
      2) GEMCO; R-150.
      3) Midwest Fasteners, Inc.; WA-150.
      4) Nelson Stud Welding; Speed Clips.
   b. Protect ends with capped self-locking washers incorporating a spring steel insert to ensure permanent retention of cap in exposed locations.
7. Nonmetal Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch- (0.41-mm-) thick nylon sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.
   a. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1) GEMCO.
2) Midwest Fasteners, Inc.

C. Staples: Outward-clinching insulation staples, nominal 3/4-inch- (19-mm-) wide, stainless steel or Monel.

D. Wire: 0.062-inch (1.6-mm) soft-annealed, stainless steel.
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
a. C & F Wire.

2.11 CORNER ANGLES

A. PVC Corner Angles: 30 mils (0.03 inch) thick, minimum 1 by 1 inch (25 by 25 mm), PVC according to ASTM D1784, Class 16354-C. White or color-coded to match adjacent surface.

B. Aluminum Corner Angles: 0.040 inch thick, minimum 1 by 1 inch (25 by 25 mm), aluminum according to ASTM B 209 (ASTM B 209M), Alloy 3003, 3005, 3105, or 5005; Temper H-14.

C. Stainless-Steel Corner Angles: 0.024 inch thick, minimum 1 by 1 inch (25 by 25 mm), stainless steel according to ASTM A 167 or ASTM A240/A240M, Type 304 or Type 316.

PART 3 EXECUTION

3.01 EXAMINATION

A. Examine substrates and conditions for compliance with requirements for installation tolerances and other conditions affecting performance of insulation application.
1. Verify that systems and equipment to be insulated have been tested and are free of defects.
2. Verify that surfaces to be insulated are clean and dry.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 PREPARATION

A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

B. Surface Preparation: Clean and prepare surfaces to be insulated. Before insulating, apply a corrosion coating to insulated surfaces as follows:
1. Stainless Steel: Coat 300 series stainless steel with an epoxy primer 5 mils (0.127 mm) thick and an epoxy finish 5 mils (0.127 mm) thick if operating in a temperature range between 140 and 300 degrees Fahrenheit. Consult coating manufacturer for appropriate coating materials and application methods for operating temperature range.
2. Carbon Steel: Coat carbon steel operating at a service temperature between 32 and 300 degrees Fahrenheit with an epoxy coating. Consult coating manufacturer for appropriate coating materials and application methods for operating temperature range.

C. Coordinate insulation installation with the trade installing heat tracing. Comply with requirements for heat tracing that apply to insulation.

D. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless-steel surfaces, use demineralized water.

3.03 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of equipment.

B. Install insulation materials, forms, vapor barriers or retarders, jackets, and thicknesses required for each item of equipment as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Keep insulation materials dry during application and finishing.

G. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

H. Install insulation with least number of joints practical.

I. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.
   1. Install insulation continuously through hangers and around anchor attachments.
   2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.
   3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.
   4. Cover inserts with jacket material matching adjacent insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.

J. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

K. Install insulation with factory-applied jackets as follows:
   1. Draw jacket tight and smooth.
   2. Cover circumferential joints with 3-inch- (75-mm-) wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.
   3. Overlap jacket longitudinal seams at least 1-1/2 inches. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 4 inches o.c.
      a. For below ambient services, apply vapor-barrier mastic over staples.
   4. Cover joints and seams with tape, according to insulation material manufacturer's written instructions, to maintain vapor seal.
   5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints.

L. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.

M. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

N. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.

O. For above ambient services, do not install insulation to the following:
   1. Vibration-control devices.
   2. Testing agency labels and stamps.
   3. Nameplates and data plates.
   4. Manholes.
   5. Handholes.
   6. Cleanouts.

3.04 INSTALLATION OF EQUIPMENT, TANK, AND VESSEL INSULATION

A. Mineral-Fiber, Pipe and Tank Insulation Installation for Tanks and Vessels: Secure insulation with adhesive and anchor pins and speed washers.
   1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 75 percent coverage of tank and vessel surfaces.

2. Groove and score insulation materials to fit as closely as possible to equipment, including contours. Bevel insulation edges for cylindrical surfaces for tight joints. Stagger end joints.
3. Protect exposed corners with secured corner angles.
4. Install adhesively attached or self-sticking insulation hangers and speed washers on sides of tanks and vessels as follows:
    a. Do not weld anchor pins to ASME-labeled pressure vessels.
    b. Select insulation hangers and adhesive that are compatible with service temperature and with substrate.
    c. On tanks and vessels, maximum anchor-pin spacing is 3 inches from insulation end joints, and 16 inches o.c. in both directions.
    d. Do not overcompress insulation during installation.
    e. Cut and miter insulation segments to fit curved sides and domed heads of tanks and vessels.
    f. Impale insulation over anchor pins and attach speed washers.
    g. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.
5. Secure each layer of insulation with stainless-steel or aluminum bands. Select band material compatible with insulation materials.
6. Where insulation hangers on equipment and vessels are not permitted or practical and where insulation support rings are not provided, install a girdle network for securing insulation. Stretch prestressed aircraft cable around the diameter of vessel and make taut with clamps, turnbuckles, or breather springs. Place one circumferential girdle around equipment approximately 6 inches from each end. Install wire or cable between two circumferential girdles 12 inches o.c. Install a wire ring around each end and around outer periphery of center openings, and stretch prestressed aircraft cable radially from the wire ring to nearest circumferential girdle. Install additional circumferential girdles along the body of equipment or tank at a minimum spacing of 48 inches o.c. Use this network for securing insulation with tie wire or bands.
7. Stagger joints between insulation layers at least 3 inches.
8. Install insulation in removable segments on equipment access doors, manholes, handholes, and other elements that require frequent removal for service and inspection.
9. Bevel and seal insulation ends around manholes, handholes, ASME stamps, and nameplates.
10. For equipment with surface temperatures below ambient, apply mastic to open ends, joints, seams, breaks, and punctures in insulation.

B. Flexible Elastomeric Thermal Insulation Installation for Tanks and Vessels: Install insulation over entire surface of tanks and vessels.
1. Apply 100 percent coverage of adhesive to surface with manufacturer's recommended adhesive.
2. Seal longitudinal seams and end joints.

C. Insulation Installation on Pumps:
1. Fabricate metal boxes lined with insulation. Fit boxes around pumps and coincide box joints with splits in pump casings. Fabricate joints with outward bolted flanges. Bolt flanges on 6-inch (150-mm) centers, starting at corners. Install 3/8-inch- (10-mm-) diameter fasteners with wing nuts. Alternatively, secure the box sections together using a latching mechanism.
2. Fabricate boxes from stainless steel, at least 0.050 inch thick.
3. For below ambient services, install a vapor barrier at seams, joints, and penetrations. Seal between flanges with replaceable gasket material to form a vapor barrier.

3.05 FIELD-APPLIED JACKET INSTALLATION

A. Where glass-cloth jackets are indicated, install directly over bare insulation or insulation with factory-applied jackets.
   1. Draw jacket smooth and tight to surface with 2-inch (50-mm) overlap at seams and joints.
   2. Embed glass cloth between two 0.062-inch- (1.6-mm-) thick coats of lagging adhesive.
   3. Completely encapsulate insulation with coating, leaving no exposed insulation.

B. Where FSK jackets are indicated, install as follows:
   1. Draw jacket material smooth and tight.
   2. Install lap or joint strips with same material as jacket.
   3. Secure jacket to insulation with manufacturer's recommended adhesive.
   4. Install jacket with 1-1/2-inch (38-mm) laps at longitudinal seams and 3-inch- (75-mm-) wide joint strips at end joints.
   5. Seal openings, punctures, and breaks in vapor-retarder jackets and exposed insulation with vapor-barrier mastic.

C. Where PVC jackets are indicated, install with 1-inch (25-mm) overlap at longitudinal seams and end joints; for horizontal applications, install with longitudinal seams along top and bottom of tanks and vessels. Seal with manufacturer's recommended adhesive.
   1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.

D. Where metal jackets are indicated, install with 2-inch (50-mm) overlap at longitudinal seams and end joints. Overlap longitudinal seams arranged to shed water. Seal end joints with weatherproof sealant recommended by insulation manufacturer. Secure jacket with stainless-steel bands 12 inches o.c. and at end joints.

3.06 FINISHES

A. Equipment Insulation with ASJ, Glass-Cloth, or Other Paintable Jacket Material: Paint jacket with paint system identified below and as specified in Section 09 9113 "Exterior Painting" and Section 09 9123 "Interior Painting."
   1. Flat Acrylic Finish: Two finish coats over a primer that is compatible with jacket material and finish coat paint. Add fungicidal agent to render fabric mildew proof.
      a. Finish Coat Material: Interior, flat, latex-emulsion size.

B. Flexible Elastomeric Thermal Insulation: After adhesive has fully cured, apply two coats of insulation manufacturer's recommended protective coating.

C. Color: Final color as selected by Engineer. Vary first and second coats to allow visual inspection of the completed Work.

D. Do not field paint aluminum or stainless-steel jackets.

3.07 FIELD QUALITY CONTROL

A. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

B. Perform tests and inspections.

C. Tests and Inspections: Inspect field-insulated equipment, randomly selected by Engineer, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location(s) for each type of equipment defined in the "Equipment Insulation Schedule" Article. For large equipment, remove only a portion adequate to determine compliance.

D. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.08 EQUIPMENT INSULATION SCHEDULE

A. Insulation materials and thicknesses are identified below. If more than one material is listed for a type of equipment, selection from materials listed is Contractor's option.

B. Insulate indoor and outdoor equipment that is not factory insulated.

C. Chilled-water pump insulation shall be one of the following:
   1. Flexible Elastomeric: 1-1/2 inches thick.

D. Chilled-water expansion/compression tank insulation shall be one of the following:
   1. Flexible Elastomeric: 1-1/2 inches thick.

E. Chilled-water air-separator insulation shall be one of the following:
   1. Flexible Elastomeric: 1-1/2 inch thick.

F. Buffer Tank insulation shall be one of the following:
   1. Cellular Glass: 4 inches thick.
   2. Mineral-Fiber Board: 3 inches thick and 3-lb/cu. ft. (48-kg/cu. m) nominal density.
   3. Mineral-Fiber Pipe and Tank: 3 inches thick.
   4. Phenolic: 3 inches thick.
   5. Polyisocyanurate (Outdoor Application Only): 3 inches thick.
   6. Polystyrene (Outdoor Application Only): 3 inches thick.

3.09 INDOOR, FIELD-APPLIED JACKET SCHEDULE

A. Install jacket over insulation material. For insulation with factory-applied jacket, install the field-applied jacket over the factory-applied jacket.

B. If more than one material is listed, selection from materials listed is Contractor's option.

C. Equipment, Concealed:
   1. None.

D. Equipment, Exposed, up to 48 Inches in Diameter or with Flat Surfaces up to 72 Inches:
   1. None.

E. Equipment, Exposed, Larger Than 48 Inches in Diameter or with Flat Surfaces Larger Than 72 Inches:
   1. None.

END OF SECTION

SECTION 23 0800
COMMISSIONING OF HVAC SYSTEMS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes commissioning process requirements for HVAC&R systems, assemblies, and equipment.

B. Related Sections:
   1. Section 01 9113 - GENERAL COMMISSIONING REQUIREMENTS for commissioning process requirements.

1.03 DEFINITIONS

A. Commissioning Plan: A document that outlines the organization, schedule, allocation of resources, and documentation requirements of the commissioning process.

B. Contractor: Contractor: May refer to project General Contractor, Mechanical Contractor, Electrical Contractor, or subcontractors thereof depending on the context with which it is used.

C. CxA: Commissioning Authority.

D. HVAC&R: Heating, Ventilating, Air Conditioning, and Refrigeration.

E. IOM Manual: Installation, Operation and Maintenance Manual

F. Systems, Subsystems, Equipment, and Components: Where these terms are used together or separately, they shall mean "as-built" systems, subsystems, equipment, and components.

1.04 CONTRACTOR'S RESPONSIBILITIES

A. Upload Requested Information to Commissioning Website including but not limited to Submittals, Performance Data, Capacity Information, Sequences of Operation, Factory Performance Tests, and IOM Manuals. If electronic versions are not available CxA, at the contractor’s option, may elect to upload information on behalf of the contractor for a nominal administration fee.

B. Complete electronic Pre-Functional Checklists (PFCs) under the direction of the CxA using either a computer with internet connection or application based tablet. Mobile devices must be synchronized nightly with the internet.

C. Perform Functional Performance Testing (PFTs) at the direction of the CxA.

D. Respond to issues generated by the Commissioning Agent using the commissioning website. Issues are to be responded to using the website commissioning tool, fax and email responses are not acceptable. After issues are generated by the CxA and assigned to contractor, contractor is to log into the website, change the issue from “open” to “addressed” and provide a brief description on how the issue has been addressed.

E. Attend construction phase controls coordination meeting.

F. Attend testing, adjusting, and balancing review and coordination meeting.

G. Participate in HVAC&R systems, assemblies, equipment, and component maintenance orientation and inspection as directed by the CxA.

H. Provide information requested by the CxA for final commissioning documentation.

1.05 CXA'S RESPONSIBILITIES

A. Provide Project-specific construction checklists and commissioning process test procedures for actual HVAC&R systems, assemblies, equipment, and components to be furnished and installed as part of the construction contract.

B. Review “Filtered” Requests for Information (RFIs) as they pertain to commissioned systems.

C. Review Contractor’s submittals.

D. Direct commissioning Functional Performance Testing.

E. Verify testing, adjusting, and balancing of Work are complete.

F. Verify and Document training occurred per the contract requirements.

G. Provide test data, inspection reports, and certificates in Systems Manual.

1.06 COMMISSIONING DOCUMENTATION

A. Provide the following information to the CxA for inclusion in the commissioning plan:
   1. Plan for delivery and review of submittals, systems manuals, and other documents and reports.
   2. Identification of installed systems, assemblies, equipment, and components including design changes that occurred during the construction phase.
   3. Process and schedule for completing construction checklists and manufacturer's prestart and startup checklists for HVAC&R systems, assemblies, equipment, and components to be verified and tested.
   4. Certificate of completion certifying that installation, prestart checks, and startup procedures have been completed.
   5. Certificate of readiness certifying that HVAC&R systems, subsystems, equipment, and associated controls are ready for testing.
   6. Test and inspection reports and certificates.
   7. Corrective action documents.
   8. Verification of testing, adjusting, and balancing reports.

1.07 SUBMITTALS

A. In addition to standard product submittals and shop drawings as defined in project specification sections Divisions 23 and 26, contractor may be required to fill out and sign off on commissioning agent system readiness matrix at the discretion of the commissioning agent. System readiness matrix will list all MEP building systems and ask for written confirmation that systems are complete and ready for functional testing. If required, system readiness matrix will be produced and provided to contractor by commissioning agent.

PART 2 - PRODUCTS (NOT USED)

PART 3 - EXECUTION

3.01 TESTING PREPARATION

A. Certify that HVAC&R systems, subsystems, equipment and controls have been installed, calibrated, and started and are operating according to the Contract Documents.

B. Certify that HVAC&R instrumentation and control systems have been completed and calibrated, that they are operating according to the Contract Documents, and that pretest set points have been recorded.

C. Certify that testing, adjusting, and balancing procedures have been completed and that testing, adjusting, and balancing reports have been submitted, discrepancies corrected, and corrective work approved.

D. Set systems, subsystems, and equipment into operating mode to be tested (e.g., normal shutdown, normal auto position, normal manual position, unoccupied cycle, emergency power, and alarm conditions).

E. Inspect and verify the position of each device and interlock identified on checklists.

F. Check safety cutouts, alarms, and interlocks with smoke control and life-safety systems during each mode of operation.

G. Testing Instrumentation: Install measuring instruments and logging devices to record test data as directed by the CxA.

3.02 TESTING AND BALANCING VERIFICATION

A. Prior to performance of testing and balancing Work, upload copies of reports, sample forms, checklists, and certificates to the commissioning website.

B. Notify the CxA at least Fifteen (15) days in advance of testing and balancing Work, and provide access for the CxA to witness testing and balancing Work.

C. Provide technicians, instrumentation, and tools to verify testing and balancing of HVAC&R systems at the direction of the CxA.
   1. The CxA will notify testing and balancing Subcontractor 10 days in advance of the date of field verification. Notice will not include data points to be verified.
   2. The testing and balancing Subcontractor shall use the same instruments (by model and serial number) that were used when original data were collected.
   3. Failure of an item includes, other than sound, a deviation of more than 10 percent. Failure of more than 10 percent of selected items shall result in rejection of final testing, adjusting, and balancing report. Remedy the deficiency and notify the CxA so verification of failed portions can be performed.

3.03 GENERAL TESTING REQUIREMENTS

A. Provide technicians, instrumentation, and tools to perform commissioning test at the direction of the CxA.

B. Scope of HVAC&R testing shall include entire HVAC&R installation, from central equipment for heat generation and refrigeration through distribution systems to each conditioned space. Testing shall include measuring capacities and effectiveness of operational and control functions.

C. Test all operating modes, interlocks, control responses, and responses to abnormal or emergency conditions, and verify proper response of building automation system controllers and sensors.

D. The CxA along with the HVAC&R Subcontractor, testing and balancing Subcontractor, and HVAC&R Instrumentation and Control Subcontractor shall prepare detailed testing plans, procedures, and checklists for HVAC&R systems, subsystems, and equipment.

E. Tests will be performed using design conditions whenever possible.

F. Simulated conditions may need to be imposed using an artificial load when it is not practical to test under design conditions. Before simulating conditions, calibrate testing instruments. Provide equipment to simulate loads. Set simulated conditions as directed by the CxA and document simulated conditions and methods of simulation. After tests, return settings to normal operating conditions.

G. The CxA may direct that set points be altered when simulating conditions is not practical.

H. The CxA may direct that sensor values be altered with a signal generator when design or simulating conditions and altering set points are not practical.

I. If tests cannot be completed because of a deficiency outside the scope of the HVAC&R system, document the deficiency and report it to the Owner. After deficiencies are resolved, reschedule tests.

J. If major problems are discovered during the test, the responsible subcontractors and General Contractor will fix the problem and the test shall be redone. If more than two functional performance tests are required, the responsible subcontractor will be back-charged for the CxA's time and expenses.

K. If the testing plan indicates specific seasonal testing, complete appropriate initial performance tests and documentation and schedule seasonal tests.

3.04 HVAC&R SYSTEMS, SUBSYSTEMS, AND EQUIPMENT TESTING PROCEDURES

A. HVAC Systems: Test all modes of operation, including occupied/unoccupied, start-up, shut down, failure, emergency, capacity, modulation, performance, and consistency tests - expected duration to be 3 days of hands on tests. This does not include setup, preparation recording, or data downloading time.

B. HVAC&R Instrumentation and Control System Testing: Field testing plans and testing requirements are specified in Division 23 Sections "Instrumentation and Control for HVAC" and "Sequence of Operations for HVAC Controls." Assist the CxA with preparation of testing plans.

C. Pipe system cleaning, flushing, hydrostatic tests, and chemical treatment requirements are specified in Division 23 piping Sections. HVAC&R Subcontractor shall prepare a pipe system cleaning, flushing, and hydrostatic testing plan. Provide cleaning, flushing, testing, and treating plan and final reports to the CxA. Plan shall include the following:
   1. Sequence of testing and testing procedures for each section of pipe to be tested, identified by pipe zone or sector identification marker. Markers shall be keyed to Drawings for each pipe sector, showing the physical location of each designated pipe test section. Drawings keyed to pipe zones or sectors shall be formatted to allow each section of piping to be physically located and identified when referred to in pipe system cleaning, flushing, hydrostatic testing, and chemical treatment plan.
   2. Description of equipment for flushing operations.
   3. Minimum flushing water velocity.
   4. Tracking checklist for managing and ensuring that all pipe sections have been cleaned, flushed, hydrostatically tested, and chemically treated.

D. Refrigeration System Testing: Provide technicians, instrumentation, tools, and equipment to test performance of chillers, cooling towers, refrigerant compressors and condensers, heat pumps, and other refrigeration systems. The CxA shall determine the sequence of testing and testing procedures for each equipment item and pipe section to be tested.

E. HVAC&R Distribution System Testing: Provide technicians, instrumentation, tools, and equipment to test performance of hydronic distribution systems.

END OF SECTION 23 0800

SECTION 23 0920.01
CONTROL SYSTEMS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. DDC System for monitoring and controlling of HVAC systems including:
      a. System Architecture
      b. Controllers
      c. Sensors and Input Devices
      d. Output Devices
      e. Power Supplies and Transformers
      f. Actuators
      g. Control Valves
      h. Low Voltage Cable
   2. Delivery of selected control devices to equipment and systems manufacturers for factory installation and to HVAC systems installers for field installation.

B. Related Requirements
   1. Section 23 0593 - TESTING, ADJUSTING, AND BALANCING FOR HVAC
   2. Division 23 Section - Commissioning of HVAC
   3. Division 26 Section - Electrical Rough-in for Low Voltage Systems

1.03 REFERENCES

A. ASHRAE 85 - Automatic Control Terminology for Heating, Ventilating, Air Conditioning

B. ASME MC85.1 - Terminology for Automatic Control

C. NEMA EMC1 - Energy Management Systems Definition

D. NFPA 70 - National Electric Code

E. NFPA 90A - Air Condition systems

F. NFPA 101 - Life Safety Code

1.04 SYSTEM DESCRIPTION

A. The facility will unify all systems into one Building Automation System. Provide a fully integrated Building Automation System (BAS) incorporating the components herein specified and indicated on the drawings. System shall include electric and electronic components to provide a complete functioning Direct Digital Control System.

B. All materials and equipment shall be standard components regularly manufactured for this and/or other systems. All components shall be thoroughly tested and proven in actual use.

C. The BAS system shall provide automatic decision-making and control for building heating, ventilating and air conditioning (HVAC) along with other functions.

1.05 SCOPE

A. Provide, assemble, mount, and wire all devices and system components.

B. Assist the Balancing Contractor and Commissioning Agent

C. Facilitate and perform the complete checkout of the entire system, verifying proper operation in accordance with design specifications.

1.06 PROGRAMMING

A. To be completed by the control contractors.

1.07 COORDINATION

A. The following incidental work shall be furnished by the designated contractor under supervision of the Controls Contractor. The Controls Contractor shall provide all required information, materials, direction, and training to the designated contractor as required to provide a complete working system.

B. The General Contractor shall:
   1. Provide access doors or other approved means of access in concealed ceiling and walls for service of control equipment.
   2. Provide all necessary cutting, patching and painting.

C. The Mechanical Contractor shall:
   1. Install control valves and sensor wells that are provided by the controls contractor.
   2. Provide necessary valved pressure taps, water, drain and overflow connections and piping.
   3. Install automatic dampers, required interconnection linkages and extend damper shafts as necessary for mounting of actuators
   4. Provide access doors or other approved means of access in ductwork for service of control equipment.

D. The Electrical Contractor shall:
   1. Provide auxiliary contacts, buttons and switches on all starters as required
   2. Provide power wiring (120V or greater) to control panels, motor starters, variable frequency drives, motors, electric actuators, electric devices and smoke detectors.

E. Control Contractor shall provide Programming Services
   1. Define BAS software programming to accomplish sequence of operation per the project specification and drawings. Load the software program into the electronic devices.
   2. Provide software program to the Controls Contractor.
   3. Provide program documentation to Controls Contractor.
   4. Incorporate final redline changes from contractor record drawings.
   5. Provide electronic graphics for the building automation system.
   6. Update system global files.

1.08 SUBMITTALS FOR REVIEW

A. Section 01 3300 - Submittal Procedures.

B. Clearly note the proposed deviations from specified sequences and equipment, and substantiate with written explanation.

C. Shop Drawings and Equipment Product Data:
   1. Manufacturer's printed data sheets for all control system components clearly identify data by component.
   2. General catalog sheets, including details of construction, ratings, and dimensions for each component.
   3. Show complete details of interconnections between temperature control systems and controls specified with equipment or systems.
   4. Provide Valve Schedules indicating size, type, design flow, valve capacity, pressure drop and CV value, identification and part number.
   5. Shop Drawings and Product Data are not required for BAS panels or specific parts included in the bill of materials in panel drawings.
   6. The Sequences provided in the document shall be reviewed by the Commissioning Agent and engineer prior to programming. The contractor shall accommodate adjustments to sequencing as dictated by the commissioning agent before programming commences.

1.09 SUBMITTALS AT CLOSEOUT

A. Section 01 7700 - Closeout Procedures - Contract Closeout; Procedures for maintenance documents and submittals.

B. Project Record Documents: Maintain documents at site during construction and submit at Contract Closeout.

C. Operation and Maintenance Data Include:
   1. Red-mark "Record" control system drawings and then turn them over to the Owner’s Representative.
   2. Provide directions for calibration, adjustment and maintenance instructions for each type of component.
   3. Complete Index of Contents
   4. Shop drawings of control system showing devices, interconnections between devices, and connections to items provided by others.
   5. Specifications data sheets on each device.

1.10 WARRANTY

A. The scope of work included in this section shall be guaranteed to be free from original defects in both material and workmanship for the period as indicated in the contract terms.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Eligibility Criteria:
   1. Subject to Geographic Limitation. Vendor shall have fully staffed office with Service Personnel, including Lead Programmer, within 100 miles of the Project Site, or no more than 2 hours drive from Project Site during summer weather condition.
   2. All buildings shall be furnished with the same manufactured equipment. Separate BAS controls manufacturers from one building to another is explicitly prohibited.
   3. Owner's Staff Training for Enhanced Programming Capability. Eligible vendor shall demonstrate Owner training protocols including Factory Training, Self-Guided tutorials, and other means to allow Owner's staff to perform Coding.
   4. Eligible vendors shall not include any software licensing fees, subscription fees, or similar cost structures. Successful vendor acknowledges Contract constitutes outright purchase of software without renewal fees that includes automatic software updates and bug fixes without any additional cost.
   5. All interested Controls Vendors shall be subject to a Pre-Qualification process prior to the Bid Date. Submit Letter of Interest to Bid project in accordance with Division 01 Section 01 2500 - Substitution Procedures. Include statement of qualifications indicating compliance with eligibility criteria, list of simliars systems in Project vicinity and surrounding territory, and contact information for three references from Facilities Manager, Physical Plant Manager, or Similar titled responsibility.
   6. Subject to compliance with Project Requirements and Eligibility Criteria, available manufacturers may include:
      a. Siemens Inc.
      b. Johnson Controls
      c. Honeywell Controls
      d. Dynamic Controls
      e. Or Approved Equal
   7. All control system integration shall adhere to BACnet open protocol standards including programming language, not just information transfer amongst the devices.

2.02 SYSTEM ARCHITECTURE

A. The BAS shall be provided with a Master Control Panel and shall function as the overall system coordinator and shall be capable of integrating multiple building functions including HVAC

equipment supervision and control, building equipment monitoring, alarm management, energy management, information management, historical data, collection and archiving. The BAS shall consist of individual DDC Control Panels, Application Specific Controllers, and Local Display Devices.

B. The system architecture shall eliminate dependence upon any single device for alarm reporting and control execution. Each DDC Control Panel shall operate independently from all other control components and will maintain control strategies when other components in system fail.

C. Communication networks between control panels shall be made without the need for a central processing controller.

2.03 CONTROLLERS

A. DDC system shall consist of a combination of network controllers, programmable application controllers and application-specific controllers to satisfy performance requirements indicated.

B. DDC controllers shall perform monitoring, control, energy optimization and other requirements indicated.

C. DDC controllers shall use a multitasking, multiuser, real-time digital control microprocessor with a distributed network database and intelligence

D. Each DDC controller shall be capable of full and complete operation as a completely independent unit and as a part of a DDC system wide distributed network.

E. Environment Requirements:
   1. Controller hardware shall be suitable for the anticipated ambient conditions.
   2. Controllers located in conditioned space shall be rated for operation at 32 to 120 deg F.

F. Network Control Unit
   1. PXC Bacnet Controller or approved equal. All equipment shall be compatible with this version.

G. Application Specific Controller
   1. Description: Microprocessor-based controllers, which through hardware or firmware design are dedicated to control a specific piece of equipment. Controllers are not fully user-programmable but are configurable and customizable for operation of equipment they are designed to control.
      a. Capable of standalone operation and shall continue to include control functions without being connected to network.
      b. Data shall be shared between networked controllers and other network devices.
   2. Communication: Application-specific controllers shall communicate with other application-specific controller and devices on network, and to programmable application and network controllers.
   3. Operator Interface: Controller shall be equipped with a service communications port for connection to a portable operator's workstation. Connection shall extend to port on space temperature sensor that is connected to controller.
   4. Serviceability:
      a. Controller shall be equipped with diagnostic LEDs or other form of local visual indication of power, communication, and processor.
      b. Wiring and cable connections shall be made to field-removable, modular terminal strips or to a termination card connected by a ribbon cable.
      c. Controller shall use nonvolatile memory and maintain all BIOS and programming information in event of power loss.
      d. Include manufacturer's logic to perform automatic calibration of terminal units at regularly scheduled intervals.

H. Local Control Panels
   1. All control panels shall be factory constructed, incorporating the FMS manufacturers standard designs and layouts. All control panels shall be UL inspected and listed as an

assembly and carry a UL 508A label listing compliance. Control panels shall be fully enclosed, with sub-panel, hinged door, and slotted flush latch. All Controls components within panels or elsewhere shall be UL 508 listed.

2. In general, the control panels shall consist of the DDC controller(s), display module, and I/O devices-such as relays, transducers, and so forth-that are not required to be located external to the control panel due to function. The display module shall be flush mounted in the panel face unless otherwise noted.
3. All I/O connections on the DDC controller shall be extended to a numbered, color-coded, and labeled terminal strip for ease of maintenance and expansion. Wiring to I/O devices shall be made from this terminal strip.
4. All other wiring in the panel, internal and external, shall be made to additional line or low voltage color-coded and labeled terminal strips. Low and line voltage wiring shall be segregated. All terminal strips and wiring shall be UL listed 300-volt service and provide adequate clearance for field wiring.
5. All wiring for every control panel shall follow a common color-coded format. All terminal strip color coding and numbering shall follow a common format. All wiring shall be neatly installed in plastic trays or tie-wrapped.
6. A convenience 120 VAC duplex receptacle, fused on/off power switch, and required transformers shall be provided in each enclosure.

2.04 SENSORS AND INPUT DEVICES

A. Temperature Sensors
1. Sensors and transmitters shall be provided, as outlined in the input/output summary and sequence of operations.
2. The temperature sensor shall be of the resistance type, and shall be either two-wire 1000 ohm nickel RTD, or two-wire 1000 ohm platinum RTD.
3. The following point types (and the accuracy of each) are required, and their associated accuracy values include errors associated with the sensor, lead wire, and A to D conversion:

Point Type Accuracy

a. Chilled Water + .5°F.

4. Room sensors shall be constructed for either surface or wallbox mounting.
5. Room sensors shall have the following options when specified:
   a. Setpoint reset slide switch providing a +3 degree (adjustable) range.
   b. Individual heating/cooling setpoint slide switches.
   c. A momentary override request push button for activation of after-hours operation.
   d. Analog thermometer.
   e. 1000 ohm thin film nickel sensor
   f. Acceptable manufacturers: Siemens

B. Thermowells
1. When thermowells are required, the sensor and well shall be supplied as a complete assembly, including well head and Greenfield fitting.
2. Thermowells shall be pressure rated and constructed in accordance with the system working pressure.
3. Thermowells and sensors shall be mounted in a threadolet or 1/2” NFT saddle and allow easy access to the sensor for repair or replacement.
4. Thermowells shall be constructed of brass.

C. Pressure Transmitters
1. General Water Pressure Transmitter Requirements:
   a. Pressure transmitters shall be constructed to withstand 100% pressure over-range without damage, and to hold calibrated accuracy when subject to a momentary 40% over-range input.

b. Pressure transmitters shall transmit a 0 to 5 VDC, 0 to 10 VDC, or 4 to 20 ma output signal.
c. Differential pressure transmitters used for flow measurement shall be sized to the flow sensing device, and shall be supplied with Tee fittings and shut-off valves in the high and low sensing pick-up lines to allow the balancing Contractor and Owner permanent, easy-to-use connection.
d. A minimum of a NEMA 1 housing shall be provided for the transmitter. Transmitters shall be located in accessible local control panels wherever possible.
e. Standalone pressure transmitters shall be mounted in a bypass valve assembly panel. The panel shall be constructed to NEMA 1 standards. The transmitter shall be installed in the panel with high and low connections piped and valved. Air bleed units, bypass valves, and compression fittings shall be provided.
f. Acceptable manufactures: Mamac

2. Medium to High Differential Water Pressure Applications (Over 21” w.c.)
a. The differential pressure transmitter shall meet the low pressure transmitter specifications with the following exceptions:
1) Differential pressure range 10” w.c. to 300 PSI.
2) Reference Accuracy: +1% of full span (includes non-linearity, hysteresis, and repeatability).
3) Standalone pressure transmitters shall be mounted in a bypass valve assembly panel. The panel shall be constructed to NEMA 1 standards. The transmitter shall be installed in the panel with high and low connections piped and valved. Air bleed units, bypass valves, and compression fittings shall be provided.
b. Acceptable manufactures: Mamac

D. Power Monitoring Devices
1. Current measurement (Amps) shall be by a combination current transformer and a current transducer. The current transformer shall be sized to reduce the full amperage of the monitored circuit to a maximum 5 Amp signal, which will be converted to a 4-20 ma DDC compatible signal for use by the Facility Management System.
2. Current Transformer - A split core current transformer shall be provided to monitor motor amps.
a. Operating frequency - 50 - 400 Hz.
b. Insulation - 0.6 Kv class 10Kv BIL.
c. UL recognized.
d. Five-amp secondary.
e. Select current ratio as appropriate for application.
f. Acceptable manufacturers: Veris Industries
3. Current Transducer - A current to voltage or current to ma transducer shall be provided. The current transducer shall include:
a. 6X input over amp rating for AC inrushes of up to 120 amps.
b. Manufactured to UL 1244.
c. Accuracy: +. 5%, Ripple +1%.
d. Minimum load resistance 30kOhm.
e. Input 0-20 Amps.
f. Output 4-20 ma.
g. Transducer shall be powered by a 24VDC regulated power supply (24 VDC +5%).
h. Acceptable manufacturers: Veris Industries

E. Status and Safety Switches
1. Switches shall be provided to monitor equipment status, safety conditions, and generate alarms at the FMS when a failure or abnormal condition occurs. Safety switches shall be provided with two sets of contacts and shall be interlock wired to shut down respective equipment.

2. Current Sensing Switches
    a. The current sensing switch shall be self-powered with solid-state circuitry and a dry contact output. It shall consist of a current transformer, a solid state current sensing circuit, adjustable trip point, solid state switch, SPDT relay, and an LED indicating the on or off status. A conductor of the load shall be passed through the window of the device. It shall accept over-current up to twice its trip point range.
    b. Current sensing switches shall be used for run status for fans, pumps, and other miscellaneous motor loads.
    c. Current sensing switches shall be calibrated to show a positive run status only when the motor is operating under load. A motor running with a broken belt or coupling shall indicate a negative run status.
    d. Acceptable manufacturers: Veris Industries

F. Water Flow Differential Pressure Switches
1. Water flow switches shall be equal to Siemens.
2. Acceptable manufactures: Siemens or approved equal.

G. Water Flow Meters
1. Turbine Flow meter with sensor, indicator, and output panel for measurement of water flowrate (GPM) in closed loop hydronic systems consisting of Dual Impeller design suited for direct insertion into pipe through gland-fitting assembly with isolation valve for positive shut-off when flow meter is removed.
2. Construction: Bronze or stainless steel body with plastic turbines or impellers.
3. Minimum pressure rating: 150 psig
4. Minimum temperature rating: 180 deg F.
5. Accuracy: Plus or minus 1.5 percent for liquids.
6. Display: For outputting Rate of Flow in GPM and analog output to BAS for minimum flow control of chiller and system bypass valve control.
7. Acceptable manufacturers: Onicon or approved equal

2.05 OUTPUT DEVICES

A. Control Relays
1. Control pilot relays shall be of a modular plug-in design with retaining springs or clips.
2. Mounting bases shall be snap-mount.
3. DPDT, 3PDT, or 4PDT relays shall be provided, as appropriate for application.
4. Contacts shall be rated for 10 amps at 120VAC.
5. Relays shall have an integral indicator light and check button.

2.06 POWER SUPPLIES AND TRANSFORMERS

A. DC Power Supplies
1. DC power supplies shall be sized for the connected device load. Total rated load shall not exceed 75% of the rated capacity of the power supply.
2. Input: 120 VAC +10%, 60Hz.
3. Output: 24 VDC.
4. Line Regulation: +0.05% for 10% line change.
5. Load Regulation: +0.05% for 50% load change.
6. Ripple and Noise: 1 mV rms, 5 mV peak to peak.
7. An appropriately sized fuse and fuse block shall be provided and located next to the power supply.
8. A power disconnect switch shall be provided next to the power supply.

B. Transformers
1. Sized to provide volts and amps as required for connected load.
2. Input voltage shall be as required for the specific application.

2.07 ACTUATORS

A. Valve actuators shall be electronic, as specified in the Sequence of Operation, Control Drawings and Wiring Diagrams.

B. Electronic Valve Actuators
   1. Electronic valve actuators shall be manufactured by the valve manufacturer.
   2. Each actuator shall have current limiting circuitry incorporated in its design to prevent damage to the actuator.
   3. Modulating and two-position actuators shall be provided as required by the sequence of operations. Actuators shall provide the minimum torque required for proper valve close-off against the system pressure for the required application. The valve actuator shall be sized based on valve manufacturer's recommendations for flow and pressure differential. All actuators shall fail in the last position unless specified with mechanical spring return in the sequence of operations. The spring return feature shall permit normally open or normally closed positions of the valves, as required. All direct shaft mount rotational actuators shall have external adjustable stops to limit the travel in either direction.
   4. Modulating Actuators shall accept 24 VAC or VDC and 120 VAC power supply and be UL listed. The control signal shall be 2-10 VDC or 4-20 ma and the actuator shall provide a clamp position feedback signal of 2-10 VDC. The feedback signal shall be independent of the input signal, and may be used to parallel other actuators and provide true position indication. The feedback signal of each valve actuator (except terminal valves) shall be wired back to a terminal strip in the control panel for trouble-shooting purposes.
   5. Two-position or open/closed actuators shall accept 24 or 120 VAC power supply and be UL listed. Butterfly isolation and other valves, as specified in the sequence of operations, shall be furnished with adjustable end switches to indicate open/closed position or be hard wired to start/stop the associated pump or chiller.
   6. Acceptable manufacturers: Siemens or approved equal.

2.08 HYDRONIC TEMPERATURE SENSORS

A. General: The RTD/Temperature Transmitter/Thermowell assembly shall come as a complete assembly from a single manufacturer. The Assembly shall be suitable for use in the accurate measurement of Chilled/Tower/Dual Water temperatures in a mechanical room environment.

B. Calibration: Each RTD must be match calibrated to the Transmitter via NIST traceable calibration standards. A calibration report for each RTD & Transmitter matched pair must be furnished to the Owner.

C. RTD:
   1. RTD type: 2-wire or 3-wire 100 ohm platinum class A or B.
   2. Outside Diameter: 0.25 inch
   3. Tolerance: +/- 0.06% Type A, +/- 0.12% Type B.
   4. Stability: +/- 0.1 % over one year.
   5. TCR: 0.00385 (ohm/ohm/oC).
   6. RTD shall be tip sensitive.
   7. Resistance vs. Temperature table for the RTD must be provided to the Owner.

D. Transmitter:
   1. Transmitter shall be match calibrated to the RTD and assembled as a matched pair.
   2. Type: 2 wire (loop powered).
   3. Input: 2 or 3 wire 100 ohm platinum class A or class B RTD.
   4. Output: Output shall be a 4-20 mA signal linear to temperature.
   5. Calibrated Span:
      a. Chilled Water: 30°F to 130°F
   6. Calibration Accuracy, including total of all errors, of the Transmitter & RTD matched pair over the entire span shall be within +/- 0.2% of the calibrated span or +/- 0.18 oF, whichever is greater.

7. Supply Voltage: 24 VDC.
8. Ambient Operating Temp: 32 to 122°F
9. Epoxy potted for moisture resistance.
10. Mounting: Transmitter shall be mounted in the RTD connection head.

E. Thermowell
1. Thermowell shall be suitable for immersion in Chilled Water.
2. Thermowell shall be stainless steel.
3. Thermowell shall have 1/2” NPT process connection to pipe thred-o-let.
4. Thermowell Insertion depth shall be no more than 2/3 and no less than 1/3 of the inside pipe diameter.

F. Assembly:
1. Assembly configuration: Spring loaded RTD with thermowell-double ended hex-connection head.
2. Connection head shall be cast aluminum with chain connecting cap to body, have 1/2” NPT process and 3/4” NPT conduit connections, and a sealing gasket between cap and body.

G. RTD/Temperature Transmitter/Thermowell assembly shall be the following or approved equal:
1. Manufacturer: Minco Products, Inc.
2. Model: AS5004-PD-[assembly length code]-Y-2-A-291-calibrated span code
3. With Thermowell: TW239U-thermowell insertion length

2.09 CONTROL VALVES

A. All automatic control valves shall be fully proportioning and provide near linear heat transfer control. The valves shall be quiet in operation and fail-safe open, closed, or in their last position. All valves shall operate in sequence with another valve when required by the sequence of operations. All control valves shall be sized by the control manufacturer, and shall be guaranteed to meet the heating and cooling loads, as specified. All control valves shall be suitable for the system flow conditions and close against the differential pressures involved. Body pressure rating and connection type (sweat, screwed, or flanged) shall conform to the pipe schedule.

B. Heating and chilled water control valves shall be modulating plug, ball, and/or butterfly, as required by the specific application. Modulating water valves shall be sized per manufacturer’s recommendations for the given application. In general, valves (2 or 3-way) serving variable flow air handling unit coils shall be sized for a pressure drop equal to the actual coil pressure drop, but no less than 5 PSI. Valves (3-way) serving constant flow air handling unit coils with secondary circuit pumps shall be sized for a pressure drop equal to 25% the actual coil pressure drop, but no less than 2 PSI. Mixing valves (3-way) serving secondary water circuits shall be sized for a pressure drop of no less than 5 PSI. Valves for terminal reheat coils shall be sized for a 2 PSIG pressure drop, but no more than a 5 PSI drop.

C. Modulating plug water valves of the single-seat type with equal percentage flow characteristics shall be used for all hot and chilled water applications, except those described hereinafter. The valve discs shall be composition type. Valve stems shall be stainless steel.

D. Butterfly valves shall be acceptable for modulating large flow applications greater than modulating plug valves, and for all two-position, open/close applications. In-line and/or three-way butterfly valves shall be heavy-duty pattern with a body rating comparable to the pipe rating, replaceable lining suitable for temperature of system, and a stainless steel vane. Valves for modulating service shall be sized and travel limited to 50 degrees of full open. Valves for isolation service shall be the same as the pipe. Valves in the closed position shall be bubble-tight.

E. Acceptable manufacturers: Siemens or approved equal.

2.10 LOW VOLTAGE CABLE

A. Analog Input, Analog Output, Binary Input, Binary Output, 24 VAC and General Purpose Cabling.
   1. Cable shall consist of copper conductors not less No. 18 AWG-stranded.
   2. Shall be 2 or 3 conductor twisted cable with a drain wire.
   3. Cable shall have a 100% overall shield.
   4. Cable shall be a plenum-rated.
   5. Cable shall meet or exceed NEC voltage rating of 300V.
   6. Cable shall be NEC type CMP.
   7. Cable shall meet or exceed UL temperature rating of +60 degrees C.
   8. Cable shall be labeled at a minimum of every 18” with the DDC System manufacturer’s name and the type of signal carried within the cable, i.e. Analog Input, Analog Output, Binary Input, Binary Output, 24 VAC.
   9. Each of the cable types specified in Item A shall be of a different color coding for easy identification and troubleshooting. Required color coding:
      a. Analog Input Cable: Yellow
      b. Analog Output Cable: Tan
      c. Binary Input Cable: Orange
      d. Binary Output Cable: Violet
      e. 24 VAC Cable: Gray
      f. General Purpose Cable: Natural

B. Primary and Secondary Communications Network Cabling.
   1. Cable shall be of type recommend by the DDC System Manufacturer.
   2. Cable shall be shielded.
   3. Cable shall be a plenum-rated.
   4. Cable shall meet or exceed NEC voltage rating of 150V.
   5. Cable shall meet or exceed UL temperature rating of +60 degrees C.
   6. Cable shall be labeled at a minimum of every 18” with the DDC System manufacturer’s name, system name and the communications network name.
   7. Each of the cable types shall be of a different color-coding for easy identification and troubleshooting and shall be of a different color than the cable specified in Item A above.

PART 3 - EXECUTION

3.01 GENERAL

A. Completely install automatic temperature control system as specified.

B. Coordinate installation of devices, wiring, and other items with other trades.

C. Current sensing relay trip levels shall be set to indicate the motor running under load vs. no load.

D. Verify wiring terminations at all devices and correct as required. Provide identification label for each termination.

E. Upon an error in reading a sensor, the control system shall indicate the last known value. The system shall then try to read the sensor again every 3 minutes (adjustable) over the next 15 minutes (adjustable). If the sensor is able to be read, then the programming shall return to normal. However if after this 15-minute period, the sensor remains unresponsive, an alarm shall be provided in the BAS allowing the facility to investigate any issue with the sensor. The last reading for the sensor shall remain to allow the system to remain functional. If the system operation is dependent on the sensor, a high-level alarm shall be indicated and if not, then a low-level alarm can be provided (typically you can provide different rated alarms).

3.02 START-UP SERVICES

A. Verify proper location of each device and point-to-point BAS integrity, correct as needed.

B. When installation is complete and automatic control system is placed in operation, adjust and calibrate all instruments and devices in system and ensure that system is operating in accord with specified sequences.

C. Diagnose component and system problems. Communicate irregularities to the appropriate contractor for correction.

D. Attend construction meetings as required to coordinate with other contractors and provide input during problem resolution.

3.03 TRENDING

A. Provide trend data for all points shown on control drawings. Trending shall be provided at a minimum of every 15 minutes and shall be backed up to a server automatically every day and kept for a period of 12 months. Each trend file shall be appended such a that a single file is provide for each month. Trend data exports shall be provided in CSV or XLS file format. The control system shall also be capable of providing custom exports based on selected points and time frames to the same file formats. This shall apply to main equipment such as Chilled Water System (inclusive of system pumps, chillers, temperatures, etc.), Hot Water System (inclusive of system pumps, boilers, temperatures, etc), etc.

B. Provide trend data for all points shown on control drawings. Trending data shall be stored using non-volatile memory within the BAS. Data shall be recorded at a minimum of every 15 minutes and kept for a period of a minimum of 12 months. The control system shall also be capable of providing custom exports based on selected points and time frames and exports shall be provided in CSV or XLS file format.

3.04 SENSORS AND INPUT DEVICES

A. All Input devices shall be installed per the manufacturer’s recommendation.

B. Veris Industry Hawkeye H8026 Series Enhanced Data Stream Electric Power Meter
   1. See electrical drawings for device location and installation details

3.05 OUTPUT DEVICES

A. All output devices shall be installed per the manufacturer recommendation.

3.06 WIRING

A. Install wiring in accordance with National Electric Code; ANSI/NFPA 70.

B. Install wiring (low and line voltage) in metal raceways or conduit unless inside control cabinet or unit enclosures. For concealed and accessible areas, plenum rated wiring and cabling may be used.

C. Class 2 wiring not installed in conduit shall be supported every 5’ from the building structure utilizing metal hangers designed for this application. Wiring shall be installed parallel to the building structural lines. All wiring shall be installed in accordance with local code requirements. Exposed wiring shall only be allowed in concealed accessible locations.

D. Low voltage control wiring and 24VAC can be run in the same conduit. Power wiring 120VAC and greater must be in a separate conduit.

E. Fastening shall be secured to walls or ceilings by means of appropriate screws, expansion screws anchors, toggle bolts, hollow wall screw anchors, nylon expansion anchors or expansion shields. All-purpose plastic anchors are not acceptable.

F. Control panels shall be mounted on vibration free walls or freestanding structural supports. Panels shall be located as indicated or approved by owner representative.

G. Locate circuits, relays, transformers or other equipment that contains or must be connected to voltages exceeding 130 volts, in separate cabinets, which may be adjacent to control panels; permanently label "DANGER 277 VOLTS" or appropriate volts.

H. All wiring in mechanical rooms shall be in conduit. Minimum control wiring conduit size 3/4”.

3.07 IDENTIFICATION STANDARDS

A. Field Devices. All field devices shall be identified by a typed (not handwritten) securely attached tag label.

B. Controller Identification: All controllers shall be identified by typed (not handwritten) securely attached tag label.

C. Panel Identification: All local control panels shall be identified by a plastic engraved nameplate securely fastened to the outside of the controller enclosure.

D. Panel Devices: All panel devices shall be identified by a typed label securely fastened to the back plate of the local control panel.

E. Raceway Identification. All the covers to junction and pull boxes of the control system raceways shall be painted blue or have identification labels stating “Control System Wiring” affixed to the covers. Labels shall be typed, not hand written.

F. Wire Identification. All low and line voltage control wiring shall be identified by a number, as referenced to the associated control diagram, at each end of the conductor or cable. Identification number shall be permanently secured to the conductor or cable and shall be typed.

3.08 TRAINING

A. Provide demonstration and training in accordance with Division 01 7900 - Demonstration and Training.

END OF SECTION 23 0920

SECTION 23 2000
PIPE JOINING

PART I - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes pipe joining methods for Hydronic, Steam and Condesnate Piping
   1. Soldering
   2. Brazing
   3. Threading
   4. Flanges
   5. Welding
   6. Heat Fusion Welding

1.03 RELATED SECTIONS

A. Section 23 3113 - Hydronic Piping

B. Section 23 2116 - Hydronic Piping Specialties

1.04 REFERENCES

A. ASME B31.1 - Power Piping

B. ASME B31.9 - Building Services Piping

C. ASME B31.5 - Refrigeration Piping and Heat Transfer Components

D. ASME BPVC-IX - Boiler and Pressure Vessel Code, Section IX - Welding and Brazing Qualifications

E. All applicable ASME Standards for welding and brazing

F. ASTM B32 - Standard Specification for Solder Metal

G. ASTM B813 - Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube

H. ASTM 828 - Standard for Making Capillary Joints by Soldering Copper and Copper Alloy Tubes and Fittings

I. AWS A5.8 - Specification for Filler Metals for Brazing and Braze Welding

J. AWS B2.1 - Specification for Welding Procedure and Performance Qualification

K. All applicable AWS D10 Standards

L. Copper Development Association - Copper Tube Handbook

M. International Mechanical Code

1.05 QUALITY ASSURANCE

A. Products and execution shall be in compliance with applicable codes and standards including those referenced above in paragraph above entitled REFERENCES.

B. Joints shall be warranted to be free from leaks and imperfections for same time period. Defective joints shall be repaired or replaced at no cost to Owner.

C. Installation shall be in compliance with Manufacturer’s recommendations.

1.06 SUBMITTALS

A. Product Data
   1. Solder and flux

2. Brazing filler metal and flux
3. Flange Gaskets: Type, material, construction, temperature/ pressure rating
4. Bolts and Nuts: Type, material, ASTM identification
5. Weld consumables

B. Documentation
1. Welder qualifications
2. Welding procedures
3. Field records
4. Field test reports

1.07 PRODUCT DELIVERY, HANDLING AND STORAGE

A. All materials shall be stored indoors protected from temperature, physical damage and exposure to fluids, dust and debris. Containers of solder, brazing and weld consumables, pastes and fluids shall remain sealed until use. Opened containers shall be kept sealed when not in use.

1.08 WARRANTY

A. All products and installation thereof shall be warranted to be free from defects in material and workmanship for period of one year from date placed into useful service or 18 months from date of delivery, whichever occurs first. Defective product shall be repaired or replaced at no cost to Owner.

B. All joints shall be warranted to be free from leaks and imperfections for same time period. Defective joint shall be repaired or replaced at no cost to Owner. Impacted piping system shall manipulated as required to make repair and fluid media shall be brought back to original condition and fill volume at no cost to Owner.

PART 2 - PRODUCTS

2.01 DISSALLOWED PIPE JOINING METHODS

A. Roll groove or cut groove joints

B. Press-connect joints

C. Mechanically formed extruded outlets

D. Piercing valves and fittings

E. Saddle connections

F. Welded branch connections

G. Reducing bushings and flanges

H. Dielectric unions or fittings
1. For exception see Section 23 3113 - Hydronic Piping

I. Other non-standard pipe joints

2.02 PIPE AND FITTINGS

A. Pipe, Fittings, Unions, Flanges
1. See Section 23 2113 - Hydronic Piping for specifications

2.03 SOLDER JOINTS

A. Approved Fill Material
1. ASTM B32 alloy E
a. 95% tin, 5% copper
2. ASTM B32 alloy HB
a. Tin-antimony-copper-zinc-silver

2.04 BRAZED JOINTS

A. Fill Material
   1. Cuprous metal to cuprous metal: 15% minimum silver content
   2. Cuprous metal to steel: 45% minimum silver content
   3. High temperature, high pressure or lab/process piping applications: 15% minimum silver content
   4. Joints difficult to access (e.g. closed chases, underground): 15% minimum silver content

2.05 THREADED JOINTS

A. Thread Sealant
   1. Paste type, non-hardening, rated for temperature.
   2. CSM specifications require use of teflon tape in addition to paste.

2.06 FLANGED JOINTS

A. See Section 23 2113 - Hydronic Piping for specifications

2.07 WELDED JOINTS

A. Materials and Methods: In compliance with Weld Procedure Specifications (WPS)

B. Welding Requirements: In compliance with section below entitled WELDING.

PART 3 - EXECUTION

3.01 SOLDERING

A. Soldering shall be in conformance with Copper Development Association - Copper Tube Handbook including:
   1. Selection of solder and flux
   2. Measuring and cutting
   3. Reaming
   4. Application of flux
   5. Assembly and support
   6. Heating
   7. Application of solder
   8. Cooling and cleaning
   9. Testing

B. Destructive Testing
   1. Note to Contractor: Given destructive testing requirements specified below, adequate quality control is required to avoid potentially substantial and costly rework. Joints will be destructively tested for quality, not just leak-tightness.
   2. Destructive testing shall be performed on up to 5% of soldered joints at Contractor's expense. Joints shall be selected by Engineer or Owner.
   3. If any joint is found to be in non-compliance with referenced standards, joint shall be repaired or replaced to satisfaction of Engineer and up to an additional 10% of joints shall be tested.
   4. If additional non-compliant joint or joints are found, all joints identified by Engineer shall be replaced.

3.02 BRAZING

A. Brazing shall be in conformance with Copper Development Association - Copper Tube Handbook including:
   1. Selection of filler material and flux
   2. Measuring, cutting, reaming
   3. Application of flux
   4. Assembly and support
   5. Applying heat and brazing

6. Cooling and removing residue
7. Testing

B. During brazing, piping shall be purged with oil-free nitrogen.

C. Joints in following system types shall be brazed:
1. Laboratory and process gas systems
2. Systems with pressures in excess of 350 PSIG
3. Systems with temperatures in excess 450 degrees C
4. Underground copper piping systems
5. Other systems as indicated in project documents

D. Destructive Testing
1. Destructive testing shall be performed as identified above for solder joints.

3.03 THREADED JOINTS

A. Tapered NPT threads shall be properly cut on piping at joints.

B. Joint sealant shall be applied.

C. Torque shall be applied as required to properly seat threads.

3.04 FLANGED JOINTS

A. Flanges shall be properly aligned with minimal application of force.

B. Gasket shall be properly positioned.

C. Bolts shall be inserted and anti-seize compound applied.

D. Bolts shall be torqued to specified value.

3.05 WELDING

A. Qualifications
1. All welders and welding procedure specifications (WPS) shall be qualified as set forth in ASME BPVC-IX Boiler and Pressure Vessel Code, Section IX
a. Welder
1) Prior to performing project welds documentation shall be submitted confirming that each welder has passed required procedure test. Welders shall be qualified as required by ASME B31.1 or ASME B31.9 as applicable.
2) Welder qualifications shall be current. If qualification test is more than six months old record of continuity shall be provided indicating welder has performed applicable and approved welding at least every six months since date of qualification test. Record of continuity shall be to satisfaction of Engineer.
b. Weld Procedure Specifications (WPS)
1) Welding procedure specifications shall be provided for project specific welding methods and materials.

B. Weld Record
1. For welding within scope of ASME B31.1 procedure for locating, monitoring, recording and maintaining quality of welds shall be submitted to Engineer for approval.
2. Welder identification shall be provided for each weld. Identification shall consist of pipe stamp and welding record. Requirement for welder identification may be relaxed with written approval of Engineer.

C. Weld Inspection and Examination
1. All welds in piping and piping components shall be carefully visually examined in accordance with ASME B31.1 or ASME B31.9 as applicable.
2. Periodically, as welding progresses, report shall be provided indicating status of project welding quality.

3. Engineer and Owner shall be provided opportunity to observe all aspects of welding prior to, during and after fabrication to assure that proper welding is provided to Owner's satisfaction. Off-site shop welding shall be included. Additionally, Owner maintains right to obtain independent weld examination.
4. Engineer and Owner shall retain right to stop in-progress welding work until resolution of any concerns are resolved to Owner's satisfaction. Such shall be at no cost to Owner.
5. Welds in piping and piping components shall be radiographically examined. 10% of welds shall be examined each month. Radiographic examination shall be in accordance with ASME B31.1. Number or percentage of welds to be examined shall be as indicated in project documents. Specific welds to be examined shall be selected by Engineer. Testing agency shall be approved by Engineer and Owner.

D. Welding Procedure
1. All welding shall comply with applicable requirements of referenced ASME and AWA Standards.
2. All fittings shall be factory standard fittings. Fabricated fittings not allowed.
3. Backing rings shall not be used with welded joints.
4. Interior of pipe and fittings shall be thoroughly cleaned prior to and after welding/assembly.
5. Welds shall be built up with stringer-bead pass followed by hot pass, followed by cover or filler pass.
6. Valleys at center or edges of welds not allowed. Unsound or unfused metal, cracks, oxidation, blow hoes or non-metallic inclusions not allowed. Any such imperfections shall be corrected in compliance with referenced standards and to satisfaction of Engineer and Owner.
7. When hot-tapping, slag, drillings or "cookies" shall be prevented from entering piping system to greatest degree possible. Any material that enters piping shall be removed by use of magnet after drilling or cutting is complete.
8. Each weld shall be painted shortly after completion to prevent corrosion. Paint shall be high temp type.

3.06 FUSION WELDING - POLYPROPYLENE

A. PP-R and PP-RCT polypropylene pipe and fittings shall be fusion welded per Manufacturer's instructions.
1. Joints in piping 4" and smaller shall be heat fusion socket welded.
2. Joints in piping 6" and larger shall be heat fusion butt welded.

END OF SECTION 23 20 00

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SECTION 23 2113
HYDRONIC PIPING

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes pipe, pipe fittings, materials, joining methods and general duty valves for the following:
   1. Chilled water piping system
   2. Air-vent piping.

1.03 RELATED REQUIREMENTS

A. Section 23 0529 - Hangers and Supports for HVAC

B. Section 23 0548 - Vibration and Seismic Control for HVAC

C. Section 23 0553 - Identification for HVAC

D. Section 23 0711 - HVAC Insulation.

E. Section 23 2114 - Hydronic Specialties.

F. Section 23 2513 - HVAC Water Treatment for Closed Systems

1.04 REFERENCE STANDARDS

A. ANSI/FCI 70-2 - Control Valve Seat Leakage; 2013.

B. ASME B1.20.1 - Pipe Threads, General Purpose, Inch; 2013 (Reaffirmed 2018).

C. ASME B16.1 - Gray Iron Pipe Flanges and Flanged Fittings: Classes 25, 125, and 250; 2015.

D. ASME B16.3 - Malleable Iron Threaded Fittings: Classes 150 and 300; 2021.

E. ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard; 2020.

F. ASME B16.10 - Face-to-Face and End-to-End Dimensions of Valves; 2017.

G. ASME B16.18 - Cast Copper Alloy Solder Joint Pressure Fittings; 2021.

H. ASME B16.21 - Nonmetallic Flat Gaskets for Pipe Flanges; 2021.

I. ASME B16.22 - Wrought Copper and Copper Alloy Solder-Joint Pressure Fittings; 2018.

J. ASME B16.34 - Valves — Flanged, Threaded, and Welding End; 2020.

K. ASME B18.2.1 - Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series); 2012 (Reaffirmed 2021).

L. ASME B31.9 - Building Services Piping; 2020.

M. ASME BPVC-IX - Qualification Standard for Welding, Brazing, and Fuzing Procedures; Welders; Brazers; and Welding, Brazing, and Fusing Operators - Welding Brazing and Fusing Qualifications; 2019.

N. ASSE 1079 - Performance Requirements for Dielectric Pipe Unions; 2012.

O. ASTM A126 - Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings; 2004 (Reapproved 2019).

P. ASTM A216/A216M - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service; 2021.

Q. ASTM A234/A234M - Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service; 2019.

R. ASTM A351/A351M - Standard Specification for Castings, Austenitic, for Pressure-Containing Parts; 2018, with Editorial Revision (2019).

S. ASTM A536 - Standard Specification for Ductile Iron Castings; 1984, with Editorial Revision (2019).

T. ASTM A733 - Standard Specification for Welded and Seamless Carbon Steel and Austenitic Stainless Steel Pipe Nipples; 2025.

U. ASTM B32 - Standard Specification for Solder Metal; 2020.

V. ASTM B61 - Standard Specification for Steam or Valve Bronze Castings; 2015 (Reapproved 2021).

W. ASTM B62 - Standard Specification for Composition Bronze or Ounce Metal Castings; 2017.

X. ASTM B88 - Standard Specification for Seamless Copper Water Tube; 2020.

Y. ASTM B88M - Standard Specification for Seamless Copper Water Tube (Metric); 2020.

Z. ASTM B306 - Standard Specification for Copper Drainage Tube (DWV); 2020.

AA. ASTM B584 - Standard Specification for Copper Alloy Sand Castings for General Applications; 2022.

BB. ASTM B813 - Standard Specification for Water Flushable Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube; 2024.

CC. ASTM B828 - Standard Practice for Making Capillary Joints by Soldering of Copper and Copper Alloy Tube and Fittings; 2023.

DD. ASTM F1545 - Standard Specification for Plastic-Lined Ferrous Metal Pipe, Fittings, and Flanges; 2015.

EE. AWS A5.8/A5.8M - Specification for Filler Metals for Brazing and Braze Welding.; 2011 and errata.

FF. AWS D1.1/D1.1M - Structural Welding Code - Steel; 2020.

GG. AWS D10.12M/D10.12 - Guide for Welding Mild Steel Pipe; 200.

HH. IAPMO PS 66 - Dielectric Fittings; 2015.

II. ISA 75.08.01 - Face-to-Face Dimensions for Integral Flanged Globe-Style Control Valve Bodies (Classes 125, 150, 250, 300, and 600); 2016.

JJ. MSS SP-67 - Butterfly Valves; 2022.

KK. MSS SP-68 - High Pressure Butterfly Valves with Offset Design; 2017.

LL. MSS SP-71 - Gray Iron Swing Check Valves, Flanged and Threaded Ends; 2018.

MM. MSS SP-80 - Bronze Gate, Globe, Angle, and Check Valves; 2019.

1.05 SUBMITTALS

A. Product Data: Include data on pipe materials, pipe fittings, valves, and accessories. Provide manufacturers catalogue information. Indicate valve data and ratings.

B. Delegated-Design Submittal:
   1. Design calculations and detailed fabrication and assembly of pipe anchors and alignment guides, hangers and supports for multiple pipes, expansion joints and loops, and attachments of the same to the building structure.
   2. Locations of pipe anchors and alignment guides and expansion joints and loops.
   3. Locations of and details for penetrations, including sleeves and sleeve seals for exterior walls, floors, basement, and foundation walls.
   4. Locations of and details for penetration and firestopping for fire- and smoke-rated wall and floor and ceiling assemblies.

C. Coordination Drawings: Piping layout, drawn to scale, on which the following items are shown and coordinated with each other, using input from installers of the items involved:
   1. Suspended ceiling components.
   2. Other building services.
   3. Structural members.

D. Qualification Data: For Installer.

E. Welding Certificates: Include welders certification of compliance with ASME BPVC-IX .

F. Field quality-control reports.

1.06 REGULATORY REQUIREMENTS

A. Conform ASME B31.9 code for installation of piping system.

B. Welding Materials and Procedures: Conform to ASME BPVC-IX and applicable state labor regulations.

C. Provide certificate of compliance from authority having jurisdiction, indicating approval of welders.

1.07 QUALITY ASSURANCE

A. Valves: Manufacturer's name and pressure rating shall be clearly marked on the outside of the valve body.

B. Welding Materials and Procedures: Conform to Chapter V, ASME B31.9 and applicable state labor regulations.

C. Welders Certification: Furnish in accordance with AWS D1.1/D1.1M and ASME B31.9.

D. Each threaded fitting shall be stamped as specified by ASME B16.3.

E. Each welded fitting shall be stamped as specified by ASME BPVC-IX ASME B31.9.

F. Installer Qualifications:
   1. Installers of Pressure-Sealed Joints: Installers shall be certified by pressure-seal joint manufacturer as having been trained and qualified to join piping with pressure-seal pipe couplings and fittings.

G. Steel Support Welding: Qualify procedures and personnel according to AWS D1.1/D1.1M.

H. Pipe Welding: Qualify procedures and operators according to ASME BPVC-IX.
   1. Comply with ASME B31.9 for materials, products, and installation.
   2. Certify that each welder has passed AWS qualification tests for welding processes involved and that certification is current.

1.08 SYSTEM DESCRIPTION

A. Where more than one piping system material is specified, ensure system components are compatible and joined to ensure the integrity of the system is not jeopardized. Provide necessary joining fittings. Ensure flanges, union, and couplings for servicing are consistently provided.

B. Use unions, flanges, and couplings downstream of valves and at equipment or apparatus connections. Do not use direct welded or threaded connections to valves, equipment or other apparatus.

C. Use non-conducting dielectric connections whenever jointing dissimilar metals.

D. Provide pipe hangers and supports in accordance with ASME B31.9 unless indicated otherwise.

E. Use ball or butterfly valves for shut-off and to isolate equipment, part of systems, or vertical risers.

F. Use globe valves for throttling, bypass, or manual flow control services.

G. Use butterfly valves in heating water systems interchangeably with gate and globe valves.

H. Use 3/4 inch ball valves with cap for drains at main shut-off valves, low points of piping, bases of vertical risers, and at equipment. Pipe to nearest floor drain.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer's product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.

B. Equipment Categories: Equipment types in this section are considered equipment categories that be furnished by one manufacturer or available through one manufacturer to provide consistency within the overall project for equipment configurations, manufacturing techniques, and to facilitate ease of maintenance for the Owner.

2.02 PERFORMANCE REQUIREMENTS

A. Hydronic piping components and installation shall be capable of withstanding the following minimum working pressure and temperature unless otherwise indicated

B. Comply with ASME B31.9 and applicable federal, state, and local regulations.

C. Piping: Provide piping, fittings, hangers and supports as required, as indicated, and as follows:
   1. Where more than one piping system material is specified, provide joining fittings that are compatible with piping materials and ensure that the integrity of the system is not jeopardized.
   2. Use non-conducting dielectric connections whenever jointing dissimilar metals.
   3. Provide pipe hangers and supports in accordance with ASME B31.9, specification 230529 unless indicated otherwise.

D. Pipe-to-Valve and Pipe-to-Equipment Connections: Use flanges or unions to allow disconnection of components for servicing; do not use direct welded, soldered, or threaded connections.

E. Valves: Provide valves where indicated and as follows:
   1. On supply and return of all equipment.
   2. At each main serving a floor.
   3. At each branch from the main piping rack in garage levels.
   4. At the base of every riser.
   5. Provide drain valves where indicated on plans. In addition to locations shown on plans, provide drain valves at main shut-off, low points of piping, bases of vertical risers, and at equipment. Use 3/4 inch ball valves with cap; pipe to nearest floor drain.
   6. Isolate equipment using butterfly valves with lug end flanges.
      a. No solder connected or grooved fitting valves shall be used on this Project.

2.03 COPPER TUBE AND FITTINGS

A. Drawn-Temper Copper Tubing: ASTM B88 (ASTM B88M), Type L (B) or Type M (C) ASTM B88 (ASTM B88M), hard drawn using one of the following joint types:
   1. Solder Joints: ASME B16.18 cast brass/bronze or ASME B16.22 solder wrought copper fittings.

a. Solder: ASTM B32 lead-free solder, HB alloy (95-5 tin-antimony) or tin and silver.
b. Braze: AWS A5.8/A5.8M BCuP copper/silver alloy.

2. Tee Connections: All tees to be wrought copper fittings with solder joints. Pulled Tee's are not allowed. Mechanically Formed Tee Option by Manufacturers such as T-Drill Industries for forming T-branch on copper water tube is not acceptable

B. Annealed-Temper Copper Tubing: ASTM B88, Type K.
1. Solder Joints: ASME B16.18 cast brass/bronze or ASME B16.22 solder wrought copper fittings.
a. Solder: ASTM B32 lead-free solder, HB alloy (95-5 tin-antimony) or tin and silver.
b. Braze: AWS A5.8/A5.8M BCuP copper/silver alloy.
2. Tee Connections: All tees to be wrought copper fittings with solder joints. Pulled Tee's are not allowed. Mechanically Formed Tee Option by Manufacturers such as T-Drill Industries for forming T-branch on copper water tube is not acceptable

C. DWV Copper Tubing: ASTM B306 , Type DWV.
1. Solder Joints: ASME B16.18 cast brass/bronze or ASME B16.22 solder wrought copper fittings.
a. Solder: ASTM B32 lead-free solder, HB alloy (95-5 tin-antimony) or tin and silver.
b. Braze: AWS A5.8/A5.8M BCuP copper/silver alloy.
2. Tee Connections: All tees to be wrought copper fittings with solder joints. Pulled Tee's are not allowed. Mechanically Formed Tee Option by Manufacturers such as T-Drill Industries for forming T-branch on copper water tube is not acceptable

D. Wrought-Copper Unions: ASME B16.22.

2.04 STEEL PIPE AND FITTINGS

A. Steel Pipe: ASTM A 53, black steel with plain ends; welded and seamless, Grade B, and Schedule 40.

B. Wrought-Steel Fittings: ASTM A234/A234M, wall thickness to match adjoining pipe.

C. Wrought Cast- and Forged-Steel Flanges and Flanged Fittings: ASME B16.5, including bolts, nuts, and gaskets of the following material group, end connections, and facings:
1. Material Group: 1.1.
2. End Connections: Butt welding.
3. Facings: Raised face.

D. Steel Pipe Nipples: ASTM A733, made of same materials and wall thicknesses as pipe in which they are installed.

2.05 JOINING MATERIALS

A. Pipe-Flange Gasket Materials: Suitable for chemical and thermal conditions of piping system contents.
1. ASME B16.21, nonmetallic, flat, asbestos free, 1/8-inch maximum thickness unless otherwise indicated.
a. Full-Face Type: For flat-face, Class 125, cast-iron and cast-bronze flanges.
b. Narrow-Face Type: For raised-face, Class 250, cast-iron and steel flanges.

B. Flange Bolts and Nuts: ASME B18.2.1, carbon steel, unless otherwise indicated.

C. Solder Filler Metals: ASTM B32, lead-free alloys. Include water-flushable flux according to ASTM B813.

D. Brazing Filler Metals: AWS A5.8/A5.8M, BCuP Series, copper-phosphorus alloys for joining copper with copper; or BAg-1, silver alloy for joining copper with bronze or steel.

E. Welding Filler Metals: Comply with AWS D10.12M/D10.12 for welding materials appropriate for wall thickness and chemical analysis of steel pipe being welded.

F. Gasket Material: Thickness, material, and type suitable for fluid to be handled and working temperatures and pressures.

2.06 UNIONS, FLANGES, AND COUPLINGS

A. Unions for Pipe 2-1/2 Inches and Under:
   1. Copper Pipe: Bronze, soldered joints, ASME B16.22.
   2. Copper Pipe: Class 150 Bronze or cast copper companion flange (all valve assemblies and final connections of equipment)

B. Flanges for Pipe 3 Inches and larger:
   1. Ferrous Piping: 150 psig forged steel, slip-on.

C. Dielectric Connections: Union or waterway fitting with water impervious isolation barrier and one galvanized or plated steel end and one copper tube end, end types to match pipe joint types used.

2.07 DIELECTRIC FITTINGS

A. General Requirements: Assembly of copper alloy and ferrous materials with separating nonconductive insulating material. Include end connections compatible with pipes to be joined.

B. Dielectric Unions: Permitted at equipment connections only and on the equipment side of the service isolation valves for the equipment. Dielectric unions not permitted on pipeline size transitions between steel and copper.
   1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
      a. A.Y. McDonald Mfg. Co.
      b. Capitol Manufacturing Company.
      c. Watts Regulator Co.
   2. Description:
      a. Standard: ASSE 1079.
      b. Pressure Rating: 125 psig minium at 180 deg F.
      c. End Connections: Solder-joint copper alloy and threaded ferrous.

C. Dielectric Flanges:
   1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
      a. Capitol Manufacturing Company.
      b. Central Plastics Company
      c. Watts Regulator Co.
   2. Description:
      a. Standard: ASSE 1079.
      b. Factory-fabricated, bolted, companion-flange assembly.
      c. Pressure Rating: 125 psig minimum at 180 deg F.
      d. End Connections: Solder-joint copper alloy and threaded ferrous; threaded solder-joint copper alloy and threaded ferrous.

D. Dielectric-Flange Insulating Kits:
   1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
      a. Advance Products & Systems, Inc.
      b. Calpico, Inc.
      c. Central Plastics Company.
   2. Description:
      a. Nonconducting materials for field assembly of companion flanges.
      b. Pressure Rating: 150 psig
      c. Gasket: Neoprene or phenolic.

d. Bolt Sleeves: Phenolic or polyethylene.
e. Washers: Phenolic with steel backing washers.

E. Dielectric Nipples:
1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
a. Grinnel Mechanical Products.
b. Precision Plumbing Products.
c. Victaulic Company.
2. Description:
a. Standard: IAPMO PS 66.
b. Electroplated steel nipple, complying with ASTM F1545.
c. Pressure Rating: 300 psig at 225 deg F.
d. End Connections: Male threaded or grooved.
e. Lining: Inert and noncorrosive, propylene.

2.08 CHILLED WATER SYSTEM VALVES

A. Isolation Service: Ball valves (2 inches IPS and smaller) and butterfly valves (2.5 inches IPS and larger) may be used for isolation service in hydronic systems. All ball valves in closed loop hydronic systems shall be full port design to minimize pressure drop.

B. Check Valves 3 Inches Diameter and Larger: Class 125 # SWP, 200 # WOG, horizontal swing check, body and cap shall be of ASTM A126 cast iron, bronze trim, bolted cap, flanged ends, including adjustable spring, lever, and weights oriented per application (horizontal or vertical piping), MSS SP-71.

C. Check Valves 2.5 Inches Diameter and Smaller: Class 150 # SWP, 300 # WOG, horizontal swing check, body and cap shall be of ASTM B62 cast bronze, TFE disc, integral bronze seats, MSS SP-80.

D. Ball Valves 3 Inches Diameter and Smaller: 600 # WOG, 150 # SWP, 2 piece body style, full port, CP brass tunneled ball, reinforced teflon seats, hex gland follower, bronze body of ASTM B584, blow-out proof stem, lever handle.

E. Butterfly Valves 2.5 Inches Diameter and Larger: 200 # WOG, ductile iron full lug body, bronze or stainless steel disc, 316 SS stem, field-replaceable phenolic-backed EPDM cartridge style seat (“boot” type seats without backing rings are not acceptable), blow-out proof stem, MSS SP-67. Valves shall be bi-directional bubble tight shut off rated for in-line service as well as dead end service with either mating flange removed. Valves 6” and smaller shall be provided with a 10 position lever handle, 8” and larger with a geared rotary hand wheel operator.

F. Silent Globe Style Check Valves 2.5 Inches Diameter and Larger: 200 # WOG minimum, body shall be of ASTM A126 Class B cast iron, center guided, stainless steel spring and screws, bronze disc, bronze seat.

G. Valves which are insulated shall have stem extensions and plastic insert/cap to keep insulation clear of the stem.

2.09 BALL-STYLE CONTROL VALVES

A. Ball Style with Single Port and Characterized Disk, 2 Inches and smaller:
1. Pressure Rating: 600 # WOG (1 Inch and smaller), 400 # WOG (1.5 Inches to 2 Inches)
2. Close-Off Pressure: 200 psig
3. Temperature Range: 0°F - 212°Fcast broze per ASTM B61 or ASTM B62 or ASTM B584,
4. Body and Tail Piece: Cast bronze per ASTM B61, ASTM B62, ASTM B584, or forged brass with nickel plating.
5. End Connections: Threaded (NPT) ends
6. Ball: Chrome-plated brass or bronze or 300 series stainless steel
7. Stem and Stem Exention:

a. Material to match ball
b. Blowout-proof design
c. Sleeve or other approved means to allow valve to be opened and closed without damaging the insulation or the vapor barrier seal.

8. Ball seats: Reinforced PTFE
9. Stem Seal: Reinforced PTFE packing ring with a threaded packing ring follower to retain the packing ring under design pressure with the linkage removed. Alternative means, such as EPDM O-rings, are acceptable if an equivalent cycle endurance can be demonstrated by testing.
10. Flow Characteristic: Equal percentage.

B. Ball Style with Two Port and Characterized Disk (3-way), 2 Inches and smaller:
1. Pressure Rating: 600 # WOG (1 Inch and smaller), 400 # WOG (1.5 Inches to 2 Inches)
2. Close-Off Pressure: 200 psig
3. Temperature Range: 0°F - 212°Fcast broze per ASTM B61 or ASTM B62 or ASTM B584,
4. Body and Tail Piece: Cast bronze per ASTM B61, ASTM B62, ASTM B584, or forged brass with nickel plating.
5. End Connections: Threaded (NPT) ends
6. Ball: Chrome-plated brass or bronze or 300 series stainless steel
7. Stem and Stem Exention:
   a. Material to match ball
   b. Blowout-proof design
   c. Sleeve or other approved means to allow valve to be opened and closed without damaging the insulation or the vapor barrier seal.
8. Ball seats: Reinforced PTFE
9. Stem Seal: Reinforced PTFE packing ring with a threaded packing ring follower to retain the packing ring under design pressure with the linkage removed. Alternative means, such as EPDM O-rings, are acceptable if an equivalent cycle endurance can be demonstrated by testing.
10. Flow Characteristics for A-Port: Equal percentage.
11. Flow Characteristics for B-Port: Modified for constant common port flow.

C. Ball Valves with Single Port and Segmented Ball:
1. Performance:
   a. Process Temperature Rating: Minus 20 to plus 450 deg F.
   b. ASME B16.34 , Class 150 or Class 300.
   c. Leakage: ANSI/FCI 70-2, Class IV.
   d. Rangeability: 300 to 1.
   e. Rotation: Zero to 90 degrees.
   f. Equal percentage flow characteristic.
2. ASME B16.10 face-to-face dimensions.
3. Valves NPS 2 and Smaller: Threaded (NPT) ends.
4. Valves NPS 2-1/2 through NPS 6: Flanged ends suitable for mating to ASME B16.5 flanges.
5. Body: Carbon or stainless steel.
6. Ball and Shaft: Stainless steel.
7. Shaft and Segmented Ball: Pinned and welded.
8. Ball Seat: Graphite.
9. Packing: PTFE V-rings and graphite packing follower.
10. Replaceable seat, ball, and shaft packing.
11. Label each valve with following:
   a. Manufacturer's name, model number, and serial number.
   b. Body size.

c. Flow directional arrow.

D. Ball Valves with Segmented Ball, Three-Way Pattern:
1. Arrangement: Two single-port valves mated to a fabricated tee with interconnecting mechanical linkage.
2. Performance:
a. Process Temperature Rating: Minus 20 to plus 450 deg F.
b. ASME B16.34, Class 150 or Class 300.
c. Leakage: ANSI/FCI 70-2, Class IV.
d. Rangeability: 300 to 1.
e. Rotation: Zero to 90 degrees.
f. Equal percentage flow characteristic.
3. Face-to-Face Dimensions: ASME B16.10.
4. Valves NPS 3through NPS 6: Flanged ends suitable for mating to ASME B16.5 flanges.
5. Body: Carbon or stainless steel.
6. Ball and Shaft: Stainless steel.
7. Shaft and Segmented Ball: Pinned and welded.
8. Ball Seat: Graphite.
9. Packing: PTFE V-rings and graphite packing follower.
10. Replaceable seat, ball, and shaft packing.
11. Label each valve with following:
a. Manufacturer's name, model number, and serial number.
b. Body size.
c. Flow directional arrow.

E. Ball Valves with Full Ball and Characterized V-Notch:
1. Performance:
a. Process Temperature Rating: Minus 20 to plus 500 deg F.
b. ASME B16.34, Class 600 for NPS 2 and smaller; Class 150 or Class 300 for larger than NPS 2.
c. Leakage: ANSI/FCI 70-2, Class VI, bi-directional.
d. Rangeability: Varies from 200 to 1 up to 800 to 1 based on notch pattern of ball.
e. Rotation: Zero to 90 degrees.
f. Equal percentage flow characteristic.
g. Full port.
2. Face-to-Face Dimension: ASME B16.10 long pattern.
3. Valves NPS 2 and Smaller: ASME B1.20.1 threaded (NPT) ends and three-piece body.
4. Valves NPS 2-1/2 through NPS 12: Flanged ends suitable for mating to ASME B16.5 flanges and two-piece body.
5. Hole in the stem slot of each ball equalizes pressure between the body cavity and the line media flow.
6. Replaceable seat, ball, and shaft packing.
7. Body: Carbon or stainless steel.
8. Ball and Shaft: Stainless steel.
9. Ball Seat: RPTFE.
10. Stem Seals for Valves NPS 2 and Smaller: Live-loaded, self-adjusting, primary and secondary sealing using belleville washers.
a. Primary Seal: Combination of thrust washer and thrust washer protector.
b. Secondary Seal: Adjustable stem packing composed of RPTFE V-rings.
11. Stem Seals for Valves Larger than NPS 2: Independent packing gland, adjusted without removing mounting hardware or operator, and contoured to uniformly distribute load across packing.
a. Primary Seal: Combination of thrust washer and thrust washer protector.

b. Secondary Seal: Adjustable stem packing composed of RPTFE V-rings.

12. Label each valve with following:
    a. Manufacturer's name, model number, and serial number.
    b. Body size.
    c. Flow directional arrow.

F. Industrial-Grade Ball Valves:
1. Performance:
    a. Process Temperature Rating: Minus 20 to plus 450 deg F.
    b. ASME B16.34, Class 150 or Class 300.
    c. Leakage: ANSI/FCI 70-2, Class VI.
    d. Rangeability: 300 to 1.
    e. Rotation: Zero to 90 degrees.
    f. Modified equal percentage flow characteristic.
2. Face-to-Face Dimensions: Comply with ASME B16.10 short pattern.
3. Body: Cast steel ASTM A216/A216M WCB.
4. Flanged Body: Suitable for mating to ASME B16.5 flanges.
5. Shaft: 316 stainless-steel ball, 17-4 PH stainless steel.
6. Ball Seat: Reinforced PTFE.
7. PTFE V-ring packing, 316 stainless-steel packing follower.
8. Replaceable seat, ball, and shaft packings.
9. Replaceable 316 stainless-steel shaft bushings with PTFE linings.
10. Corrosion-resistant nameplate indicating the following:
    a. Manufacturer's name, model number, and serial number.
    b. Body size.
    c. Body and trim materials.
    d. Trim type.
    e. Body and flange rating.
    f. Arrow indicating direction of flow.

G. Pressure-Independent Ball Valves NPS 2 and Smaller:
1. Performance:
    a. Pressure Rating: 600 psig for NPS 1 and 400 psig for NPS 1-1/2 and NPS 2.
    b. Close-off pressure of 200 psig.
    c. Process Temperature Range: Between zero to 212 deg F.
    d. Rangeability: 100 to 1.
2. Integral Pressure Regulator: Located upstream of ball to regulate pressure, to maintain a constant pressure differential while operating within a pressure differential range of 5 to 50 psig.
3. Body: Forged brass, nickel plated, and with threaded ends.
4. Ball: Chrome-plated brass.
5. Stem and Stem Extension: Chrome-plated brass, blowout-proof design.
6. Stem sleeve or other approved means to allow valve to be opened and closed without damaging field-applied insulation and insulation vapor barrier seal.
7. Ball Seats: Reinforced PTFE.
8. Stem Seal: Reinforced PTFE packing ring stem seal with threaded packing ring follower to retain the packing ring under design pressure with the linkage removed. Alternative means, such as EPDM O-rings, are acceptable if equivalent cycle endurance can be achieved.
9. Flow Characteristic: Equal percentage.

2.10 BUTTERFLY-STYLE CONTROL VALVES

A. Commercial-Grade, Two-Way Butterfly Valves:
1. Performance:

a. Bi-directional bubble tight shutoff at 250 psig.
b. Comply with MSS SP-67 or MSS SP-68.
c. Rotation: Zero to 90 degrees.
d. Linear or modified equal percentage flow characteristic.

2. Body: Cast iron ASTM A126, Class B, ductile ironASTM A536 or cast steel ASTM A216/A216M WCB fully lugged, suitable for mating to ASME B16.5 flanges.
3. Disc: 316 stainless steel.
4. Shaft: 316 or 17-4 PH stainless steel.
5. Seat: Reinforced EPDM or reinforced PTFE with retaining ring.
6. Shaft Bushings: Reinforced PTFE or stainless steel.
7. Replaceable seat, disc, and shaft bushings.
8. Corrosion-resistant nameplate indicating:
   a. Manufacturer's name, model number, and serial number.
   b. Body size.
   c. Body and trim materials.
   d. Flow arrow.

B. Commercial-Grade, Three-Way Butterfly Valves:
1. Arrangement: Two valves mated to a fabricated tee with interconnecting mechanical linkage.
2. Performance:
   a. Bi-directional bubble tight shutoff at 250 psig.
   b. Comply with MSS SP-67 or MSS SP-68.
   c. Rotation: Zero to 90 degrees.
   d. Linear or modified equal percentage flow characteristic.
3. Body: Cast iron ASTM A126, Class B, ductile iron ASTM A536 or cast steel ASTM A216/A216M WCB fully lugged, suitable for mating to ASME B16.5 flanges.
4. Disc: 316 stainless steel.
5. Shaft: 316 or 17-4 PH stainless steel.
6. Seat: Reinforced EPDM or reinforced PTFE seat with retaining ring.
7. Shaft Bushings: Reinforced PTFE or stainless steel.
8. Replaceable seat, disc, and shaft bushings.
9. Corrosion-resistant nameplate indicating:
   a. Manufacturer's name, model number, and serial number.
   b. Body size.
   c. Body and trim materials.
   d. Flow arrow.

C. Industrial-Grade Butterfly Valves:
1. Performance:
   a. Process Temperature Rating: Minus 200 to plus 849 deg F.
   b. ASME B16.34, Class 150 or Class 300 for larger sizes.
   c. Complies with MSS SP-68.
   d. Leakage: FCI 70-2, Class VI, bi-directional.
   e. Rangeability: 100 to 1.
   f. Rotation: Zero to 90 degrees.
   g. Linear or modified equal percentage flow characteristic.
2. Body: Cast steel ASTM A216/A216M WCB, fully lugged, suitable for mating to ASME B16.5 flanges.
3. Disc: ASTM A351/A351M, CF3M or CF8M stainless steel.
4. Shaft: 17-4 PH stainless steel.
5. Seat: Reinforced PTFE with retaining ring.
6. Shaft Bushings: Reinforced PTFE or stainless steel.

7. Replaceable seat, disc, and shaft bushings.
8. Corrosion-resistant nameplate indicating:
    a. Manufacturer's name, model number, and serial number.
    b. Body size.
    c. Body and trim materials.
    d. Body rating.
    e. Arrow indicating direction of flow.

2.11 GLOBE-STYLE CONTROL VALVES

A. General Globe-Style Valve Requirements:
1. Globe-style control valve body dimensions shall comply with ISA 75.08.01**.**
2. Construct the valves to be serviceable from the top.
3. For cage guided valves, trim shall be field interchangeable for different valve flow characteristics, such as equal percentage, linear, and quick opening.
4. Reduced trim for one nominal size smaller shall be available for industrial valves NPS 1 and larger.
5. Replaceable seats and plugs.
6. Furnish each control valve with a corrosion-resistant nameplate indicating the following:
    a. Manufacturer's name, model number, and serial number.
    b. Body and trim size.
    c. Arrow indicating direction of flow.

B. Two-Way Globe Valves NPS 2 and Smaller:
1. Globe Style: Single port.
2. Body: Cast bronze or forged brass with ASME B16.5, Class 250 rating.
3. End Connections: Threaded.
4. Bonnet: Screwed.
5. Packing: PTFE V-ring.
6. Plug: Top guided.
7. Plug, Seat, and Stem: Brass or stainless steel.
8. Process Temperature Range: 35 to 248 deg F.
9. Ambient Operating Temperature: 35 to 150 deg F.
10. Leakage: ANSI/FCI 70-2 , Class IV.
11. Rangeability: 25 to 1.
12. Equal percentage flow characteristic.

C. Three-Way Globe Valves NPS 2 and Smaller:
1. Globe Style: Mix flow pattern.
2. Body: Cast bronze or forged brass with ASME B16.5, Class 250 rating.
3. End Connections: Threaded.
4. Bonnet: Screwed.
5. Packing: PTFE V-ring.
6. Plug: Top guided.
7. Plug, Seat, and Stem: Brass or stainless steel.
8. Process Temperature Range: 35 to 248 deg F.
9. Ambient Operating Temperature: 35 to 150 deg F.
10. Leakage: ANSI/FCI 70-2, Class IV.
11. Rangeability: 25 to 1.
12. Linear flow characteristic.

D. Two-Way Globe Valves NPS 2-1/2 to NPS 6:
1. Globe Style: Single port.
2. Body: Cast iron complying with ASME B16.1, Class 125.
3. End Connections: Flanged, suitable for mating to ASME B16.5, Class 150 flanges.

4. Bonnet: Bolted.
5. Packing: PTFE cone-ring.
6. Plug: Top or bottom guided.
7. Plug, Seat, and Stem: Brass or stainless steel.
8. Process Temperature Rating: 35 to 281 deg F.
9. Leakage: 0.1 percent of maximum flow.
10. Rangeability: Varies with valve size between 6 and 10 to 1.
11. Modified linear flow characteristic.

E. Three-Way Globe Valves NPS 2-1/2 to NPS 6:
1. Globe Style: Mix flow pattern.
2. Body: Cast iron complying with ASME B16.1, Class 125.
3. End Connections: Flanged suitable for mating to ASME B16.5, Class 150 flanges.
4. Bonnet: Bolted.
5. Packing: PTFE cone-ring.
6. Plug: Top or bottom guided.
7. Plug, Seat, and Stem: Brass or stainless steel.
8. Process Temperature Rating: 35 to 281 deg F.
9. Leakage: 0.1 percent of maximum flow.
10. Rangeability: Varies with valve size between 6 and 10 to 1.
11. Modified linear flow characteristic.

F. Industrial-Grade Straight-Through Globe Valves NPS 3/4 and Smaller:
1. Globe Style: Single port.
2. Body: ASTM B62 bronze complying with ASME B16.5, Class 250.
3. End Connections: Threaded.
4. Bonnet: Screwed or bolted.
5. Packing: PTFE V-ring.
6. Plug: Top or cage guided; balanced or unbalanced.
7. Plug, Seat, and Stem: 316 stainless steel, 17-4 PH stainless-steel cage.
8. Process Temperature Range: Minus 20 to plus 400 deg F.
9. Ambient Operating Temperature: Minus 20 to plus 150 deg F.
10. Leakage: ANSI/FCI 70-2, Class IV.
11. Equal percentage flow characteristic.

G. Industrial-Grade Straight-Through Globe Valves NPS 1 and Larger:
1. Globe Style: Single port.
2. Body: Cast iron or cast steel.
3. End Connections for NPS 2: Threaded.
4. End Connections for NPS 2-1/2 and Larger: Raised face flanged.
5. Bonnet: Bolted.
6. Packing: PTFE V-ring.
7. Plug: Cage guided and unbalanced.
8. Plug, Seat, and Stem: 416 stainless-steel plug and seat, 17-4 PH stainless-steel cage and 316 stainless-steel stem.
9. Valve Stem: Thread and pin stem to plug.
10. Valve Stem Finish: Polished to 5 microinches rms or less.
11. Plug and Seat Surfaces: Hardened facing.
12. Process Temperature Range: Zero to 450 deg F.
13. Ambient Operating Temperature: Minus 20 to plus 150 deg F.
14. Leakage: ANSI/FCI 70-2, Class IV
15. Flow Characteristic: Equal percentage

PART 3 - EXECUTION

3.01 PIPING APPLICATIONS

A. Where more than one piping type is specified, selection shall be Contractor's option.

B. Chilled-water piping, NPS 2-1/2" and smaller shall be the following:
   1. Type L drawn-temper copper tubing, wrought-copper fittings, and with soldered joints.

C. Chilled-water piping, NPS 3" and larger shall be the following:
   1. Standard Weight (schedule 40) steel pipe, wrought-steel fittings and wrought-cast or forged-steel flanges and flange fittings, and welded and flanged joints.
   2. Mechanical grooved coupling systems not permitted on chilled water piping systems.

D. Air-Vent Piping:
   1. Inlet: Same as service where installed with metal-to-plastic transition fittings for plastic piping systems according to piping manufacturer's written instructions.
   2. Outlet: Type L, annealed-temper copper tubing with soldered or flared joints.

3.02 INSTALLATIONS

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems. Install piping as indicated unless deviations to layout are approved on Coordination Drawings.

B. Pipe Installation:
   1. All the various piping systems shall be made up straight and true and run in orderly manner, plumb and parallel to building structural. Install piping to conserve building space. Coordinate location with other trades and do not interfere with use of space for other work.
   2. Piping shall follow as closely as possible the routes shown on Drawings which take into consideration conditions to be met at the Site.
   3. Should any unforeseen conditions arise, lines shall be changed or rerouted after proper approval has been obtained.
   4. All piping shall be installed with due regard to expansion and contraction and so as to prevent excessive strain and stress in the piping, in connections, or in equipment to which the lines are connected.
   5. Group piping whenever practical at common elevations.
   6. Slope piping and arrange system to drain at low points. Use eccentric reducers where applicable to maintain the bottom of pipe level.
   7. Branch tap connections are to be from the top to horizontal position of pipe run.
   8. Where pipe support members are welded to structural building framing, scrape, brush clean, and apply one coat of zinc rich primer to welding.
   9. Provide and install Pete's plugs adjacent to thermo wells for electronic temperature sensors, to electronic pressure sensors and install Pete's plugs adjacent where shown or noted on piping drawings or drawing details.
   10. Provide clearance for installation of insulation, and access to valves and fittings.
   11. Prepare pipe, fittings, supports, and accessories for finish painting. Chilled water piping insulated with cellular glass does not require finish painting.
   12. All piping shall be clean when it is installed. Before installation it shall be checked, upended, swabbed if necessary, and all rust or dirt from storage or from laying on the ground shall be removed.
   13. Where leaks occur, the pipe shall be repaired and the tests repeated. No leaks shall be corrected by peening. Defective piping and joints shall be removed and replaced.
   14. Procedure of Assembling Screw Pipe Fittings: All screw joints shall be made with taper threads, properly cut. Joints shall be made tight with Teflon tape or Teflon-based compound appropriate to the medium, material and temperature range of the system. Compound shall be applied to the pipe threads only and not to fittings. When threads

are cut on pipes, the ends shall be carefully reamed to remove any burrs. Before installing pipe that has been cut and threaded, the lengths of pipe shall be upended and hammered to remove all shavings and foreign material.

C. Valve Installation:
   1. Locate all valves such that the removal of their bonnets is possible. All flanged valves shown in horizontal lines with the valve stem in a horizontal position shall be positioned so the valve stem is inclined one bolt hole above the horizontal position.
   2. Screw pattern valves placed in horizontal lines shall be installed with their valve stems include at an angle of a minimum of 30 degrees above the horizontal position.
   3. All valves must be true and straight at the time the system is tested and inspected for final acceptance.
   4. Valves shall be installed as nearly as possible to the locations indicated in the Drawings. Any change in valve location must be so indicated on the Record As-Built Drawings.
   5. All valves must be of threaded or flanged type. No solder connected or grooved fitting valves shall be used on this Project.
   6. Equipment, valves, expansion joints, relief devices, strainers, etc., must be removed or isolated during the test if the pressure/force ratings of the devices are not as high as that specified for the test. Piping shall be drained and protected any time ambient temperature is below freezing.
   7. Where leaks occur, the pipe shall be repaired and the tests repeated. No leaks shall be corrected by peening. Defective piping and joints shall be removed and replaced.
   8. All threaded valves.installed in copper piping shall be provided with copper or bronze male adapters on each side of valves. Sweat solder adapters to pipe before installing valves.
   9. Provide access where valves and fittings are not exposed. Coordinate size and location of access doors with engineering drawings.
   10. Install valves with stems upright or horizontal, not inverted.
   11. All manually operated shutoff valves 3" and larger located 8 feet (Bottom of pipe) or higher above finished floor or stationary platform in mechanical rooms, accessible pipe chases or as noted on Project Drawings shall be chain wheel operated. Chains shall be installed and secured to allow clear passage at walk through areas.

D. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.

E. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.

F. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.

G. Install piping to permit valve servicing.

H. Install piping at indicated slopes.

I. Install manual air vent with isolation valve at all ends and high points in system that could otherwise air bind.

J. Install piping free of sags and bends.

K. Install fittings for changes in direction and branch connections.

L. Install piping to allow application of insulation.

M. Select system components with pressure rating equal to or greater than system operating pressure.

N. Install groups of pipes parallel to each other, spaced to permit applying insulation and servicing of valves.

O. Install drains, consisting of a tee fitting, NPS 3/4 ball valve, and short NPS 3/4 threaded nipple with cap, at low points in piping system mains and elsewhere as required for system drainage.

P. Install piping at a uniform grade of 0.2 percent upward in direction of flow.

Q. Reduce pipe sizes using eccentric reducer fitting installed with level side up.

R. Install branch connections to mains using mechanically formed fittings (pulled-tees are not acceptable) in main pipe, with the branch connected to the top of the main pipe. For up-feed risers, connect the branch to the top of the main pipe.

S. Bullhead tees shall not be utilzied on piping systems.

T. Install unions in piping, NPS 2 and smaller, adjacent to valves, at final connections of equipment, and elsewhere as indicated.

U. Install flanges in piping, NPS 2-1/2 and larger, at all valve assemblies and final connections of equipment and elsewhere as indicated.

V. Install shutoff valve immediately upstream of each dielectric fitting.

W. Comply with requirements in Section 230516 "Expansion Fittings and Loops for HVAC" for installation of expansion loops, expansion joints, anchors, and pipe alignment guides.

X. Comply with requirements in Section 230553 "Identification for HVAC" for identifying piping.

Y. Install sleeves for piping penetrations of walls, ceilings, and floors. Comply with requirements for sleeves specified in Section 230517 "Sleeves and Sleeve Seals for HVAC."

Z. Install sleeve seals for piping penetrations of concrete walls and slabs. Comply with requirements for sleeve seals specified in Section 230517 "Sleeves and Sleeve Seals for HVAC."

AA. Install escutcheons for piping penetrations of walls, ceilings, and floors. Comply with requirements for escutcheons specified in Section 23 0518 "Escutcheons for HVAC Piping."

3.03 DIELECTRIC FITTING INSTALLATION

A. Install dielectric fittings in piping at connections of dissimilar metal piping and tubing.

B. Dielectric Fittings for NPS 2 and Smaller: Use dielectric nipples or unions. Dielectric Unions permitted only at equipment connections on the equipment side of service isolation valves. At copper pipe size branch connections to steel pipe size mains and material size transition fittings not protected by service isolation, use dielectric nipple only.

C. Dielectric Fittings for NPS 2-1/2 to NPS 4: Use dielectric flanges or nipples.

D. Dielectric Fittings for NPS 5 and Larger: Use dielectric flange kits.

3.04 HANGERS AND SUPPORTS

A. Comply with requirements in Section 230529 "Hangers and Supports for HVAC Piping and Equipment" for hanger, support, and anchor devices. Comply with the following requirements for maximum spacing of supports.

B. Comply with requirements in Section 23 0548 "Vibration Controls for HVAC" for seismic restraints.

3.05 PIPE JOINT CONSTRUCTION

A. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.

B. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

C. Soldered Joints: Apply ASTM B813, water-flushable flux, unless otherwise indicated, to tube end. Construct joints according to ASTM B828 or CDA's "Copper Tube Handbook," using lead-free solder alloy complying with ASTM B32.

D. Brazed Joints: Construct joints according to AWS's "Brazing Handbook," "Pipe and Tube" Chapter, using copper-phosphorus brazing filler metal complying with AWS A5.8/A5.8M.

E. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:
   1. Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is specified.
   2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.

F. Welded Joints: Construct joints according to AWS D10.12M/D10.12, using qualified processes and welding operators according to "Quality Assurance" Article.

G. Flanged Joints: Select appropriate gasket material, size, type, and thickness for service application. Install gasket concentrically positioned. Use suitable lubricants on bolt threads.

H. Mechanically Formed, Copper-Tube-Outlet Joints: Not Permitted.

3.06 FIELD QUALITY CONTROL

A. Prepare hydronic piping according to ASME B31.9 and as follows:
   1. Leave joints, including welds, uninsulated and exposed for examination during test.
   2. Provide temporary restraints for expansion joints that cannot sustain reactions due to test pressure. If temporary restraints are impractical, isolate expansion joints from testing.
   3. Flush hydronic piping systems with clean water; then remove and clean or replace strainer screens.
   4. Isolate equipment from piping. If a valve is used to isolate equipment, its closure shall be capable of sealing against test pressure without damage to valve. Install blinds in flanged joints to isolate equipment.
   5. Install safety valve, set at a pressure no more than one-third higher than test pressure, to protect against damage by expanding liquid or other source of overpressure during test.

3.07 TESTING

A. All welds are subject to inspection, visual and/or x-ray, for compliance with Specifications. The Owner will, at the Owner's option, provide employees or employ a testing laboratory for the purposes of performing said inspections and/or x-ray testing. Initial visual and x-ray inspections will be provided by the Owner. The Contractor shall be responsible for all labor, material and travel expenses involved in the re-inspection and retesting of any welds found to be unacceptable. In addition, the Contractor shall be responsible for the costs involved in any and all additional testing required or recommended by ASME B31.9 due to the discovery of poor, unacceptable or rejected welds.

B. Welds lacking penetration, containing excessive porosity or cracks, or are found to be unacceptable for any reason, must be removed and replaced with an original quality weld as specified herein. All qualifying tests, welding and stress relieving procedures shall, moreover, be in accord with Standard Qualification for Welding Procedures, Welders and Welding Operators, Appendix A, Section 6 of the Code, current edition.

C. System Pressure Tests: All Hydronic Systems shall be tested at 1.5 time the operating pressure. Minimum 125 psi. The test shall be done with water.

D. Perform the following tests on hydronic piping:
   1. Use ambient temperature water as a testing medium unless there is risk of damage due to freezing. Another liquid that is safe for workers and compatible with piping may be used.
   2. While filling system, use vents installed at high points of system to release air. Use drains installed at low points for complete draining of test liquid.
   3. Isolate expansion tanks and determine that hydronic system is full of water.
   4. Subject piping system to hydrostatic test pressure that is not less than 1.5 times the system's working pressure. Test pressure shall not exceed maximum pressure for any vessel, pump, valve, or other component in system under test. Verify that stress due to

pressure at bottom of vertical runs does not exceed 90 percent of specified minimum yield strength or 1.7 times the "SE" value in Appendix A in ASME B31.9.

5. After hydrostatic test pressure has been applied for at least 10 minutes, examine piping, joints, and connections for leakage. Eliminate leaks by tightening, repairing, or replacing components, and repeat hydrostatic test until there are no leaks.
6. Prepare written report of testing.

E. Perform the following before operating the system:
1. Open manual valves fully.
2. Inspect pumps for proper rotation.
3. Set makeup pressure-reducing valves for required system pressure.
4. Inspect air vents at high points of system and determine if all are installed and operating freely (automatic type), or bleed air completely (manual type).
5. Set temperature controls so all coils are calling for full flow.
6. Inspect and set operating temperatures of hydronic equipment, such as boilers, chillers, and fluid coolers to specified values.
7. Verify lubrication of motors and bearings.

END OF SECTION 23 2113

SECTION 23 2116
HYDRONIC PIPING SPECIALTIES

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section Includes:
   1. Air vents.
   2. Air/dirt separators.
   3. Pressure relief valves
   4. Strainers.
   5. Balancing valves
   6. Expansion tanks
   7. Connectors
   8. Buffer Tanks

B. Related Requirements:
   1. Division 23 Section - Hydronic Piping.
   2. Division 23 Sections for HVAC Water Treatment.

1.03 REFERENCE STANDARDS

A. ASME ASME BPVC-VIII-1 - Boiler and Pressure Vessel Code, Section VIII, Division 1 - Rules for Construction of Pressure Vessels; The American Society of Mechanical Engineers; 2025.

1.04 SUBMITTALS

A. Product Data: Provide product data for manufactured products and assemblies required for this project. Include component sizes, rough-in requirements, service sizes, and finishes. Include product description, model and dimensions.

B. Certificates: Inspection certificates for pressure vessels from authority having jurisdiction.

C. Project Record Documents: Record actual locations of flow controls.

D. Maintenance Data: Include installation instructions, assembly views, lubrication instructions, and replacement parts list.

1.05 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing the type of products specified in this section, with minimum three years of documented experience.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Accept valves on site in shipping containers with labeling in place. Inspect for damage.

B. Provide temporary end caps and closures on piping and fittings. Maintain in place until installation.

C. Protect piping systems from entry of foreign materials by temporary covers, completing sections of the work, and isolating parts of completed system.

1.07 MAINTENANCE SERVICE

A. Furnish service and maintenance of glycol system for one year from date of substantial completion.

B. Perform monthly visit to make glycol fluid concentration analysis on site with refractive index measurement instrument. Detail findings with maintenance personnel in writing of corrective actions needed including analysis and amounts of glycol or water added.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer's product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.

B. Equipment Categories: Equipment types in this section are considered equipment categories that be furnished by one manufacturer or available through one manufacturer to provide consistency within the overall project for equipment configurations, manufacturing techniques, and to facilitate ease of maintenance for the Owner.

2.02 AIR VENTS

A. Manufacturers:
   1. Amtrol Inc: www.amtrol.com.
   2. Armstrong International, Inc: www.armstronginternational.com.
   3. ITT Bell & Gossett: www.bellgossett.com.
   4. Taco, Inc: www.taco-hvac.com.

B. Float Type:
   1. 125 psig pressure rating, brass or semi-steel body, copper, polypropylene, or solid non-metallic float, stainless steel valve and valve seat; suitable for system operating temperature and pressure; with isolating valve.
   2. Locate all air vents and their discharge lines in accessible locations, preferably clustered
   3. Route discharge lines to nearest floor drain without air traps.

2.03 COALESCING-TYPE AIR AND DIRT SEPARATORS:

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
   1. Armstrong Pumps, Inc.
   2. Bell & Gossett; a Xylem brand.
   3. Spirotherm, Inc.
   4. Amtrol

B. Tank: Fabricated steel tank; ASME constructed and stamped for 125-psig (862-kPa) working pressure and 270 deg F (130 deg C) maximum operating temperature.

C. Coalescing Medium: Stainless steel.

D. Air Vent: Threaded to the top of the separator.

E. Inline Inlet and Outlet Connections: Threaded for NPS 2 (DN 50) and smaller; Class 150 flanged connections for NPS 2-1/2 (DN 65) and larger.

F. Blowdown Connection: Threaded to the bottom of the separator.

G. Size: Match system flow capacity.

2.04 STRAINERS

A. Manufacturers:
   1. Armstrong International, Inc: www.armstronginternational.com.
   2. WEAMCO: www.weamco.com.

B. Size 2 inch and Under:
   1. Screwed brass or iron body for 175 psi working pressure, Y pattern with 1/32 inch stainless steel perforated screen.

C. Size 2-1/2 inch to 4 inch:
   1. Flanged iron body for 175 psi working pressure, Y pattern with 3/64 inch stainless steel perforated screen.

D. Size 5 inch and Larger:
   1. Flanged iron body for 175 psi working pressure, basket pattern with 1/8 inch stainless steel perforated screen. Lifting lugs for lid

E. Coordinate with Hydronic Pumps for requirements

2.05 HYDRONIC BALANCING VALVES

A. Description: Multi-Turn Globe Style Valve: Each valve shall be a multi-turn globe style valve capable of bubble-tight shut-off with a memory device and indicator scale

B. Manufacturers: Provide products by one of the following.
   1. Tour & Anderson
   2. Armstrong International, Inc.
   3. Wheatley
   4. Nexus
   5. Patterson Pump Company

C. Scope: Scope includes valves used for general balancing and isolation service, 1.5 to 6 inch, up to 125 psig maximum.

D. Return Piping Installation: Balancing valves shall be installed in the return piping, NOT in supply piping. Provide adequate straight pipe upstream and downstream of balancing valves as required by the manufacturer.

E. Flow Measurement Ports: Each valve shall incorporate flow measurement ports to allow balancing. Ports shall be within 45 degrees of valve stem to facilitate access to handle and ports.

F. Infinite Setting Devices: Each valve shall incorporate infinite setting devices to prevent reopening past established balancing position when valve is used for isolation purposes.

G. Sizing: Each balance valve shall be sized for the specified flow rate (gpm) of the associated equipment. Balance valves shall not be simply specified as “line size”.

2.06 CONNECTORS

A. Stainless-Steel Bellow, Flexible Connectors:
   1. Body: Stainless-steel bellows with woven, flexible, bronze, wire-reinforcing protective jacket.
   2. End Connections: Threaded or flanged to match equipment connected.
   3. Performance: Capable of 3/4-inch (20-mm) misalignment.
   4. CWP Rating: 150 psig (1035 kPa).
   5. Maximum Operating Temperature: 250 deg F (121 deg C).

2.07 DIAPHRAGM-OPERATED SAFETY VALVES: ASME LABELED.

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
   1. AMTROL, Inc.
   2. Apollo Valves; Conbraco Industries, Inc.
   3. Armstrong Pumps, Inc.
   4. Bell & Gossett; a Xylem brand.
   5. Spence Engineering Company, Inc.
   6. Watts; a Watts Water Technologies company.

7. [Insert manufacturer's name].

B. Body: Bronze or brass.

C. Disc: Glass and carbon-filled PTFE.

D. Seat: Brass.

E. Stem Seals: EPDM O-rings.

F. Diaphragm: EPT.

G. Wetted, Internal Work Parts: Brass and rubber.

H. Inlet Strainer: removable without system shutdown.

I. Valve Seat and Stem: Noncorrosive.

J. Valve Size, Capacity, and Operating Pressure: Comply with ASME Boiler and Pressure Vessel Code: Section IV, and selected to suit system in which installed, with operating pressure and capacity factory set and field adjustable.

2.08 DIAPHRAGM-TYPE EXPANSION TANKS:

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
1. AMTROL, Inc.
2. Armstrong Pumps, Inc.
3. Bell & Gossett; a Xylem brand.
4. Flo Fab Inc.
5. TACO Comfort Solutions, Inc.

B. Tank: Welded steel, ASME rated for 125-psig (860-kPa) working pressure and 375 deg F (191 deg C) maximum operating temperature. Factory test after taps are fabricated and supports installed and are labeled according to ASME Boiler and Pressure Vessel Code: Section VIII, Division 1.

C. Diaphragm: Securely sealed into tank to separate air charge from system water to maintain required expansion capacity.

D. Air-Charge Fittings: Schrader valve, stainless steel with EPDM seats.

E. Size per system volume and working temperature

2.09 BUFFER TANKS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer’s product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.
1. Niles
2. Alternate manufacturers may be considered provided equivalent product and tank can fit through existing doorway to mehcanical room.

B. The tank must be designed, constructed and stamped 125 psi @ -20F to 450F in accordance with section VIII, Division I of the A.S.M.E. Boiler and Pressure Vessel code and registered with the National Board of boiler and pressure vessel inspections. The buffer tank shall be painted with red oxide shop primer paint. A manufacturers data report for pressure vessels, Form U-1A

as required by the provisions of the A.S.M.E. Boiler and Pressure Vessel code, shall be furnished for each buffer tank upon request.

C. Tank shall be provided with a center baffle plate with inlet and outlet on upper shell.

D. The top of the tank shall be provided with an FNPT (Female National Pipe Threads) port for the air vent

E. The bottom of the tank shall be provided with an FNPT (Female National Pipe Threads) drain port.

F. Provide with flanged pipe connections.

G. Provide with manway.

H. Provide with Niles Defender Epoxy internal coating/lining.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Install specialties in accordance with manufacturer's instructions.

B. Provide manual air vents at all system high points realtive to significant branch piping and as indicated.

C. For automatic air vents in ceiling spaces or other concealed locations, provide vent tubing to nearest drain.

D. Install expansion tanks on the floor. Vent and purge air from hydronic system, and ensure that tank is properly charged with air to suit system Project requirements.

E. Provide air separator on suction side of system circulation pump.

F. Provide relief valves on pressure tanks, low pressure side of reducing valves, heat exchangers, and expansion tanks.

G. Select system relief valve capacity so that it is greater than make-up pressure reducing valve capacity. Select equipment relief valve capacity to exceed rating of connected equipment.

H. Pipe relief valve outlet to nearest floor drain.

I. Install in-line air separators in pump suction. Install drain valve on air separators NPS 2 (DN 50) and larger. Coordinate with Hyronic Pumps Specifictions

J. All strainers 4-inch and under to have 1/2-inch ball valve and hose bib adapter at drain. All strainers 5-inch and larger to have 3/4-inch ball valve and hose bib adapter at drain.

K. Install piping from boiler air outlet, air separator, or air purger to expansion tank with a 2 percent upward slope toward tank.

L. All hydronic system balancing valves to be sized based on flow, per manufacturer's guidelines, not pipe size.

END OF SECTION 23 21146

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SECTION 23 2123
HYDRONIC PUMPS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. This Section includes the following:
   1. Separately coupled, base-mounted, end-suction centrifugal pumps.
   2. Pump specialty fittings.

1.03 SUBMITTALS

A. Product Data: Include certified performance curves and rated capacities, operating characteristics, furnished specialties, final impeller dimensions, and accessories for each type of product indicated. Indicate pump's operating point on curves.

B. Shop Drawings: Show pump layout and connections. Include setting drawings with templates for installing foundation and anchor bolts and other anchorages.
   1. Wiring Diagrams: Power, signal, and control wiring.

C. Operation and Maintenance Data: For pumps to include in emergency, operation, and maintenance manuals.

1.04 QUALITY ASSURANCE

A. Source Limitations: Obtain hydronic pumps through one source from a single manufacturer.

B. Product Options: Drawings indicate size, profiles, and dimensional requirements of hydronic pumps and are based on the specific system indicated.

C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction, and marked for intended use.

D. UL Compliance: Comply with UL 778 for motor-operated water pumps.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Manufacturer's Preparation for Shipping: Clean flanges and exposed machined metal surfaces and treat with anticorrosion compound after assembly and testing. Protect flanges, pipe openings, and nozzles with wooden flange covers or with screwed-in plugs.
   1. Store pumps in dry location.
   2. Retain protective covers for flanges and protective coatings during storage.
   3. Protect bearings and couplings against damage from sand, grit, and other foreign matter.
   4. Comply with pump manufacturer's written rigging instructions.

1.06 COORDINATION

A. Coordinate size and location of concrete bases. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified in Division 03.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Available Manufacturers: Subject to compliance with requirements, Available Manufacturers offering products that may be incorporated into the Work include, but are not limited to, the listed manufacturers in the following sections. Where a specific manufacturer’s product has been contemplated as the basis of design, that manufacturer has been listed in the equipment schedules on the Drawings or other Sections in this Specification. Where a basis of design is contemplated but other Available Manufacturer desired to be utilized, the proposed material or

equipment shall be identical to the specified material or equipment in all characteristics of quality, function, capacity, efficiency, spatial criteria and serviceability, regardless of application in the Project and, in addition, when the Engineer deems that aesthetic significance is important, the equal material or equipment must be identical in all characteristics of visual appearance, design, color and texture.

B. Equipment Categories: Equipment types in this section are considered equipment categories that be furnished by one manufacturer or available through one manufacturer to provide consistency within the overall project for equipment configurations, manufacturing techniques, and to facilitate ease of maintenance for the Owner.

2.02 GENERAL REQUIREMENTS

A. Pump motors shall be 1750 rpm maximum and sized for non-overloading service.

B. Pumps shall have stainless steel shafts and sleeves, bronze wear rings, and cast iron bases. Bases shall be designed for grouting in place.

C. Mechanical seals shall be John Crane Type XP662D1, with Viton bellows, tungsten carbide and carbon seal faces.

D. In special applications where packed pumps are used, packing shall be Chesterton 328 braided teflon, Crane, Durametallic or equal.

E. Pumps shall have Woods Dura-Flex or Rexnord Omega elastomeric couplings.

F. Pumps, in general, shall have a grease lubricated, heavy duty, deep groove ball bearings with a certified rating design of 200,000 hours of average bearing life. Acceptable manufacturers are S.K.F., Fafnir, and New Departure.

2.03 SEPARATELY COUPLED, BASE-MOUNTED, END-SUCTION CENTRIFUGAL PUMPS

A. Manufacturers: Subject to compliance with the requirements, provide products by one of the following:
   1. Armstrong Pumps
   2. Aurora Pumps
   3. Buffalo Pumps, Inc.; an Ampco Pittsburgh Co.
   4. PACO Pumps.
   5. Taco, Inc.
   6. Bell & Gossett; Div. of ITT Industries.

B. Description: Factory-assembled and -tested, centrifugal, overhung-impeller, separately coupled, end-suction pump as defined in HI 1.1-1.2 and HI 1.3; designed for base mounting, with pump and motor shafts horizontal. Rate pump for 175-psig (1204-kPa) minimum working pressure and a continuous water temperature of 225 deg F (107 deg C).

C. Pump Construction:
   1. Casing: Radially split, cast iron, with replaceable bronze wear rings, threaded gage tappings at inlet and outlet, drain plug at bottom and air vent at top of volute, and flanged connections. Provide integral mount on volute to support the casing, and attached piping to allow removal and replacement of impeller without disconnecting piping or requiring the realignment of pump and motor shaft.
   2. Impeller: ASTM B584, cast bronze; statically and dynamically balanced, keyed to shaft, and secured with a locking cap screw. Trim impeller to match specified performance.
   3. Pump Shaft: Stainless steel.
   4. Coupling Guard: Dual rated; ANSI B15.1, Section 8; OSHA 1910.219 approved; steel; removable; attached to mounting frame.
   5. Mounting Frame: Welded-steel frame and cross members, factory fabricated from ASTM A36/A36M channels and angles. Fabricate to mount pump casing, coupling guard, and motor.

D. Motor: Single speed, with grease-lubricated ball bearings, unless otherwise indicated; secured to mounting frame, with adjustable alignment. Comply with requirements in Division 23 Section "Common Motor Requirements for HVAC Equipment."

2.04 PUMP SPECIALTY FITTINGS

A. Suction Diffuser:
   1. Manufacturers: Subject to compliance with the requirements, provide products by one of the following:
      a. Taco.
      b. Victaulic.
   2. Cast iron, angle-type body with flow straightening device, removable strainer (bronze for start-up, stainless steel for permanent), tapped suction gage port, blow down connection, and adjustable support foot.
   3. Rated for maximum working pressure of 175 psi at 250 degree F.

PART 3 - EXECUTION

3.01 EXAMINATION

A. Examine equipment foundations and anchor-bolt locations for compliance with requirements for installation tolerances and other conditions affecting performance of work.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.02 PUMP INSTALLATION

A. Comply with HI 1.4.

B. Install all pumps in strict accordance with manufacturer’s instructions. Provide service space around pumps as recommended by the pump manufacturer.

C. Independently support pumps and piping so weight of piping is not supported by pumps and weight of pumps is not supported by piping.

D. Install continuous-thread hanger rods and spring hangers of sufficient size to support pump weight. Vibration isolation devices are specified in Division 23 Section "Vibration Controls for HVAC Piping and Equipment." Fabricate brackets or supports as required. Hanger and support materials are specified in Division 23 Section "Hangers and Supports for HVAC Piping and Equipment."

E. Set base-mounted pumps on concrete foundation. Disconnect coupling before setting. Do not reconnect couplings until alignment procedure is complete.

F. Support pump baseplate on rectangular metal blocks and shims, or on metal wedges with small taper, at points near foundation bolts to provide a gap of 3/4 to 1-1/2 inches (19 to 38 mm) between pump base and foundation for grouting.

G. Adjust metal supports or wedges until pump and driver shafts are level. Check coupling faces and suction and discharge flanges of pump to verify that they are level and plumb.

H. Do not mount pumps on walls that are common to critical areas such as offices, conference rooms, classrooms, etc. In-line pumps shall be installed directly in the piping system, and supported independently from the piping.

3.03 ALIGNMENT

A. Align pump and motor shafts and piping connections after setting on foundation, grout has been set and foundation bolts have been tightened, and piping connections have been made.

B. Comply with pump and coupling manufacturers' written instructions.

C. Adjust pump and motor shafts for angular and offset alignment by methods specified in HI 1.1-1.5, "Centrifugal Pumps for Nomenclature, Definitions, Application and Operation."

D. After alignment is correct, tighten foundation bolts evenly but not too firmly. Completely fill baseplate with nonshrink, nonmetallic grout while metal blocks and shims or wedges are in place. After grout has cured, fully tighten foundation bolts.

E. Grout pump mounting base full after piping is connected but before pump drive is aligned. After grouting, align pump drive shaft to 5 mils, even if pump is factory aligned, and conduct vibration test.

F. Realignment after installation prior to start up will be performed by Owner.

3.04 CONNECTIONS

A. Piping installation requirements are specified in other Division 23 Sections. Drawings indicate general arrangement of piping, fittings, and specialties.

B. Install piping adjacent to machine to allow service and maintenance.

C. Connect piping to pumps. Install valves that are same size as piping connected to pumps.

D. Install suction and discharge pipe sizes equal to or greater than diameter of pump nozzles.

E. Install check valve, balancing valve and shutoff valve on discharge side of pumps. Triple duty valve is not acceptable.

F. Install Y-type strainer and shutoff valve on suction side of pumps. Suction diffusers can be used in lieu of in-line strainers, long radius elbow and spool piece.

G. Install flexible connectors on suction and discharge sides of base-mounted pumps between pump casing and valves.

H. Install pressure gages on pump suction and discharge, at integral pressure-gage tapping, or install single gage with multiple input selector valve.

I. Install check valve and ball valve on each condensate pump unit discharge.

J. Install electrical connections for power, controls, and devices.

3.05 STARTUP SERVICE

A. Engage a factory-authorized service representative to perform startup service.
   1. Complete installation and startup checks according to manufacturer's written instructions.

B. Check piping connections for tightness.

C. Clean strainers on suction piping.

D. Perform the following startup checks for each pump before starting:

E. Verify bearing lubrication.

F. Verify that pump is free to rotate by hand and that pump for handling hot liquid is free to rotate with pump hot and cold. If pump is bound or drags, do not operate until cause of trouble is determined and corrected.

G. Verify that pump is rotating in the correct direction.

H. Prime pump by opening suction valves and closing drains, and prepare pump for operation.

I. Start motor.
   1. Open discharge valve slowly.

3.06 DEMONSTRATION

A. Engage a factory-authorized service representative to train Owner's maintenance personnel to adjust, operate, and maintain hydronic pumps.

END OF SECTION 232123

SECTION 23 2513
WATER TREATMENT FOR CLOSED LOOP HYDRONIC SYSTEMS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General Conditions and Division 01 Specification Sections, apply to this Section.

B. Specifications throughout all Divisions of the Project Manual are directly applicable to this Section, and this Section is directly applicable to them.

1.02 SUMMARY

A. Products specified in this section include the following:
   1. Chemical Inhibitors for Water
   2. Chemicals for Inhibitors and cleaning
   3. Shot Feeder
   4. Corrosion Coupon Rack
   5. Glycol Feed System
   6. Propylene Glycol

B. Provide a Chemical Services Provider (CSP) to perform the services described below.

C. Provide water treatment products and services including:
   1. Chemicals for cleaning, passivation, and treatment of each system.
   2. On-going treatment, chemicals, testing, certification, and reports to demonstrate effective water treatment is maintained at all times.
   3. Start-up and operation of chemical treatment equipment.

D. Provide a complete chemical water treatment program during construction for all new and reused piping networks. This program shall begin from the point each system is filled during construction and extend until Substantial Completion or final Owner acceptance of each system, whichever occurs later. The program shall include water analysis, chemicals, testing, equipment, consulting and service for the following systems:
   1. Chilled Water System
   2. Glycol contained in each of the above systems

E. Attend project meetings as required to plan, schedule and coordinate above activities with other project contractors and the Owner.

F. Provide recommended testing procedures and chemical treatment schedule for Owner's personnel. This information shall be submitted to the owner in a Program Administration Manual.

G. Related Sections:
   1. Section 23 0591 - HVAC System Startup
   2. Section23 0800 - Commissioning of HVAC
   3. Section 23 0920.01 - Control Systems
   4. Section 23 2113 - Hydronic Piping
   5. Section 23 2116 - Hydronic Piping Specialties

1.03 SUBMITTALS

A. Product Data: Include rated capacities; water-pressure drops; shipping, installed, and operating weights; and complete data on furnished products listed below:
   1. Shot Feeders
   2. Coupon Racks
   3. Glycol Feed System
   4. Flow Indicators
   5. Valves

6. Product specifications and MSDS's for each chemical used
7. Cleaning Procedures
8. Passivation Procedures
9. Chemical Treatment Procedures
10. Shop Drawings

B. Shop Drawings to include detailed equipment assemblies indicating dimensions, weights, loads, required clearances, method of field assembly, components, and the location and size of each field connection as necessary to assist the mechanical contractor with proper system installations

C. Submit resume of water treatment personnel and name of Chemical Services Provider.

D. Service Reports as indicated under Part 2 below.

1.04 QUALITY ASSURANCE

A. Manufacturers and Products: The products and manufacturers specified in this Section establish the standard of quality for the Work. Subject to compliance with all requirements, provide specified products from the manufacturers named in Part 2.

B. Products in this section shall be built, tested, and installed in compliance with the specified quality assurance standards; latest editions, unless noted otherwise.
1. American Society for Testing and Materials (ASTM)
   a. D859 - Test Method for Silica in Water
   b. D1067 - Test Methods for Acidity/Alkalinity in Water
   c. D1068 - Test Methods for Iron in Water
   d. D1126 - Test Method for Hardness in Water
   e. D3370 - Practices for Sampling Water from Closed Conduits
   f. D4012 - Test Method for Adenosine Triphosphate (ATP) Content of Microorganisms in Water
   g. D5465 - Practice for Determining Microbial Colony Counts from Waters Analyzed by Plating Methods

C. Conform to all applicable Codes, Regulations, and Municipal requirements for the use and disposal of chemicals (including cleaning compounds) and waste to public sewer systems.
1. Glycols (of any type) shall not be discharged to the sanitary sewer. Dispose of in accordance with local, state and federal EPA requirements.

D. Wastewater containing any chemical or sediment is prohibited from discharge to the storm water system. No chemical shall contain Chromates or Inorganic Phosphates.

1.05 DELIVERY, STORAGE, AND HANDLING

A. Package for delivery to best protect finished surfaces while using the least amount of single-use packaging as possible. Deliver equipment and components adequately packaged for lifting, skidding, or rolling into final position, according to manufacturer's instructions. If possible, package and ship products using reusable blankets and fabrics, or reusable cardboard and crate systems.

B. Store materials and equipment raised off the floor on pallets and protected with coverings to prevent damage due to weather and construction activities. Store in areas that prevent damage due to freezing and extreme temperatures or sunlight. Arrange coverings to provide air circulation to avoid damage from condensation or chemical build-up. Protect from damage, dirt and debris at all times.

C. Store chemicals in curb protected area. Such secondary containment areas must have the capacity to hold the volume of the largest container or 10% of the combined containers, whichever larger. If room has no floor drains, then the room itself may be considered sufficient secondary containment. If the room is considered the secondary containment ensure there is a

lip at the door so no liquids can exit the room in the event of a leak. Verify field conditions before storing any chemicals

1. Provide temporary containment areas when permanent containment areas do not exist. Remove temporary containment at the end of construction.

1.06 WARRANTY

A. Provide a complete warranty for parts and labor for a minimum of one year from the date of Substantial Completion. This warranty does not include ongoing chemical treatment or monitoring.

1.07 ACCEPTABLE CHEMICAL SERVICE PROVIDERS (CSPS)

A. CSP shall be regularly engaged in HVAC water treatment services for other facilities within a 150 mile radius of project site.

B. Subject to compliance with project requirements available CSPs may include, but are not necessarily limited to the following vendors:
   1. Industrial Solutions, Inc.
   2. Innovative Water Treatment, LLC.
   3. Nalco Water, an Ecolab Company
   4. Rocky Mountain Aquatech, Inc

PART 2 - PRODUCTS

2.01 GENERAL

A. CSP Services:
   1. The CSP shall provide all cleaning and treatment chemicals, basic services, testing, equipment and materials necessary for a complete water treatment program.
   2. The CSP shall perform an analysis of the supply water to determine the type and quantities of chemical treatment needed to maintain the required water quality to prevent corrosion, scaling, and biological growth. The CSP will at minimum provide weekly site visits to verify proper water treatment for the first month after any system or part of a system is treated. Provide monthly visits thereafter, or more often if required to assure performance requirements are being met, to analyze water samples, inspect equipment, and add additional chemicals as required to maintain proper water treatment, until final written Owner acceptance of the respective system.
   3. At each site visit CSP shall analyze each system for corrosion inhibitors, pH, total iron, total copper, bacteria levels (provide monthly analytical laboratory analysis), and conductivity; inspect loss from corrosion coupons (provide analytical laboratory analysis every three months), record make-up meter readings, and perform any other tests necessary to validate that corrosion, scale, and organic growth is being prevented.
   4. Reports:
      a. Submit a written startup test report for each system placed into service.
      b. A service report shall be prepared on site by the CSP, submitted at the time of each service visit (with copies immediately provided to the Owner and Commissioner), which shall include all required test results and recommendations.
      c. Additionally, provide final reports for approval to the Owner and Commissioner regarding each site service visit, certified by an Officer of the CSP, within one week of any water treatment activity. Such reports shall include the results of any field or lab tests. Reports shall clearly state if the required water quality and maximum corrosion rates are being achieved.
      d. At a minimum, each report shall include the following information:
         1) System Treated:
         2) Date
         3) Conductivity
         4) pH

5) Total Iron
6) Total Copper
7) Bacteria(cfu) (monthly analytical laboratory analysis)
8) Coupon Corrosion Rates (three month analytical laboratory analysis)
9) Make-up Water Quantity since Last Visit
10) Corrosion Inhibitor level (ppm)
11) Silica level (ppm)

5. Conduct final on-site system turn over meeting with Owner and Commissioner. Present final validation report demonstrating that performance requirements have been achieved and that each system is currently properly treated.
6. When required by Part 3, train Owner's maintenance personnel in water treatment procedures.
7. Program Administration Manual:
   a. The CSP shall provide the Owner with two (2) Program Administration Manuals including sections covering program control, testing requirements, in plant logs, safety data and system information. Each section shall include the following:
      1) Program Control: Include a complete program outline with chemical descriptions, control ranges, and required action for “out of range” situations. Also include complete ordering instructions with applicable account numbers and phone numbers.
      2) Testing Requirements: This section will include basic testing practices as well as detailed test instructions for each test to be performed.
      3) In-Plant Logs: Include detailed testing log sheets for each system to be tested including room to record one (1) month’s data on one (1) sheet and a section to log chemical inventory.
      4) Safety Data: Include the CSP’s 24-hour toll free safety hotline number as well as MSDS sheets for each chemical used.
      5) System Information: Include any equipment manuals and related information.

2.02 TREATMENT PROGRAM - FLUSHING AND CLEANING

A. Cleaning chemicals shall be non-phosphate and non-acidic.

B. Flush and clean all new and reused piping.

C. Provide a flush/clean plan for approval. Provide and later remove all temporary bypasses, drains, vents, etc. required to flush and clean the system. Temporary components for flushing and cleaning are not indicated on the drawings.

D. Flush and clean systems per the general procedures specified in Related Sections; clean systems per the approved cleaning procedures submitted under this specification section. Provide and install all required cleaning chemicals.

E. Verify that adequate cleaning chemical was added, that cleaning was effective, and test that cleaning chemicals were properly rinsed from the system. Provide a report certifying that cleaning and rinsing, was properly executed.

F. Provide passivation and chemical treatment immediately after each system has been cleaned and flushed. Systems shall not stand filled with fluids for periods longer than 48 hours without beginning flushing and cleaning which shall immediately be followed by the passivation and chemical treatment procedures.

2.03 TREATMENT PROGRAM - CHEMICAL TREATMENT AND PASSIVATION

A. Provide passivation and chemical treatment per the approved chemical treatment procedures. Chemically treat new and reused piping networks using chemicals and treatment procedures compatible with the system being cleaned and the chemical treatment chemicals in existing piping systems being connected to. Test existing piping systems or otherwise determine

chemical treatment in existing piping and validate compatibility. Note that existing systems may be silica based, adjust water treatment program accordingly.

B. When connecting into active existing systems, treat piping networks installed or reused by the project and verify by lab analysis that the treatment levels per the treatment plan have been attained prior to connecting into the existing active system. Submit validating report for approval.
   1. Do not connect into existing active system until report is approved by Owner.
   2. Provide temporary valves, piping, and accessories as required to treat the project's piping prior to connection to the active system.

C. Provide passivation/chemical treatment at system startup or immediately upon operation of a system for temporary cooling and heating, whichever comes first.

D. Provide chemical treatment immediately after each system has been cleaned and flushed. Thereafter immediately begin the approved water treatment maintenance program to passivate and prevent corrosion, scale, and organic growth and to maintain treatment chemical levels. Note that systems or parts of systems will not typically be started at the same time; adjust treatment strategy accordingly.

E. Provide chemicals that comply with State and Federal regulations.

F. Chemical inhibitor shall be polymer based and contain azole in an amount appropriate for the percentage of internal copper surface area of the system piping. Provide products with PTSA (p-Toluenesulfonic acid) florescent tracer dyes. Molybdenum shall not be used as a tracer. Optionally, for hot water heating systems only, nitrite/azole based inhibitors may be used.

G. No chemicals shall contain chromates or inorganic phosphates.

H. Utilize non-oxidizing biocides in the event sterilization is required.

I. Adjust chemical concentrations as required to achieve the required performance

2.04 PERFORMANCE REQUIREMENTS

A. The water treatment programs proposed by the CSP shall maintain corrosion rates below 0.5 mils per year for mild steel and 0.1 mils per year for copper. These rates shall be verified with corrosion coupons provided by the CSP Coupons shall be analyzed in the CSP's analytical laboratory every three months, more often if required, to assure these levels are not exceeded.

B. Microbiological growth levels shall be maintained below 10,000 cfu's aerobic bacteria and 10 cfu's anaerobic bacteria. The CSP shall perform detailed microbiological culturing at the CSP's company's analytical laboratory monthly, more often if required, to assure these levels are not exceeded

2.05 EQUIPMENT

A. Provide all components required for a fully operational water treatment system. Provide components that are not adversely affected by the treatment chemicals employed.

B. Chemical Shot Feeders:
   1. Acceptable Manufacturers:
      a. Griswold Controls.
      b. J.L. Wingert Company
      c. Neptune, a brand of the Dover Company.
      d. Veris Industries
      e. Wessels.
   2. Provide shot feeders for each system. Provide steel 5-gallon minimum tank with interior and exterior epoxy coating, removable screwed and gasketed top (4" diam. minimum), pipe threaded inlet and outlet, dish bottom style with floor skirt, rated at 200 psi, 200°F.

C. Corrosion Coupon Rack:
   1. Acceptable Manufacturers:

a. Cannon Water Technology
b. J.L. Wingert
c. Vector Industries

2. Supply for each corresponding shot feeder:
   a. Chilled water: 3/4 inch stainless steel coupon rack, with two coupon holders, inlet/outlet 316 stainless steel full port ball valves, and a 1/4 inch sample valve. Include a 0-5 gpm nominal range variable area flow meter, rated 150 psig at 70 deg F, with graduated polysulfone or acrylic cylinder, with an accuracy of +/-5%. Provide mild steel and copper coupons.

D. Glycol Feed System:
   1. Acceptable Manufacturers:
      a. J.L. Wingert Company.
      b. Neptune Chemical Pump Co.
      c. Advantage Controls Inc.
      d. Wessels Co.
      e. Axiom.
   2. Provide a floor mounted packaged glycol feed system that is pre-wired and pre-plumbed, and includes a storage tank, control panel, controls, pump, piping, and valves, all supported on a steel frame.
   3. Provide a 50-gallon minimum translucent polyethylene single wall tank with polyethylene cover. Minimum 1/4-inch-thick walls.
   4. Provide a welded steel stand suitable for floor mounting and painted with two coat system consisting of oxide primer and alkyd enamel finish. Provide holes in the base for the installation of anchor bolts and leveling of equipment.
   5. Provide a bronze gear pump, 120 V, single phase, and rated for a minimum of 1.5 gpm at 90 psig.
   6. Valves and Piping: Pump suction shall include ball valve and Y-strainer. Pump discharge shall include silent check valve, pressure switch, pressure gauge, and 150 psi relief valve piped back to tank. All piping shall be Type L Copper. Connect pump suction and discharge with braided stainless steel or bronze flex connectors. All piping, valves and accessories shall conform to Related Sections.
   7. Provide a NEMA 4X control panel with 8 foot, 120 V grounded power cord. Equip the panel with motor starter, 2-position main power switch with light, hand /off/ auto switch with light for gear pump, low level red light, low level audible alarm with push button silence switch, dry contacts for remote monitoring of alarm, and a 15 amp fuse.
   8. Provide an adjustable pressure switch, selected with a range to match system operating pressure.
   9. Provide electrical components that conform to Division 26 requirements. Provide all wiring in conduit. Loose, exposed or unprotected wire is not acceptable

2.06 INHIBITED PROPYLENE GLYCOL

A. Acceptable Manufacturer and Product:
   1. Dow Chemical - Dowfrost TM

B. Provide for closed loop heating and chilled water hydronic systems as indicated on the drawings.

C. Provide inhibited propylene glycol, mixed with softened water at a concentration of 35%~~-40%~~ by volume for both the heating and chilled water loops for burst protection down to 60 deg F below zero. Provide charts, tables, and/or graphs indicating freeze and burst protection for all glycol percentages as part of product submittal data sheets.

3 EXECUTION

3.01 EXAMINATION AND PREPARATION

A. Pre-Treatment Conference: Prior to treatment activities, meet with the Project Engineer, Commissioner and contractors to verify treatment procedures, discuss coordination with existing piping networks, and to coordinate treatment activities with construction schedule.

B. Flush and clean systems per Related Sections and Part 2 above.

3.02 INSTALLATION

A. Install all required components, and chemically passivate/treat and maintain the system as described in Part 2.

B. Use and dispose of chemicals and wastewater (including from existing piping networks) per the Quality Assurance section of this specification. All costs of disposal shall be borne by the contractor.

C. Provide and install all chemicals and all incidental components. Install equipment at locations shown on the Drawings. Follow the manufacturer's installation and startup instructions.

D. Passivate and chemically treat each system immediately after it has been cleaned and flushed.

E. Install shot feeders with openings no higher than 4' above finished floor. Pipe shot feeder drain to nearest floor drain. Install top coupon of corrosion coupon rack no higher than 6' above finished floor.

F. The treatment program shall begin at system start-up or immediately upon first operation of the system by the contractor for temporary heating and cooling, whichever comes first. Continue the program until Substantial Completion.

3.03 INSTALLATION OF INHIBITED PROPYLENE GLYCOL SYSTEM

A. Following system flushing and cleaning, drain the entire piping system. Add propylene glycol/inhibitor solution into the system. Circulate system fluid, adding additional propylene glycol/inhibitor as required until all air is purged from the system.

B. Sample fluid, measure concentration of glycol and inhibitor. Drain system fluid as required and add additional solution to achieve the design concentration of glycol and the manufacturer's recommended concentration of inhibitor. Submit report of activities, including volumes of all fluids used.

3.04 INSTALLATION OF GLYCOL FEED SYSTEM

A. Level and bolt Glycol Feed System to the shop keeping pad. Support all piping and conduits independent of the Glycol Feed Unit. Provide initial fill of Feed unit reservoir.

3.05 FIELD QUALITY CONTROL:

A. The CSP shall:
   1. Verify that systems and equipment have been properly cleaned, flushed, and filled with water, and are fully operational before introducing operational chemicals for water-treatment systems.
   2. Introduce and maintain chemical treatment to achieve the specified performance requirements.
   3. Submit a written startup test report for each system placed into service.
   4. Provide validation testing and required reports.
   5. Attend final on-site system turn over meeting with Owner and Commissioner. Present final validation report demonstrating that performance requirements have been achieved and that each system is currently properly treated.

3.06 COMMISSIONING:

A. Perform Commissioning activities per Related Sections above.

3.07 OWNER TRAINING

A. CSP shall train Owner's maintenance personnel on equipment operation, adding and maintaining proper chemical levels, trouble-shooting, servicing and preventative maintenance procedures. Do not conduct training until after start up and commissioning is completed. Provide training using documents including the Program Administration Manual and the approved Operations and Maintenance manuals. Provide these documents to the Owner at least two weeks prior to the training to allow sufficient time for review.
   1. Provide 2 hours training minimum.

B. Schedule training at least 7 days in advance, at date and time approved by Owner.

END OF SECTION 232513

SECTION 23 6426.01
AIR-COOLED LIQUID CHILLER

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.02 SUMMARY

A. Section includes design, performance criteria, refrigerants, controls, and installation requirements for air-cooled chiller.

B. Microprocessor controlled, air-cooled liquid chiller for outdoor installation, utilizing digital scroll compressors, low sound fans, and electronic expansion valve.

C. All fans are controlled with variable speed fan drive motors. Chiller software shall be specifically developed to coordinate optimal fan speed for application conditions and provide refrigerant circuit optimization, resulting in higher part load efficiency and reduced acoustic levels.

1.03 REFERENCE STANDARDS

A. ASTM D1654 - Standard Test Method for Evaluation of Painted or Coated Specimens Subjected to Corrosive Environments; 2024.

B. ASTM D3359 - Standard Test Methods for Rating Adhesion by Tape Test; 2023.

C. ASTM D3363 - Standard Test Method for Film Hardness by Pencil Test; 2022.

1.04 QUALITY ASSURANCE

A. Unit shall be rated in accordance with AHRI (Air-Conditioning, Heating and Refrigeration Institute) Standard 550/590, latest edition (U.S.A.) and all units shall be ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) 90.1-2019 compliant.

B. Unit construction shall comply with ASHRAE 15 Safety Code, UL (Underwriters Laboratories) latest edition, and ASME (American Society of Mechanical Engineers) applicable codes (U.S.A. codes).

C. The management system governing the manufacture of this product is ISO (International Organization for Standardization) 9001:2015 certified.

D. An operational test, in which the chiller is run under load, is performed at the factory. This test checks for proper operation of fans as well as various controls and safeties, and a Certificate of Unit Testing, indicating successful end-of-line testing, is provided with the unit.

1.05 DELIVERY, STORAGE AND HANDLING

A. Chiller shall be capable of withstanding -40°F (-40°C) to 158°F (70°C) storage temperatures for an indefinite period of time.

B. Unit shall be stored and handled per unit manufacturer's recommendations.

C. Ship each chiller with a full charge of refrigerant. Charge each chiller with nitrogen if refrigerant is shipped in containers separate from chiller.

D. Comply with manufacturer's installation instructions for rigging, unloading, and transporting chillers.

1.06 PERFORMANCE REQUIREMENTS

A. Site Altitude: Chiller shall be suitable for altitude in which installed without affecting performance indicated. Make adjustments to affected chiller components to account for site altitude.

1.07 SUBMITTALS

A. Product Data: For each type of product indicated. Include refrigerant, rated capacities, operating characteristics, furnished specialties, and accessories.
   1. Performance at ARI standard conditions and at conditions indicated.
   2. Performance at ARI standard unloading conditions.
   3. Minimum evaporator flow rate.
   4. Refrigerant capacity of chiller.
   5. Oil capacity of chiller.
   6. Fluid capacity of evaporator.
   7. Characteristics of safety relief valves.
   8. Fluid capacity of condenser and heat-reclaim condenser.
   9. Minimum entering condenser-fluid temperature.
   10. Performance at varying capacities with constant-design entering condenser-fluid temperature. Repeat performance at varying capacities for different condenser-fluid temperatures from design to minimum in 5 deg F increments.
   11. Minimum entering condenser-air temperature.
   12. Maximum entering condenser-air temperature.
   13. Performance at varying capacities with constant-design entering condenser-air temperature. Repeat performance at varying capacities for different entering condenser-air temperatures from design to minimum in 10 deg F increments.

B. Submit dimensional plan and elevation view drawings, weights and loadings, required clearances, location and size of all field connections, electrical requirements and wiring diagrams.

C. Submit product data indicating rated capacities, accessories and any special data.

D. Submit manufacturer's installation instructions.

E. Cordination Drawings: See Section 23 0010 - HVAC (Heating, Ventilation and Air Conditioning) General Provisions for requirements

1.08 VERIFICATION OF CAPACITY AND EFFICIENCY

A. All proposals for chiller performance must include an AHRI approved selection method. Verification of date and version of computer program selection or catalog is available through AHRI.

1.09 WARRANTY

A. Provide a full parts warranty for one year from start-up or 18 months from shipment, whichever occurs first.

B. Labor - whole unit (1st year)

PART 2 PRODUCTS

2.01 ACCEPTABLE MANUFACTURERS

A. This specification is based on the Carrier 30RC series air-cooled screw chiller platform. Equivalent units and manufacturers must meet all performance criteria, and will be considered only upon prior approval. Subject to compliance with capacities, characteristics, performance and dimensional criteria, of the Basis of Design product, alternate available manufacturers shall be:
   1. York, by Johnson Controls
   2. McQuay, by Daikin
   3. Smardt Chiller Group.

2.02 EQUIPMENT

A. General:

1. Factory assembled, single-piece chassis, air-cooled liquid chiller. Contained within the unit cabinet shall be all factory wiring, piping, controls, refrigerant charge (R-32), and special features required prior to field start-up.

B. Materials of Construction:
   1. Frame shall be of heavy-gauge, galvanized steel.
   2. Exterior panels shall be galvanized steel with a baked enamel powder or pre-painted finish.
   3. Painted parts shall withstand 1000 hours in constant neutral salt spray under ASTM (American Society for Testing and Materials) B117 conditions with a 0.04 inch scribe per ASTM D1654. After test, painted parts shall show no signs of wrinkling or cracking, no loss of adhesion, no evidence of blistering, and the mean creepage shall not exceed 1/4 inch. (Rating ³ per ASTM D1654) on either side of the scribe line.

C. Fans:
   1. Condenser fans shall be 9-blade airfoil cross-section, reinforced polymer construction, shrouded-axial type, and shall be statically and dynamically balanced with inherent corrosion resistance.
   2. The variable speed drives for the condenser fans shall include a DC link reactor.
   3. Fan operation shall allow reduced sound levels during scheduled unoccupied operating periods. Manufacturers without unoccupied reduced sound capability shall submit 1/3 octave band data and sound power data as measured according to AHRI 370 as confirmation of unit sound characteristics.
   4. Air shall be discharged vertically upward.
   5. Fans shall be protected by coated steel wire safety guards.
   6. Fan blades shall have serrated edges to minimize the sound that is produced.

D. Compressor/Compressor Assembly:
   1. Fully hermetic, direct-drive, scroll-type compressors.
   2. Compressor motors shall be cooled by refrigerant gas passing through motor windings and shall have either internal line break thermal and current overload protection or external current overload modules with compressor temperature sensors.
   3. Compressors shall be mounted on rubber in shear vibration isolators.
   4. Staging of compressors shall provide unloading capability.
   5. The employment of digital compressors (when selected) provides many additional stages of unloading.
   6. Each compressor shall be equipped with crankcase heaters to minimize oil dilution.
   7. Crankcase heaters shall not be required when microchannel condenser coils are utilized.

E. Brazed Plate Evaporator:
   1. Evaporator shall be rated for a refrigerant working-side pressure of 710 psig and shall be tested for a maximum water-side pressure of 300 psig.
   2. Shall be single-pass, ANSI (American National Standards Institute) type 316 stainless steel, brazed plate construction.
   3. Shell shall be insulated with 3/4-in. 0.75 inch closed-cell, polyvinyl-chloride foam with a maximum K factor of 0.28.
   4. Shall incorporate 1 independent refrigerant circuits.
   5. Evaporator shall have factory-installed heater, to protect Evaporator from ambient temperature freeze down to –20°F (–29°C). Unit shall be provided with a factory-installed flow switch.
   6. All connections shall use standard Victaulic-type fittings.
   7. Evaporator shall be supplied with a 20 mesh strainer.

F. Condenser:

1. Coil shall be air-cooled Novation heat exchanger technology with microchannel (MCHX) coils and shall have a series of flat tubes containing a series of multiple, parallel flow microchannels layered between the refrigerant manifolds.
2. Coils shall consist of a two-pass arrangement. Coil construction shall consist of aluminum alloys for fins, tubes, and manifolds in combination with a corrosion-resistant coating.
3. Tubes shall be cleaned, dehydrated, and sealed.
4. Assembled condenser coils shall be leak tested and pressure tested at 656 psig.

G. Refrigeration Components:
1. Refrigerant circuit components shall include filter drier, moisture indicating sight glass, electronic expansion device and complete operating charge of both refrigerant R-32 and compressor oil.

H. Controls, Safeties, and Diagnostics:
1. Unit controls shall include the following minimum components:
   a. Microprocessor with non-volatile memory. Battery backup system shall not be accepted.
   b. Separate terminal block for power and controls.
   c. Control transformer to serve all controllers, relays, and control components.
   d. ON/OFF control switch.
   e. Replaceable solid-state controllers.
   f. Pressure sensors shall be installed to measure suction and discharge pressure for each circuit. Thermistors shall be installed to measure evaporator entering and leaving fluid temperatures, outdoor ambient temperature, and discharge temperature.
2. Unit controls shall include the following functions:
   a. Automatic circuit primary/secondary for dual circuit chillers.
   b. Hermetic scroll compressors are maintenance free and protected by an auto-adaptive control that minimizes compressor wear.
   c. Capacity control based on leaving chilled fluid temperature and compensated by rate of change of return-fluid temperature with temperature set point accuracy to 0.1°F (0.06°C).
   d. Limiting the chilled fluid temperature pull-down rate at start-up to an adjustable range of 0.2°F to 2°F (0.11°C to 1.1°C) per minute to prevent excessive demand spikes at start-up.
   e. Seven-day time schedule.
   f. Leaving chilled fluid temperature reset from return fluid and outside air temperature.
   g. Chilled water pump start/stop control and primary/standby sequencing to ensure equal pump run time.
   h. Timed maintenance scheduling to signal maintenance activities for pumps, condenser coil cleanings, strainer maintenance and user-defined maintenance activities.
   i. Low ambient protection to energize evaporator and hydronic system heaters.
   j. Periodic pump start to ensure pump seals are properly maintained during off-season periods.
   k. Single step demand limit control activated by remote contact closure.
   l. Nighttime sound mode to reduce the sound of the machine by a user-defined schedule.
3. Diagnostics:
   a. The control panel shall include, as standard, a display:
      1) Color touch screen display with stylus.
      2) Display shall allow a user to navigate through menus, select desired options, and modify data.
   b. Features of the display shall include:
      1) BACnet integration

2) Automatic reporting of alarms over email.
3) Ability to graphically plot trends of system performance and conditions over time.
4) Graphical summary display of current chiller operation and water conditions.
5) Display shall allow access to configuration, maintenance, service, set point, time schedules, alarm history, and status data.
6) Password protection against unauthorized access to configuration and maintenance information, and display set up parameters.
7) Full compatibility to provide email alarm notification and to provide network capability to fully monitor and control chiller.
8) Display shall be capable of displaying the last 50 alarms, with clear full text description and time and date stamp, and will store a snapshot of operating conditions before and after the 10 most recent alarms.
9) Display run hours and number of starts for machine and individual compressors.
10) The control system shall allow software upgrade without the need for new hardware modules.

4. Safeties:
   a. Unit shall be equipped with thermistors and all necessary components in conjunction with the control system to provide the unit with the following protections:
      1) Reverse rotation.
      2) Low chilled fluid temperature.
      3) Motor overtemperature.
      4) High pressure.
      5) Electrical overload.
      6) Thermal overload.
      7) Loss of refrigerant charge.
   b. Condenser fan motors shall have internal overcurrent protection.

I. Operating Characteristics:
   1. Unit shall be capable of operating down to –20°F (–29°C).
   2. Unit shall be capable of starting and running at outdoor ambient temperatures up to 120°F (50°C). Unit shall additionally be able to stay online when running with a 125°F (52°C) ambient temperature.
   3. Unit shall be capable of starting up with 95°F (35°C) entering fluid temperature to the evaporator.

J. Fan Motors:
   1. Condenser fan motors shall be totally enclosed, air over, 3-phase type with permanently lubricated bearings and Class F insulation.

K. Electrical Requirements:
   1. Unit/module primary electrical power supply shall enter the unit at a single electrical box.
   2. Unit shall operate on 3-phase power at the voltage shown in the equipment schedule.
   3. Control points shall be accessed through terminal block.
   4. Unit shall be shipped with factory control and power wiring installed.
   5. Unit shall have a standard SCCR (short circuit current rating) of 10 kA.

L. Chilled Water Circuit:
   1. Chilled water circuit shall be rated for 300 psig.
   2. Solid-state flow monitor with integral relay shall be factory installed and wired.
   3. Brass body strainer with 20 mesh screen and ball type blow down.

M. Special Features:
   1. High-efficiency variable condenser fans: All fans on the unit shall have variable speed fan motors to provide higher part load efficiency and reduced acoustic levels. Each fan circuit shall have a factory-installed, independent variable speed drive. Variable speed drives are rated IP-55 enclosures and UL Listed. The use of this option or accessory, with the

addition of antifreeze in the evaporator circuit and wind baffles or hail guards, shall allow running with outdoor ambient temperatures down to –20°F (–28.9°C).

2. Low-Ambient Operation:
   a. Unit shall be capable of starting as low as and operating down to –20°F (–29°C) with the use of high-efficiency variable condenser fans. In addition, adequate field-supplied antifreeze with suitable corrosion inhibitor protection shall be field-installed in the evaporator circuit and factory-installed hail guards shall be provided.
3. Unit-Mounted Non-Fused Disconnect:
   a. Unit shall be supplied with factory-installed, non-fused electrical disconnect for main power supply. This option shall be included with the high SCCR option.
4. Energy Management Module: A factory or field-installed module shall provide the following energy management capabilities
   a. Cooling set point and demand limit control
   b. 2-point demand limit control (from 15% to 100%) activated by a remote contact closure
5. Hail Guards: Unit shall be supplied with factory or field-installed, louvered, sheet metal panels which securely fasten to the chiller and provide condenser coil protection against hail and other physical damage. Hail guards shall facilitate operation down to –20°F (–29°C) in conjunction with high-efficiency variable condenser fans.
6. Vibration Isolation: Vibration isolation pads shall be supplied for field installation at unit mounting points.
7. BACnet (MS/TP or IP): Shall provide a factory-installed interface between the chiller and a BACnet Local Area Network (LAN) and allows integration with third-party BACnet building automation system.
8. GFI Convenience Outlet: Shall be factory nstalled to provide the chiller with a 4 amp GFI receptacle. The receptacle shall have independent fuse protection. The convenience outlet shall be a 115-v female receptacle.
9. Compressor Blankets: Provide factory installed sound blankets around each compressor in conjunction with low-sound AeroAcoustic fans to provide chiller sound reduction.
10. Digital Compressor Option: Provide a factory-installed digital compressor to provide incremental steps for tighter temperature control.
11. Suction Line Insulation: Shall provide a tubular, closed-cell form of insulation that is factory-installed on the chiller suction piping.
12. Phase Loss Monitor: Shall be factory installed to provide the unit with protection against phase loss.

PART 3 - EXECUTION

3.01 INSTALLATION

A. Install in accordance with manufacturer’s requirements.
   1. Level the chiller using the base rail as a reference. The chiller must be level within ½ in over the entire length and width. Use shims as necessary to level the chiller.

B. Install chillers on support structure indicated on Structural Drawings.

C. Equipment Mounting:
   1. Comply with requirements for vibration isolation devices specified in Section 23 0548 - Vibration and Seismic Controls for HVAC

D. Charge chiller with refrigerant and fill with oil if not factory installed.

3.02 CONNECTIONS

A. Comply with requirements for piping specified in Section 232113 "Hydronic Piping," Section 232116 "Hydronic Piping Specialties,". Drawings indicate general arrangement of piping, fittings, and specialties.

B. Install piping adjacent to chiller to allow service and maintenance.

C. Evaporator Fluid Connections: Connect to evaporator inlet with flex connector shutoff valve, thermometer, and plugged tee with pressure gage. Connect to evaporator outlet with shutoff valve, flexible connector, flow switch, thermometer, plugged tee with shutoff valve and pressure gage, and drain connection with valve. Make connections to chiller with a mechanical coupling.

D. Connect each chiller drain connection with a union and drain pipe, and extend pipe, full size of connection, to floor drain. Provide a shutoff valve at each connection.

3.03 SERVICE AND START-UP

A. Startup - Provide all labor and materials to perform startup. Startup shall be performed by a factory-trained technician from the original equipment manufacturer (OEM). Technician shall confirm that equipment has been correctly installed and passes specification checklist prior to equipment becoming operational and covered under OEM warranty. This shall be done in strict accordance with manufacturer’s specifications and requirements. Third-party service agencies are not permitted.

B. A start-up log shall be furnished by the factory approved start-up technician to document the chiller's start-up date and shall be signed by the owner or his authorized representative prior to commissioning the chillers.

C. Chiller manufacturers shall maintain service capabilities no more than 100 miles from the jobsite.

D. Provide local service agent with direct access to factory support on equipment.

E. The service provider shall employ a minimum of (1) full time, competent HVAC and automation system servicepersons on staff, whose office in which they operate from is within (100) miles of the job site and who have been within their employment for a minimum of (10) years.

F. During the first 12 months of operation, a factory-trained technician from the original equipment manufacturer (OEM) shall perform quarterly on-site operating inspections to confirm the chiller’s operational performance. The manufacturer shall provide the owner with a report describing the condition of the equipment, current operating log, any issues found needing to be addressed, and recommended corrective actions.

END OF SECTION

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SECTION 26 0010
GENERAL ELECTRICAL REQUIREMENTS

PART 1 GENERAL

1.01 GENERAL

A. Electrical layout drawings are diagrammatic. Refer to Mechanical HVAC, as well as existing conditions from contractor required site observation for guidance on dimensions, ceiling heights, locations of ducts, pipes, structural members or other systems. Locate lighting fixtures symmetrically in proper relation to finished areas except where dimensioned on the drawings or located on reflected ceiling plans. Mounting details, fittings, offsets, and slopes are not necessarily shown. Prior to commencement of work, conclusively coordinate the detailed installation of the conduit and equipment with all ductwork, piping, cable trays, structural, structural elements, and other work of other trades. Utilize 3-dimensional modeling and clash detection software to produce clash free shop coordination drawings of all disciplines at scale of 1/4" = 1’-0”. No additional costs will be considered for changes because of interference with the work of others, or with other work within the same division. This includes any additional material required to correct minor job conflicts for normally accepted procedures of all trades.

B. Workmanship Warranty: Upon completion of the work the contractor shall furnish to the owner a written guarantee stating that the contract warrants all new material and equipment, and all associated work and components shall be free from defects and/or failure for a period of one (1) year from the date of project substantial completion or turnover of the building to owner, whichever occurs last. Duration in which equipment is placed into operation for start-up and functional testing that occurs prior to substantial completion or building turnover shall not count as time against the warranty period. Any and all materials, equipment, workmanship and labor found to be defective or that fails in any way is to be replaced or corrected at no additional cost to the owner during the warranty period.

C. Prior to submitting a bid, each bidder shall carefully examine the bidding documents. Bring to the attention of the Engineer, in writing; any discrepancies, ambiguities, inconsistencies, or errors therein that are discovered during examination of drawings. Further, all bidders shall examine the site, condition and all associated scopes of work that fall under this contract and determine all costs for the work are fully covered in their bid submission. Failure to request clarification during the bid period for any inadequacy, omission, or conflict shall not relieve contractor of responsibilities. Signing of the contract shall be considered to implicitly denote that the contractor has a thorough understanding of the full intent and scope of these construction contract drawings and specifications. The contractor shall become familiar with the building, structure and the works of the other trades involved. This contractor shall work and cooperate with the other trades to ensure a quality installation and minimize conflicts.

D. The Contractor shall obtain, at the Contractor’s expense, from the Engineer a set of AutoCAD or compatible format engineering drawings on electronic media where desired by the Contractor and/or required by the Specifications for use in preparing the shop drawings, coordination drawings, and record drawings. The Contractor shall provide to the Architect and Engineer a written release of liability acceptable to the Engineer prior to receiving the electronic media.

1.02 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section. Specifications throughout all Divisions of the Project Manual are directly applicable to this Section, and this Section is directly applicable to them.

B. This Section specifies the basic requirements for electrical installations and includes requirements common to more than one section of Division 26, Division 27, and Division 28. It expands and supplements the requirements specified in other sections of Division 01 through 50.

C. All drawings and specifications are to be read in coordination with one another such that work shown on one set is binding to the other. Where a discrepancy exists within these contract documents, the specifications shall take precedence. Where a discrepancy exists within the specification, Contractor shall price the more expensive approach, and the details shall be worked out during construction.

D. This specification is part of a comprehensive set of design drawings and specifications that include other engineering and design disciplines. Installing contractor shall be responsible to review the full set of contract documents; including all drawings, specifications, and addenda, in conjunction with this drawing package to provide an efficiently sequenced and fully coordinated installation. Provide Coordination Drawings per specifications. Contractor bears the responsibility to provide any additional labor and materials that result from a lack of interdisciplinary coordination.

E. Refer to Divisions 07 and 09 for sealing and firestopping requirements where raceways penetrate smoke, fire, and sound rated walls.

F. Refer to Division 08 for electrified door hardware and equipment provided by other trades.

1.03 REFERENCE STANDARDS

A. The latest editions of the following standards (including supplements and official interpretations) are minimum requirements (refer to juridictional authority for latest adopted editions):
1. NFPA 70 - National Electrical Code (NEC).
2. Conform to all applicable State and Local Codes.
3. American National Standards Institute (ANSI).
4. National Electrical Safety Code (NESC).
5. Americans with Disabilities Acts (ADA) and American National Standards Institute (ANSI) 117.
6. National Electrical Manufacturer’s Association (NEMA).
7. A.I.A. “Guidelines for Design and Construction of Hospital and Healthcare Facilities.”
8. Underwriter’s Laboratories (UL).
9. Insulated Cable Engineers Association (ICEA).
10. International Building Code (IBC).
11. International Mechanical Code (IMC).
12. Institute of Electrical and Electronic Engineers (IEEE).

B. The complete installation shall comply with requirements of the utility (electrical and telephone) companies furnishing service to this installation. Applicable codes, ordinances, standards and statutes take precedence when they are more stringent or conflict with the drawings and specifications

1.04 DEFINITIONS

A. Definitions: “Provide” shall mean “furnish and install”. “Furnish” means to supply all materials, labor, equipment, testing apparatus, controls, tests, accessories and all other items customarily required for the proper and complete application. “Install” means to join, unit, fasten, link, attach, set up or otherwise connect together before testing and turning over to Owner, complete and ready for regular operation. The words “accept” or “acceptable” denote only that the equipment items are in general conformance with the design concept of the project

B. The term "supply", "provide", "install", or “furnish” shall mean to supply and install all labor, materials, tools and equipment as may be necessary to install same for the complete and functional operation of the electrical systems for this project.

C. Concealed: Concealed areas are those areas that cannot be seen by building occupants.

D. Exposed: Exposed areas are all areas that are exposed to view by building occupants, including areas below counter tops, inside cabinets and closets, inside all equipment rooms, and areas outside the building exterior envelope, exposed to the outdoors.

1.05 QUALITY ASSURANCE

A. Contractor Qualifications: An acceptable Contractor for the Work under this Division must have personnel with experience, training and skill to provide a practical working system. The Contractor shall furnish acceptable evidence of having installed not less than three systems of size and type comparable to this Project. All personnel installing equipment under this Division shall possess valid city, and State licenses for their skill level. Each Journeyman shall supervise no more than two apprentice helpers. Refer also to Owner's Special Conditions.

B. The contractor shall keep fully informed on all Federal and State laws, local bylaws and regulations, and all ordinances and decrees of bodies or tribunals having any jurisdiction or authority which in any manner affects those engaged or employed on the work, or which in any way affects the conduct of the work. The contractor shall at all times observe and comply with all such laws, bylaws, ordinances, regulations, orders and decrees in force at the time of award.

C. Discrepancies: All drawings and specifications are to be read in coordination with one another such that work shown on one set is binding to the other. Documents are intended to comply with listed codes, ordinances, regulations and standards. Where discrepancies occur, immediately notify the Owner's Project Manager in writing and ask for an interpretation. Any item/issue not clarified prior to bidding is understood to be priced as the more expensive option. Should installed materials or workmanship fail to comply, the Contractor is responsible for correcting the improper installation at no additional cost to the Owner.

D. Conflicting Requirements: Where compliance with two or more standards is specified, and the standards establish different or conflicting requirements for minimum quantities or quality levels, clarify uncertainties with the Architect/Engineer prior to quotation.
   1. Clarification methods: At the time of bidding, bidders shall familiarize themselves with the drawings and specifications. Any questions, misunderstandings, conflicts, deletions, discontinued products, catalog number discrepancies, discrepancies between the equipment supplied and the intent or function of the equipment, etc., shall be submitted to the Architect/Engineer in writing for clarification prior to issuance of the final addendum and bidding of the project. Where discrepancies or multiple interpretations occur, the most stringent (which is generally recognized as the most costly) that meets the intent of the documents shall be enforced.

E. Should any dispute arise as to the quality or fitness of material or workmanship, the decision shall rest solely with the engineer, and shall be based on the requirements that all materials shall meet or exceed specification requirements, and workmanship shall be of the best quality, and what is usual and customary in erecting other buildings shall in no way enter into consideration.

1.06 REQUIRED SUBMITTALS

A. The Design Professional's review and appropriate action on all submittals and shop drawings is only for the limited purpose of checking for conformance with the design concept and the information expressed in the contract documents. This review shall not include:
   1. Accuracy or completeness of details, such as quantities, dimensions, weights or gauges, fabrication processes
   2. Construction means or methods
   3. Coordination of the work with other trades
   4. Construction safety precautions

B. The Design Professional shall not be responsible for any deviations from the contract documents not brought specifically to the attention of the Design Professional in writing by the Contractor. This shall clearly identify the design and the specific element which vary from the Design. The Contractor shall be responsible for all remedy for lack of strict conformance associated with this criteria.

C. Coordination Drawings

1. Work out all installation conditions in advance of installation. The Contractor shall be responsible for preparing coordination drawings, showing all work, in all areas. The Contractor shall be responsible for providing all labor and materials necessary to overcome congested conditions at no increase in contact sum. The Contractors base bid shall include any and all time and manpower necessary to develop such coordination efforts and drawings. Increases to contract sum or schedule shall not be considered for such effort.
2. Coordination drawings shall be prepared by the Contractor for his utilization and are his responsibility to assure systems will be installed in a manner to allow all systems to function properly.
3. Prepare and submit required coordination drawings showing major elements, components, and systems of equipment and materials in relationship with other building components. Prepare drawings to an accurate scale, large enough to indicate required detail, and showing the necessary dimensions. Indicate the locations of all equipment and materials, including clearances for servicing and maintaining equipment. Indicate movement and positioning of large equipment into the building during construction.
4. Coordination drawings are informational submittals. Submit coordination drawings to Engineer for information only to document proper coordination of all portions of work and that coordination issues have been identified and resolved prior to submitting to the Engineer and prior to commencing construction in each affected area. The review of the coordination drawings by the Engineer does not constitute a relief of responsibility of the Contractor or a change to the contract documents. The Contractor shall have sole responsibility in developing a fully coordinated and integrated ceiling cavity.
5. Clearly indicate solutions to space problems. Identification of space problems without solutions is not acceptable. Only areas clearly identified will be reviewed.
6. No release of any shop drawings or product submittals will proceed until a coordination drawing, meeting the requirements within these specifications, has been submitted and approved for every area of construction indicated on the Electrical and Technology plans.
7. Start of work constitutes a material requirement for the aforementioned Coordination Drawings. Contractor shall bear any and all costs associated with changes that are not specifically addressed on the Coordination Drawings that were reviewed by the Engineer of Record.
8. Modifications required as result of failure to resolve interferences, provide correct coordination drawings or call attention to changes required in other work as result of modifications shall be paid for by responsible Contractor/Subcontractor.

D. Product Data and Shop Drawings
   1. See Section 26 0005 for additional requirements.
   2. Provide submittal data and shop drawings as required for all major components. (Listed in individual sections.
   3. Schedule of Values: Provide Preliminary Schedule of Values to Engineer with product data submittal. Provide a final Schedule of Values at close-out of project including updated values based on actual installation. Provide according to the following descriptions.
      a. General Construction (total)
      b. Mobilization/Demobilization
      c. Demolition
      d. Basic Materials/Devices/Equipment Connections
      e. Testing, Acceptance, and Commissioning
   4. Coordination Drawings: See above.

E. Substantial Completion
   1. See Section 26 0005 for additional requirements.
   2. Equipment Settings Report: Where specifically identified or within individual Division 26 Sections or necessary for confirmation of products, submit Equipment Settings Report indicating final configurations and settings.

3. Test Reports. (Commissioning as well as Listed in individual sections): Submit test reports which have been signed and dated by the accredited firm or testing agency performing the test. Prepare test reports in the manner specified in the standard or regulation governing the test procedure (if any) as indicated.

F. Closeout Submittals
1. See Section 26 0005 for additional requirements.
2. Software Licenses: Provide documentation of ownership under the owner's corporate name (coordinate with owner's representative for exact ownership wording) for Software Licenses provided as part of the work included under Divisions 26, 27, and 28. Include information for updates, subscription requirements if applicable, backup, support, login, passwords, date when purchased, expiration date if applicable, version, etc. Include in the O & M Manual after review and "No Exceptions Taken" has been accomplished.
3. Owner maintenance materials (Listed in individual sections).
4. O&M Manuals
5. Signed off Substantial Completion Punchlist.
6. Signed off Test Reports.
7. As-built markups for Record Drawing conversion. Revisions to the Contract Documents shall be legible and shall be prepared using the following color scheme:
   a. Red shall indicate new items, deviations and routing.
   b. Green shall indicate items removed or deleted.
   c. Blue shall be used for relevant notes and descriptions.
8. Electrical System and Equipment Warranties.

PART 2 PRODUCTS

2.01 GENERAL

A. All work, materials, and trades shall follow the latest adopted editions of the National Fire Protection Administration (NFPA) and International Code Council (ICC) standards for building construction and energy use. Refer to Architectural code sheet for additional information. It is the contractor's responsibility to perform work in conformance with all applicable codes, ordinances, and life safety measures as required, by local, state, and national authorities. Consult literature and websites of jurisdictional authorities for any amendments or additional requirements.

B. All materials shall be new and listed for the application unless otherwise noted. Equipment shall be set straight and square with the structure. Conduit shall be installed parallel or at right angles to all walls and structural members. Hanger rods, equipment bases, supports and isolation shall be set plumb and square and in vertical alignment where applicable. All work shall be neat and clean upon completion. All work shall be performed in a workmanlike manner to the satisfaction of the owner and engineer.

C. NEC and UL: Products shall conform to requirements of the National Electrical Code. Where Underwriters' Laboratories have set standards, listed products and issued labels, products used shall be listed and labeled by UL.

D. Basis of Design: Items specified herein by manufacturer and model number are to be considered the minimum quality acceptable for the specified application. For substitutions, it shall be the responsibility of the contractor to provide sufficient information to the Design Team in order to establish "or equal" status and to coordinate installation of same into the available space. Increased costs to the project associated with contractor substitutions shall be the responsibility of the contractor.

E. Substitutions: The burden of proof that proposed equipment is equal in size, capacity, performance, and other pertinent criteria for this specific installation, or superior to that specified is up to the Contractor. Substituted equipment will only be allowed where specifically listed in a written addendum. If substitutions are not granted, the specified materials and equipment must be installed. Where substituted equipment is allowed, it shall be the

Contractor's responsibility to notify all related trades of the accepted substitution and to assume full responsibility for all costs caused as a result of the substitution.

F. Owner Furnish Contractor Installed (OFCI): This contractor shall be responsible for the receiving and installation of all Owner supplied equipment unless noted otherwise. This scope requires coordination with the owner supplied equipment, and not all details of that equipment are necessarily included in these drawings. Refer to manufacturer's drawings, equipment submittal data sheets, and equipment manufacturer's selections and refurbishment guidelines for new project altitude, and meet the requirements defined therein.

G. Owner shall pay all fees required for connection to municipal and public utility facilities through their service contract with each respective service provider.

H. Contractor shall arrange for and pay for all permits, inspections, licenses and certificates required in connection with the work.

I. Existing: Equipment and control systems should match, integrate, communicate or cooperate with Owner's existing systems, such as power monitoring systems, building automation, fire alarm, motor control centers, switchgears, breakers, transformers, and lighting dimming systems.

2.02 DELIVERY, STORAGE AND HANDLING

A. All equipment and materials shall be delivered to the Project Site clean and sealed for protection. Deliver products to project properly identified with names, model numbers, types, grades, compliance labels, and similar information needed for distinct identifications; adequately packaged and protected to prevent damage during shipment, storage, and handling.

B. Moisture: During construction, protect switchgear, transformers, motors, control equipment, and other items from insulation moisture absorption and metallic component corrosion by appropriate use of strip heaters, lamps or other suitable means. Apply protection immediately upon receiving the products and maintain continually.

C. Damage: Take such precautions as are necessary to protect apparatus and materials from damage. Failure to protect materials is sufficient cause for rejection of the apparatus or material in question.

D. Finish: Protect factory finish from damage during construction operations until acceptance of the Project. Restore any finishes that become stained or damaged to Owner's satisfaction.

E. Check delivered equipment against contract documents and submittals. Store equipment and materials at the site, unless off-site storage is authorized in writing. Protect stored equipment and materials from damage and weather.

F. Coordinate deliveries of electrical materials and equipment to minimize construction site congestion. Limit each shipment of materials and equipment to the items and quantities needed for the smooth and efficient flow of installations.

2.03 DELEGATED AND/OR DEFERRED DESIGN BY CONTRACTOR

A. The construction of this project requires the Contractor to include the detailing and design of select systems and/or subsystems. All such design work associated with the development of the coordination drawings shall be the complete responsibility of the Contractor.

B. The Contractor shall take the full responsibility to develop and complete routing strategies which will allow fully coordinated system to be installed in a fully functional manner. The Engineers' contract drawings shall be for system design intent and general configurations.

C. Systems or subsystems which require design responsibility by the contractor include but are not limited to:
   1. Equipment supports, hangers, anchors

PART 3 EXECUTION

3.01 INSTALLATION

A. Installation shall meet or exceed all applicable federal, state and local requirements, referenced standards and conform to codes and ordinances of authorities having jurisdiction.

B. Installation shall be in accordance with manufacturer's published recommendations.

C. Cooperation with Other Trades: Cooperate with trades of adjacent, related or affected materials or operations, and with trades performing continuations of this Work in order to effect timely and accurate placing of Work and to coordinate, in proper and correct sequence, the Work of such trades.

D. Workmanship: Work shall be performed by competent workers skilled in their trade. This installation must be complete.

E. Motors: Electrical Work includes the electrical connection of all motors, except those that are wired as a part of equipment.

F. Provide final electrical connections to equipment furnished under other divisions and by the Owner. Furnish detailed Shop Drawings of equipment indicating the exact number and location of rough-in points. Such final Shop Drawings may indicate adjustments in total number and exact location of rough-in points, and in equipment dimensions. Making adjustments to field conditions is considered a part of the Work required.

1. Roughing-in: When roughing-in electrical branch circuits to various items of equipment, terminate at proper points as indicated on detailed equipment Shop Drawings or as directed by Owner. Do not rely on Drawings accompanying these Specifications for rough-in locations, only for general routing of circuiting.

G. Final Connections: Laboratory casework, medical equipment, and food service equipment will include service fittings such as switches, duplex receptacles, lighting fixtures, etc., on the casework or equipment. Provide branch circuit connections to meet service fitting requirements.

H. All unused openings such as but not limited to, knockouts on panels and boxes, surface wireway openings, busway openings, circuit breaker empty slots shall be covered with approved cover plates.

I. Space Limitations: Equipment selected shall conform to the building features and shall be coordinated with all components. Do not provide equipment that will not meet arrangement and space limitations. Contractor shall submit room layouts with submitted items shown drawn to scale. Submittals will be rejected without floor plan Drawings showing submitted items.

J. Where mounting heights are not detailed or dimensioned, install electrical conduits, boxes, and overhead equipment to provide the maximum headroom possible. In general, keep installations tight to structure. Make allowance for expansion and contraction for all building electrical components and conduit systems that are subject to such.

K. Provide all necessary scaffolding, ladders, hoists, rigging, etc. As required for the installation of the work within the scope of Division 26, 27, and 28. All hoisting, supports, delivery, etc of equipment and material for this project shall be the responsibility of this contractor.

L. Extend all conduits so that junction and pull boxes are in accessible locations. Provide access panel or doors where equipment or boxes are concealed behind finished surfaces.

3.02 TESTING

A. Testing as required by these specifications shall pertain to all equipment, wiring, devices, etc. installed under this contract as well as all equipment, wiring, devices, etc being reused within the scope of this project.

B. Verify operation and functional performance of electrical power delivery systems, lighting system and digital lighting controls for compliance with "Design Intent", as defined by the Contract Documents. Furnish labor and material to accomplish complete mechanical and

electrical system commissioning as specified herein. Complete interim commissioning of electrical systems during initial season operation.

C. Job Conditions: The commissioning contractor shall become familiar with the contract documents all Addenda and change orders issued for this project prior to commencing the commissioning work.

D. Qualifications: The "Commissioning Authority" shall be defined as a company or agency of experienced personnel, qualified to plan & carry out the overall commissioning progress. The Commissioning Authority shall submit for owner review, an outline of the organization's personnel qualification resources, commissioning, documentation process and commissioning plan specifically prepared for this project.

E. Contractors Specialty: Individual sub-contractors for the appropriate mechanical and electrical divisions - will be responsible for providing labor, material, equipment, etc., required within the scope of commissioning to facilitate the commissioning process. The listed Sub-Contractor will perform tests and verification procedures required by the commissioning process when requested by the Commissioning Authority and directed by the General Contractor.

F. Required Testing:
   1. Required studies to be performed by the Contractor include:
      a. Short-Circuit Study
   2. Required inspections and tests to be performed by the Contractor include:
      a. Low Voltage Cables (24V Max) - NETA ATS Section 7.3.1
      b. Low Voltage Cables (600V Max) - NETA ATS Section 7.3.2
      c. Low Voltage Switches (Air) - NETA ATS Section 7.5.1.1

G. Test Conditions:
   1. Place circuits and equipment into service under normal conditions, collectively and separately, as may be necessary to determine satisfactory operation. Perform specified tests in the presence of the Owner's representative(s). Furnish all instruments, wiring, equipment and personnel required for conducting tests. Demonstrate that the equipment operates in accordance with requirements of the Contract Documents. Special tests on certain items are specified hereinafter.
   2. Prior to testing, Contractor shall submit to Owner for approval, installation verification Prefunctional Checklists and Functional Performance Test procedures. These shall be used for documentation as part of the commissioning process.

H. Test Dates: Schedule final acceptance sufficiently in advance of the Contract date to permit completion of any necessary adjustment or alterations within the number of days allotted for completion of the Contract. Provide written notification to Owner at least fourteen (14) calendar days in advance of Functional Performance Test dates.

I. Retests: If retesting is required due to initial failure, conduct retests of such time duration as may be necessary to assure proper functioning of adjusted or altered parts or items of equipment. Any resultant delay as a result of such necessary retests does not relieve the Contractor of Contractor’s responsibility under this Contract.

J. Test Report: Submit three copies of the completed report no later than fifteen (15) days after completion of test unless directed otherwise. The test report shall be bound and its contents certified. The test report shall include the following:
   1. Summary of project.
   2. Description of equipment tested.
   3. Description of test.
   4. List of test equipment used in calibration and calibration date.
   5. Test results.
   6. Conclusions and recommendations.
   7. Appendix, including appropriate test forms

K. Circuit Verification: All 120-volt single-phase circuits shall be verified to match the Drawings and panel schedules by “ringing out" each circuit in the presence of the Owner’s representative(s).

L. Refer to Commissioning Specification Sections for additional start-up, pre-functional and operational checkout, and for functional performance test procedures.

M. This contract will not be considered completed nor will final payment be made until all specified material, including test reports, settings reports, and final Schedule of Values with all Electrical and Information Technology change order costs included and identified is provided and the manual is reviewed by the Architect/Engineer

3.03 SUBSTANTIAL COMPLETION AND CLOSEOUT

A. Substantial Completion
   1. See requirements in Section 26 0005

B. Closeout Requirements
   1. See requirements in Section 26 0005

END OF SECTION

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SECTION 26 0519
ELECTRICAL CONDUCTORS AND CABLES

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Single conductor building wire.

B. Metal-clad cable.

C. Wiring connectors.

D. Electrical tape.

E. Heat shrink tubing.

F. Wire pulling lubricant.

G. Cable ties.

1.02 RELATED REQUIREMENTS

A. Section 07 8400 - Firestopping.

B. Section 26 0553 - Identification for Electrical Systems: Identification products and requirements.

1.03 REFERENCE STANDARDS

A. ASTM B3 - Standard Specification for Soft or Annealed Copper Wire; 2013 (Reapproved 2018).

B. ASTM B8 - Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft; 2011 (Reapproved 2017).

C. ASTM B33 - Standard Specification for Tin-Coated Soft or Annealed Copper Wire for Electrical Purposes; 2010, with Editorial Revision (2020).

D. ASTM B787/B787M - Standard Specification for 19 Wire Combination Unilay-Stranded Copper Conductors for Subsequent Insulation; 2004 (Reapproved 2020).

E. ASTM D3005 - Standard Specification for Low-Temperature Resistant Vinyl Chloride Plastic Pressure-Sensitive Electrical Insulating Tape; 2017.

F. NECA 1 - Standard for Good Workmanship in Electrical Construction; 2015.

G. NECA 120 - Standard for Installing Armored Cable (AC) and Type Metal-Clad (MC) Cable; 2018.

H. NEMA WC 70 - Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy; 2021.

I. NETA ATS - Standard For Acceptance Testing Specifications For Electrical Power Equipment And Systems; 2021.

J. NFPA 70 - National Electrical Code; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.

K. UL 44 - Thermoset-Insulated Wires and Cables; Current Edition, Including All Revisions.

L. UL 83 - Thermoplastic-Insulated Wires and Cables; Current Edition, Including All Revisions.

M. UL 267 - Outline of Investigation for Wire-Pulling Compounds; Current Edition, Including All Revisions.

N. UL 486A-486B - Wire Connectors; Current Edition, Including All Revisions.

O. UL 486C - Splicing Wire Connectors; Current Edition, Including All Revisions.

P. UL 486D - Sealed Wire Connector Systems; Current Edition, Including All Revisions.

Q. UL 510 - Polyvinyl Chloride, Polyethylene, and Rubber Insulating Tape; Current Edition, Including All Revisions.

R. UL 1569 - Metal-Clad Cables; Current Edition, Including All Revisions.

1.04 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Coordinate sizes of raceways, boxes, and equipment enclosures installed under other sections with the actual conductors to be installed, including adjustments for conductor sizes increased for voltage drop.
   2. Coordinate with electrical equipment installed under other sections to provide terminations suitable for use with the conductors to be installed.
   3. Notify Engineer of any conflicts with or deviations from Contract Documents. Obtain direction before proceeding with work.

1.05 SUBMITTALS

A. See Section 01 3000 - Administrative Requirements, for submittal procedures.

B. Wire Pulling Lubricant: Certification of compatibility with conductors/cables where used with the following insulation/jacket types:
   1. Low-smoke zero halogen (LSZH).

C. Field Quality Control Test Reports.

D. Project Record Documents: Record actual installed circuiting arrangements. Record actual routing for underground circuits and all panel feeders.

1.06 QUALITY ASSURANCE

A. Comply with requirements of NFPA 70.

1.07 DELIVERY, STORAGE, AND HANDLING

A. Receive, inspect, handle, and store conductors and cables in accordance with manufacturer's instructions.

1.08 FIELD CONDITIONS

A. Do not install or otherwise handle thermoplastic-insulated conductors at temperatures lower than 14 degrees F, unless otherwise permitted by manufacturer's instructions. When installation below this temperature is unavoidable, notify Engineer and obtain direction before proceeding with work.

PART 2 PRODUCTS

2.01 CONDUCTOR AND CABLE APPLICATIONS

A. Do not use conductors and cables for applications other than as permitted by NFPA 70 and product listing.

B. Provide single conductor building wire installed in suitable raceway unless otherwise indicated, permitted, or required.

C. Nonmetallic-sheathed cable is not permitted.

D. Underground feeder and branch-circuit cable is not permitted.

E. Service entrance cable is not permitted.

F. Armored cable is not permitted.

G. Metal-clad cable is permitted only as follows:
   1. Where not otherwise restricted, may be used:
      a. For motor connections in dry locations.
         1) Maximum Length: 6 feet.
   2. In addition to other applicable restrictions, may not be used:
      a. Where not approved for use by the authority having jurisdiction.
      b. Where exposed to view, except in dedicated electrical, communications, and mechanical rooms where not subject to damage.

c. Where exposed to damage.
d. For damp, wet, or corrosive locations, unless provided with a PVC jacket listed as suitable for those locations.
e. For isolated ground circuits, unless provided with an additional isolated/insulated grounding conductor.

2.02 CONDUCTOR AND CABLE GENERAL REQUIREMENTS

A. Provide products that comply with requirements of NFPA 70.

B. Provide products listed, classified, and labeled as suitable for the purpose intended.

C. Unless specifically indicated to be excluded, provide all required conduit, boxes, wiring, connectors, etc. as required for a complete operating system.

D. Comply with NEMA WC 70.

E. Thermoplastic-Insulated Conductors and Cables: Listed and labeled as complying with UL 83.

F. Thermoset-Insulated Conductors and Cables: Listed and labeled as complying with UL 44.

G. Conductors and Cables Installed Where Exposed to Direct Rays of Sun: Listed and labeled as sunlight resistant.

H. Conductors and Cables Installed Exposed in Spaces Used for Environmental Air (only where specifically permitted): Plenum rated, listed and labeled as suitable for use in return air plenums.

I. Conductor Material:
1. Provide copper conductors only. Aluminum conductors are not acceptable for this project. Conductor sizes indicated are based on copper.
2. Copper Conductors: Soft drawn annealed, 98 percent conductivity, uncoated copper conductors complying with ASTM B3, ASTM B8, or ASTM B787/B787M unless otherwise indicated.
3. Tinned Copper Conductors: Comply with ASTM B33.

J. Minimum Conductor Size:
1. Branch Circuits: 12 AWG.
a. Exceptions:
1) 20 A, 120 V circuits longer than 75 feet: 10 AWG, for voltage drop.
2) 20 A, 120 V circuits longer than 150 feet: 8 AWG, for voltage drop.
3) 20 A, 277 V circuits longer than 150 feet: 10 AWG, for voltage drop.
2. Control Circuits: 14 AWG.

K. Where conductor size is not indicated, size to comply with NFPA 70 but not less than applicable minimum size requirements specified.

L. Conductor Color Coding:
1. Color code conductors as indicated unless otherwise required by the authority having jurisdiction. Maintain consistent color coding throughout project.
2. Color Coding Method: Integrally colored insulation.
a. Conductors size 4 AWG and larger may have black insulation color coded using vinyl color coding electrical tape.
3. Color Code:
a. 208Y/120 V, 3 Phase, 4 Wire System:
1) Phase A: Black.
2) Phase B: Red.
3) Phase C: Blue.
4) Neutral/Grounded: White.
b. Equipment Ground, All Systems: Green.
c. For control circuits, comply with manufacturer's recommended color code.

2.03 SINGLE CONDUCTOR BUILDING WIRE

A. Manufacturers:
   1. Copper Building Wire:
      a. Cerro Wire LLC: www.cerrowire.com/#sle.
      b. Encore Wire Corporation: www.encorewire.com/#sle.
      c. General Cable Technologies Corporation: www.generalcable.com/#sle.
      d. Service Wire Co: www.servicewire.com/#sle.
      e. Southwire Company: www.southwire.com/#sle.
      f. Substitutions: See Section 01 6000 - Product Requirements.

B. Description: Single conductor insulated wire.

C. Conductor Stranding:
   1. Feeders and Branch Circuits:
      a. Size 10 AWG and Smaller: Solid.
      b. Size 8 AWG and Larger: Stranded.

D. Insulation Voltage Rating: 600 V.

E. Insulation:
   1. Copper Building Wire: Type THHN/THWN or THHN/THWN-2.
      a. Size 4 AWG and Larger: Type XHHW-2.
      b. Installed Underground: Type XHHW-2.

2.04 METAL-CLAD CABLE

A. Manufacturers:
   1. AFC Cable Systems Inc: www.afcweb.com/#sle.
   2. Encore Wire Corporation: www.encorewire.com/#sle.
   3. Service Wire Co: www.servicewire.com/#sle.
   4. Southwire Company: www.southwire.com/#sle.
   5. Substitutions: See Section 01 6000 - Product Requirements.

B. Description: NFPA 70, Type MC cable listed and labeled as complying with UL 1569, and listed for use in classified firestop systems to be used.

C. Conductor Stranding:
   1. Size 10 AWG and Smaller: Solid.
   2. Size 8 AWG and Larger: Stranded.

D. Insulation Voltage Rating: 600 V.

E. Insulation: Type THHN, THHN/THWN, or THHN/THWN-2.

F. Grounding: Full-size integral equipment grounding conductor.

G. Armor: Steel, interlocked tape.

H. Provide PVC jacket applied over cable armor.

2.05 WIRING CONNECTORS

A. Description: Wiring connectors appropriate for the application, suitable for use with the conductors to be connected, and listed as complying with UL 486A-486B or UL 486C as applicable.

B. Wiring Connectors for Splices and Taps:
   1. Copper Conductors Size 8 AWG and Smaller: Use twist-on insulated spring connectors.
   2. Copper Conductors Size 6 AWG and Larger: Use mechanical connectors or compression connectors.

C. Wiring Connectors for Terminations:

1. Provide terminal lugs for connecting conductors to equipment furnished with terminations designed for terminal lugs.
2. Provide compression adapters for connecting conductors to equipment furnished with mechanical lugs when only compression connectors are specified.
3. Where over-sized conductors are larger than the equipment terminations can accommodate, provide connectors suitable for reducing to appropriate size, but not less than required for the rating of the overcurrent protective device.

D. Do not use insulation-piercing or insulation-displacement connectors designed for use with conductors without stripping insulation.

E. Do not use push-in wire connectors as a substitute for twist-on insulated spring connectors.

F. Twist-on Insulated Spring Connectors: Rated 600 V, 221 degrees F for standard applications and 302 degrees F for high temperature applications; pre-filled with sealant and listed as complying with UL 486D for damp and wet locations.
   1. Manufacturers:
      a. 3M: www.3m.com/#sle.
      b. Ideal Industries, Inc: www.idealindustries.com/#sle.
      c. NSI Industries LLC: www.nsiindustries.com/#sle.
      d. Substitutions: See Section 01 6000 - Product Requirements.

G. Mechanical Connectors: Provide bolted type or set-screw type.
   1. Manufacturers:
      a. Burndy LLC: www.burndy.com/#sle.
      b. Ilsco: www.ilsco.com/#sle.
      c. Thomas & Betts Corporation: www.tnb.com/#sle.
      d. Substitutions: See Section 01 6000 - Product Requirements.

H. Compression Connectors: Provide circumferential type or hex type crimp configuration.
   1. Manufacturers:
      a. Burndy LLC: www.burndy.com/#sle.
      b. Ilsco: www.ilsco.com/#sle.
      c. Thomas & Betts Corporation: www.tnb.com/#sle.
      d. Substitutions: See Section 01 6000 - Product Requirements.

2.06 ACCESSORIES

A. Electrical Tape:
   1. Manufacturers:
      a. 3M: www.3m.com/#sle.
      b. Plymouth Rubber Europa: www.plymouthrubber.com/#sle.
      c. Substitutions: See Section 01 6000 - Product Requirements.
   2. Vinyl Color Coding Electrical Tape: Integrally colored to match color code indicated; listed as complying with UL 510; minimum thickness of 7 mil; resistant to abrasion, corrosion, and sunlight; suitable for continuous temperature environment up to 221 degrees F.
   3. Vinyl Insulating Electrical Tape: Complying with ASTM D3005 and listed as complying with UL 510; minimum thickness of 7 mil; resistant to abrasion, corrosion, and sunlight; conformable for application down to 0 degrees F and suitable for continuous temperature environment up to 221 degrees F.

B. Heat Shrink Tubing: Heavy-wall, split-resistant, with factory-applied adhesive; rated 600 V; suitable for direct burial applications; listed as complying with UL 486D.

C. Wire Pulling Lubricant:
   1. Manufacturers:
      a. 3M: www.3m.com/#sle.
      b. American Polywater Corporation: www.polywater.com/#sle.
      c. Ideal Industries, Inc: www.idealindustries.com/#sle.

d. Substitutions: See Section 01 6000 - Product Requirements.

2. Listed and labeled as complying with UL 267.
3. Suitable for use with conductors/cables and associated insulation/jackets to be installed.
4. Suitable for use at installation temperature.
5. Products:
   a. American Polywater Corporation; Polywater J Cable Pulling Lubricant: www.polywater.com/#sle.
   b. American Polywater Corporation; Polywater LZ Cable Pulling Lubricant: www.polywater.com/#sle.
   c. Substitutions: See Section 01 6000 - Product Requirements.

D. Cable Ties: Material and tensile strength rating suitable for application.
1. Manufacturers:
   a. Burndy LLC: www.burndy.com/#sle.
   b. Substitutions: See Section 01 6000 - Product Requirements.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that interior of building has been protected from weather.

B. Verify that work likely to damage wire and cable has been completed.

C. Verify that raceways, boxes, and equipment enclosures are installed and are properly sized to accommodate conductors and cables in accordance with NFPA 70.

D. Verify that field measurements are as indicated.

E. Verify that conditions are satisfactory for installation prior to starting work.

3.02 INSTALLATION

A. Circuiting Requirements:
1. Unless dimensioned, circuit routing indicated is diagrammatic.
2. When circuit destination is indicated without specific routing, determine exact routing required.
3. Arrange circuiting to minimize splices.
4. Include circuit lengths required to install connected devices within 10 ft of location indicated.
5. Maintain separation of Class 1, Class 2, and Class 3 remote-control, signaling, and power-limited circuits in accordance with NFPA 70.
6. Maintain separation of wiring for emergency systems in accordance with NFPA 70.
7. Circuiting Adjustments: Unless otherwise indicated, when branch circuits are indicated as separate, combining them together in a single raceway is permitted, under the following conditions:
   a. For 20A/1P circuits: Provide no more than six current-carrying conductors in a single raceway. Dedicated neutral conductors are considered current-carrying conductors.
8. Common Neutrals: Unless otherwise indicated, sharing of neutral/grounded conductors among single phase branch circuits of different phases installed in the same raceway is not permitted. Provide dedicated neutral/grounded conductor for each individual branch circuit.

B. Install products in accordance with manufacturer's instructions.

C. Perform work in accordance with NECA 1 (general workmanship).

D. Install metal-clad cable (Type MC) in accordance with NECA 120.

E. Installation in Raceway:
1. Tape ends of conductors and cables to prevent infiltration of moisture and other contaminants.

2. Pull all conductors and cables together into raceway at same time.
3. Do not damage conductors and cables or exceed manufacturer's recommended maximum pulling tension and sidewall pressure.
4. Use suitable wire pulling lubricant where necessary, except when lubricant is not recommended by the manufacturer.

F. Paralleled Conductors: Install conductors of the same length and terminate in the same manner.

G. Secure and support conductors and cables in accordance with NFPA 70 using suitable supports and methods approved by the authority having jurisdiction. Provide independent support from building structure. Do not provide support from raceways, piping, ductwork, or other systems.

H. Terminate cables using suitable fittings.
1. Armored Cable (Type AC):
   a. Use listed fittings and anti-short, insulating bushings.
   b. Cut cable armor only using specialized tools to prevent damaging conductors or insulation. Do not use hacksaw or wire cutters to cut armor.
   c. Do not use direct-bearing set-screw type fittings for cables with aluminum armor.
2. Metal-Clad Cable (Type MC):
   a. Use listed fittings.
   b. Cut cable armor only using specialized tools to prevent damaging conductors or insulation. Do not use hacksaw or wire cutters to cut armor.

I. Install conductors with a minimum of 12 inches of slack at each outlet.

J. Neatly train and bundle conductors inside boxes, wireways, panelboards and other equipment enclosures.

K. Group or otherwise identify neutral/grounded conductors with associated ungrounded conductors inside enclosures in accordance with NFPA 70.

L. Make wiring connections using specified wiring connectors.
1. Make splices and taps only in accessible boxes. Do not pull splices into raceways or make splices in conduit bodies.
2. Remove appropriate amount of conductor insulation for making connections without cutting, nicking or damaging conductors.
3. Do not remove conductor strands to facilitate insertion into connector.
4. Clean contact surfaces on conductors and connectors to suitable remove corrosion, oxides, and other contaminates. Do not use wire brush on plated connector surfaces.
5. Mechanical Connectors: Secure connections according to manufacturer's recommended torque settings.
6. Compression Connectors: Secure connections using manufacturer's recommended tools and dies.

M. Insulate splices and taps that are made with uninsulated connectors using methods suitable for the application, with insulation and mechanical strength at least equivalent to unspliced conductors.

N. Insulate ends of spare conductors using vinyl insulating electrical tape.

O. Field-Applied Color Coding: Where vinyl color coding electrical tape is used in lieu of integrally colored insulation as permitted in Part 2 under "Color Coding", apply half overlapping turns of tape at each termination and at each location conductors are accessible.

P. Install firestopping to preserve fire resistance rating of partitions and other elements, using materials and methods specified in Section 07 8400.

Q. Unless specifically indicated to be excluded, provide final connections to all equipment and devices, including those furnished by others, as required for a complete operating system.

3.03 FIELD QUALITY CONTROL

A. See Section 01 4000 - Quality Requirements, for additional requirements.

B. Inspect and test in accordance with NETA ATS, except Section 4.

C. Correct deficiencies and replace damaged or defective conductors and cables.

END OF SECTION

SECTION 26 0529
HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Support and attachment requirements and components for equipment, conduit, cable, boxes, and other electrical work.

1.02 RELATED REQUIREMENTS

A. Section 03 3000 - Cast-in-Place Concrete: Concrete equipment pads.

B. Section 05 5000 - Metal Fabrications: Materials and requirements for fabricated metal supports.

C. Section 26 0533 - Conduit for Electrical Systems: Additional support and attachment requirements for conduits.

1.03 REFERENCE STANDARDS

A. ASTM A123/A123M - Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products; 2017.

B. ASTM A153/A153M - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware; 2023.

C. ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel; 2023.

D. MFMA-4 - Metal Framing Standards Publication; 2004.

E. NECA 1 - Standard for Good Workmanship in Electrical Construction; 2015.

F. NFPA 70 - National Electrical Code; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.

1.04 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Coordinate sizes and arrangement of supports and bases with actual equipment and components to be installed.
   2. Coordinate work to provide additional framing and materials required for installation.
   3. Coordinate compatibility of support and attachment components with mounting surfaces at installed locations.
   4. Coordinate arrangement of supports with ductwork, piping, equipment and other potential conflicts.
   5. Notify Engineer of conflicts with or deviations from Contract Documents. Obtain direction before proceeding with work.

B. Sequencing:
   1. Do not install products on or provide attachment to concrete surfaces until concrete has cured; see Section 03 3000.

1.05 SUBMITTALS

A. See Section 01 3000 - Administrative Requirements for submittal procedures.

B. Product Data: Provide manufacturer's standard catalog pages and data sheets for channel/strut framing systems, nonpenetrating rooftop supports, and post-installed concrete/masonry anchors.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Receive, inspect, handle, and store products in accordance with manufacturer's instructions.

PART 2 PRODUCTS

2.01 SUPPORT AND ATTACHMENT COMPONENTS

A. General Requirements:
   1. Comply with the following. Where requirements differ, comply with most stringent.
      a. NFPA 70.
      b. Requirements of authorities having jurisdiction.
   2. Provide required hangers, supports, anchors, fasteners, fittings, accessories, and hardware as necessary for complete installation of electrical work.
   3. Provide products listed, classified, and labeled as suitable for purpose intended, where applicable.
   4. Do not use products for applications other than as permitted by NFPA 70 and product listing.
   5. Steel Components: Use corrosion-resistant materials suitable for environment where installed.
      a. Zinc-Plated Steel: Electroplated in accordance with ASTM B633.
      b. Galvanized Steel: Hot-dip galvanized after fabrication in accordance with ASTM A123/A123M or ASTM A153/A153M.

B. Conduit and Cable Supports: Straps and clamps suitable for conduit or cable to be supported.
   1. Manufacturers:
      a. ABB: www.electrification.us.abb.com/#sle.
      b. Eaton Corporation: www.eaton.com/#sle.
      c. Emerson Electric Co; O-Z/Gedney: www.emerson.com/#sle.
      d. HoldRite, a brand of Reliance Worldwide Corporation: www.holdrite.com/#sle.
      e. nVent; Caddy: www.nvent.com/#sle.
      f. Substitutions: See Section 01 6000 - Product Requirements.
   2. Conduit Straps: One-hole or two-hole type; steel or malleable iron.
   3. Conduit Clamps: Bolted type unless otherwise indicated.
   4. Products:
      a. Gripple, Inc; Universal Bracket: www.gripple.com/#sle.
      b. Gripple, Inc; Fast Trak: www.gripple.com/#sle.
      c. Gripple, Inc; Universal Clamp (Threaded): www.gripple.com/#sle.
      d. Gripple, Inc; Low Profile Bracket Kits: www.gripple.com/#sle.
      e. Substitutions: See Section 01 6000 - Product Requirements.

C. Outlet Box Supports: Hangers and brackets suitable for boxes to be supported.
   1. Manufacturers:
      a. ABB: www.electrification.us.abb.com/#sle.
      b. Eaton Corporation: www.eaton.com/#sle.
      c. Emerson Electric Co; O-Z/Gedney: www.emerson.com/#sle.
      d. HoldRite, a brand of Reliance Worldwide Corporation: www.holdrite.com/#sle.
      e. nVent; Caddy: www.nvent.com/#sle.
      f. Substitutions: See Section 01 6000 - Product Requirements.

D. Metal Channel/Strut Framing Systems:
   1. Manufacturers:
      a. ABB: www.electrification.us.abb.com/#sle.
      b. Atkore International Inc; Unistrut: www.unistrut.us/#sle.
      c. Custom Strut and Roll Forming, LLC: www.customstrut.com/#sle.
      d. Eaton Corporation: www.eaton.com/#sle.
      e. Elgen Manufacturing Company, Inc: www.elgenmfg.com/#sle.
      f. Substitutions: See Section 01 6000 - Product Requirements.

2. Description: Factory-fabricated, continuous-slot, metal channel/strut and associated fittings, accessories, and hardware required for field assembly of supports.
3. Comply with MFMA-4.

E. Hanger Rods: Threaded, zinc-plated steel unless otherwise indicated.

F. Anchors and Fasteners:
1. Manufacturers - Mechanical Anchors:
a. Dewalt: anchors.dewalt.com/#sle.
b. Hilti, Inc: www.hilti.com/#sle.
c. ITW Red Head, a division of Illinois Tool Works, Inc: www.itwredhead.com/#sle.
d. Simpson Strong-Tie Company Inc: www.strongtie.com/#sle.
e. Substitutions: See Section 01 6000 - Product Requirements.
2. Unless otherwise indicated and where not otherwise restricted, use anchor and fastener types indicated for specified applications.
3. Powder-actuated fasteners are not permitted.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that field measurements are as indicated.

B. Verify that mounting surfaces are ready to receive support and attachment components.

C. Verify that conditions are satisfactory for installation prior to starting work.

3.02 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install hangers and supports in accordance with NECA 1.

C. Provide independent support from building structure. Do not provide support from piping, ductwork, or other systems.

D. Unless specifically indicated or approved by Engineer, do not provide support from suspended ceiling support system or ceiling grid.

E. Unless specifically indicated or approved by Engineer, do not provide support from roof deck.

F. Do not penetrate or otherwise notch or cut structural members without approval of Structural Engineer.

G. Equipment Support and Attachment:
1. Use metal, fabricated supports or supports assembled from metal channel/strut to support equipment as required.
2. Use metal channel/strut secured to studs to support equipment surface mounted on hollow stud walls when wall strength is not sufficient to resist pull-out.
3. Use metal channel/strut to support surface-mounted equipment in wet or damp locations to provide space between equipment and mounting surface.
4. Securely fasten floor-mounted equipment. Do not install equipment such that it relies on its own weight for support.

H. Secure fasteners in accordance with manufacturer's recommended torque settings.

I. Remove temporary supports.

3.03 FIELD QUALITY CONTROL

A. See Section 01 4000 - Quality Requirements for additional requirements.

B. Inspect support and attachment components for damage and defects.

C. Repair cuts and abrasions in galvanized finishes using zinc-rich paint recommended by manufacturer. Replace components that exhibit signs of corrosion.

D. Correct deficiencies and replace damaged or defective support and attachment components.

END OF SECTION

SECTION 26 0533
CONDUIT FOR ELECTRICAL SYSTEMS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Galvanized steel rigid metal conduit (RMC).

B. Stainless steel rigid metal conduit (RMC).

C. PVC-coated galvanized steel rigid metal conduit (RMC).

D. Flexible metal conduit (FMC).

E. Liquidtight flexible metal conduit (LFMC).

F. Galvanized steel electrical metallic tubing (EMT).

G. Aluminum electrical metallic tubing (EMT).

H. Rigid polyvinyl chloride (PVC) conduit.

1.02 RELATED REQUIREMENTS

A. Section 03 3000 - Cast-in-Place Concrete: Concrete encasement of conduits.

B. Section 07 8400 - Firestopping.

C. Section 26 0526 - Grounding and Bonding for Electrical Systems.

D. Section 26 0529 - Hangers and Supports for Electrical Systems.

E. Section 26 0553 - Identification for Electrical Systems: Identification products and requirements.

1.03 REFERENCE STANDARDS

A. ANSI C80.1 - American National Standard for Electrical Rigid Steel Conduit (ERSC); 2020.

B. ANSI C80.3 - American National Standard for Electrical Metallic Tubing -- Steel (EMT-S); 2020.

C. NECA 1 - Standard for Good Workmanship in Electrical Construction; 2015.

D. NECA 101 - Standard for Installing Steel Conduits (Rigid, IMC, EMT); 2020.

E. NECA 111 - Standard for Installing Nonmetallic Raceways (RNC, ENT, LFNC); 2017.

F. NEMA FB 1 - Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing, and Cable; 2014.

G. NEMA RN 1 - Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Metal Conduit and Intermediate Metal Conduit; 2018.

H. NEMA TC 2 - Electrical Polyvinyl Chloride (PVC) Conduit; 2020.

I. NEMA TC 3 - Polyvinyl Chloride (PVC) Fittings for Use with Rigid PVC Conduit and Tubing; 2021.

J. NFPA 70 - National Electrical Code; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.

K. UL 1 - Flexible Metal Conduit; Current Edition, Including All Revisions.

L. UL 6 - Electrical Rigid Metal Conduit-Steel; Current Edition, Including All Revisions.

M. UL 6A - Electrical Rigid Metal Conduit-Aluminum, Red Brass, and Stainless Steel; Current Edition, Including All Revisions.

N. UL 360 - Liquid-Tight Flexible Metal Conduit; Current Edition, Including All Revisions.

O. UL 514A - Metallic Outlet Boxes; Current Edition, Including All Revisions.

P. UL 514B - Conduit, Tubing, and Cable Fittings; Current Edition, Including All Revisions.

Q. UL 651 - Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings; Current Edition, Including All Revisions.

R. UL 797 - Electrical Metallic Tubing-Steel; Current Edition, Including All Revisions.

S. UL 797A - Electrical Metallic Tubing - Aluminum and Stainless Steel; Current Edition, Including All Revisions.

T. UL 2419 - Outline of Investigation for Electrically Conductive Corrosion Resistant Compounds; Current Edition, Including All Revisions.

1.04 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Coordinate minimum sizes of conduits with actual type and quantity of conductors to be installed, including adjustments for conductor sizes increased for voltage drop.
   2. Coordinate arrangement of conduits with structural members, ductwork, piping, equipment, and other potential conflicts.
   3. Verify exact conduit termination locations required for boxes, enclosures, and equipment.
   4. Coordinate work to provide roof penetrations that preserve integrity of roofing system and do not void roof warranty.
   5. Notify Engineer of conflicts with or deviations from Contract Documents. Obtain direction before proceeding with work.

B. Sequencing:
   1. Do not begin installation of conductors and cables until installation of conduit between termination points is complete.

1.05 SUBMITTALS

A. See Section 01 3000 - Administrative Requirements for submittals procedures.

B. Project Record Documents: Record actual routing for conduits installed underground, conduits embedded within concrete slabs, and conduits 2-inch (53 mm) trade size and larger.

PART 2 PRODUCTS

2.01 CONDUIT APPLICATIONS

A. Do not use conduit and associated fittings for applications other than as permitted by NFPA 70, manufacturer's instructions, and product listing.

B. Unless otherwise indicated and where not otherwise restricted, use conduit types indicated for specified applications. Where more than one listed application applies, comply with most restrictive requirements. Where conduit type for particular application is not specified, use galvanized steel rigid metal conduit.

C. Exposed, Interior, Not Subject to Physical Damage: Use galvanized steel electrical metallic tubing (EMT) or aluminum electrical metallic tubing (EMT).

D. Exposed, Interior, Subject to Physical Damage: Use galvanized steel rigid metal conduit (RMC).
   1. Locations subject to physical damage include, but are not limited to:
      a. Where exposed below 8 feet, except within electrical and communication rooms or closets.
      b. Where exposed below 20 feet in warehouse areas.

E. Exposed, Exterior, Not Subject to Severe Physical Damage: Use galvanized steel rigid metal conduit (RMC), stainless steel rigid metal conduit (RMC), galvanized steel intermediate metal conduit (IMC), stainless steel intermediate metal conduit (IMC), galvanized steel electrical metallic tubing (EMT), or stainless steel electrical metallic tubing (EMT).

F. Flexible Connections to Vibrating Equipment:
   1. Dry Locations: Use flexible metal conduit (FMC).
   2. Damp, Wet, or Corrosive Locations: Use liquidtight flexible metal conduit (LFMC).

3. Maximum Length: 6 feet unless otherwise indicated.

2.02 CONDUIT - GENERAL REQUIREMENTS

A. Comply with NFPA 70.

B. Provide conduit, fittings, supports, and accessories required for complete raceway system.

C. Provide products listed, classified, and labeled as suitable for purpose intended.

D. Minimum Conduit Size, Unless Otherwise Indicated:
   1. Branch Circuits: 3/4-inch trade size.
   2. Branch Circuit Homeruns: 3/4-inch trade size.
   3. Control Circuits: 1/2-inch trade size.

E. Where conduit size is not indicated, size to comply with NFPA 70 but not less than applicable minimum size requirements specified.

2.03 GALVANIZED STEEL RIGID METAL CONDUIT (RMC)

A. Manufacturers:
   1. Allied Tube & Conduit, a division of Atkore International: www.alliedeg.com/#sle.
   2. Nucor Tubular Products: www.nucortubular.com/#sle.
   3. Western Tube, a division of Zekelman Industries: www.westerntube.com/#sle.
   4. Wheatland Tube, a division of Zekelman Industries: www.wheatland.com/#sle.
   5. Substitutions: See Section 01 6000 - Product Requirements.

B. Description: NFPA 70, Type RMC galvanized steel rigid metal conduit complying with ANSI C80.1 and listed and labeled as complying with UL 6.

C. Fittings:
   1. Manufacturers:
      a. ABB; T&B: www.electrification.us.abb.com/#sle.
      b. Allied Tube & Conduit, a division of Atkore International: www.alliedeg.us/#sle.
      c. Bridgeport Fittings Inc: www.bptfittings.com/#sle.
      d. Emerson Electric Co; O-Z/Gedney: www.emerson.com/#sle.
      e. Substitutions: See Section 01 6000 - Product Requirements.
   2. Nonhazardous Locations: Use fittings complying with NEMA FB 1 and listed and labeled as complying with UL 514B or UL 6.
   3. Material: Use steel or malleable iron.
      a. Where not subject to severe corrosive influence, stainless steel or aluminum fittings may be used.
      b. Do not use die cast zinc fittings.
   4. Connectors and Couplings: Use threaded type fittings only. Threadless fittings, including set screw and compression/gland types, are not permitted.

2.04 STAINLESS STEEL RIGID METAL CONDUIT (RMC)

A. Manufacturers:
   1. Calbrite, a division of Atkore International: www.calbrite.com/#sle.
   2. Gibson Stainless & Specialty Inc: www.gibsonstainless.com/#sle.
   3. Patriot Industries, a division of Patriot Aluminum Products LLC: www.patriotsas.com/#sle.
   4. Rymco USA: www.rymcousa.com/#sle.
   5. Substitutions: See Section 01 6000 - Product Requirements.

B. Description: NFPA 70, Type RMC stainless steel rigid metal conduit complying with ANSI C80.1 and listed and labeled as complying with UL 6A.

C. Fittings:
   1. Manufacturers:
      a. Calbrite, a division of Atkore International: www.calbrite.com/#sle.
      b. Eaton: www.eaton.com/#sle.

c. Gibson Stainless & Specialty Inc: www.gibsonstainless.com/#sle.
d. Patriot Industries, a division of Patriot Aluminum Products LLC: www.patriotsas.com/#sle.
e. Substitutions: See Section 01 6000 - Product Requirements.

2. Nonhazardous Locations: Use fittings complying with NEMA FB 1 and listed and labeled as complying with UL 514B or UL 6A.
3. Material: Use stainless steel with corrosion resistance equivalent to conduit.
4. Connectors and Couplings: Use threaded type fittings only. Threadless fittings, including set screw and compression/gland types, are not permitted.

2.05 FLEXIBLE METAL CONDUIT (FMC)

A. Manufacturers:
1. AFC Cable Systems, a division of Atkore International: www.afcweb.com/#sle.
2. Electri-Flex Company: www.electriflex.com/#sle.
3. International Metal Hose: www.metalhose.com/#sle.
4. Substitutions: See Section 01 6000 - Product Requirements.

B. Description: NFPA 70, Type FMC standard-wall steel flexible metal conduit listed and labeled as complying with UL 1, and listed for use in classified firestop systems.

C. Fittings:
1. Manufacturers:
a. ABB; T&B: www.electrification.us.abb.com/#sle.
b. Bridgeport Fittings, LLC: www.bptfittings.com/#sle.
c. Emerson Electric Co; O-Z/Gedney: www.emerson.com/#sle.
d. Substitutions: See Section 01 6000 - Product Requirements.
2. Description: Fittings complying with NEMA FB 1 and listed and labeled as complying with UL 514B.
3. Material: Use steel or malleable iron.

2.06 LIQUIDTIGHT FLEXIBLE METAL CONDUIT (LFMC)

A. Manufacturers:
1. AFC Cable Systems, a division of Atkore International: www.afcweb.com/#sle.
2. Electri-Flex Company: www.electriflex.com/#sle.
3. International Metal Hose: www.metalhose.com/#sle.
4. Substitutions: See Section 01 6000 - Product Requirements.

B. Description: NFPA 70, Type LFMC polyvinyl chloride (PVC) jacketed steel flexible metal conduit listed and labeled as complying with UL 360.

C. Fittings:
1. Manufacturers:
a. ABB; T&B: www.electrification.us.abb.com/#sle.
b. Bridgeport Fittings, LLC: www.bptfittings.com/#sle.
c. Emerson Electric Co; O-Z/Gedney: www.emerson.com/#sle.
d. _____.
e. Substitutions: See Section 01 6000 - Product Requirements.
2. Description: Fittings complying with NEMA FB 1 and listed and labeled as complying with UL 514B.
3. Material: Use steel or malleable iron.
a. Do not use die cast zinc fittings.

2.07 GALVANIZED STEEL ELECTRICAL METALLIC TUBING (EMT)

A. Manufacturers:
1. Allied Tube & Conduit, a division of Atkore International: www.alliedeg.com/#sle.
2. Nucor Tubular Products: www.nucortubular/#sle.

3. Rymco USA: www.rymcousa.com/#sle.
4. Western Tube, a division of Zekelman Industries: www.westerntube.com/#sle.
5. Wheatland Tube, a division of Zekelman Industries: www.wheatland.com/#sle.
6. Substitutions: See Section 01 6000 - Product Requirements.

B. Description: NFPA 70, Type EMT galvanized steel electrical metallic tubing complying with ANSI C80.3 and listed and labeled as complying with UL 797.

C. Fittings:
1. Manufacturers:
a. ABB; T&B: www.electrification.us.abb.com/#sle.
b. Allied Tube & Conduit, a division of Atkore International: www.alliedeg.us/#sle.
c. Bridgeport Fittings, LLC: www.bptfittings.com/#sle.
d. Emerson Electric Co; O-Z/Gedney: www.emerson.com/#sle.
e. Substitutions: See Section 01 6000 - Product Requirements.
2. Description: Fittings complying with NEMA FB 1 and listed and labeled as complying with UL 514B.
3. Material: Use steel or malleable iron.
a. Do not use die cast zinc fittings.
4. Connectors and Couplings: Use compression/gland type.
a. Do not use indenter type connectors and couplings.

2.08 ACCESSORIES

A. Corrosion Protection Tape: PVC-based, minimum thickness of 20 mil, 0.020 inch.

B. Conduit Joint Compound: Corrosion-resistant, electrically conductive compound listed as complying with UL 2419; suitable for use with conduit to be installed.

C. Pull Strings: Use nylon or polyester tape with average breaking strength of not less than 1,250 lbf.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that field measurements are as indicated.

B. Verify that mounting surfaces are ready to receive conduits.

C. Verify that conditions are satisfactory for installation prior to starting work.

3.02 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install conduit in accordance with NECA 1.

C. Install galvanized steel rigid metal conduit (RMC) in accordance with NECA 101.

D. Conduit Routing:
1. Unless dimensioned, conduit routing indicated is diagrammatic.
2. Conceal conduits unless specifically indicated to be exposed.
3. Conduits in the following areas may be exposed, unless otherwise indicated:
a. Electrical rooms.
b. Mechanical equipment rooms.
4. Unless otherwise approved, do not route exposed conduits:
a. Across floors.
b. Across roofs.
c. Across top of parapet walls.
d. Across building exterior surfaces.
5. Conduits installed underground or embedded in concrete may be routed in shortest possible manner unless otherwise indicated. Route other conduits parallel or

perpendicular to building structure and surfaces, following surface contours where practical.

6. Arrange conduit to maintain adequate headroom, clearances, and access.
7. Arrange conduit to provide no more than equivalent of four 90-degree bends between pull points.
8. Arrange conduit to provide no more than 150 feet between pull points.
9. Route conduits above water and drain piping where possible.
10. Arrange conduit to prevent moisture traps. Provide drain fittings at low points and at sealing fittings where moisture may collect.
11. Maintain minimum clearance of 6 inches between conduits and piping for other systems.
12. Maintain minimum clearance of 12 inches between conduits and hot surfaces. This includes, but is not limited to:
    a. Heaters.
    b. Hot water piping.
    c. Flues.
13. Group parallel conduits in same area on common rack.

E. Conduit Support:
1. Secure and support conduits in accordance with NFPA 70 using suitable supports and methods approved by authorities having jurisdiction; see Section 26 0529.
2. Provide independent support from building structure. Do not provide support from piping, ductwork, or other systems.
3. Use metal channel/strut with accessory conduit clamps to support multiple parallel surface-mounted conduits.
4. Use conduit clamp to support single conduit from beam clamp or threaded rod.
5. Use trapeze hangers assembled from threaded rods and metal channel/strut with accessory conduit clamps to support multiple parallel suspended conduits.
6. Use of wire for support of conduits is not permitted.

F. Connections and Terminations:
1. Use approved zinc-rich paint or conduit joint compound on field-cut threads of galvanized steel conduits prior to making connections.
2. Where two threaded conduits must be joined and neither can be rotated, use three-piece couplings or split couplings. Do not use running threads.
3. Use suitable adapters where required to transition from one type of conduit to another.
4. Provide drip loops for liquidtight flexible conduit connections to prevent drainage of liquid into connectors.
5. Terminate threaded conduits in boxes and enclosures using threaded hubs or double lock nuts for dry locations and raintight hubs for wet locations.
6. Provide insulating bushings, insulated throats, or listed metal fittings with smooth, rounded edges at conduit terminations to protect conductors.
7. Secure joints and connections to provide mechanical strength and electrical continuity.

G. Penetrations:
1. Do not penetrate or otherwise notch or cut structural members, including footings and grade beams, without approval of Structural Engineer.
2. Make penetrations perpendicular to surfaces unless otherwise indicated.
3. Provide sleeves for penetrations as indicated or as required to facilitate installation. Set sleeves flush with exposed surfaces unless otherwise indicated or required.
4. Conceal bends for conduit risers emerging above ground.
5. Where conduits penetrate waterproof membrane, seal as required to maintain integrity of membrane.
6. Make penetrations for roof-mounted equipment within associated equipment openings and curbs where possible to minimize roofing system penetrations. Where penetrations are

necessary, seal as indicated or as required to preserve integrity of roofing system and maintain roof warranty.

7. Install firestopping to preserve fire resistance rating of partitions and other elements; see Section 07 8400.

H. Identify conduits; see Section 26 0553.

3.03 FIELD QUALITY CONTROL

A. See Section 01 4000 - Quality Requirements for additional requirements.

B. Repair cuts and abrasions in galvanized finishes using zinc-rich paint recommended by manufacturer. Replace components that exhibit signs of corrosion.

C. Correct deficiencies and replace damaged or defective conduits.

3.04 PROTECTION

A. Immediately after installation of conduit, use suitable manufactured plugs to provide protection from entry of moisture and foreign material and do not remove until ready for installation of conductors.

END OF SECTION

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SECTION 26 0553
IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Electrical identification requirements.

B. Identification nameplates and labels.

C. Voltage markers.

D. Underground warning tape.

E. Floor marking tape.

F. Warning signs and labels.

1.02 RELATED REQUIREMENTS

A. Section 09 9113 - Exterior Painting.

B. Section 09 9123 - Interior Painting.

C. Section 26 0519 - Electrical Conductors and Cables: Color coding for power conductors and cables 600 V and less; vinyl color coding electrical tape.

D. Section 26 2300 - Low-Voltage Switchgear: Factory-installed mimic bus.

E. Section 27 1000 - Structured Cabling: Identification for communications cabling and devices.

1.03 REFERENCE STANDARDS

A. NFPA 70 - National Electrical Code; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.

B. NFPA 70E - Standard for Electrical Safety in the Workplace; 2027.

C. UL 969 - Marking and Labeling Systems; Current Edition, Including All Revisions.

1.04 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Verify final designations for equipment, systems, and components to be identified prior to fabrication of identification products.

B. Sequencing:
   1. Do not conceal items to be identified, in locations such as above suspended ceilings, until identification products have been installed.
   2. Do not install identification products until final surface finishes and painting are complete.

1.05 SUBMITTALS

A. See Section 01 3000 - Administrative Requirements for submittals procedures.

B. Product Data: Provide manufacturer's standard catalog pages and data sheets for each product.

C. Shop Drawings: Provide schedule of items to be identified indicating proposed designations, materials, legends, and formats.

1.06 FIELD CONDITIONS

A. Do not install adhesive products when ambient temperature is lower than recommended by manufacturer.

PART 2 PRODUCTS

2.01 IDENTIFICATION REQUIREMENTS

A. Identification for Equipment:

1. Use identification nameplate to identify each piece of electrical distribution and control equipment and associated sections, compartments, and components.
    a. Switchgear / Switchboards:
        1) See electrical plans for tag requirements and details.
        2) Identify power source and circuit number. Include location when not within sight of equipment.
        3) Use identification nameplate to identify load(s) served for each branch device.
        4) See Section 26 2300 for factory-installed mimic bus.
2. Available Fault Current Documentation: Use identification nameplate to identify the available fault current and date calculations were performed at locations indicated.
    a. Service equipment.

B. Identification for Conductors and Cables:
1. Color Coding for Power Conductors 600 V and Less: Comply with Section 26 0519.

C. Identification for Raceways:
1. Use voltage markers to identify highest voltage present for accessible conduits at maximum intervals of 20 feet.
2. Use voltage markers, color-coded bands, or factory-painted conduits to identify specified systems for accessible conduits.
    a. Maximum Intervals: 20 feet.
    b. Color-Coded Bands: Use vinyl color coding electrical tape to mark bands 3 inches wide.
        1) Vinyl Color Coding Electrical Tape: Comply with Section 26 0519.

D. Identification for Boxes:
1. Use voltage markers or color coded boxes to identify systems other than normal power system.
    a. For exposed boxes in public areas, do not color code.
2. Use identification labels or handwritten text using indelible marker to identify circuits enclosed.

E. Identification for Devices:
1. Identification for Communications Devices: Comply with Section 27 1000.
2. Use identification label to identify fire alarm system devices.

2.02 IDENTIFICATION NAMEPLATES AND LABELS

A. Identification Nameplates:
1. Manufacturers:
    a. Brimar Industries, Inc: www.brimar.com/#sle.
    b. Kolbi Pipe Marker Co: www.kolbipipemarkers.com/#sle.
    c. Seton Identification Products: www.seton.com/#sle.
    d. Substitutions: See Section 01 6000 - Product Requirements.
2. Materials:
    a. Indoor Clean, Dry Locations: Use plastic nameplates.
    b. Outdoor Locations: Use stainless steel or aluminum nameplates suitable for exterior use.
3. Plastic Nameplates: Two-layer or three-layer laminated acrylic or electrically non-conductive phenolic with beveled edges; minimum thickness of 1/16 inch; engraved text.
4. Mounting Holes for Mechanical Fasteners: Two, centered on sides for sizes up to 1 inch high; Four, located at corners for larger sizes.

B. Identification Labels:
1. Manufacturers:
    a. Brady Corporation: www.bradyid.com/#sle.
    b. Brother International Corporation: www.brother-usa.com/#sle.

c. Panduit Corp: www.panduit.com/#sle.
d. Substitutions: See Section 01 6000 - Product Requirements.

2. Materials: Use self-adhesive laminated plastic labels; UV, chemical, water, heat, and abrasion resistant.
3. Text: Use factory pre-printed or machine-printed text. Do not use handwritten text unless otherwise indicated.

C. Format for Equipment Identification:
1. See electrical plans for tag requirements and details.
2. Color:
a. Normal Power System: White text on black background.

2.03 VOLTAGE MARKERS

A. Manufacturers:
1. Brady Corporatio: www.bradyid.com/#sle.
2. Brimar Industries, Inc: www.brimar.com/#sle.
3. Seton Identification Products: www.seton.com/#sle.
4. Substitutions: See Section 01 6000 - Product Requirements.

B. Markers for Conduits: Use factory pre-printed self-adhesive vinyl, self-adhesive vinyl cloth, or vinyl snap-around type markers.

C. Markers for Boxes and Equipment Enclosures: Use factory pre-printed self-adhesive vinyl or self-adhesive vinyl cloth type markers.

D. Minimum Size:
1. Markers for Conduits: As recommended by manufacturer for conduit size to be identified.
2. Markers for Pull Boxes: 1 1/8 by 4 1/2 inches.
3. Markers for Junction Boxes: 1/2 by 2 1/4 inches.

E. Legend:
1. Markers for Voltage Identification: Highest voltage present.
2. Markers for System Identification:

F. Color: Black text on orange background unless otherwise indicated.

2.04 WARNING SIGNS AND LABELS

A. Manufacturers:
1. Brimar Industries, Inc: www.brimar.com/#sle.
2. Clarion Safety Systems, LLC: www.clarionsafety.com/#sle.
3. Insite Solutions, LLC: www.stop-painting.com/#sle.
4. Seton Identification Products: www.seton.com/#sle.
5. Substitutions: See Section 01 6000 - Product Requirements.

B. Comply with ANSI Z535.2 or ANSI Z535.4 as applicable.

C. Warning Signs:
1. Materials:
2. Minimum Size: 7 by 10 inches unless otherwise indicated.

D. Warning Labels:
1. Materials: Use factory pre-printed or machine-printed self-adhesive polyester or self-adhesive vinyl labels; UV, chemical, water, heat, and abrasion resistant; produced using materials recognized to UL 969.
2. Machine-Printed Labels: Use thermal transfer process printing machines and accessories recommended by label manufacturer.
3. Minimum Size: 2 by 4 inches unless otherwise indicated.

PART 3 EXECUTION

3.01 PREPARATION

A. Clean surfaces to receive adhesive products according to manufacturer's instructions.

3.02 INSTALLATION

A. Install products in accordance with manufacturer's instructions.

B. Install identification products to be plainly visible for examination, adjustment, servicing, and maintenance. Unless otherwise indicated, locate products as follows:
   1. Surface-Mounted Equipment: Enclosure front.
   2. Flush-Mounted Equipment: Inside of equipment door.
   3. Free-Standing Equipment: Enclosure front; also enclosure rear for equipment with rear access.
   4. Elevated Equipment: Legible from the floor or working platform.
   5. Branch Devices: Adjacent to device.
   6. Interior Components: Legible from the point of access.
   7. Conduits: Legible from the floor.
   8. Boxes: Outside face of cover.
   9. Conductors and Cables: Legible from the point of access.
   10. Devices: Outside face of cover.

C. Install identification products centered, level, and parallel with lines of item being identified.

D. Secure nameplates to exterior surfaces of enclosures using stainless steel screws and to interior surfaces using self-adhesive backing or epoxy cement.

E. Install self-adhesive labels and markers to achieve maximum adhesion, with no bubbles or wrinkles and edges properly sealed.

F. Mark all handwritten text, where permitted, to be neat and legible.

3.03 FIELD QUALITY CONTROL

A. See Section 01 4000 - Quality Requirements, for additional requirements.

B. Replace self-adhesive labels and markers that exhibit bubbles, wrinkles, curling or other signs of improper adhesion.

END OF SECTION

SECTION 26 0583
WIRING CONNECTIONS

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Electrical connections to equipment.

1.02 RELATED REQUIREMENTS

A. Section 26 0519 - Electrical Conductors and Cables.

B. Section 26 0533 - Conduit for Electrical Systems.

C. Section 26 0534 - Boxes for Electrical Systems.

D. Section 26 2726 - Wiring Devices.

1.03 REFERENCE STANDARDS

A. NEMA WD 1 - General Color Requirements for Wiring Devices; 1999 (Reaffirmed 2020).

B. NEMA WD 6 - Wiring Devices - Dimensional Specifications; 2021.

C. NFPA 70 - National Electrical Code; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.

1.04 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Obtain and review shop drawings, product data, manufacturer's wiring diagrams, and manufacturer's instructions for equipment furnished under other sections.
   2. Determine connection locations and requirements.

B. Sequencing:
   1. Install rough-in of electrical connections before installation of equipment is required.
   2. Make electrical connections before required start-up of equipment.

1.05 SUBMITTALS

A. See Section 01 3000 - Administrative Requirements, for submittal procedures.

B. Product Data: Provide wiring device manufacturer’s catalog information showing dimensions, configurations, and construction.

1.06 QUALITY ASSURANCE

A. Comply with requirements of NFPA 70.

PART 2 PRODUCTS

2.01 MATERIALS

A. Wiring Devices: As specified in Section 26 2726.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that equipment is ready for electrical connection, wiring, and energization.

3.02 ELECTRICAL CONNECTIONS

A. Division of responsibility is suggested below. However, all works remains a responsibility of the Prime Contract holder.
   1. Legend is understood as folows:
      a. GC - General Contractor
      b. MC - Mechanical and/or Plumbing Contractor
      c. EC - Electrical Contractor
      d. F - Furnished: Procure Equipment and/or Materials

e. I - Installed: Set in Place, Equipment and/or Materials
f. W - Wired: Land condctors on maufacture's dicated terminals
g. P - Provide: Shall mean Furnished, Installed, and Wired
h. CPW - Control Power and Wiring
i. INT - Integrration: Programming, Testing, and/or Certificaiton

2. Equipment motors
   a. F=MC, I=MC, W=EC
3. Magnetic motor starters
   a. Auto Control w/ or w/o HOA
      1) F=MC, I=EC, W=EC, CPW=MC
   b. Auto Control w/ or w/o HOA but supplied as a part of factory-wired equipment
      1) F=MC, I=MC, W=MC, CPW=MC
   c. Manually controlled
      1) F=MC, I=MC, W=EC, CPW=MC
   d. Manually controlled but supplied as a part of factory-wired equipment
      1) F=MC, I=MC, W=MC, CPW=MC
4. Motors/ Valves/ Damper motor
   a. F=MC, I=MC, W=EC, CPW=MC
5. Boiler and water heater controls/boiler burner control panels
   a. F=MC, I=MC, W=EC, CPW=MC
6. Power wiring for DDC device (valves, damper motors, air valves, VAV boxes, actuators, etc.
   a. F=MC, I=MC, W=EC

B. Make electrical connections in accordance with equipment manufacturer's instructions.

C. Make conduit connections to equipment using flexible conduit. Use liquidtight flexible conduit with watertight connectors in damp or wet locations.

D. Connect heat producing equipment using wire and cable with insulation suitable for temperatures encountered.

E. Install disconnect switches, controllers, control stations, and control devices to complete equipment wiring requirements.

F. Install terminal block jumpers to complete equipment wiring requirements.

G. Install interconnecting conduit and wiring between devices and equipment to complete equipment wiring requirements.

END OF SECTION

SECTION 26 2726
WIRING DEVICES

PART 1 GENERAL

1.01 SECTION INCLUDES

A. Receptacles.

B. Wall plates and covers.

1.02 RELATED REQUIREMENTS

A. Section 26 0534 - Boxes for Electrical Systems.

B. Section 26 0553 - Identification for Electrical Systems: Identification products and requirements.

C. Section 26 0583 - Wiring Connections: Cords and plugs for equipment.

1.03 REFERENCE STANDARDS

A. FS W-C-596 - Connector, Electrical, Power, General Specification for; 2014h (Validated 2022).

B. FS W-S-896 - Switches, Toggle (Toggle and Lock), Flush Mounted (General Specification); 2017g (Validated 2023).

C. NECA 1 - Standard for Good Workmanship in Electrical Construction; 2015.

D. NECA 130 - Standard for Installing and Maintaining Wiring Devices; 2016.

E. NEMA WD 1 - General Color Requirements for Wiring Devices; 1999 (Reaffirmed 2020).

F. NEMA WD 6 - Wiring Devices - Dimensional Specifications; 2021.

G. NFPA 70 - National Electrical Code; Most Recent Edition Adopted by Authority Having Jurisdiction, Including All Applicable Amendments and Supplements.

H. UL 20 - General-Use Snap Switches; Current Edition, Including All Revisions.

I. UL 943 - Ground-Fault Circuit-Interrupters; Current Edition, Including All Revisions.

1.04 ADMINISTRATIVE REQUIREMENTS

A. Coordination:
   1. Coordinate wiring device ratings and configurations with the electrical requirements of actual equipment to be installed.
   2. Coordinate the installation and preparation of uneven surfaces, such as split face block, to provide suitable surface for installation of wiring devices.
   3. Notify Owner of any conflicts or deviations from Contract Documents to obtain direction prior to proceeding with work.

B. Sequencing:
   1. Do not install wiring devices until final surface finishes and painting are complete.

1.05 SUBMITTALS

A. See Section 01 3000 - Administrative Requirements, for submittal procedures.

B. Product Data: Provide manufacturer's catalog information showing dimensions, colors, and configurations.

C. Project Record Documents: Record actual installed locations of wiring devices.

D. Maintenance Materials: Furnish the following for Owner's use in maintenance of project.
   1. See Section 01 6000 - Product Requirements, for additional provisions.

1.06 QUALITY ASSURANCE

A. Comply with requirements of NFPA 70.

B. Products: Listed, classified, and labeled as suitable for the purpose intended.

PART 2 PRODUCTS

2.01 WIRING DEVICE APPLICATIONS

A. Provide wiring devices suitable for intended use and with ratings adequate for load served.

B. Provide weather resistant GFCI receptacles with specified weatherproof covers for receptacles installed outdoors or in damp or wet locations.

C. Provide tamper resistant receptacles for receptacles installed in dwelling units.

D. Provide GFCI protection for receptacles installed within 6 feet of sinks.

E. Provide GFCI protection for receptacles installed in kitchens.

F. Provide GFCI protection at breakers for receptacles serving electric drinking fountains.

2.02 WIRING DEVICE FINISHES

A. Provide wiring device finishes as described below unless otherwise indicated.

B. Wiring Devices, Unless Otherwise Indicated: White with galvanized steel wall plate.

2.03 RECEPTACLES

A. Manufacturers:
   1. Hubbell Incorporated: www.hubbell.com/#sle.
   2. Leviton Manufacturing Company, Inc: www.leviton.com/#sle.
   3. Lutron Electronics Company, Inc; Designer Style: www.lutron.com/#sle.
   4. Pass & Seymour, a brand of Legrand North America, Inc: www.legrand.us/#sle.
   5. Substitutions: See Section 01 6000 - Product Requirements.

B. Receptacles - General Requirements: Self-grounding, complying with NEMA WD 1 and NEMA WD 6, and listed as complying with UL 498, and where applicable, FS W-C-596; types as indicated on the drawings.
   1. Wiring Provisions: Terminal screws for side wiring or screw actuated binding clamp for back wiring with separate ground terminal screw.
   2. NEMA configurations specified are according to NEMA WD 6.

C. Convenience Receptacles:
   1. Standard Convenience Receptacles: Commercial specification grade, 20A, 125V, NEMA 5-20R; single or duplex as indicated on the drawings.
   2. Weather Resistant Convenience Receptacles: Commercial specification grade, 20A, 125V, NEMA 5-20R, listed and labeled as weather resistant type complying with UL 498 Supplement SE suitable for installation in damp or wet locations; single or duplex as indicated on the drawings.

D. GFCI Receptacles:
   1. GFCI Receptacles - General Requirements: Self-testing, with feed-through protection and light to indicate ground fault tripped condition and loss of protection; listed as complying with UL 943, class A.
      a. Provide test and reset buttons of same color as device.
   2. Standard GFCI Receptacles: Commercial specification grade, duplex, 20A, 125V, NEMA 5-20R, rectangular decorator style.
   3. Weather Resistant GFCI Receptacles: Commercial specification grade, duplex, 20A, 125V, NEMA 5-20R, rectangular decorator style, listed and labeled as weather resistant type complying with UL 498 Supplement SE suitable for installation in damp or wet locations.

2.04 WALL PLATES AND COVERS

A. Manufacturers:
   1. Hubbell Incorporated: www.hubbell-wiring.com/#sle.
   2. Intermatic, Inc: www.intermatic.com/#sle.

3. Leviton Manufacturing Company, Inc: www.leviton.com/#sle.
4. Lutron Electronics Company, Inc: www.lutron.com/#sle.
5. Pass & Seymour, a brand of Legrand North America, Inc: www.legrand.us/#sle.
6. Substitutions: See Section 01 6000 - Product Requirements.

B. Wall Plates: Comply with UL 514D.
1. Configuration: One piece cover as required for quantity and types of corresponding wiring devices.
2. Size: Standard.
3. Screws: Metal with slotted heads finished to match wall plate finish.

C. Galvanized Steel Wall Plates: Rounded corners and edges, with corrosion resistant screws.

D. Weatherproof Covers for Wet Locations: Gasketed, cast aluminum, with hinged lockable cover and corrosion-resistant screws; listed as suitable for use in wet locations while in use with attachment plugs connected and identified as extra-duty type.

PART 3 EXECUTION

3.01 EXAMINATION

A. Verify that branch circuit wiring installation is completed, tested, and ready for connection to wiring devices.

B. Verify that conditions are satisfactory for installation prior to starting work.

3.02 INSTALLATION

A. Perform work in accordance with NECA 1 (general workmanship) and, where applicable, NECA 130, including mounting heights specified in those standards unless otherwise indicated.

B. Install wiring devices in accordance with manufacturer's instructions.

C. Where required, connect wiring devices using pigtails not less than 6 inches long. Do not connect more than one conductor to wiring device terminals.

D. Connect wiring devices by wrapping conductor clockwise 3/4 turn around screw terminal and tightening to proper torque specified by the manufacturer. Where present, do not use push-in pressure terminals that do not rely on screw-actuated binding.

E. Unless otherwise indicated, connect wiring device grounding terminal to branch circuit equipment grounding conductor and to outlet box with bonding jumper.

F. Provide GFCI receptacles with integral GFCI protection at each location indicated. Do not use feed-through wiring to protect downstream devices.

G. Install wiring devices plumb and level with mounting yoke held rigidly in place.

H. Install vertically mounted receptacles with grounding pole on top and horizontally mounted receptacles with grounding pole on left.

I. Install wall plates to fit completely flush to wall with no gaps and rough opening completely covered without strain on wall plate. Repair or reinstall improperly installed outlet boxes or improperly sized rough openings. Do not use oversized wall plates in lieu of meeting this requirement.

J. Install blank wall plates on junction boxes and on outlet boxes with no wiring devices installed or designated for future use.

K. Identify wiring devices in accordance with Section 26 0553.

3.03 FIELD QUALITY CONTROL

A. See Section 01 4000 - Quality Requirements, for additional requirements.

B. Inspect each wiring device for damage and defects.

C. Operate each wall switch, wall dimmer, and fan speed controller with circuit energized to verify proper operation.

D. Test each receptacle to verify operation and proper polarity.

E. Test each GFCI receptacle for proper tripping operation according to manufacturer's instructions.

F. Correct wiring deficiencies and replace damaged or defective wiring devices.

3.04 ADJUSTING

A. Adjust devices and wall plates to be flush and level.

3.05 CLEANING

A. Clean exposed surfaces to remove dirt, paint, or other foreign material and restore to match original factory finish.

END OF SECTION